



All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without prior permission of NB Medical Education.

Declaration

The information contained in this booklet is of a general nature. Although we seek to provide accurate and up to date information, there can be no guarantee that the information is accurate on the day it is received, or that it will continue to be accurate in the future. We do not accept responsibility for any errors. This is particularly so with regard to drug indications, suggested doses, side-effects and interactions, which should always be guided by an appropriate national formulary.

No pharmaceutical influence and no conflict of interest guarantee

We can assure all delegates that any sponsors or exhibitors have NO influence on any course content, either in the written book or the presentations. We are completely independent and all the course content and material is 100% independent from, and not influenced by, any external organisations.

The course is externally accredited by the Royal College of General Practitioners.



CONTENTS

Introduction

Presenters and Acknowledgements	1
Welcome to the Course	3
The NB Dashboard & App	4
NB Plus	5

Cardiovascular Disease

Hypertension	6
Orthostatic Hypotension	18
Lipids and Statins	22
Atrial Fibrillation	33
Chronic Heart Failure	41
Ischaemia with Non-Obstructed Arteries (INOCA)	55
Stable Angina	58
Acute Coronary Syndromes & Secondary Prevention	61
Stroke and TIA	63
Which Antiplatelet When?	69
Heart Valve Disease	70
Venous Thromboembolism	71
Peripheral Arterial Disease	76
Familial Hypercholesterolaemia	81

Prescribing Hot Topics

Safe Use of SGLT2 Inhibitors	84
Direct Oral Anticoagulants (DOACs)	88
Risks of Long Term Nitrofurantoin and Amiodarone	92
Montelukast, Neuropsychiatric Reactions	94
Serotonin Syndrome	95
Opiates and Benzodiazepines	96
Medicines Associated with Dependence or Withdrawal	97
Prescribing in Renal Impairment	98
Medical Cannabis	99
Disease Modifying Drugs (DMARDs)	103
Biologic Agents	106
Aspirin, Bleeding Risk & PPI	108

Diabetes & Endocrine

Type 2 Diabetes - Pharmacological Management	111
Diabetes - Guidelines	116
Steroid-Induced Diabetes and Hyperglycaemia	124
Diabetes in the Older Person	125
Pre-Diabetes	129
Remission of Type 2 Diabetes	132
Hypothyroidism	136
Subclinical Hypothyroidism	138
Clinical & Subclinical Hyperthyroidism	141
Thyroid, Pregnancy & Fertility	144

CONTENTS

Cancer

Abdominal Pain and Cancer	148
FIT Tests	149
Pancreatic Cancer	152
Ovarian Cancer	154
Myeloma	157
PSA Testing and Prostate Cancer	161
Early Diagnosis of Cancer	166
Lung Cancer	174
Bladder Cancer	178
Platelets and Cancer	180
Microcytosis & Cancer Risk	183
Lynch Syndrome	185
Long-Term Cancer Care	188
NICE Suspected Cancer Guidelines	192
Scottish Cancer Referral Guidelines	204
Colorectal Cancer in Younger Adults	206
Familial Breast Cancer	208
Breast Cancer Survivorship	209

Kidney Disease

Chronic Kidney Disease	212
Acute Kidney Injury	218

Urology

Gynaecomastia	222
Lower Urinary Tract Symptoms	223
Renal Colic	228
Low Testosterone in Men (Hypogonadism)	230
Prostatitis	233
Haemospermia	238

Neurology

Migraine	240
Headaches	246
Suspected Neurological Conditions in Adults Over 16	253
Suspected Neurological Conditions in Children Under 16: Recognition and Referral	258
Functional Neurological Disorders	262
Restless Legs Syndrome	264
Parkinson's Disease	266
Facial Palsy & Bells Palsy	272
Epilepsy	274
Epilepsy, Contraception and Pregnancy	279
Multiple Sclerosis	283
Essential Tremor	287

CONTENTS

Pain

Neuropathic Pain	289
Chronic Pain	291
Gabapentinoids for Pain	297
Fibromyalgia	300

Mental Health

Depression in Adults	304
Antidepressants - Side Effects and Prescribing Issues	309
Self Harm	320
Eating Disorders	324
Suicide Prevention	332
Insomnia	336
Panic Disorder	338
Bipolar Disorder	340
Personality Disorder	343
Post Traumatic Stress Disorder	354
Burnout	357
Obsessive Compulsive Disorder	361
Generalised Anxiety Disorder	362
Mental Illness in Children & Young People	364
ADHD in Adults	368
Health Anxiety	369
Psychosis and Schizophrenia	371

Musculoskeletal

Gout	373
Osteoarthritis	375
Shoulder Pain	380
Osteoporosis	384
Giant Cell Arteritis	391
Polymyalgia Rheumatica	395
Joint and Soft Tissue Injections	397
Rheumatoid Arthritis	400
Back Pain & Sciatica	405
Spondyloarthritis	411
Metastatic Spinal Cord Compression	414
Greater Trochanteric Pain Syndrome	416
Elbow Pain	417
Knee Pain	419
Iliotibial Band Syndrome	420
Carpal Tunnel Syndrome	421

Women's Health

Endometriosis	422
Menopause	425
Urinary Incontinence	432
Contraception	437
Heavy Menstrual Bleeding	443

CONTENTS

Women's Health

Miscarriage and Ectopic Pregnancy	445
Common Problems in Pregnancy	447
Vaginal Discharge	449
Breast Pain	453
Perinatal Mental Health	455
Hypertension in Pregnancy	459
Mesh Implants	461
Polycystic Ovarian Syndrome	463
Prenatal Testing	465

Infectious Diseases

Group A Streptococcus	468
Sepsis	471
Minor Illness	476
Urinary Tract Infections	482
Monkeypox (Mpox)	489
Influenza	491
Clostridioides Difficile	493
Antibiotic Prescribing	494
Cellulitis	497
Vaccinations	500
Pertussis	502
Impetigo	505
Mumps	507
Measles	509
Lower Respiratory Tract Infection	511
Shingles	513
Chickenpox	517
Infective Diarrhoea	519
Lyme Disease	524
HIV	528

COVID-19

Acute COVID-19 Infection	535
Long COVID	541
Mental Health and COVID	548
VITT: Vaccine Induced Thrombocytopenia & Thrombosis	552

Respiratory

Asthma	553
Chronic Obstructive Pulmonary Disease	563
Obstructive Sleep Apnoea / Hypopnoea Syndrome	569
Macrolides in Respiratory Disease	575
Chronic Cough	579
Bronchiectasis	582

CONTENTS

Gastroenterology

Dyspepsia and GORD	584
Irritable Bowel Syndrome	591
Chronic Diarrhoea	594
Bowel Obstruction	598
Inflammatory Bowel Disease	600
Coeliac Disease	604
Eosinophilic Oesophagitis	607
Barrett's Oesophagus	609
Diverticular Disease	611
Liver Disease	614
Gilberts Syndrome	618
Anal Fissure	620
Haemochromatosis	621

ENT

Anosmia	624
Meniere's Disease	627
Throat Symptoms	630
Tinnitus	632
Hearing Loss	636
Vertigo	640
Temporomandibular Disorders	645

Dermatology

Acne Vulgaris	648
Hidradenitis Suppurativa	653
Topical Steroid Withdrawal	656
Bites and Stings	657
Eczema	660
Acne Rosacea	667
Maskne	670
Urticaria	673
Chilblains	675
Scabies	677
Psoriasis	679

Paediatrics

Feverish Illness in Children	682
Tic Disorder	685
Viral Induced Wheeze	688
Respiratory Illness in Children	691
Bronchiolitis	693
Croup	697
Constipation in Children	698
Recurrent Cough in Children	700
Abdominal Migraine	704

CONTENTS

Paediatrics

Faltering Growth	706
Enuresis	709
GORD in Infants	712
Obstructive Sleep Apnoea in Children	715

Ophthalmology

Infective Conjunctivitis	717
Blepharitis	719
Corneal Ulcers and Acute Red Eye	720
Dry Eye Syndrome	722
Allergic Eye Disease	723

Oral Health

Dental Conditions in General Practice	725
Burning Mouth Syndrome, Geographic Tongue & Oral Candidiasis	728

Medicine for the Elderly

Dementia	731
Frailty and Sarcopenia	738
Multimorbidity	743
Falls Prevention	749

End of Life Care

End of Life Care	752
------------------------	-----

Allergy

Anaphylaxis	757
Food Allergy in Children	759
Atopy and Allergic Rhinitis	764
Penicillin Allergy	767

General

REM Sleep Behaviour Disorder	769
Malnutrition	770
Relative Energy Deficiency	772
Myalgic Encephalomyelitis/Chronic Fatigue Syndrome	775
NEWS: National Early Warning Score	778
The Environment and GP	781
Learning Disability	786
Overweight / Obesity	789
Smoking	797
Alcohol	801
Exercise	807
Novel Psychoactive Substances	809

The Last Pages

Being a GP in 2023	812
Takeaways	813
Glossary	817

PRESENTERS AND ACKNOWLEDGEMENTS

This course is researched, written and presented by the following team of GPs, led by Simon Curtis. All our medical educators on this course are practising GPs and they all work in the NHS.

Simon Curtis Simon is a GP in Oxford and an Honorary Senior Clinical Lecturer in General Practice at Oxford University. Passionate about General Practice and the need for GP-to-GP education he co-founded the Hot Topics course 25 years ago (!!) in 1998 and has taught on the course and led its development ever since. He trained at the Middlesex Hospital in London, qualifying in 1988. He keeps sane by listening to music and riding his road bike, trying to keep up with...

Neal Tucker Neal is a GP in Oxford and the **e-learning lead** of NB Medical. He has developed the 'Hot Topics' material as online, interactive learning modules. He also presents the hugely popular [Hot Topics podcasts](#), that have received rave reviews, especially during the COVID-19 pandemic. He trained at Southampton, qualifying in 2003. Often running around after small children, he escapes on his bike although SC slows him down.

Siobhan Becker Siobhan is a senior partner in a large practice in Oxford and an Honorary **Senior Clinical Lecturer** at Oxford University. She also leads our **Hot Topics for Primary Care Nurses course** along with the **Hot Topics Mental Health course**. She qualified in London in 1995. Passionate about General Practice, she brings many years of teaching and clinical experience to the team. Outside of work she runs, surfs, skis and is never still!

Kate Digby Kate is a GP partner in Gloucestershire and is immersed in the world of GP Education and Training. She's a GP Training Programme Director for the Severn Deanery and also teaches at Bristol University. She is our **Cancer lead** and developed our **Hot Topics in Cancer** course. She's also a Tough Mudder!

Zoe Norris Zoe is a GP in East Yorkshire. As well as medical education, she is also passionate about GP politics and well-being and is a great flag bearer for our profession! She is the Medical Director of Humberside LMC and has developed the Regaining Control course with Stephanie.

Rob Walker Rob is a GP in a large training practice in Amazingstoke...sorry, Basingstoke. He trained in Oxford and qualified in 2001. He has previously been the education and research lead at North Hampshire CCG and now is a member of the NICE GP Reference Panel. He joins our team as a researcher and writer, as well as a regular 'wingman', answering your questions during our live Webinars. When not elbow-deep in BMJs or NICE guidelines he can be found running in the Hampshire countryside or playing the ancient game of Real Tennis in Oxford, well away from the bikes of SC and NT...

Stephanie de Giorgio Stephanie trained in London at Kings and qualified in 2000. She is a GP in Kent. She originally trained in Obstetrics and Gynaecology before moving into GP. She has continued her interest in Women's Health, and she leads our new **Women's Health** course. She has significant experience in education having worked as a VTS programme director, trainer and appraiser. The issue of GP peer support and preventing burnout is of key importance to her and she co-founded the hugely successful Resilient GP with colleagues to support the profession and has developed our own **Regaining Control** course with Zoe. She has a big social media presence ([@DrSdeG](#)). She lives by the seaside with her husband, two children and two cats.

Sarah Davies is a GP in Wales, and as well as a Hot Topics presenter she is our new **Diabetes** lead. She qualified in Cardiff and initially trained in hospital medicine, especially diabetes, before making the excellent move into GP! She has continued her interest in diabetes completing the postgraduate diploma, and regularly presents at national meetings. She is a Diabetes UK Clinical Champion. Outside of work she is surrounded by ballet costumes and football boots and loves walking her dog in the countryside.



INTRODUCTION

Duncan Hill is a GP in Manchester. He has a special interest in **MSK and minor surgery** and has built extensive national experience as a presenter on minor surgery and joint injection courses. He is passionate about trying to bridge the gap between primary and secondary care and works as part of the Frailty service at Manchester University foundation trust. When not working he will be found running in the hills or spending time with his young family.

Mark Hadden is a GP in Dollar, Scotland and is our Scottish lead. He has a wealth of general, acute and emergency medicine experience and continues to work in a variety of urgent care environments as well as being a GP partner. He qualified from Glasgow in 2001. When he isn't negotiating with toddlers he is out in the fresh air on bikes or skis.

Will Duffin is a portfolio GP in Exeter. He has a passion for education & adventure and has worked in many overseas environments from working with UNICEF in Myanmar to providing medical cover for high altitude treks. His education roles include being a teaching associate at Bristol University and education lead and conference director for World Extreme Medicine.

Nick Jones is a GP in central Oxford and a GP Cardiologist at Oxford University Hospitals. He is also a Wellcome Trust Doctoral Fellow at Oxford University, with an interest in Cardiovascular Disease in primary care. Nick has been enthusiastic about teaching since his undergraduate days at Newcastle, where he completed an intercalated Masters in Clinical Education. He also works as an author for the RCGP Essential Knowledge Update team. After a busy day in the practice, Nick loves a swim in the Thames. Outside of work he loves all sport, anything that gets him outside (cycling, running, camping...) and sleeping. [@dmrjones](#)

Andy Ward is a GP in Leicester for Inclusion Healthcare, providing primary care for asylum seekers and homeless patients. He has a long history in medical education and is a Senior Clinical Educator at Leicester Medical School. When not at work, he is likely to be riding his unsuitable bike over unsuitable terrain or walking his unsuitable dogs.

Rachel Brettell is a GP in Oxfordshire, and spends her time trying to persuade medical students that GP is still the best medical career out there! When she isn't working she is usually busy trying to stop her two small people from falling out of trees, accumulating craft projects and learning to speak Norwegian badly but enthusiastically!

Philippa Davies is a GP based in Birmingham and has a special interest in dermatology. She pursued her passion for dermatology by completing the Cardiff Uni postgraduate diploma in practical dermatology and now assists as a course tutor. Philippa is our new dermatology lead. When not working she enjoys playing the piano, time with family and friends and being taken for walks by her dog!

Acknowledgements: with many thanks to the following researchers and academics for their willingness to kindly advise us (and help clarify!) regarding their research and papers discussed in this course:

Andrea Cipriani, Willie Hamilton, Catherine Millington-Sanders, Rory Collins, Julia Hippisley-Cox, Andrew Moore, Peter Rothwell, Richard Hobbs, Scott Murray, Sir Mike Richards, Susan Smith, John Robson, Bernard Prendergast, Richard Haynes, Paul Glasziou, Ahmed Rashid, Liliana Risi, Adrian Baker, Jonathan Hill, Matt Hoghton, Chris Clarke, Karl Gaffney, Rachel Pryke, Francine Toye, Sarah Purdy, Brian Johnson, Bianca Hemmingsen, Roman Romero-Ortuno, Louisa Griffiths and any others we may have forgotten to mention. Thank you!

Many thanks to our GP friends in the **Republic of Ireland** Drs Shane McKeogh and Darach O'Ciardha from GPBuddy.ie for their invaluable advice on issues in the Republic.

And finally thanks to our great team, who make it all happen: Dr Phil Nichols and Janette, Alyson, Craig, Daniel, Susan, Julie, Georgia, Louise, Jordan, Gary, Glenn, Paul & Alan – thanks guys!



WELCOME TO THE SPRING & SUMMER 2023 'HOT TOPICS' GP UPDATE COURSE!

A warm welcome to all our regular friends. And to our new delegates, thank you for doing our course whether you are with us via live webinar, 'on-demand' or face to face.

As well as learning lots to help you to help your patients, we really hope that you enjoy this course. You have a stressful job but when doing this course *you can relax*. We've done all the hard work for you and will bring you bang up to date with what's been published over the last year, merging the literature with clinical cases as we go through the course to bring the evidence to life and illustrate the learning points. We hope that you can imagine the course as a 'virtual clinic'. We highlight **learning points** and **Quality Improvement Ideas** which you may want to discuss in your appraisal, put in your personal development plan and take away and work on over the next year. Everything in this book is bang up to date, but today we present just the most recent material which we think is most clinically useful.

We have some significant new developments for the course:

- The new, improved **NB Dashboard!** You can now access our [NB Dashboard](#). Course delegates will be able to access the electronic version of the book via the Dashboard, as well as to view the webinar version of the course. This means you can always have the knowledge you need at your fingertips. You can also use it to write your reflective notes and learning points from the course, which will synch with your appraisal toolkit.
- We have a '**paper short, electronic long-format**' of the coursebook. The paper book contains the Hot Topics we discuss on the course and which have most recently had significant updates. The **digital version** of the book contains many more additional topics, is instantly searchable and is hyperlinked to the original sources and patient resources. This is available via the App, online and as a PDF and you can download it via **the 'NB Dashboard' section on our website**.
- For those delegates who subscribe to **NB Plus**, you now have full access to our complete range of webinars, interactive CPD modules and learning materials 24/7. This means you can access ALL of our courses, either live for the full interactive experience or on-demand whenever is convenient for you [NB Plus subscription service](#)

We really hope you enjoy the course and that you find the coursebook, dashboard, app and supporting materials a good resource to help you practise what we call 'patient-based, evidence-informed medicine' on a daily basis. And we hope you will attend the course every year, as we ensure the presented topics are all changed every 12 months. Enjoy the course, and do keep in touch with us via email, [Facebook](#) and [Twitter](#)

Simon, Neal, Siobhan, Kate, Zoe, Rob, Stephanie, Sarah, Duncan, Mark, Will, Nick, Andy, Philippa and Rachel

Simon Curtis, Neal Tucker, Siobhan Becker, Kate Digby, Zoe Norris, Rob Walker, Stephanie de Giorgio, Sarah Davies, Duncan Hill, Mark Hadden, Will Duffin, Nick Jones, Andy Ward, Philippa Davies and Rachel Brettell

NB Medical Education, March 2023

Email addresses for all presenters are firstname@nbmedical.com e.g. simon@nbmedical.com

Facebook: [Hot Topics Facebook](#) page Twitter: [@GPHotTopics](#)



THE NB DASHBOARD & APP

To complement our courses and the updated [NB Dashboard](#), and at no additional cost, we introduce our [NB Medical App](#), (downloads available for [Apple](#) and [Android](#) devices), allowing you access to your current course material from your mobile phone, with the ability to sync your CPD to your FourteenFish account or upload it to your Clarity Appraisal Toolkit.



The **NB Dashboard and App** gives you access to all of your learning resources, in your practice or “on the go”. You can:

- Access and view all your **current Hot Topics reference books** for courses you have attended. These are broken down by topic with the ability to search across your books for topics and with rapid access to your ‘Keep it Simple Summaries’ for quick evidence-based answers to clinical questions
- Access your **Hot Topics webinars**, which you can view (broken down by subject or session) on your portable device
- Access your **certificate of attendance** (automatically uploaded to your CPD tracker) for all NB courses attended
- Access your **CPD Learning modules** within the NB Medical App & Dashboard
- Track your reading progress in the NB Medical App & Dashboard and produce a CPD summary to forward onto your appraiser
- Record reflective learning notes and save them to the CPD Tracker
- Export all your CPD evidence to a single, printable document
- If you are a **FourteenFish** member, you can sync all your NB CPD to your FourteenFish account
- For **Clarity Appraisal Toolkit** users, connect directly to your Clarity account, then upload your CPD and NB certificates of attendance into your Appraisal toolkit
- View **NB Medical Blogs** and our **‘KISS from NB’** (Keep it Simple Summaries)
- Book upcoming course days from within the NB Medical App & Dashboard

The **NB Dashboard and App** are **free for all healthcare professionals**. When you attend a course or purchase an NB product, your coursebook and webinars will be made available within your account to view on mobile, tablet, laptop and PC.

Instructions explaining how to set up your account will be emailed to you when your subscriptions become available on the day of the course or when you purchase your webinar. **Your login details will be the same across all devices.**

Any problems with access see our website www.nbmedical.com, talk to our staff during the breaks, email hottopics@nbmedical.com or call our office on 0191 5197293.



NB PLUS provides you with instant access to NB Medical's complete digital library which is continually updated throughout the year.

Your NB Plus subscription includes :

- **ALL LIVE** and on-demand Hot Topics course webinars, choose between live webinar events providing the opportunity for interaction and immediate Q&A or on-demand that you can do in your own time at your own pace
- ALL online course reference books for all courses
- One **FREE** hard copy course reference book of your choice - exclusive to annual subscribers
- Exclusive hard copy course reference book discounts – all books £20 each
- Access to NB Learning – **over 100 hours of interactive CPD modules** for quick 'on the go' learning
- Appraisal Essentials including Safeguarding and Child Protection and Basic Life Support updates
- All material is available to view via the NB Dashboard & App on your phone/tablet or laptop/desktop

All your annual CPD in one place!



Available now from: www.nbmedical.com/nb-plus

HYPERTENSION

No one questions the importance of blood pressure lowering in principle, but **how low to set BP targets remains an area of significant debate**. The final report of the SPRINT trial was published in 2021 [N Engl J Med 2021; 384:1921-1930](#) which advocates lower BP targets and informed many international guidelines. There were many concerns around the methodology of SPRINT, as well as the higher side effect profile, leading NICE to continue with its more conservative targets.

The **debate around BP targets for people with cardiovascular disease is an especially Hot Topic** - in many guidelines people with established CVD are recommended to have lower BP targets, but the evidence underpinning this is not convincing. A new meta-analysis in [Lancet 2021;397:1625-36](#) suggests that the benefits of BP lowering are the same irrespective of CVD status. Shortly after this evidence was published, **NICE updated its guidance in March 2022** ([NICE NG136 March 2022](#)) confirming that we should *'use the same blood pressure targets for people with and without cardiovascular disease'*. This was **supported by a recently updated Cochrane review** in November 2022 [CD010315](#) that concluded there was **insufficient evidence to justify lower BP targets in people with established CVD**.

Other Hot Topics in the world of hypertension include important data, that has practical implications for us in General Practice, suggesting our old friend **paracetamol might be a contributory factor to high blood pressure** [Drug and Therapeutics Bulletin 2022;60:147](#). We also have an important UK based RCT (the TIME trial [Lancet 2022; 400: 1417-25](#)) that contradicts the previous HYGIA study and concludes **timing of antihypertensive therapy does not matter for CVD outcomes**.

1 BLOOD PRESSURE TARGETS

► **BP Targets - How low should we go?** [NEJM May 2021](#) [NICE NG136 March 2022](#)

One of the key pieces of evidence pushing for lower BP targets is the SPRINT trial ([N Engl J Med 2021; 384:1921-1930](#)). It was initially published in 2015, having been stopped early due to significant improvements in the composite outcome of CVD events and CVD death in the intervention arm with intensive BP lowering, and at the time was described as a 'landmark' study. However, as time went by there was increasing [controversy over the methodology](#) for measuring blood pressure, with many hypertension experts concluding that the BP measurements were so 'idealised' that the BP levels reported could not be extrapolated to day-to-day practice, and the intensive systolic BP target of 120 mm/Hg was probably more like 130 mm/Hg in routine practice. However, despite its controversies, it did cause a shifting of the sands, with a **push towards lower BP targets than previously recommended**, and certainly influenced both the US and European guidelines, with the [European guidelines](#) recommending a BP target of 130/80 for many people 'if tolerated', especially those < 65 years old. **Recently the SPRINT trial published a final review of the data, including the post-intervention period** after the trial had been stopped, which included a further year of data until the end of July 2016. **Summary:**

- 9361 patients aged > 50 with a systolic BP of 130-180 and with raised cardiovascular risk (aged >75, CKD, CVD or 10-year CVD risk ≥ 15%).
- Patients were randomized to standard treatment (target of <140mmHg) or intensive treatment (target of <120mmHg) with a composite (primary) outcome of Acute coronary syndrome/MI, stroke, decompensated heart failure or CVD death.
- Including the post-intervention period did not substantially alter the initial results (i.e. benefits were sustained over the following post-intervention year) apart from the improvements seen initially in heart failure no longer differed between the 2 groups.
- Blood pressure inevitably increased post-trial in both groups, and not surprisingly went up more in the intervention group (mean increase 6.9 mm/Hg) than in the control group (mean increase 2.6 mm/Hg).
- But there was still a **significant improvement in the primary outcome in the intensive group (1.77%/yr) vs standard group (2.40%/yr) giving a 25% decrease in events (HR = 0.75, CI 0.61-0.92)**.
- There were however **significant increases in serious adverse events** in the intensive group including hypotension, electrolyte abnormalities, acute kidney injury and syncope.



However, when NICE published its guidance in 2019 they did not advocate lower BP targets, and stuck to the previous (clinic) target of 140/90, with a more lenient target of 150/90 for those aged ≥ 80 . There was clearly significant concern about the methodology used by SPRINT, the demographics of the trial participants, and the applicability of the trial data to usual UK populations. IN SPRINT most of the participants were younger, and less frail than many of our usual patients. Although 28% of participants were over-aged 75 (and likely 'fit'), the **mean age of participants was only 68 years old**, and those with previous strokes or dementia were excluded. NICE also highlighted the 'flip' side to the benefits of intensive BP control with increased adverse events, which are likely to be higher still in the more frail, less fit older person.

► Should people with existing CVD have lower BP targets?

[Lancet 2021;397:1625–36](#) [NICE NG136](#) [Cochrane CD010315](#) [Nov 2022](#)

But what about people with existing cardiovascular disease? Given they are a group with the highest risk of CV events should we consider lower BP levels for them compared to those without CVD, as advocated in many international guidelines? Two recent pieces of research suggest not. First a large [meta-analysis in the Lancet](#) last year which was [recently reviewed in the DTB](#):

- The data included 344,716 patients across 48 RCTs, and assessed the BP-lowering effects across various baseline BP thresholds and CVD status, with 45% having underlying CVD.
- The primary outcome was the incidence of MACE (major CV events defined as a composite of fatal and non-fatal stroke, fatal and non-fatal MI, or IHD, or heart failure causing death or hospitalisation).
- There was **no significant difference in the BP-lowering effects between those with and without CVD** - A reduction of 5 mm Hg was associated with a hazard ratio for MACE of 0.89 (95% CI 0.86 to 0.92) in people with CVD and 0.91 (95% CI 0.89 to 0.94) in people without CVD.
- **Conclusion: Reducing BP by 5 mm Hg reduces the risk of MACE by ~10% irrespective of whether there is established cardiovascular disease.**

Then in [November 2022 Cochrane](#) updated it's 2017 review on this subject:

- They included 7 RCTs of people with hypertension and cardiovascular disease totalling 9595 people.
- They compared total mortality and cardiovascular mortality in those with lower BP targets ($\leq 135/85$) and those with standard BP targets (140/90-160/100).
- Blood pressure ranges were confirmed to be lower in the lower target group (by 8.8/4.5 mm/Hg) but there was **no difference in either total or CVD mortality between those in the lower vs standard BP groups** (moderate certainty evidence):
 - Total mortality RR 1.05, CI = 0.91-1.23
 - CVD mortality RR 1.03, CI = 0.82-1.29
- There was a non-significant trend towards lower total CVD events in the lower BP group (RR 0.89, CI = 0.80-1.00) but this was based on low certainty evidence.
- There was a substantial trend to higher participation withdrawals due to adverse events in the lower BP group, but again the evidence certainty was low.
- **Conclusion: There is insufficient evidence to justify lower BP targets ($\leq 135/85$) in people with hypertension and established cardiovascular disease.**

These results (particularly the Cochrane review) very much support the updated NICE hypertension guideline which does not advocate lower BP targets for people with established CVD, although the absolute BP target is still a matter of debate. In [March 2022 NICE](#) updated its hypertension guideline (see KISS below) Key points in the evidence review:



- NICE concluded that after reviewing the totality of the evidence *'the committee agreed that the evidence did not support a recommendation for lower blood pressure targets in adults with hypertension and established cardiovascular disease, compared to those without established cardiovascular disease'*.
- There was a discussion regarding those with **previous stroke/TIA** as a systolic target of 130 has been advocated in previous guidelines but NICE concluded that *'although the evidence did also show a signal for benefit for lower blood pressure targets in the stroke population, there was also some evidence of harm'* and they went on to highlight the absolute benefits were likely to be small.
- Regarding **harms** NICE highlighted that there was evidence of clinically important harms of a lower BP targets with increased numbers of participant withdrawals and acute kidney injury events. They also noted the evidence of **increased resource use** with intensive blood pressure targets, with higher drug burdens and increased numbers of GP and practice nurse visits and emergency admissions.

However in this update NICE have slightly altered (and strengthened) their wording on targets to include the line that we should reduce BP below the target level and *'ensure that it is maintained below that level'* i.e. **aim to get below the BP target and not tolerate BP levels at or slightly above the target level** - their rewording is aimed to ensure *'neither patients nor practitioners will be accepting of clinic blood pressure values' above target levels*.

KISS: Updated NICE guidance on BP targets [NICE NG136 March 2022](#)

NICE have made some **small but important changes/clarifications** to their hypertension guideline regarding **BP targets**, after a further review of the evidence on targets:

- For adults with hypertension aged under 80, reduce clinic blood pressure to below 140/90 mmHg and *ensure that it is maintained below that level*.
- For adults with hypertension aged 80 and over, reduce clinic blood pressure to below 150/90 mmHg and *ensure that it is maintained below that level*. Use clinical judgement for people with frailty or multimorbidity.
- Use the same blood pressure targets for people with and without cardiovascular disease.

NICE wanted *'to emphasise the importance of reducing clinic blood pressure below 140/90 mmHg in those aged <80 years such that neither patients nor practitioners will be accepting of clinic blood pressure values of 140/90 mmHg or higher. Similarly, in those aged 80 years and over the importance of reducing clinic blood pressure below 150/90 mmHg was emphasised by rewording the recommendation.'*

► **NICE BP targets for people with diabetes and CKD**

Back in 2019, the updated NICE hypertension guideline started the process of aligning blood pressure targets across different groups with other co-morbidities. Until that point there seemed to be a different BP target for every chronic disease, leaving a confusing landscape that was difficult to keep up with! **Since that point the BP targets have been aligned** across the [type 2 diabetes](#), [CKD](#) and most recently in August 2022 the [type 1 diabetes](#) guidelines:

- For people with type 1 diabetes BP target is now the same at <140/90
- Unless the ACR is ≥ 70 mg/mmol, in which case target is <130/80 (same as CKD guideline)
- If aged ≥ 80 target <150/90 irrespective of ACR level

For those with **type 2 diabetes** the BP targets as per the main hypertension guideline: (see KISS below)

- <140/90
- Unless aged ≥ 80 then BP target <150/90

2 SECONDARY CAUSES OF HYPERTENSION

► Medicines that may raise BP in adults [Drug and Therapeutics Bulletin 2022;60:70](#)

When we diagnose and monitor people with hypertension we are well aware of some of the key factors that can contribute to raised blood pressure - lack of exercise, high salt intake, smoking, obesity etc. **But how often do we review medications that may be contributing to the high BP?** A recent [DTB review](#) suggests we should probably take more note of this potential factor based on data from an observational study in the US recently published in [JAMA Intern Med. 2022;182\(1\):90-93](#). This cohort study of 27,599 adults assessed the prevalence and use of medications that can raise BP and the association with BP control and use of antihypertensive medications:

- Of those with hypertension 18.5% (95% CI = 17.5-19.5) reported using medications that can raise BP.
- **Taking medications that can raise BP was associated with an increased risk of uncontrolled hypertension** in those not taking antihypertensives (OR 1.24, 95% CI = 1.08-1.43).
- In those taking antihypertensives, concomitant use of medicines that can raise BP was not associated with uncontrolled hypertension, but **use of medication that may raise BP was associated with greater use of antihypertensives**.
- The most **commonly reported groups of medicines** were antidepressants (8.7%), NSAIDs (6.5%), steroids (1.9%) and oestrogens (1.7%).

There are obviously a lot of caveats to this data, with potential confounding and the fact that we don't know what size of BP effect the medicines had and whether it is clinically significant. However, the implications of the data are that **use of these medicines is common**, and they may be associated with uncontrolled hypertension in those not taking antihypertensives and that **greater use of antihypertensives are needed to control BP in those taking these medicines**. NICE does not address this issue but the [CKS guideline March 2022](#) lists a number of drugs and other substances that can contribute to hypertension.

KISS: Drugs and other substances that can contribute to hypertension [CKS March 2022](#)

List not exhaustive, but includes:

- Alcohol - CKS notes that variable hypertension that is resistant to commonly used drugs may be seen with excess alcohol and that high blood pressure can resolve within a week or two of complete abstinence.
- Ciclosporin.
- Cocaine and other substances of abuse.
- Combined oral contraceptive.
- Corticosteroids.
- Erythropoietin.
- Leflunomide.
- Liquorice - also present in some herbal medicines.
- NSAIDS.
- Sympathomimetics - may be found in over-the-counter cough and cold remedies (for example ephedrine and phenylpropanolamine).
- Venlafaxine.
- Mirabegron - should be used with caution in people with known hypertension and needs regular BP monitoring for all people on this drug due to risks of causing high BP [MHRA 2015](#)

Obviously some of these medications we can do nothing about and will be essential treatments for other conditions, but some (notably NSAIDS given the prevalence estimates in the original data) we can address. This list also **highlights the importance of asking about over-the-counter medicines, herbal remedies** and how many Bertie Bassett's Granny is tucking in to...

► Could paracetamol cause worsening hypertension?

PATH-BP trial [Circulation Feb 2022](#) & [Drug and Therapeutics Bulletin 2022;60:147](#)

The one notable absentee from the medications mentioned above is paracetamol. Long thought of as a benign and very safe drug, there has been increasing observational evidence that paracetamol, like NSAIDS, may be a contributory drug for high blood pressure. This small neat RCT investigated this:

- 110 participants, mean age 62 years, with hypertension either treated on stable doses of antihypertensives (~2/3 of total), or untreated (~1/3), all with ambulatory BPs <150/95.
- This was a cross-over RCT so participants were allocated to either paracetamol 4g daily or placebo for 2 weeks, then had a washout period of 2 weeks then switched to the alternative arm for a further 2 weeks.
- People already taking regular paracetamol were excluded as were people with heart disease and cerebrovascular disease, liver disease, and those on NSAIDS and anticoagulants.
- BP was measured both by standard clinic measurement as well as 24 hr ambulatory BP.
- There was a **significant placebo-corrected rise in BP in the paracetamol arm**:
 - Clinic readings = systolic 4.6 mm/Hg (95% CI = 2.4-6.7), diastolic 1.6 mm/Hg (95% CI = 0.1-3.0)
 - Ambulatory BP = systolic 4.2 mm/Hg (95% CI = 2.4-6.0), diastolic 1.4 mm/Hg (95% CI = 0.3-2.5)
- The **rise in blood pressure was seen as early as day 4 and was sustained at day 14**.
- **Conclusion:** *The authors conclude that a standard dose of paracetamol could cause a significant rise in blood pressure in people with known hypertension, with potential increase in CVD risk.*

There are certainly a few caveats to this data we need to bear in mind:

1. This was a very short-term study - we don't know if these rises in BP are sustained, although previous observational data suggests it probably is.
2. We don't know if these rises in BP will translate to rises in CVD.
3. The study may not be generalisable - all were white, they were predominantly male (>75%) and they all had confirmed hypertension (so may not be applicable to those who are normotensive).

However, if true and sustained, this is a significant finding. The **rise in BP seen is similar to that seen with NSAIDS which (as discussed above) are a known and common contributor to high BP**. If the rise seen is sustained **the authors estimate the level of BP rise seen might translate to ~20% more CV events** - given the number of people taking regular paracetamol this could be a substantial CVD burden.

► And what about sodium-containing paracetamol products?

[Eur Heart J. 2022;43:1743–55](#) [Drug and Therapeutics Bulletin 2022;60:181](#)

As we all know, **soluble, effervescent and dispersible paracetamol-containing analgesics contain a substantial amount of salt**, and the concern over these products and CVD was highlighted again recently in [observational data](#) and a recent [DTB review in December 2022](#):

- These two observational cohort studies reviewed people aged 60-90 (mean age 72), totalling about 300,000 participants in all, and compared CVD and all-cause mortality after 1 year in those prescribed sodium-containing paracetamol products with those prescribed non-sodium containing paracetamol. About half of those studied had hypertension.

- Irrespective of hypertension status there was a **significant increase in CVD events (MI, stroke, heart failure) in those taking sodium-containing paracetamol**:
 - In the hypertension group there was a 1 year absolute risk increase of 1% (weighted hazard ratio 1.59, 95% CI = 1.32-1.92)
 - In the non-hypertension group there was a 1 year absolute risk increase 0.7% (weight hazard ratio 1.45, 95% CI = 1.18-1.79)
 - CVD risks were dose-dependent with higher CVD rates with increased numbers of sodium-containing prescriptions.
- **All-cause mortality was also substantially higher in those taking sodium-containing paracetamol** with a 1.5% absolute increase in the hypertensive group (weighted HR 2.05, 95% CI = 1.92-2.19), and similar results in the non-hypertensive group.
- **Conclusion:** *sodium containing paracetamol analgesics substantially increase CVD events and all-cause mortality, irrespective of hypertension status.*

These results are not surprising. We know salt intake is as strong risk factor for hypertension and some of these soluble or effervescent paracetamol-containing analgesics contain as much as 400mg salt per tablet, which if taken at maximum doses would equate to >3g salt/day, well above the recommended daily amount simply from the tablets. These results are using non-sodium-containing paracetamol as the comparator - if the BP effects of paracetamol in its own right, seen in the previous study, are sustained the absolute CVD burden of sodium-containing paracetamol products would be even higher. Given >1 million of these products were dispensed in England in the year to July 2022 the **DTB conclude** that '*healthcare professionals should avoid prescribing effervescent, dispersible and soluble medicines that contain sodium wherever possible*'.

► Investigating hypertension in younger adults [BMJ 2022;376:e067924](https://doi.org/10.1136/bmj-2022-067924)

Whilst on the subject of secondary causes of hypertension, a very useful [BMJ review article](#) helps us to **clarify when we should consider a secondary cause for hypertension and importantly what tests we should (and should not) do in primary care**. This is especially welcome as there are no national guidelines that really address this issue (or certainly not from a primary care point of view), and indeed the only NICE comment on hypertension in young adults and potential secondary causes is the somewhat limited note that 'For adults aged under 40 with hypertension, consider seeking specialist evaluation of secondary causes of hypertension and a more detailed assessment of the long-term balance of treatment benefit and risks'.

We summarise the paper in the KISS below but first some background points:

- NICE defines young-onset hypertension as **hypertension in adults under the age of 40**.
- Data from England estimates the prevalence of hypertension in adults aged 35-44 as 12.2%.
- **Secondary causes for hypertension are much more common in young adults**, accounting for between 5-30% of cases of young-onset hypertension.
- The **commonest causes of secondary hypertension in young adults** are thyroid dysfunction, renal parenchymal disease, renal artery stenosis, and primary aldosteronism.
- Other rarer causes include coarctation of the aorta and pheochromocytoma.
- In the general adult population, the commonest cause of secondary hypertension is **obstructive sleep apnoea** (prevalence estimates of 5-15% in hypertensive patients).
- Treating young-onset hypertension and diagnosing secondary causes is important due to the disproportionate burden of CVD and mortality risk, the fact that some secondary causes have additional risks beyond the hypertension risk, and some secondary causes can be treated mitigating the long-term risks.

KISS: Hypertension in young adults and secondary causes of hypertension [BMJ 2022;376:e067924](#)

Initial primary care assessment in young-onset hypertension:

- **Confirm the diagnosis** with home or ambulatory BP measurements; if available **ambulatory BP may be preferable** as it can identify patients with no nocturnal drop in BP - if no nocturnal dip in BP is seen there is a higher risk of a secondary cause and there is increased risk of CKD and CVD.
- Check for symptoms and risk factors for **obstructive sleep apnoea** - consider [STOP-BANG](#) screen (see our chapter on sleep apnoea in the respiratory section for further detail).
- Check for **drugs/other substances** that can cause secondary hypertension (see KISS above).
- **Potential Examination findings:**
 - Renal artery stenosis - may have **renal bruits** - auscultate 2-3cm laterally and above umbilicus on both sides; bruits may be present in as many as 78-87% of cases of renal artery stenosis.
 - Renal parenchymal disease - ballotable kidneys.
 - Coarctation of aorta (rare) - radio femoral delay, faint femoral pulses or discrepancy >15mm/Hg in BP between arms.
 - Hypertensive retinopathy on fundal exam (may be best to send to optician for assessment).
- **Investigations in primary care:**
 - **Urine** - dip and send for ACR.
 - **U&E:**
 - Assess for hypertension-induced renal disease (reduced eGFR).
 - **Hypokalaemia** - May be present in primary aldosteronism although only in the minority of cases; also note **diuretic-induced hypokalaemia** is suggestive of primary aldosteronism.
 - Also, watch for **significant ACEI/ARB induced renal deterioration** - eGFR fall of >25% or creatinine rise of >30% suggests possible renal artery stenosis.
 - **Thyroid function** - both hypo and hyperthyroidism can cause secondary hypertension.
 - HbA1c/lipids - for overall CVD risk assessment.
 - FBC - HB changes may be seen in secondary hypertension; polycythaemia is a rare cause.
 - **12 lead ECG** - mainly looking for signs of LVH, may be a sign of hypertrophic cardiac disease.
 - **Renal USS** - look for structural renal disease and renal artery stenosis (but note normal USS does NOT rule out renal artery stenosis and confirmation needed with secondary care investigations).
 - Echo (if available) if signs on exam or ECG to suggest underlying cardiac disease e.g. valve disease, coarctation, hypertrophic cardiomyopathy.
- **Investigations NOT to be done in primary care:**
 - **Urinary metanephrines** for phaeochromocytoma. Refer if feature present (**the '5 Ps'** - paroxysmal hypertension, palpitations, perspiration, pallor, pounding headache).
 - **Random serum cortisol** for Cushing's - unreliable. Refer if Cushingoid features present.
 - **Aldosterone-renin ratio (ARR)** for primary aldosteronism - needs careful pre-test medication verification/interpretation and strict lab processing time frame (within 1 hour of sample). Refer if spontaneous or diuretic-induced hypokalaemia.

When to refer?

- Any symptoms/signs/investigations **suggesting secondary cause** - phaeochromocytoma, coarctation of the aorta, renal parenchymal disease, renal artery stenosis, Cushing's disease, sleep apnoea.
- The **exception is thyroid disease-induced hypertension** - hypothyroidism can be managed in primary care, but hyperthyroidism may need secondary care input.
- **Resistant hypertension to ≥3 drugs at max tolerated dose** - this is strongly associated with secondary hypertension, and it is **particularly important not to miss cases of primary aldosteronism** - as discussed in a recent BMJ article ([BMJ 2022;377:e065250](#)) hypokalaemia (traditionally thought of as a hallmark of the disease) is often not present so **absence of hypokalaemia is not reassuring**; however, if hypokalaemia or diuretic-induced hypokalaemia is present in the context of hypertension this is strongly suggestive and referral is indicated.



3 OTHER HOT TOPICS FROM THE JOURNALS

► What time should patients take their anti-hypertensive medications?

[Hygia Chronotherapy Trial](#) & [TIME trial](#) [Lancet 2022; 400: 1417–25](#) with [NB Blog Jan 2023](#)

Why the interest in the timing of when you take your blood pressure medications? Well, it all boils down to chronotherapy. This is the theory that you adjust when you take medications to complement the body's natural biological rhythms to improve outcomes. As discussed in the recent [Lancet editorial](#) *'Neurohormonal variations result in diurnal changes in blood pressure, with peak blood pressure typically being recorded after awakening and lowest blood pressure during the night-time, a phenomenon commonly referred to as dipping. The absence of diurnal blood pressure change is generally associated with adverse cardiovascular outcomes. This association has led to a suggestion that night-time administration of antihypertensive medications to mimic the physiological dipping of blood pressure could result in improved cardiovascular outcomes.'*

This theory was original tested in the [MAPEC trial](#) back in 2010 and indeed the trial suggested that evening dosing of antihypertensives did reduce CVD events. This was followed up by the [Hygia Chronotherapy trial](#) in 2020, which caused quite a stir as it supported the MAPEC findings with significant results:

- The Hygia Trial was a large primary care based prospective study in Spain with 19,084 patients with hypertension, mean age 60 years old, randomly assigned to take all BP medications either in the morning or at bedtime, median follow-up 6.3 years.
- BP was monitored at least annually with 48-hour ambulatory BP monitoring (ABPM).
- Primary outcome was a composite of major cardiovascular events (MACE = CVD death, MI, coronary revascularisation, heart failure and stroke).
- **Results:**
 - Improved mean ABPM in bedtime group with significant 45% RRR in primary outcome (HR 0.55; 95% CI = 0.50-0.61).
 - In absolute terms, for total events there was a 9% absolute risk reduction and NNT = 11, for CVD events ARR is 5.4 with an NNT of 18.6
 - No difference in adverse effects or concordance between the two groups.
- **Conclusion:** *Taking antihypertensive medications before bed, rather than in the morning, is a simple, risk-free intervention and significantly reduces the risk of CVD events.*

The evidence from the Hygia Trial seemed convincing, and unusually for an intervention with such potential for CVD reduction, it comes at no extra cost. However, as discussed in the latest [Lancet editorial](#) both the **MAPEC and Hygia trials have come under criticism with several concerns surrounding the design, conduct, and results of these studies** resulting in a formal investigation into the content and conduct of the Hygia study. Although the investigation identified no clear grounds for factual or ethical concerns, the **reliability of these studies continues to be questioned.**

Recently the **Treatment In Morning versus Evening (TIME) study** was published, adding further to the debate on this subject. What was it and what did it show?

- It was a pragmatic, prospective RCT from the UK and included 21 104 adults with hypertension who were randomly assigned to morning (0600–1000 h) or evening (2000–0000 h) administration of all antihypertensive medications with a median follow-up of 5.2 years.
- Mean age was ~65 years, 57.5% were men and the vast majority were white (>90%).
- The primary outcome was the composite cardiovascular endpoint of vascular death or hospitalisation for non-fatal MI or non-fatal stroke.
- There was **no significant difference between the two groups for the primary outcome** - primary outcome event occurred in 3.4% in the evening group vs 3.7% in the morning treatment group with unadjusted HR 0.95, 95% CI = 0.83–1.10.
- **Conclusion:** *Evening dosing of antihypertensive medication was not different from morning dosing in terms of major cardiovascular outcomes.*



As always we need to be aware of **caveats with this data**. First, the trial was conducted remotely using questionnaires and uploaded data so adherence (or otherwise) to the trial dose timings is at risk of recall bias. Secondly, there were higher numbers reporting non-adherence in the evening dosing group compared to the morning which could dilute any potential benefits of the evening dosing, although as the authors rightly state 'the findings probably reflect what would happen if patients in usual care were allocated to evening or morning dosing.' Finally, the study population was almost exclusively white, which questions the generalisability to other ethnic populations.

So **where does that leave us with the question of what time to take antihypertensives?** The TIME trial suggests timing of antihypertensives doesn't matter and given the concerns over the Hygia trial, we probably should not recommend people routinely take their BP medications before bed. Further studies may help answer this question, in the form of the Canadian [BedMed trial](#) and the associated BedMed-frail trial, which will look in more detail at side effects and safety concerns in frail populations. But until then **the focus for us should be on concordance - probably the most important time to take BP medications is the time the patient will remember to do so**; as the TIME authors conclude '*Patients can be advised that they can take their regular antihypertensive medications at a convenient time that minimises any undesirable effects*'.

► **Reverse dipping and a call for 24-hour ambulatory BP monitoring?** [BJGP 2023; 73 \(726\)](#)

Whilst on the subject of diurnal BP variation and 'non-dippers', what about people whose BP actually goes UP overnight? This group of people are known colloquially as 'reverse-dippers', and like 'non-dippers' they have a higher association of cardiovascular disease than those people who maintain their normal diurnal BP variation. But **how common is this 'reverse dipping' scenario?** This was investigated in a very recent paper in the 2023 New Year edition of the BJGP. The study looked at 2 cohorts - one a hospital cohort during acute admissions, the other a community cohort - all of which had 24-hour ambulatory BP monitoring (ABPM). We shall just focus on the community cohort:

- The community cohort comprised 770 people across 28 UK General Practices recruited between 2010-2012 who had had both clinic and 24-hour ABPM as part of the BP-Eth study investigating the presence of white coat hypertension in different ethnic groups.
- People were aged between 40-75 and ~60% had diagnosed hypertension.
- People were classified according to their day to night time BP levels - **dippers** (normal diurnal variation with a drop in BP overnight), **non-dippers** (similar BP levels between day and night) and **reverse dippers** (BP levels consistently higher overnight).
- In the community cohort **>10% were reverse dippers, and their BP was on average 8.5mm/Hg higher overnight.**
- The **reverse dippers had lower daytime BP levels than dippers, and also had higher levels of associated cardiometabolic co-morbidities.**
- **Conclusion:** *Not measuring nighttime BP puts reverse-dippers at risk of failure to diagnose hypertension.*

Why does this matter? Well, the authors make a good case that by **not doing nighttime BPs we will miss a significant proportion of people with masked hypertension** - they may have normal BPs in the daytime (when done with clinic or home BP monitoring) but an ~8.5 mm/Hg higher level overnight may well tip many into confirmed hypertension and the need for treatment, especially given this group have higher rates of CVD. As discussed in an [associated editorial](#), the **NICE guidance only recommends daytime ABPM which would also miss this group of reverse dippers.**

The **authors call for much wider use of 24-hour ABPM in General Practice**, and as discussed in the [editorial](#) this would have the added benefit of picking up people who are getting asymptomatic overnight hypotension - this would help redress the balance of under and over-treatment in hypertension. **Is this realistic?** Let's be honest, no it isn't currently. Many practices don't even have access to daytime 24-hour ABPM, let alone 24-hour ABPM, and given the current pressure on both primary care and the wider NHS this is not going to happen any time soon. Currently, for most of us, diagnosis/confirmation of hypertension is still likely to revolve around home BP monitoring which we summarise in the KISS below:

KISS: Home BP monitoring

[Drug and Therapeutics Bulletin 2021;59:119-123](#) & [Drug and Therapeutics Bulletin 2021;59:179](#)

■ **Is HBPM effective/what's the evidence?**

- Clinic BP measurements (CBPM) are notoriously variable, and by comparison ABPM (ambulatory BP measurements) and HBPM have higher reproducibility and prognostic significance, and indeed **HBPM is superior to CBPM in prediction future CVD events**.
- HBPM has the added advantage over CBPM that it will pick up white coat hypertension (and masked hypertension when BP normal in clinic but high at home) - ABPM is considered the gold standard for **diagnosing hypertension** as it has better evidence but access is limited so HBPM is a good alternative.
- The **evidence is a bit mixed on long-term management of hypertension with HBPM** and it's worth noting that NICE do not specifically endorse the use of HBPM for monitoring but note it is an option for people who choose remote monitoring (and it is supported by the [NHS BP@home scheme](#)), but if using HBPM education, ongoing support and engagement with patients appears to be important.

■ **What practical issues do we need to be aware of when considering HBPM?**

- **Correction factor** - it's important to remember to add 5/5mm/Hg to HBPM readings to get a clinic 'equivalent' to guide both diagnosis and treatment targets as guidelines are based on clinic levels.
- **AF** - Most international guidelines recommend HBPM should not be used for people with AF due to concerns about inaccuracies, however, there is evidence that if patients use 3 readings each time and use the average, inaccuracies can be minimised.

■ **What are the important factors to tell patients when they do HBPM?**

- Avoid smoking, eating, caffeine intake and exercise 30 min before measurements.
- Ensure the room is quiet and free of distractions (eg, mobile phone or television).
- The person should sit upright comfortably with the arm supported at the level of the heart, legs uncrossed, both feet flat on the floor and avoid talking.
- Do readings over 7 consecutive days (minimum 4 days).
- For each BP recording, take two consecutive measurements at least 1 minute apart. Additional measurements should be taken if the first two measurements are >10 mm Hg different.
- Measurements should be recorded two times per day in the morning (06:00–12:00) (before antihypertensive treatment if appropriate) and evening (18:00–00:00) at similar time intervals.
- The average of the readings should be calculated, ignoring the first day as first-day BP measurements have a tendency for higher variability.
- **Excellent patient info on HBPM** from [Blood Pressure UK](#).

■ **What about wrist BP monitors?**

- If a wrist cuff device is used by the patient, it is important to recommend correct arm positioning at the level of the heart to minimise inaccuracy and variability in BP measurements.
- Some hypertension societies do not recommend wrist cuff devices due to a higher tendency for inaccurate measurements mainly from incorrect arm positioning.
- But they can be useful for patients with very large upper arms or when upper arm cuff measurements are too painful or distressing for the patient.

■ **Device accuracy/which ones to use/when should they be changed or re-calibrated?**

- Evidence from the BJGP in 2020 suggests that generally, home BP monitors are pretty accurate, but the evidence (reviewed in DTB) showed that:
 - **Validated devices should be used** (96% of the validated devices passed all tests compared with 64% of non-validated devices) - [click here](#) for BIHS list of validated devices.
 - **Risk of inaccuracy increases with time** - 5% of monitors failed testing for the first 4 years but 26% of older models failed testing.
 - Although patients do not need to spend large sums of money on expensive devices (vast majority were <£30), **avoid the really cheap ones (<£10)** as they have a much higher failure rate.
- The conclusion is that **most home BP monitors are fine as long as they are validated, in the mid-range price bracket £10-30 and they are recalibrated every 2 yrs or changed every 4 yrs**.



KISS: Hypertension [NICE NG136 Updated 2022](#) [Click here for a visual summary](#)

■ **Diagnosis/assessment/monitoring of hypertension:**

- Measure BP in both arms & if a difference of ≥ 15 mmHg, base management on the higher reading
- Hypertension should be confirmed ideally by use of ambulatory BP monitoring (ABPM).
- However, if ABPM unsuitable **home BP monitoring (HBPM) may be used**; if using HBPM record BP morning and evening for 7 days (minimum 4), discard readings on the first day and use an average of the remaining readings.
- Hypertension diagnosis confirmed in those with clinic BP $\geq 140/90$ AND ABPM/HBPM $\geq 135/85$
 - Stage 1 = clinic BP 140/90-159/99 and ABPM/HBPM 135/85-149/94.
 - Stage 2 = clinic BP 160/100-179/119 and ABPM/HBPM $\geq 150/95$.
- Either clinic readings or HBPM may be used for monitoring hypertension.
- For adults <40 with hypertension consider specialist referral for secondary causes/treatment.

■ **Antihypertensive drug treatment** (always together with lifestyle advice):

- **Offer** drug treatment to those (any age) with **stage 2 hypertension**.
- **Discuss** starting drug treatment in those < 80 yrs old with **stage 1 hypertension AND**:
 - Target organ damage, CVD, renal disease, diabetes, OR **10 year CVD risk $\geq 10\%$** .
 - **Consider** drug treatment in younger adults <60 years old even if CVD risk $<10\%$.
 - *Discuss individual CVD risk and preferences for treatment, including no treatment, and explain risks and benefits before starting drug treatment - [click here for patient decision aid](#).*
- Consider drugs if ≥ 80 yrs old & clinic BP $\geq 150/90$ (once hypertension confirmed on ABPM/HBPM).

■ **Targets** (based on clinic readings so add 5/5 to HBPM):

- Aged < 80 years old $<140/90$ and *ensure it is maintained below that level*
- Aged ≥ 80 years old $<150/90$ (but use clinical judgement in those with frailty or multimorbidity).
- Use the **same BP targets for people with and without cardiovascular disease**.
- Use **STANDING BP readings** if postural drop ≥ 20 mmHg or symptoms of postural hypotension.

■ **Drug treatment:**

- **Step 1:**
 - ACEI or ARB (if <55 and not African or Caribbean family origin, OR diabetic of ANY family origin).
 - CCB for all others - if CCB not tolerated use thiazide-like diuretic (e.g. indapamide).
- **Step 2:**
 - If on ACEI/ARB add either CCB or thiazide-like diuretic.
 - If on CCB add ACE/ARB or thiazide-like diuretic.
- **Step 3:**
 - Before considering triple therapy check drugs at optimal doses and review concordance.
 - If further treatment required combine ACE/ARB + CCB + thiazide.
- **Step 4:**
 - Before considering further treatment confirm elevated readings with ABPM or HBPM assess for postural hypotension and check concordance.
 - If further treatment required options include spironolactone (if $K \leq 4.5$ mol/l), further diuretic, alpha or beta blocker; if still not controlled on 4 drugs seek expert advice.
- Patients of **African or Caribbean** origin should have **ARB in preference to ACEI**.

■ **Severe hypertension** (stage 3 hypertension = clinic BP $\geq 180/120$):

- Refer for **same-day specialist assessment** if BP $\geq 180/120$ AND
 - Signs of retinal haemorrhage/papilloedema or suspected pheochromocytoma or life-threatening symptoms (e.g. new-onset confusion, chest pain, heart failure or new renal failure).
- If BP $\geq 180/120$ and no signs above:
 - Investigate for target organ damage ASAP.
 - If target organ damage presents start drug treatment without waiting for ABPM/HBPM readings.
 - If no target organ damage, repeat BP readings within 7 days.



NB Learning Points: Hypertension

- The **debate on BP targets** continues and culminated in a review of the evidence by NICE; **NICE have stuck with previous targets** of <140/90 for those aged <80 and <150/90 for those aged ≥80 but they have slightly altered (and strengthened) their wording on targets to include the line that we should reduce BP below the target level and *'ensure that it is maintained below that level'*.
- Recent Cochrane data has concluded that **lower BP targets (<135/85) for people with known CVD did not improve either total or CVD-specific mortality** and thus there is **insufficient evidence to justify lower BP targets in this group**; this supports the updated NICE guideline which says we should *'Use the same blood pressure targets for people with and without cardiovascular disease.'*
- We mustn't forget to think about causes of **secondary hypertension in our younger patients**, especially primary aldosteronism which is often missed - a very useful BMJ review article helps 'fill in the gaps' as to how we should assess hypertension in younger people and what investigations we should (and should not) do in primary care.
- Tied in with secondary causes of hypertension we have important data suggesting that **paracetamol may cause worsening hypertension** and that **sodium-containing (e.g. soluble, effervescent) analgesics are associated with higher CVD events and all-cause mortality**.
- The recently published TIME trial contradicts previous data from the HYGIA trial, by concluding that **timing of antihypertensive medication does not alter CVD outcomes**. The HYGIA trial has had a number of criticisms and this latest data suggests that when to take BP medications should be informed more by convenience and any potential side effects of the drugs.
- We have had 2 timely reviews from the DTB on **home BP monitoring (HBPM)** given the push to remote monitoring in the COVID pandemic:
 - **HBPM better predicts CVD outcomes than clinic BP monitoring** - it is widely accepted as an option for diagnosing hypertension, and although the evidence for its use in monitoring BP is less certain good patient education and engagement are important to implementation.
 - Most home BP monitors are fine as long as they are **validated, in the mid-range price bracket £10-30 and that they are recalibrated every 2 years or changed every 4 years**.
 - **Wrist monitors** are probably more inaccurate due to the risk of incorrect arm positioning, but can be considered for people with very large upper arms.
- We've had interesting recent research suggesting that as many as 1:10 people may be 'reverse-dippers' where their blood pressure rises overnight (rather than falls) - these people are at higher CVD risk and are at risk of under-diagnosis of hypertension using clinic or HBPM as their BP levels are more normal in the day time; better access to 24 hour ambulatory BP monitoring has been called for, but realistically for most GPs we remain reliant on HBPM for confirmation of hypertension.

NB Quality Improvement Ideas: Hypertension

- When considering the option of drug management (especially for people with Stage 1 hypertension) direct patients to the **NICE patient decision aid** to help them make informed decisions about whether, or not, they want to consider drug therapy - [click here](#).
- When thinking about risk factors for/contributory factors for high blood pressure, **have you considered drug causes, including paracetamol?** If patients have uncontrolled hypertension are they taking regular paracetamol? If they are deriving minimal benefit from paracetamol considering stopping it.
- **Audit patients with hypertension being prescribed effervescent, soluble or dispersible analgesics and aim to stop these medications**, or switch to standard (non-sodium-containing) formulations. These analgesics contain high sodium levels and are associated with worse CVD outcomes.
- **Audit patients diagnosed with hypertension <40 years old** - have they had appropriate investigations for secondary causes of hypertension as outlined above +/- referral?
- Do you have a **protocol for Home BP monitoring** in your practice with information to give to your patients? If not use the information in the KISS box above to formulate one, together with the **excellent patient information** from [Blood Pressure UK](#) and the [list of validated BP machines](#) from BIHS.

ORTHOSTATIC HYPOTENSION

We see a lot of older people complaining of a symptom that can be variously described as dizzy, light-headed, unsteady, muzzy... Assessment of dizziness in primary care is difficult, and we often tend to focus on the vestibular system but *many patients complaining of these symptoms have undiagnosed orthostatic or postural hypotension*.

Two recent reviews of this condition have been published. A '*pragmatic guide to diagnosis and treatment*' published by the Drugs & Therapeutics Bulletin [DTB 2020;58:166](#) is very useful, and this includes an excellent, simple step-wise approach to management. The BMJ has also published a clinical update on postural hypotension [BMJ 2021;373:n922](#) and we have had a new systematic review on the evidence for fludrocortisone [Cochrane 2021](#) which was reviewed in the [DTB Dec 2021](#). Finally, we have recent data [BJGP 2023; 73 \(726\)](#) suggesting that postural hypotension is significantly under-reported in GP records, and an [associated editorial](#) highlights the implications of this together with the importance of identifying people with asymptomatic postural hypotension who are still at risk of the complications discussed below.

Chronic OH is an incredibly common and disabling condition. Looking at the list of symptoms and potential complications of OH, we can see how important it is to look for and appropriately manage.

KISS: Symptoms & Complications caused by Orthostatic Hypotension (OH)

[BMJ 2021;373:n922](#) and [DTB 2020;58:166](#)

- **Symptoms of OH**
 - Dizziness
 - Light-headedness, vertigo, unsteadiness
 - Syncope and pre-syncope
 - Visual disturbance
 - Transient loss, blurring or dimming of vision
 - Shortness of breath
 - Chest pain
 - Fatigue
 - Cognitive impairment
- **Complications of OH**
 - Falls
 - Stroke
 - Dementia
 - Increased mortality

► How is orthostatic hypotension defined, and diagnosed?

- A sustained drop in blood pressure of $\geq 20\text{mmHg}$ systolic and/or $\geq 10\text{mmHg}$ diastolic occurring within 3 minutes of standing
- The gold standard for diagnosis is a tilt table test but in primary care, an 'active stand' test of lying and standing BP is needed - see KISS box below
- This is a test that can be done by a trained Healthcare Assistant in a 10 to 15-minute consultation
- **Supine hypertension** is when patients supine have hypertension ($\text{SBP} \geq 140$ and/or $\text{DBP} \geq 90$) and yet orthostatic hypotension; it is common in people with neurogenic OH due to diabetes or Parkinson's, and it may be missed as a diagnosis as patients often have an acceptable BP when seated
 - Management of supine hypertension is challenging. Treatment of supine hypertension is still recommended to reduce cardiovascular risk, but of course, it carries a risk of exacerbating OH so shared decision-making is essential
 - It may be necessary to accept a supine BP $> 160\text{ mmHg}$ to be able to achieve a standing BP that will support an upright posture

KISS: The Active Stand Test for Postural Hypotension

[BMJ2021;373:n922](#) & [DTB 2020;58:166](#)

- Test ideally done in the morning
- The patient lies supine for ≥ 5 minutes
- Measure baseline BP and HR
- Patient stands up
 - Measure BP and HR at 30 to 60 seconds
 - Measure BP and HR at 2 mins and 3 mins
- Interpretation
 - *Orthostatic hypotension is diagnosed if there is a postural drop of ≥ 20 mmHg systolic and/or ≥ 10 mmHg diastolic at any point within 3 minutes of standing*

► How common is OH, and what are the implications? [BJGP 2023; 73 \(726\)](#)

- OH is incredibly common, affecting as many as 1 in 4 older people.
- A recent [large meta-analysis](#) suggests that the prevalence of OH is 17-19% in community and primary care populations and 31% in residential and nursing home residents.
- Symptoms are generally worse in the morning and exacerbated by prolonged sitting or lying, warm environments and eating.
- However, despite the high likely prevalence, reporting of OH remains low in GP records, as reported in this [recent study in the BJGP](#) and an [associated editorial](#):
 - This retrospective cohort study reviewed GP records and recorded diagnoses of postural hypotension in people aged ≥ 50 - only ~25,000 people out of a total of almost 3 million had a recorded diagnosis - **equivalent to a cumulative 10-year prevalence of only 1%.**
 - The authors and editor discuss the **myriad reasons for this significant discrepancy between predicted prevalence of OH and recoded diagnosis** - some may be having this actively managed without a formal code, but many people with OH (as many as 2/3) may be asymptomatic, and rising workload pressures are all likely to be factors.
 - The implications are fairly obvious - a large group of people with undiagnosed and unmanaged OH with the consequent risks of OH complications, especially falls.
 - They call for a proactive approach to recognition of OH and wider use of ambulatory BP monitoring; this sounds good in theory, but given the aforementioned workload pressures and lack of access to ambulatory BP monitors in primary care, this is a way off being a reality.
- However, one important implication for us, which is more immediately practical, is to at least **try to identify people who have, or at risk of OH who have hypertension**:
 - The [NICE hypertension guideline](#) advises that we should assess for OH in people at high risk (those aged >80 , those with diabetes and those with possible symptoms of OH)
 - **If either a significant postural drop is found or they have symptoms of postural hypotension, treat to a blood pressure target based on standing blood pressure.**



► *What causes OH?*

- Acute OH can, of course, be as a result of a sudden loss of blood volume and BP (e.g. GI haemorrhage) or it may be physiological (e.g. pregnancy)
- We tend to remember drugs as a common cause, but chronic OH may be caused by many other conditions, especially Parkinson's disease, dementia and diabetes
- Ageing-related physiology is a major contributor in a lot of patients and will be exacerbated by polypharmacy and other factors such as dehydration
- Medication as a cause: as well as antihypertensives other drugs can commonly cause OH; it is estimated that 250 medications can cause OH, but the DTB has compiled a pragmatic list of 'common culprits' - see box below
- Further investigation will be guided by individual clinical assessment and likely aetiologies, but likely to include ECG and blood tests

KISS: Causes of Chronic Orthostatic Hypotension [BMJ2021;373:n922](#) & [DTB 2020;58:166](#)

- **Alcohol**
- **Medication**
 - **Cardiovascular Drugs**
 - Diuretics; ACEi and ARB; CCB; Beta Blockers; Alpha blockers
 - Nitrates, and other vasodilators such as sildenafil
 - **CNS Drugs**
 - Tricyclics; Antipsychotics; Anti-Parkinson's drugs
 - **Antimuscarinic drugs** e.g. oxybutynin, solifenacin, tolterodine
- **Neurogenic OH**
 - Neurodegenerative diseases, such as Parkinson's and dementia with Lewy bodies
 - Diabetes mellitus
- **Cardiovascular disease**
 - Heart failure, aortic stenosis, arrhythmia including paroxysmal AF
- **Endocrine disease**
 - Addison's and adrenal insufficiency, thyroid disease, chronic liver disease, diabetes insipidus

► *How should we manage chronic OH?*

The DTB paper proposes a simple 4-step plan (see KISS box below). They state the evidence base for this is poor due to lack of evidence, and it is a guideline that is based on low-quality evidence and expert recommendations. The evidence base for fludrocortisone is weak, and a recent systematic review [Cochrane 2021](#) which was reviewed in the [DTB Dec 2021](#) found very low certainty evidence of benefit. Importantly the aim of management is to reduce symptoms and falls risk, not necessarily to correct the drop in BP.

KISS: Management of Chronic Orthostatic Hypotension

Based on [DTB 2020;58:166](#) and [BMJ2021;373:n922](#)

- **Step One, further investigation**
 - Exclude acute or reversible cause e.g. alcohol
 - Investigations (ECG, bloods including FBC, U&E, LFT, glucose, TSH, B12, folate)
 - Other investigations as per clinical picture e.g. echo if structural heart disease suspected
- **Step Two, medication review**
 - Review of likely 'culprit' medication including anti-hypertensives, diuretics, tricyclics, antipsychotics, antiparkinsonian drugs and antimuscarinics
 - Stop alpha-blockers, beta-blockers and diuretics in preference to other antihypertensives
 - Substitute any of the above with ACEi, ARB or CCB
- **Step Three, non-pharmacological measures**
 - Change position slowly (!) and get up slowly and in stages from lying to standing
 - Avoid provocative factors e.g. increased temperature, alcohol, large meals etc
 - Lifestyle modification e.g. increased fluid intake, physical activity, sleep with head of bed elevated
 - Maintain good hydration and consider volume expansion e.g. bolus drinking of 500mls water on waking; increase water consumption to a total of 2-3L day; increase salt consumption by 1-2 teaspoons; caution in patients with hypertension, heart failure or oedema
 - Physical counter manoeuvres; compression stockings have weak evidence
- **Step Four, pharmacological management**
 - Only consider when all of the above has not been effective
 - Fludrocortisone 50 - 300 mcg per day
 - Weak evidence, but most commonly used treatment and infrequent adverse events
 - Check serum K is in normal range and 2 to 4 weeks after starting, thereafter 6 monthly
 - Start at 50 mcg daily in frailty, otherwise 100 mcg daily
 - Assess response at 1 to 2 weeks and increase by dose increments of 50-100 mcg
 - If bothersome oedema occurs, reduce dose or stop; do not commence diuretic
- **Step five, referral**
 - If symptom control inadequate despite above measures and/or recurrent falls and syncope
 - In specialist setting may also be prescribed Midodrine (a prodrug of an alpha agonist)

NB Learning Points & Quality Improvement Ideas

- Chronic orthostatic hypotension is common, affecting up to 17-19% of older adults and as many as ~30% living in care settings.
- It is associated with common symptoms e.g. dizziness, lightheadedness and fatigue, but may be asymptomatic in as many as 2/3.
- For diagnosis in primary care, arrange an appointment for an Active Stand Test (see KISS above).
- For patients with hypertension at risk of postural hypotension (those >80, with diabetes or any possible symptoms) do supine/sitting BPs and standing BPs - if either a significant postural drop is found or they have symptoms of postural hypotension, treat to BP target based on standing blood pressure.
- Management (see KISS) revolves around medication review and lifestyle guidance.
- Fludrocortisone may be considered as a trial in the community if simple measures failure.

LIPIDS AND STATINS

In January 2023 NICE produced the long-awaited DRAFT version of their lipid modification guideline [NICE 2023, CG181 DRAFT](#). As by now you will well know, they have lowered the risk threshold to consider offering statins below the 10% level of 10-year CVD risk. It has been estimated this could lead to 15 million more people being potentially eligible, with massive workload implications for us. See the key recommendations in the box below. There are a number of controversial points in them, so it may well be that they change when the definitive guidance is published (scheduled for May 2023).

KISS: NICE Draft Guidelines 2023 [NICE 2023, CG181 DRAFT](#)

- **Estimate CVD risk** of the practice population using CVD risk factors already recorded in their records
 - Use the **QRISK3 tool** to estimate risk
 - Recognise that CVD risk may be underestimated in people already taking meds for CVD prevention, HIV, people who have recently stopped smoking and people with systemic autoimmune and systemic inflammatory disorders
 - Consider using a lifetime risk tool for people aged under 40 with risk factors
 - Prioritise people for a formal assessment if their 10-year risk is $\geq 10\%$
 - Do not use QRISK3 for people with familial hypercholesterolaemia, type 1 diabetes or CKD with eGFR < 60 and/or microalbuminuria - consult the individual guidelines for these conditions
- **Advise and support lifestyle changes for primary and secondary prevention of CVD**
- **Statins should be *offered* for primary prevention for all patients with a 10-year CVD risk $> 10\%$**
 - NICE estimates that a 10-year CVD risk of 10%, the NNT to prevent major CVD event would be 25
 - Offer atorvastatin 20mg or 40mg
- **Statins can be *considered* for primary prevention as part of a shared decision-making process, even if their 10-year CVD risk is $< 10\%$**
 - NICE estimate that a 10-year CVD risk of 5%, the NNT to prevent a major CVD event would be 50
 - Consider atorvastatin 20mg if CVD risk is $< 10\%$, and the person is happy to take a statin or there is concern that the patient's risk of CVD may be underestimated
- **For people aged 85 and older**
 - *Consider* atorvastatin 20mg as statins *may* reduce CV events
- **Secondary prevention**
 - Start atorvastatin 80mg in people with CVD
 - Use a lower dose if patient preference, potential drug interactions or adverse effects
- **Baseline tests**
 - Advise women of childbearing potential to stop taking statins if pregnancy is suspected, and if planning a pregnancy to stop 3 months before they attempt to conceive
 - Before starting statins do full clinical assessment (e.g. smoking, alcohol, BP, BMI) and perform baseline bloods (full lipid profile including TC, non-HDL, HDL and Tg plus diabetes status, renal function, ALT or AST and TSH)
 - Before offering a statin, ask if patient has had generalised unexplained muscle pain, tenderness or weakness and if they have measure creatinine kinase
 - If CK is raised but < 5 times upper limit of normal, start statin at lower dose
 - If CK raised > 5 times upper limit of norm, rpt CK after 7 days and if still $> 5x$ norm don't start statin
 - Advise patients that the risk of muscle pain, tenderness or weakness with statin use is small; if they develop unexplained muscle symptoms, advise them to seek medical advice and check CK
 - Do not measure CK in asymptomatic people
- **Follow-up and targets**
 - Measure lipids at 3 months of treatment, and aim for a $> 40\%$ reduction in non-HDL
 - If a $> 40\%$ reduction in non-HDL is not achieved, then discuss adherence, optimise diet and lifestyle measures and consider increasing dose to 80 mg if judged to be at higher risk
 - Measure ALT or AST at 3 months, and at 12 months but not again unless clinically indicated
- **For other treatment options including novel drugs and combination therapy**, see [NICE TA guidance for lipids page](#) and discussion below

Other Hot Topics in lipid management include a **flurry of novel lipid-lowering drug options that have appeared in the last couple of years**. In 2021 we had NICE approval for **bempedoic acid** ([NICE TA694](#)) and **inclisiran**, the new LDL-lowering drug given by 6 monthly subcutaneous injections ([NICE TA733 Oct 2021](#)). Then in 2022, a new triglyceride-lowering drug **icosapent ethyl** came to the party ([TA805 July 2022](#)). The introduction of these drugs has not been without controversy, especially with regard to inclisiran, and the blurring of the boundaries and relationships between drug companies, NHSE and NICE has been questioned as a worrying trend ([Drug and Therapeutics Bulletin 2022;60:102-107](#)).

One of the wider implications of these new drugs is the aim to reach lower lipid levels in high-risk patients, especially those with established CVD, in line with the latest evidence and guidelines. Pending the new NICE guideline, **NHS England have produced a recent summary guideline** incorporating all lipid guidance in one document - it was initially published in 2020 but had a recent review in December 2022 ([NHSE Summary of lipid Guidelines](#)).

However, despite all the excitement about these new lipid drugs, there is no doubt that one of the main drivers for their introduction has been the **issue over side effects (or perceived side effects) of statins**, leading to underuse and poor concordance with this class of drug. It has been a perennial topic in primary care, but new research adds important evidence to this debate in the form of two meta-analyses ([Lancet 2022; 400: 832–45](#)) & ([BMJ 2021;374:n1537](#)), and a fascinating no-of-1 trial ([BMJ 2021;372:n135](#)). They all suggest that **most of the reported statin side effects are not down to the drug themselves but other factors**. If we could convince more people to stay on their statins there may be less need for these expensive new lipid agents, which still have somewhat limited evidence bases...

1 Do statins cause muscle-related or other side effects?

► **Statin therapy and muscle symptoms** [Lancet 2022; 400: 832–45](#) & [NB Blog Sept 2022](#)

Whether, or how much, statins cause muscle side effects has remained a divisive topic for years, so this latest meta-analysis helps move the debate on in a positive way. The latest offering from the CTTC (Cholesterol Treatment Trialists' Collaboration) is both independent and rigorous in its data collection. **Importantly they sought new individual patient-level data and so were able to analyse more detail on both the type of muscle symptoms reported, as well as the timing of any symptoms in different types of patient, and for different statin regimes**. So what did they find?

- The main meta-analysis reviewed high-quality double-blind RCTs (19 trials = 123,940) and the authors also reviewed a further 4 RCTs (n=30,724) looking at intensive vs less intensive starting regimes.
- In the main meta-analysis the mean age was 63 years, 27.9% were female, about ~50% had known CVD and 18.5% had diabetes - mean follow-up was 4.3 years.
- The authors looked at a wide range of reported muscle symptoms including pain, cramps, spasms, fatigue and weakness.
- **A huge number of people (>25%) reported muscle-related symptoms whether they were taking statins (27.1%) or placebo (26.6%).**
- This equates to a **small 3% increase in reported muscle symptoms in the statin vs placebo group** (RR = 1.03, CI = 1.01-1.06) and there was **no significant difference between statin type**.
- Muscle-related symptoms were reported at a **relatively higher level in the first year** in the statin vs placebo group (RR = 1.07, CI = 1.04-1.10) and after year 1 there was no difference in reported symptoms between the two groups.
- Regarding **more intensive** (e.g. 40-80mg atorvastatin or 20-40mg rosuvastatin) vs **less intensive statin regimes** there was a slightly higher rate of muscle-reported symptoms: RR = 1.08, CI = 1.04-1.13 for intensive regimes vs. RR = 1.03, CI = 1.00-1.05 for less intensive regimes compared to placebo; this excess continued in the intensive regimes beyond year one, but to a lesser degree.
- Regarding **serious muscle side effects** there was a **very small excess risk in the statin groups** - myopathy was reported in 0.08% in the statin groups and 0.04% in the placebo groups which equates to a very small absolute excess of only 0.08 per 1000 person-years.

- In the small number of trials that reported CK levels only 3.3% of cases had CK levels > 3x the upper limit of normal ([NHS guidance](#) recommends continuing on statins if CK levels <4x upper limit of normal and symptoms are tolerable).
- **Conclusions:** Statins do cause a **very small increase in muscle-related side effects, mostly in the first year**, with a slightly higher risk if on higher intensity regimes; however **>90% of muscle-related symptoms reported are NOT due to the statin.**

This is important research and confirms that the majority of muscle-related symptoms reported by patients are not caused by the statin, especially if they have tolerated the statin OK for a year and then develop symptoms later. **Importantly the authors found no evidence of incomplete adverse event reporting in the trials, and also no evidence of 'bias by exclusion'** - one of the criticisms cited in the past is that previous data/results have excluded people with significant co-morbidities, but this data showed **similar risks of muscle symptoms even in those with severe co-morbidities** e.g. CKD on dialysis, Stage II-IV heart failure.

To put this in context regarding a risk/benefit assessment of statins - If you put 1000 people on a moderate intensity statin (e.g. atorvastatin 20mg) for 5 years you would expect:

- 11 episodes of statin-induced muscle pain or weakness.
- But you will prevent 50 major CVD events if you do have pre-existing CVD (secondary prevention) or 25 major CVD events if on the statin for primary prevention.
- If you are on a higher-intensity statin (e.g. atorvastatin 40-80mg) you will double the number of CVD events prevented in both primary and secondary prevention groups, with only a slightly higher risk of muscle-related side effects.

There is useful guidance on how to manage statin intolerance from the [NHS here](#).

► ***Do statins cause myalgia or other side effects?*** [BMJ 2021;374:n1537](#)

Previous to the study above this meta-analysis looked at a wider range of potential statin side effects, as well as the benefits from statins. Although it looked predominantly at primary prevention (the study above had about ~50% of patients on statins for secondary prevention) the overall themes/results were very similar:

- 62 RCTs in total with >120,000 patients, average follow-up 3.9 years.
- The majority of patients were on statins for primary prevention (just 6% had confirmed CVD).
- There was a **small increased risk of self-reported muscle symptoms in the statin groups** OR 1.06, CI = 1.01-1.13), but **no increased risk of serious/clinically confirmed muscle disorders.**
- There was also a small but significant increased risk of liver (OR 1.33, CI = 1.12-1.58) and renal dysfunction (OR 1.14, CI = 1.01-1.28).
- There was also an associated increased risk of eye conditions (OR 1.23, CI = 1.04-1.47) - a mixture of cataracts and other eye conditions.
- *There was no increased risk of diabetes.*
- Comparisons between type of statin did not show substantial differences.
- Importantly the results showed a **significant CVD benefit in the statin groups**:
 - Reduction in MI (OR 0.72, CI = 0.66-0.78)
 - Reduction in stroke (OR 0.80, CI = 0.72-0.89)
 - Reduction in death from CVD (OR 0.83, CI = 0.76-0.91)

So as with the more recent meta-analysis, this **data concludes that statins do cause side effects** (which drug doesn't?), including non-serious muscle symptoms, liver dysfunction (defined as rises in transaminases >3x baseline, but we know that is rarely serious and often transient), renal dysfunction (decline in renal function or presence of proteinuria), and some eye conditions. But the **majority of these side effects are not serious and may be transient.**

To put this in context if treating 10,000 people for 1 year with statins, you would expect:

- **Side effects:** 15 extra people with self-reported muscle symptom, 8 extra people with liver dysfunction, 12 extra people with renal dysfunction and 14 extra people with eye conditions.
- **Benefits:** But in contrast you **would prevent** 19 MIs, 9 strokes and 8 deaths from CVD.

The study concludes that although risks are present with statins the *'benefit-to-harm balance of statins for primary prevention of cardiovascular disease is generally favourable'*. Regarding the specific issue of muscle symptoms, the authors suggest we highlight to patients that in most cases the symptoms are not attributable to the statin and that we need to explore patients misconceptions of intolerance (easier said than done!), and they suggest that in the future we might consider behavioural interventions including *'no-of-1 trials to minimise unnecessary withdrawal of treatment'*....

► **Statins and the nocebo effect** [BMJ 2021;372:n135](#)

As discussed in the [associated editorial](#) to this study, what happens when someone sitting next to you says they have nits? You start feeling itchy! This is an **example of the nocebo effect** and is thought to be a large reason why people taking statins experience muscle side effects - *'people expect to experience muscle symptoms, perhaps because of media reports, and therefore any small ache or pain, which might have happened anyway, is considered by the patient to be attributable to their medication'*. We often try to take people off their statin then try again, but crucially they always know when they are on the statin or not, meaning the nocebo effect continues to play out. **How can we disentangle this and remove the nocebo effect for the individual?** Step forward the **placebo-controlled n-of-1 trials** published in the BMJ recently - possibly the trial of the year, both clever and powerful in equal measure.

So what is a no-of-1 trial? The premise is simple. You give individual patients active drug or placebo for blocks of time, but the timing of when you have active drug or placebo is randomised and fully blinded throughout. Then you monitor side effects whilst on the active drug or placebo, building up a profile of whether side effects (in the individual) are more prominent when on the active drug or not. Brilliant.

- The above study recruited 200 patients from **UK primary care who had recently stopped or were considering stopping their statins due to side effects**.
- They then put all of them through a series of no-of-1 randomised placebo-controlled trials.
- They all had **6 treatment periods of 2 months each of either placebo or atorvastatin 20mg**.
- The results confirmed what the majority of RCTs have shown - statins did not cause any significant increase in muscle symptoms - ***there was no difference in mean muscle symptom scores between the statin and placebo periods***.
- Withdrawals due to intolerable side effects were similar between statin (9%) and placebo (7%) periods.
- Importantly **2/3 of people completing the trial reported restarting long-term statin treatment**.

The associated editorial discussed the results, reporting that many participants found the trial helpful in proving to them (as an individual) that any ***muscle symptoms were not necessarily due to the statin and gave them the confidence to carry on with treatment***.

2 Novel Lipid Lowering Drugs

► Inclisiran

Inclisiran is a novel, cholesterol-lowering drug given by a 6 monthly injection. NICE took everyone by surprise in October 2021 when they recommended its use in selected secondary prevention patients, with the intention that it be given by us in primary care [NICE TA733 Oct 2021](#). This was widely reported in the media, who dubbed it the new 'anti-cholesterol vaccine'. However, it is a decision which has caused disquiet in many sections of the medical community with a recent BMJ editorial [BMJ 2021;375:n2462](#) and a [RCGP and BMA Statement](#) saying that the guidance should be reconsidered.

Inclisiran is not a vaccine! it is a novel drug - a 'small interfering RNA' drug (siRNA) - that interferes with synthesis of PCSK9, leading to up-regulation of LDL receptors which increases LDL clearance from the blood and lowers LDL-C levels.

The evidence is based on 2 key trials - [ORION-10](#) (atherosclerotic CVD and LDL-C ≥ 1.8 mmol/mol) and [ORION-11](#) (mixed secondary prevention as per ORION-10 plus some primary prevention with risk factors and LDL-C ≥ 2.6). Most patients were on statins (89% in ORION-10 and 95% in ORION-11) with the majority on high intensity statins (69% in ORION-10 and 78% in ORION-11), although only small numbers were on ezetimibe (11% and 9% respectively) **Both placebo controlled RCTs showed a reduction of LDL-C of approx 50% maintained by 18 months.** It's important to note **two important caveats** to the data so far: We have NO hard CVD outcome data yet on this drug (all studies so far have used LDL-C as a surrogate marker) and we have NO comparative data comparing this drug to any current lipid-lowering therapy. The lack of data on clinically meaningful outcomes is a major concern [BMJ 2021;375:n2462](#). A UK-based study [ORION-4](#) is planned to report in 2026 with CVD outcomes.

Thus far no significant difference in any side effects vs placebo has been noted, apart from local injection site reactions (8.2% Inclisiran vs 1.8% placebo). Although this safety data is reassuring as the BMJ editorial points out the trials have only been 18 months in duration and for a drug intended for lifelong use there remains considerable uncertainty over long-term risk-benefit [BMJ 2021;375:n2462](#).

Who can have inclisiran, how is it administered and why have NICE recommended it?

- Inclisiran is recommended by NICE [NICE TA733](#) as an option for primary hypercholesterolaemia or mixed dyslipidaemia (as an adjunct to diet) if:
 - **Secondary cardiovascular prevention** with a history of any of the following CVD events: ACS, coronary or other arterial revascularisation, CHD, ischaemic stroke, PAD, *and*
 - **LDL-C is persistently ≥ 2.6 mmol/mol despite maximal lipid-lowering therapy:**
 - That is max tolerated statin with or without other lipid-lowering therapies (e.g. ezetimibe), or
 - Other lipid-lowering therapies when statins are not tolerated or contraindicated.
- It is NOT recommended for primary prevention (including those with familial hypercholesterolaemia), unless part of a research trial.
- It is administered via subcutaneous injection, first 2 doses 3 months apart then twice yearly.
- Inclisiran has also been accepted for use in Scotland [SMC 2021](#), but is restricted to specialist use only.
- NICE clearly feel there is a 'gap' in treatment for those at high risk of CVD (i.e. secondary prevention) who still have elevated LDL-C ≥ 2.6 despite statins (or due to intolerance of statins), yet who are not eligible for the PCSK9 inhibitors (need LDL-C ≥ 3.5 or 4.0 depending on high risk/very high risk). NICE state in their accompanying press release that the drug is *'a potential game-changer in preventing thousands of deaths and can be given in primary care settings'*
- They also note **significant barriers to PCSK9 inhibitors** even if patients are eligible. They are very expensive, have to be prescribed by specialists and for injections every 2-4 weeks.
- They found the drug to be cost-effective for secondary prevention, but crucially this is dependent on a **cost-reduction agreement with Novartis**. This agreement is commercially sensitive and has not been put into the public domain, but the uncertainty regarding cost is another concern raised in the BMJ editorial [BMJ 2021;375:n2462](#)

What are the potential implications for primary care?

- NICE are clear *'Inclisiran is likely to be used in a primary care setting'*.
- We would need to check LDL-C levels in those with established CVD and [NHS data](#) suggests 300,000 patients will be eligible for this, with potential significant workload implications for us in primary care
- How this rollout will actually happen in practice is still very uncertain, with many questions still to be answered, including how the extra work will be funded. A feasibility study looking at delivering inclisiran in a primary care setting The SPIRIT trial is underway in Manchester and due to report later in 2022
- A broader implication, however, is the need to look more critically at LDL levels in high risk patients, and to escalate treatment to high-dose statins +/- ezetimibe to reach lower targets (see guidelines section below) in high risk patients. Both the [European Lipid Guidelines 2019](#) and also the [NHSE Lipid Guidelines 2020](#) recommend trying to reach a target LDL of < 1.8 mmol/L in secondary prevention patients.
- The [RCGP and BMA Statement](#) also raises concerns about the 'black triangle' status of a new drug being widely rolled out in primary care.

So, should we prescribe inclisiran now?

It would be fair to say that the **inclisiran recommendations have been far from universally supported!** There have been considerable concerns raised, not only about the lack of underlying hard data and long-term safety profile, but also the lack of transparency of the 'deal' between NHSE and Novartis as well as concerns about the suitability of the relationship between drug company, Government, regulatory bodies and recommending bodies (NICE). An extensive review in the [Drug and Therapeutics Bulletin 2022;60:102-107](#) *'Inclisiran: more hope less hype'* discussed a number of the issues:

- **Lack of proper outcome data** - all regulatory bodies have highlighted that although inclisiran reduces LDL in the short term, we have no evidence yet as to whether this will translate into improved CVD outcomes - *'the trials' primary and secondary outcomes were surrogate endpoints and did not include any patient-centred measures such as effect on mortality or CVD.'*
- **Safety** - we still have no long-term safety data and the DTB reminds us of **two cautionary tales** of previous lipid drugs that were shown to reduce lipids levels significantly in early trials but were then withdrawn due to significant increases in rhabdomyolysis (cerivastatin) and increased CVD events and death (torcetrapib).
- **Effect attenuation over time** - given the nature of the drug there is a risk that antibodies may develop over time attenuating treatment effect. The DTB cites a PCSK9 inhibitor (bococizumab) whose development was stopped due to nearly half of the participants developing anti-drug antibodies and nearly a third neutralising antibodies. Indeed over the 18 months of treatment in the trials rates of **anti-drug antibodies to inclisiran went up from 1.8% to 4.9%.**
- **In summary**, the final paragraph from the DTB sums up *'The introduction of inclisiran has been actively facilitated through a commercial arrangement between the company and NHS England and Improvement. We share the concerns that have been raised about the lack of transparency over the commercial arrangement with the company and the enthusiasm with which a drug with no data on its impact on clinical outcomes or its long-term safety is being adopted by NHS England. We consider that it might be more appropriate to maximise the use of other lipid-lowering options that have more evidence behind them. We also suggest that it would be better to wait for the results of the ORION-4 study before raising expectations of the unproven long-term clinical benefits and unknown long-term adverse effects of a drug that has been hyped as 'revolutionary' and 'cutting-edge' on the UK Government's website.'*

The [RCGP and BMA Statement](#) also **advises we should not be prescribing this drug for now** on the basis of lack of long-term data on cardiovascular outcomes and safety, uncertainty over the additional capacity and resources which would be needed to deliver this programme and also because it is a 'black triangle' drug. Whilst waiting for clarification on these issues, they advise **focusing treatment on existing guidelines with escalation to maximal dose of high-intensity statins plus ezetimibe where appropriate.** The BMJ editorial [BMJ 2021;375:n2462](#) states that *'the drug may prove to be a game-changer, or it may not'* but given that we do not know if it will improve cardiovascular outcomes, how much it will cost and that we have insufficient safety data that NICE should 'reconsider' its decision. Watch this space...!

► **Bempedoic acid** [NICE TA694](#) [Drug and Therapeutics Bulletin 2022;60:120-124](#)

Another novel lipid-lowering drug developed to lower LDL, bempedoic acid, was given the green light by NICE back in 2021, and was recently reviewed by the DTB, with a less than glowing commendation...

Bempedoic acid is a prodrug, that when activated reduces cholesterol synthesis in the liver by blocking an enzyme involved in cholesterol biosynthesis, and also reduces LDL in the blood by up-regulating LDL receptors. The evidence base for its use is currently, at best, thin. Like inclisiran, there is no data on CVD outcomes or mortality, and no head-to-head data with other lipid-lowering drugs. The data is very short term indeed, mostly only 3 months:

- Pooled data showed LDL-C reduction for bempedoic acid vs placebo of ~25% (absolute reduction of -0.9 mmol/L)
- Pooled data for bempedoic acid with maximum tolerated statin vs placebo showed LDL-C reductions of ~18% (absolute reduction -0.5 mmol/L)
- To give context atorvastatin monotherapy reduces LDL-C by 37-55% depending on dose, and inclisiran and the PCSK9 inhibitors reduce LDL-C by ~50% either alone or in combination with statins or ezetimibe.
- Notable **side effects and interactions** include an **increased risk of hyperuricaemia and anaemia**, a **small possible increased risk of tendon rupture**, and increase statin exposure, particularly simvastatin, leading to **SPC recommendation that simvastatin dose be limited to 20mg if used concurrently with bempedoic acid**.

Given the very limited data (and at best modest effects) it was a surprise that NICE approved this drug in 2021. NICE recommend that **Bempedoic acid is now an option *in combination with ezetimibe* for both primary and secondary prevention** (as separate tablets or fixed-dose combination) for those in whom statins are contraindicated or not tolerated AND who don't reach LDL target with stand-alone ezetimibe. Like inclisiran, the true cost-effectiveness of this drug is unknown as the drug cost is hidden behind another 'commercial agreement', and although not cheap at ~£55/month it is considerably cheaper than the eye-waveringly expensive PCSK9 inhibitors (~£340/month to >£1000/month), although to put in context the monthly cost of 80mg atorvastatin is £1.50...The DTB conclude that 'it is difficult to see a major role for bempedoic acid in the primary and secondary prevention of CVD.'

► **Do PCSK9 inhibitors and ezetimibe have clinical outcome data to support their use?**

This is an important question given that we are asking these questions about the newest LDL-lowering drugs above. A recent systematic review and network meta-analysis [BMJ 2022;377:eo69116](#) has found that in patients who are either taking maximally tolerated statins or who are statin intolerant, the addition of ezetimibe or PCSK9 inhibitors may reduce non-fatal MI or stroke in patients at high/very high CVD risk but not in those with moderate/low risk.

► **Icosapent ethyl**

So the latest lipid-lowering kid on the block, icosapent ethyl (Vazkepa), is another novel treatment, this time focussing on lowering triglyceride levels. Although the main focus on reducing CVD risk revolves around non-HDL-C or LDL, it is also known that raised triglycerides are a risk factor for poor outcomes and until this point, we have had no triglyceride-lowering drugs that improve outcomes (from a CVD point of view). Back in 2019, The **REDUCE-IT trial** was published ([N Engl J Med 2019; 380:11-22](#)), and on the back of that the ESC added it to its recommendations that year. Summary of that trial:

- Double-blind placebo controlled RCT comparing **icosapent ethyl 2g BD to placebo** in 8179 patients with established CVD or diabetes with other risk factors.
- **All patients were on statins**, although only a small percentage were on ezetimibe as well (6.4%).
- Overall ~70% had established CVD and the remaining 30% were for primary prevention.
- **Primary outcome** was a composite of CV death, nonfatal MI or stroke, coronary revascularisation, or unstable angina, and the median follow-up was 4.9 years.

- There was a **significant reduction in the primary outcome** in the icosapent ethyl group 17.2% vs placebo 22.0% (HR 0.75, 95% CI = 0.68-0.83), as well as a number of secondary outcomes (including CVD death 4.3% vs 5.2%, HR 0.80, 95% CI = 0.66-0.98)
- There was an **increase in rates of hospitalisation for AF in the icosapent ethyl group** (3.1% vs 2.1%) as well as serious bleeding events (2.7% vs 2.1%).
- **Conclusion:** *Among patients with elevated triglyceride levels despite the use of statins, the risk of ischemic events, including cardiovascular death, was significantly lower among those who received 2 g of icosapent ethyl twice daily than among those who received placebo.*

In July 2022 NICE published guidance ([TA805](#)) recommending **Icosapent ethyl** as an option:

- **If they have established CVD** (i.e. secondary prevention only) AND
- **Are taking statins** AND
- **Have raised FASTING triglycerides of ≥ 1.7 mmol AND**
- **LDL between 1.04 mmol/L-2.6 mmol/L**
- There were some concerns raised by NICE on the applicability of the trial population to usual UK populations, and the drug was clearly close to the cost-effectiveness threshold of £20,000-£30,000 per QALY, as it is not cheap (~£144/month) but overall NICE felt there was enough evidence for use and that it is **likely to be cost-effective in the secondary prevention setting.**

3 Lipid and Statin Guidelines

► NHS England Summary of Guidance for lipid management

Pending the new NICE guideline, we have had multiple lipid-related TAs from NICE relating to combinations and novel drugs [NICE Lipid TAs page](#) making for a very complicated and unworkable situation. So in 2020, NHS England published a very useful document ([Summary of national guidance for lipid management](#)) combining all the lipid recommendations into 2 pages. This was recently reviewed and updated in Dec 2022. This is summarised in the KISS below.

► Triglycerides - current guidance [CKS 2022](#) [NICE CG181 2016](#) [ESC 2019](#)

- Triglyceride (TG) levels are strongly associated with CVD, however when you control/adjust for non-HDL levels the association is significantly diminished, although CKS state elevated TGs are still an independent risk factor for CVD.
- Although **lifestyle factors** can affect non-HDL/LDL levels, they have **much more of an effect on TGs**. Whereas weight reduction has limited effect on LDL, it has substantial effects on TGs, and exercise improves TGs above and beyond the weight-lowering effect. Alcohol is a strong driver for elevated TGs, as is refined carbohydrate and sugars.
- Generally, non-fasting lipid profiles are recommended but if raised TGs are found a repeat **fasting sample is needed**.
- The CKS recommends the following regarding **raised TGs**:
 - **TG > 20 mmol/L** not due to excess alcohol/poor glycaemic control - refer for urgent specialist review.
 - **TG 10–20 mmol/L** - repeat with a fasting test (after an interval of 5 days, but within 2 weeks), review for secondary causes, and seek specialist advice if the triglyceride concentration remains > 10 mmol/L.
 - **TGs 4.5–9.9 mmol/L** — be aware CVD risk may be underestimated by risk assessment tools and optimize the management of other CVD risk factors present.
- **Drug options?**
 - **Statins remain the most effective treatment for raised TGs** and lower TG levels by 10-20% and lower global CVD risk; they are not recommended specifically for elevated TGs by NICE but the ESC recommend them for high risk patients with TGs > 2.3 mmol/L.
 - **Fibrates** have limited evidence for CVD protection; the ESC recommend them as an option for people with ongoing elevated TGs despite statins, but they are actively **not recommended by NICE**.
 - **Icosapent ethyl** - now recommended by NICE for high risk people with elevated TGs - see above.

NB Learning Points & Quality Improvement Ideas: Lipids and Statins

- New **NICE 2023 DRAFT guidance** recommends considering statins for patients with a < 10% 10-year risk; this may change, pending definitive guidance later in 2023
- **Inclisiran** is a novel LDL-lowering drug which works on the PCSK9 pathway. It is given by a twice-yearly injection and has been approved by NICE as a drug to be given in primary care to secondary prevention patients with a persistent LDL-C ≥ 2.6 mmol/L despite maximal lipid-lowering therapy.
- Although there is evidence that inclisiran lowers LDL by about 50%, **we lack clinical outcome data**. This and uncertainty over long-term safety and costs have led to a BMJ editorial, the BMA and RCGP advising that NICE should re-consider their position pending further research and data.
- **Icosapent ethyl** is a new triglyceride-lowering drug which is now recommended in the UK following NICE guidance for people with **Established CVD** (i.e. secondary prevention only) **AND Are taking statins** **AND Have raised FASTING triglycerides of ≥ 1.7 mmol** **AND LDL between 1.04 mmol/L-2.6 mmol/L**.
- A trio of research papers in 2021/2022 have concluded that statins only rarely cause muscle symptoms, and much of the reported muscle side effects may be due to the 'nocebo' effect or other causes. The latest meta-analysis concludes that **>90% of reported muscle symptoms are NOT due to the statin**. This is particularly true for people who have tolerated statins well for >1 year then developed muscle-related symptoms.
- This evidence confirms that the benefits in terms of CVD reduction of statins, vastly outweigh the small number of side effects and the latest meta-analysis concludes that if treating 1000 people for 5 years with 20mg atorvastatin, you would expect:
 - 11 episodes of statin-induced muscle pain or weakness.
 - But you will prevent 50 major CVD events if you do have pre-existing CVD (secondary prevention) or 25 major CVD events if on the statin for primary prevention.
- A **collaborative lipid guideline** has been produced by NHSE and NICE which nicely summarises the totality of lipid management.

Quality Improvement Ideas:

- **Review and circulate the joint lipid guideline** to your colleagues, which is an excellent summary of lipids management for both primary and secondary prevention - [click here](#)
- Discuss the latest Lancet meta-analysis on statins and muscle-related side effects with your colleagues and aim to **bring that data into discussions with patients who develop muscle-related symptoms on statins** - reassure them that in >90% of cases the symptoms are not related to the statin and highlight the CVD benefits from the study above. If struggling with patients and statin intolerance use the [NHS guidance on statin intolerance here](#).
- Focus CVD risk reduction on your highest-risk patients who have established CVD:
 - **Audit patients with established CVD** aim for a reduction of non-HDL-C of > 40% from baseline; consider target of non-HDL-C < 2.5 mmol/l or LDL < 1.8 mmol/l if baseline non-HDL-C not available.
 - If they are not at target are they on maximal statin dose - if not titrate to maximum dose? If so are they on ezetimibe? If they remain above target despite maximal statin and ezetimibe do they fulfil criteria for referral to the lipid clinic for consideration of an injectable lipid-lowering drug?

KISS: Lipids and Statins

Based on [NHSE Lipid Guidelines](#)

Risk assessment

- **Assess CV risk** for primary prevention e.g. using the [QRISK3](#) score.
- QRISK can be used up to the age of 84, after that consider patients to be at risk on the basis of age alone particularly people who smoke or who have raised BP.
- QRISK3 can also be used in people with CKD and diabetes, although NICE recommend we don't use QRISK score for those with type 1 diabetes.
- In Scotland SIGN recommend the [ASSIGN score](#) - this is specifically tailored to the Scottish population, but does not include some of the risk factors in QRISK3 (e.g. migraine).
- Do NOT use QRISK if severe hyperlipidaemia and familial hypercholesterolaemia suspected (TC > 7.5 mmol/l and/or LDL > 4.9 mmol/L and a personal or FH of confirmed CHD aged < 60).

Baseline assessment

- Before starting treatment, assess background risk factors and perform baseline bloods:
 - Full lipid profile inc. triglycerides (non-fasting sample), HbA1c, renal function, ALT or AST, TSH.
- If total cholesterol (TC) is >7.5 consider familial hypercholesterolaemia (see separate chapter); if TC >9 or non-HDL cholesterol is > 7.5 refer for specialist assessment irrespective of family history.
- NB - **Non-HDL cholesterol** = TC - HDL.

Initial Treatment

- **Lifestyle 'treatment' should remain the cornerstone of CVD reduction in all patients:**
 - Stop smoking, exercise (ideally 150 mins moderate-intensity aerobic activity/week as well as some muscle-strengthening activities), reduce alcohol (max 14 units/week for men and women, avoid binge drinking), diet using principles of the ['Mediterranean' diet](#)
- Do not recommend plant stanols or sterols.
- **Primary prevention:**
 - Assess and correct all secondary causes of dyslipidaemia before starting statins.
 - Offer statin treatment to those with a ≥10% 10-year risk (NICE recommendation England/Wales).
 - in Scotland SIGN recommend to offer statin if 10-year CVD risk ≥20%.
 - **Atorvastatin 20mg** is recommended as the primary prevention drug of choice.
 - Consider statin for those aged over 85 based on clinical judgement & patient choice.
 - Statins can be used if the upper limit of transaminases is <3 times upper limit of normal.
- **Secondary prevention:**
 - Patients with **established CVD** should be offered high-dose 80mg of atorvastatin.
 - Use a lower dose if high risk of adverse events or interactions or patient preference.
 - Do not delay statin treatment in secondary prevention.
- **Primary prevention for people with Diabetes:**
 - Type 1 diabetes - offer atorvastatin 20mg if aged >40, have had diabetes for >10 years, have established nephropathy or have other CVD risk factors.
 - Type 2 diabetes - assess/treat as for primary prevention above.
- **CKD:**
 - Offer atorvastatin 20mg for primary or secondary prevention of CVD with CKD.
 - If eGFR is <30, specialist advice on higher doses is needed, otherwise titrate as usual.
- The following are **NOT routinely recommended for primary or secondary prevention of CVD:**
 - Fibrates.
 - Bile acid sequestrants.
 - Omega-3 fatty acid compounds.



KISS: Lipids and Statins continued

Targets/further treatment: ([click here](#) for NHSE flowchart summary)

- **NICE/SIGN:** check lipid levels after 3/12, and if a **>40% reduction in non-HDL cholesterol** has not been achieved to discuss adherence, take at night, optimise lifestyle advice and consider a dose increase (up to a maximum of atorvastatin 80mg) based on clinical judgement.
- **NHSE Guidance for secondary prevention** - aim for reduction of non-HDL-C of > 40% from baseline; consider target of non-HDL-C < 2.5 mmol/l or LDL < 1.8 mmol/l if baseline non-HDL-C not available.
- **Consider combination therapy if goals are not reached with statins:**
 - If target not achieved with max tolerated statin add ezetimibe 10mg OD.
 - If **statin intolerance confirmed** ([click here](#) for statin intolerance pathway):
 - **Ezetimibe monotherapy** can be considered.
 - **Ezetimibe/bempedoic acid** combination can be considered if not at target (subject to local pathways) - see NICE [TA694](#).
 - **Inclisiran** recommended by NICE/NHSE as an option for **secondary prevention** if LDL-C ≥ 2.6mmol/L despite max tolerated doses of lipid-lowering drugs above (subject to local pathways).
 - **PCSK9 inhibitors** recommended by [SIGN](#) and NICE ([TA393](#) & [TA394](#)) for **secondary prevention**:
 - LDL-C > 4.0 mmol/L, or
 - LDL-C > 3.5 mmol/L if CV events in ≥ 1 vascular bed (i.e. polyvascular disease).

Monitoring

- Advise patients to report **muscle pain or weakness** and if this occurs check CK levels; if CK normal explore with patient other causes of symptoms and that evidence ([BMJ 2021;372:n135](#) & [BMJ 2021;374:n1537](#) & [Lancet 2022; 400: 832–45](#)) suggests only small numbers get genuine muscle side effects from statins; be aware that >90% of reported muscle symptoms are not caused by statins; however if intolerance unacceptable consider ([click here](#) for more detailed guidance):
 - 'De-challenge' and 're-challenge' approach.
 - Reducing dose within same group or switch to alternate e.g. atorva 10-20mg or rosuva 5-10mg.
 - Consider alternate day or twice weekly regime initially and titrate frequency of dosing.
 - If CK levels >4x upper limit of normal follow [statin intolerance pathway](#).
- **LFTs:**
 - NICE still recommend doing ALT or AST at baseline, 3 months, 12 months, and only again if clinically indicated.
 - However, LFT monitoring is being questioned with some advocating [BJGP2017;67:e194](#) a baseline ALT and then repeated only if clinically indicated or on very high dosage.

High Triglycerides (TGs)

- If **TGs > 20 mmol/L** (not caused by high alcohol or poor glycemic control) refer for urgent specialist review (high risk of pancreatitis).
- If **TGs 10-20 mmol/L**: Repeat fasting levels within 2 weeks; review for secondary causes - particularly alcohol, but also obesity, hyperglycaemia, dietary fructose and sucrose; seek specialist advice if TGs remains > 10 mmol/L.
- If **TGs 4.5-9.9 mmol/L**: Modify risk factors where possible and be aware CVD risk is likely to be underestimated - **consider statins** as standard CVD risk calculators will underestimate true CVD risk in this group; European guidelines recommend statins for people at high risk (which they define as anyone with significant risk factors e.g. diabetes, CKD) with TGs >2.3 mmol/L.
- [NICE TA 805](#) recommends **Icosapent ethyl** as an option for people who have:
 - **Established CVD** (i.e. secondary prevention only) AND
 - **Are taking statins** AND
 - Have raised **FASTING triglycerides of ≥1.7 mmol** AND LDL between 1.04 mmol/L-2.6 mmol/L



ATRIAL FIBRILLATION

In 2021 we have important new guidelines on Atrial Fibrillation [NICE, NG 196 June 2021](#), summarised in the [BMJ2021;373:n1150](#) with a useful infographic. In 2020 we also had new guidelines from the European Society of Cardiology ESC [ESC AF Management Aug 2020](#). The European guideline is broadly similar to NICE, but is broader in scope and takes a more holistic approach recommending an 'ABC' strategy to reduce overall cardiovascular risk.

Some things in AF management have become clear since previous guidelines 4 or 5 years ago, namely the **pre-eminence of DOACs for stroke reduction**. DOACs should be used now in preference to warfarin for most patients, but **which DOAC to use has become a Hot Topic**. The evidence base for which DOAC to use in preference is uncertain, but [new recommendations](#) (in England at least), which have subsequently been incorporated into the [2022/23 Impact and Investment Fund](#) component of the PCN Network Contract DES, **incentivising edoxaban over other DOACs** have certainly caused some discussion.

Yet some questions remain, notably around how we assess and manage bleeding risks with anticoagulation and also the age-old argument of rate vs rhythm control. In addition to the guidelines, a recent study in the NEJM [N Engl J Med 2020; 383:1305-1316](#) is **building the evidence for earlier rhythm control**. On the back of this, a new large observational trial was published in the [BMJ 2021;375:e066450](#) which also supports early rhythm control for new-onset AF.

1 NICE Atrial Fibrillation 2021, NG 196 [NICE, NG 196 June 2021](#)

The new NICE guidance on AF was published in 2021. There are a number of changes to the previous 2014 guidance, and whilst most of that is consolidation of our current management practices, there were a few surprises. The guidance is summarised in the KISS below, but **what are the important learning points, take-home messages and changes to previous guidance?**

► Diagnosis of AF

- NICE re-iterates that suspected AF should still be confirmed on a 12-lead ECG
- For those with suspected **paroxysmal AF**, diagnosis should be made with either a 24-hour ambulatory ECG (if suspected episodes <24 hours apart) or an event recorder (if suspected episodes >24 hours apart)
- **New devices?** (e.g. KardiaMobile, Microlife BP monitors, iPhone apps etc.). In the NICE AF guideline they conclude that generally there is still **not enough evidence to support the use of these devices for diagnosis**. However, more recently [NICE MTG64 Jan 2022](#) recommends the **KardiaMobile as an option for people with suspected paroxysmal AF as an alternative to traditional ambulatory ECG devices**, and this may be particularly useful for people with infrequent symptoms. Read our [NB blog](#) for more info.

► Does my patient need an echocardiogram?

- NICE advise an echocardiogram 'if it is important for long-term management'. This essentially means if we suspect there may be functional heart disease (i.e. heart failure) or structural heart disease (i.e. valvular disease if a significant murmur) or if referral for a rhythm control strategy is being advised
- So, if we do not suspect heart failure or significant valve disease and will not be referring for a rhythm approach (which is probably most of our patients with AF) then it is not essential, and certainly should not delay anticoagulation

► Assessing bleeding risk

- NICE recommend we now use the **ORBIT tool**, which slightly came from left-field. Previously we had used the HAS-BLED score.
- NICE assessed that the ORBIT tool was the most appropriate bleeding risk score to use, as it had the best evidence for assessing bleeding risk in people on DOACs and NICE state has higher accuracy in predicting absolute bleeding risk.

- **The ORBIT tool** uses clinical and blood test parameters to assess the risk of a major bleed (defined as fatal bleeding, symptomatic bleeding in critical organ or drop in Hb requiring >2 units blood):
 - Low Hb or haematocrit = +2 points
 - Age >74 years = +1 point
 - Bleeding history (any GI bleed, intracranial bleed or haemorrhage stroke) = +2 points
 - eGFR <60 = +1 point
 - On antiplatelets = +1 point
 - Score = ≤2 low risk, score 3 = medium risk, score ≥4 = high risk
 - [Click here](#) for an easy to use an online calculator
- However, the *key message on assessing bleeding risk remains that these tools are there to identify people at higher risk of bleeding and to modify risk factors* (eg uncontrolled hypertension, concurrent medication e.g. NSAIDs, harmful alcohol use, correcting reversible causes of anaemia) **rather than denying them anticoagulation.**
- **Should we add a PPI to provide additional gastro-protection?** This is an important clinical question that is not addressed in national guidelines. Recent evidence [Heart 2021;0:1-6](#) suggests that **PPI cover may be protective, especially in elderly patients with AF.**
- As NICE state '*for most people, the benefit of anticoagulation outweighs the bleeding risk*' and also that we should '*not withhold anticoagulation solely based on age or risk of falls*'.

► Clarification on when to use anticoagulation

- This is not new but is an important reminder and is clarified in the guidance, on **when to offer/consider anticoagulation**, based on the [CHA2DS2VASc score](#):
 - Score of ≥2 OFFER anticoagulation (a high level of evidence/recommendation)
 - Men with a score of 1 CONSIDER anticoagulation (evidence-less certain/more balanced)
 - In the < 65-year-olds with no other risk factors apart from their sex (i.e. score 0 for men or score 1 for women) DO NOT offer anticoagulation.
- **Assessment for anticoagulation should be done for people with ALL modalities of AF**, whether symptomatic or asymptomatic, permanent, persistent or paroxysmal and in people with atrial flutter. This important point was mirrored in the ESC guidance.
- Another key message reinforced in the ESC guidelines is that anticoagulation should be offered to the majority of people with AF and that there are few absolute contraindications to anticoagulation (e.g. active severe bleeding, severe thrombocytopenia plts <50, severe anaemia, intracranial bleed).

► Which anticoagulant?

- The highlight message is as predicted, that DOACs are the drugs of choice for most people with AF; the move from warfarin to DOAC has been accelerated by the COVID pandemic with the push for less blood monitoring, although **warfarin should still be used for people with moderate/severe mitral stenosis or prosthetic valves.**
- Regarding DOAC choice, apixaban, dabigatran, edoxaban and rivaroxaban are all recommended by NICE as first line options so that we can select the best choice for our patient in line with their co-morbidities and local formularies; but note below new potential **recommendations in England to use edoxaban as first line.**
- *NB see our DOACS specific chapter in the Prescribing section for more on DOACs, including safe prescribing in renal dysfunction and whether we should co-prescribe PPIs*
- **When should we use warfarin?**
 - **CKS/UK recommendations remain that DOACs should not be used in people with moderate/severe mitral stenosis or a prosthetic heart valve.** This is broadly the advice from ESC as well, although which is the best anticoagulant for people with AF and BIO-prosthetic valves (i.e. tissue prosthetic valves, not there mechanical ones) is less certain. A recent study in the NEJM ([N Engl J Med 2020; 383:2117-2126](#)) showed **non-inferiority of Rivaroxaban compared to warfarin in people with AF and bio-prosthetic mitral valves**, with respect to death/major CVD events/major bleeding at 12 months. Ultimately we should be guided by our cardiology colleagues here, but it remains universal advice to use warfarin in people who have moderate/severe mitral stenosis or mechanical heart valves.

► Why am I being asked to prescribe edoxaban for patients with AF? [IIF 2022/23](#)

Back in January 2022 when the UK health system was somewhat busy with a massive surge in COVID cases, not to mention a backlog of general work, a [commissioning statement](#) was published that probably missed most peoples inboxes. NHS England had done a deal with DOAC manufacturers which was [announced in November 2021](#) recommending that:

- Edoxaban should be used first line *'where this is clinically appropriate'*.
- Or if not clinically appropriate *'clinicians should then consider rivaroxaban first, then apixaban or dabigatran'*.
- The recommendations went one stage further though to recommend *'For patients already prescribed a DOAC for the treatment of AF: subject to the criteria specified in the relevant NICE technology appraisal guidance, commissioners may wish to consider developing local policy to review patients currently prescribed apixaban, rivaroxaban or dabigatran, where clinically appropriate'*.

The aim of this deal is to **widen access to more affordable DOACs**, and the money saved from these deals will get put back into Detect, Protect and Perfect schemes to promote better detection and treatment of AF, ultimately in the hope it will lead to fewer strokes.

Subsequently, these recommendations were incorporated into the [2022/23 Impact and Investment Fund \(IIF\)](#), which forms part of the Network Contract DES, and is a financial incentive scheme that focuses on supporting PCNs to deliver health improvements. The NHS Long Term Plan has boldly committed to preventing 150,000 strokes, heart attacks and dementia cases by 2029 through earlier detection of CVD. One of the domains in the IIF is CVD prevention focusing on the 'ABC' of AF, Blood pressure and Cholesterol. Within the AF section, **PCNs are incentivised for getting more people with AF onto DOACs (based on appropriate CHA2DS2Vasc scores) but also for using a higher percentage of DOACs as edoxaban.**

A procurement process happened in 2021 when NHSE gave all DOAC suppliers an opportunity to effectively bid for lower tariffs to be recommended as a preferred NHS drug and Daiichi Sankyo offered the most significant discount for their drug edoxaban. One can see the reason why NHSE is so keen to try to incentivise a more cost-effective DOAC - **DOACs are the highest spend drug in primary care accounting for >£500 million/yr** and if no action is taken DOAC expenditure is expected to double. But the increasing number of deals done by the NHS with drug companies is drawing criticism as the nature of the deals (and therefore cost-effectiveness) is hidden behind commercially sensitive agreements.

The IIF will incentivise PCNs to prescribe more edoxaban, as a % of the DOACs used for new AF but also **supports the (potential) switch to edoxaban for people already with AF and on another DOAC**. The IIF does note that *'It is important that switching patients who are currently prescribed a different brand of DOAC to Edoxaban is done in a clinically appropriate way and as the result of a shared decision-making conversation.'* It's worth noting that the NICE AF guidance which was only published last year, highlighted that we have no head-to-head studies between the different DOACs, so we don't really know which is the most clinically effective option yet, but that *'When all these outcomes were combined in the cost-effectiveness analysis, apixaban was the most clinically effective and cost-effective anticoagulant based on UK drug tariff prices at the time'*. So if locally DOACs other than edoxaban have been used first line e.g. apixaban, making these thresholds/switches could be difficult. Also, there is a risk that future data may show edoxaban to be inferior to other DOACs, and if so further headaches may be in store...

► What if anticoagulation is contraindicated or not tolerated?

- In this situation, NICE advise cardiology referral for consideration of a left atrial appendage occlusion procedure

► *Rate or rhythm control?*

- There is essentially no change to previous guidance from NICE, but we will touch on this ongoing debate via the ESC guidance and some new research.
- **NICE recommend rate control as first-line strategy except in people:**
 - Who have a reversible cause
 - Who have heart failure where the AF is thought to be the primary cause
 - With new-onset AF
 - With atrial flutter who are suitable for ablation strategy
 - For whom rhythm control strategy is more suitable based on clinical judgement
 - Who have ongoing symptoms once heart rate controlled or where rate control strategy has been unsuccessful
- We should remember to *consider referring for possible intervention if drug treatment is unsuccessful, unsuitable or not tolerated*; there are now a range of interventional options for left atrial ablation which can be considered, although NICE do caution that the procedures are *'not always effective and the resolution of symptoms may not be long-lasting'*

► Which drugs should we use for rate control?

- Again this is broadly the same as previously:
 - 1st line = mono-therapy with standard beta-blocker (i.e. not sotalol) OR rate-limiting calcium-channel blocker/CCB (diltiazem or verapamil).
 - 2nd line = dual therapy with 2 of beta-blocker/diltiazem/digoxin.
 - Digoxin monotherapy can be considered in those who are sedentary or other rate-limiting options are unsuitable due to co-morbidities.
 - We need to remember to AVOID CCBs if the person has heart failure.

► Should we ever stop anticoagulation?

- This is an important question, and one addressed in the new guidance.
- **For people who have not had an intervention to stop AF (e.g. catheter ablation or surgery), we should NOT stop anticoagulation if the AF becomes clinically undetectable.**
- The rationale for this is that the person may still be having runs of paroxysmal AF and still have a high stroke risk so we should continue to base anticoagulation decisions on the CHA2DS2VASc score.

2 ESC Guideline for diagnosis and management of AF [August 2020](#)

Much of the ESC guidance mirrors the draft NICE guidance, albeit in a bit more detail and with a few variations. However, one big difference between the guidelines is that NICE are very much focused on the specifics related to AF, whereas the **ESC encourage us to look at CVD outcomes more widely in people presenting with AF**. Patients with atrial fibrillation have a broad range of adverse cardiovascular outcomes, and this guideline encourages a more holistic approach using the ABC pathway.

► Holistic management of AF - the ABC pathway

The really key message from this updated ESC guidance is for us to *think more widely and holistically about AF* and that for all the exciting new advances in AF management by doing there basics well we can improve both quality of life AND CVD outcomes. [RCT data](#) comparing a systematic ABC approach (see below) compared to usual care suggests this **strategy can significantly reduce composite CVD outcomes** (MACE, death and re-hospitalisation) HR 0.36, CI 0.22-0.67.

A Anticoagulation/Avoid stroke

- This aspect of AF management is really where most of our focus has been over the last 5 years or so, and the data suggests we continue to improve our anticoagulation rates
- As discussed above the ESC highlight that the absolute contraindications to anticoagulation are limited and that **age and falls risk certainly are not contraindications in their own right**.
- One interesting study (ELDERCARE-AF) published in the NEJM recently ([N Engl J Med 2020; 383:1735-1745](#)) randomised very elderly people with AF (>age 80, of whom over half were >85) who were considered unsuitable for standard anticoagulation to either placebo or low dose edoxaban (15mg daily):
 - All-cause mortality rates were similar but stroke or systemic embolisation rates were significantly lower in the edoxaban group (HR 0.34, CI 0.19-0.61), albeit with an inevitable increase in major bleeds (3.3% in edoxaban group vs 1.8% in placebo), but the reduction in stroke rates would be associated with substantial improvements in QOL.
 - There are some caveats in that the population were Japanese only, and some of the people deemed 'unsuitable' for anticoagulation probably would be considered for usual DOAC doses now.
 - This is not a practice-changing study, but it does introduce the prospect that low-dose DOACs, for those at very high risk of bleeding complications, maybe an option in the future.

B Better symptom control

- **Rate control strategy - what HR target should we aim for?**
 - The current evidence suggests no difference in clinic events whether aiming for a lenient HR target (≤ 110 bpm) or a strict HT target (≤ 80 bpm).
 - Based on this the ESC advise initially aiming for HR target ≤ 110 bpm, and if still symptomatic or there is LV function deterioration then aiming for HR ≤ 80 bpm.
- **Rhythm vs Rate Control strategy?**
 - At the time of the ESC publication the RCT data suggested that the only reason for using a rhythm over rate control strategy was to reduce AF-related symptoms and improve QoL, not to improve clinical outcomes, however, the EAST-AFNET4 and GARFIELD-AF trials (below) question that premise, or at least in those with new-onset AF.
 - The decision on whether to offer rhythm over rate control will inevitably be based on patient-specific factors, but the ESC give some useful **guiding principles as to when rhythm control may be more favourable to rate control**:
 - Younger age
 - 1st AF episode or short history
 - No/few co-morbidities or heart disease
 - Rate control difficult to achieve
 - AF precipitated by temporary event (e.g. acute illness)

C Cardiovascular risk factor and Comorbidity optimisation

The message from the ESC is to take a global overview of people with AF and address all potential underlying/associated factors. AF often results from a combination of underlying interacting factors leading to atrial remodelling/cardiomyopathy. Thus to effect change we need to address these lifestyle factors and risk factors as a whole - the evidence suggests that targeting individual factors (e.g. hypertension) makes little difference to AF rates and burdens, but **comprehensive risk factor modification has been shown to improve outcomes**. In summary:

- **Cardio-metabolic risk factors:**
 - Addressing obesity together with other CVD risk factors (e.g. diabetes) has been shown to reduce AF recurrence and symptom burden.
 - Optimal glycemic control in diabetic patients after catheter ablation significantly ↓ AF recurrence.
- **Alcohol** - We have RCT evidence that reducing alcohol in regular drinkers reduces AF recurrence.
- **Caffeine?** Probably does not contribute to or cause AF.

- **Physical activity** is likely to help overall cardiovascular disease burden, but AF incidence may be increased in elite athletes or those doing long-term or endurance sports.
- **Hypertension - Target BP $\leq 130/80$:**
 - The commonest etiological factor in AF and crucial to address, not just from an AF-specific point of view, but it is also a factor in AF-related complications (stroke, heart failure and bleeding risk).
- **Sleep apnoea** - often not considered, but AF is a common associated condition with sleep apnoea; considering, diagnosing and treating sleep apnoea with CPAP in patients with AF may reduce AF recurrence and reduces the overall CVD burden.

► **What about atrial flutter?** Should broadly be treated in exactly the same way as AF, but cardioversion is more likely to be effective in atrial flutter, especially electrical cardioversion.

► **Early rhythm control in AF - EAST-AFNET 4 trial** [N Engl J Med 2020; 383:1305-1316](#)

Shortly after the ESC guidance came out this potentially important trial was published in the NEJM in October 2020. As discussed in the associated editorial [N Engl J Med 2020; 383:1383-1384](#) previous studies looking at rhythm vs rate control had not shown any improvements in mortality or CVD outcomes in the rhythm control groups, hence the board recommendation for rate control initially in most people. However, rhythm control management options have advanced significantly, and previous studies have looked at a wide range of AF longevity. **The defining factor in this new study was that patients had been recently diagnosed with AF** (within 1 year) and it is the *first study to shown improvements in CVD outcomes as well as mortality in those treated with a rhythm control strategy over a rate control strategy*.

- 2789 patients diagnosed with AF within 1 yr (median time to diagnosis 36 days), who had, or were at high risk of CVD, randomly assigned to rhythm control (antiarrhythmic drugs or AF ablation) or usual care.
- Median age ~70 yrs old, median time to enrolment 36 days after diagnosis, median F/U 5.1 years.
- Primary outcome was a composite of death from CVD cause, stroke, hospitalisation with heart failure or acute coronary syndrome.
- Rhythm control included treatment by ablation or with antiarrhythmic drugs
- There were **significantly fewer primary outcome events in the rhythm-control group** (HR = 0.79, CI 0.66-0.94), as well as a lower risk of individual components of death from CVD cause and stroke; in absolute terms, the primary outcome was reduced from 5 events per 100 person-years to 3.9
- **Conclusion:** *early rhythm control in people with AF and CVD conditions reduces mortality and adverse CVD outcomes.*

► **Cardioversion for newly diagnosed AF - GARFIELD-AF trial** [BMJ 2021;375:e066450](#)

Further data supporting early rhythm control for (new onset) AF came late in 2021 in the form of the GARFIELD-AF trial. This was a **very large observational global trial** based on prospectively collected data for people with new diagnoses of non-valvular AF:

- >44,000 patients with new AF (diagnosed within 6 weeks) were analysed, comparing those that underwent cardioversion (at baseline or within 48 days) vs those that did not.
- ~15% underwent cardioversion, with 2/3 having drug cardioversion and 1/3 DC cardioversion.
- **All-cause mortality was significantly lower in the cardioversion group at 1 year** (adjusted HR = 0.74, 95% CI 0.63-0.86), and **this benefit was maintained at 2 years follow-up**.
- **Conclusion:** early cardioversion for new-onset AF within 6 weeks significantly reduces all-cause mortality at 1 year with benefits maintained at 2 years.

There are caveats to this study. The main one is that baseline risks were (not surprisingly) lower in the group that underwent cardioversion - the authors adjusted for as many confounding variables as possible but the **risk of residual confounding and bias remains**. The other significant caveat is that **these patients had their cardioversions done very promptly** (the vast majority within 48 days of enrolment) - given the significant pressures on our system in the UK, it is questionable how many of our patients can expect to be referred, assessed and cardioverted in that sort of time frame.

There are inevitably significant resource implications to referring more people for rhythm control strategies. If these results are reproduced in further trials, it will be interesting to see if NICE changes its current stance to recommend more people with early AF be referred for rhythm control. In the meantime, this study lowers the threshold for referral for a rhythm control strategy for patients in line with current NICE guidance i.e. patients

- *Who have a reversible cause*
- *Who have heart failure where the AF is thought to be the primary cause*
- *With new-onset AF*
- *With atrial flutter who are suitable for ablation strategy*
- *For whom rhythm control strategy is more suitable based on clinical judgement*
- *Who have ongoing symptoms once heart rate controlled or where rate control strategy has been unsuccessful*

NB Learning Points: Atrial Fibrillation

- NICE now recommend we use the **ORBIT tool to assess bleeding risk** in those being considered for anticoagulation, but the aim of this is to identify those at higher risk of bleeding and address risk factors (e.g. uncontrolled hypertension, alcohol abuse, concurrent medications e.g. NSAIDs), not to necessarily deny them anticoagulation. [Click here](#) for online ORBIT calculator.
- The **benefits of anticoagulation outweigh risks in the vast majority of people**, and age and falls risk are not contraindications to anticoagulation in their own right.
- Anticoagulation should be offered to people with ALL forms of AF (based on the CHA2DS2VASc score), **including those with paroxysmal AF**.
- We should not stop anticoagulation if AF becomes clinically undetectable - these people are still likely to be having paroxysmal AF and so anticoagulation should be continued.
- NICE recommend DOACs first line, a choice of **apixaban, dabigatran, rivaroxaban or edoxaban based on clinical judgement and patient choice**, although in England new recommendations may mean that **edoxaban is incentivised/prioritised over other DOACs**.
- Rate control is still likely to be the first-line strategy in most people, but there is increasing evidence (both RCT and observational) that **early rhythm control in people with new-onset AF improves both CVD outcomes and mortality**.
- Rhythm control strategy is more likely to be favourable in younger people, those with new-onset AF and those with no/few CVD co-morbidities.
- Initial **heart rate target** with rate control strategy ≤ 110 bpm, aiming for a more stringent target of ≤ 80 bpm if they remain symptomatic (from the AF).
- We need to look at AF more holistically as part of a CVD risk 'package' adopting the **ABC strategy** (Anticoagulate/Avoid stroke, Better symptom control, CVD/Comorbidity risk factor optimisation) as advocated by the ESC can **improve CVD outcomes**.

NB Quality Improvement Ideas: Atrial Fibrillation

- **Review your patients with paroxysmal AF:**
 - Have they been assessed for anticoagulation based on CHA2DS2VASc?
 - If not assess their risk and offer/consider anticoagulation based on the score, as we would for people with permanent/persistent AF.
- Aim to adopt a global CVD risk assessment when reviewing people with AF and **adopt the ABC approach to management**.
- For those on rate control strategy, a **lenient initial HR target of ≤ 110 bpm is appropriate** and we only need to aim for a more stringent target of ≤ 80 bpm if they remain symptomatic from the AF.

KISS: Atrial Fibrillation [NICE, NG 196 June 2021](#) [ESC AF Guideline Aug 2020](#)

Diagnosis/Assessment:

- Undertake opportunistic screening esp in those with risk factors or any suggestive symptoms e.g. dyspnoea, palpitations etc. **Irregular pulse? Confirm with ECG.**
- If **paroxysmal AF** suspected use 24-hour ECG (if frequent symptoms <24 hours apart) or event recorder if symptoms more sporadic or >24 hours apart. KardiaMobile now an option for PAF [MTG64](#)
- If AF confirmed, check baseline **bloods** inc. TSH, FBC, U&E, LFT.
- **Echo?** Not needed for most people, but consider if rhythm control strategy being considered or suspicion of structural or functional heart disease (e.g. heart failure, valve disease).

Management: Use the ABC approach:

- **A = Anticoagulation/Avoid stroke (including those with paroxysmal AF):**
 - Assess stroke risk using CHA2DS2-VASc score [click here](#)
 - Score ≥ 2 : Offer anticoagulation taking bleeding risk into account.
 - Score = 1 in men consider anticoagulation taking bleeding risk into account.
 - Score = 0 in men or ≤ 1 in women do not offer anticoagulation, but review again aged 65 or if develops diabetes or other cardiovascular comorbidities
 - **Offer a DOAC as first line choice** of apixaban, rivaroxaban, edoxaban or dabigatran based on best individual choice for patient and local formularies.
 - **Warfarin still preferred** for those with mechanical heart valves & moderate-severe mitral stenosis.
 - **Assess bleeding risk using the ORBIT tool** ([click here](#) for online calculator):
 - Focus is to identify those at higher risk and modify risk factors (e.g. uncontrolled hypertension, alcohol excess, concurrent drugs e.g. NSAIDs, [consider PPI](#) esp. in older people).
 - For most people benefits of anticoagulation will outweigh risks; age and falls risk are not contraindications to anticoagulation in their own right.
- **B = Better symptom control:**
 - **Rate control** first line for most people:
 - Initial **HR target** ≤ 110 bpm, aiming for lower ≤ 80 bpm if ongoing symptoms (from the AF).
 - **1st line:** mono-therapy with standard BB (i.e. not sotalol) OR rate-limiting CCB (diltiazem or verapamil); remember to avoid CCBs in heart failure.
 - **2nd line:** dual therapy with 2 of beta-blocker/diltiazem/digoxin.
 - **Digoxin** monotherapy can be considered in those who are sedentary or other rate-limiting options are unsuitable due to co-morbidities.
 - **Rhythm control** may be more favourable if: Younger age; 1st AF episode or short history; No/few co-morbidities or heart disease; AF precipitated by temporary event (e.g. acute illness).
- **C = CVD/Co-morbidity risk factor modification:**
 - Comprehensive risk factor modification has been shown to improve CVD outcomes.
 - Address **cardiometabolic risk factors** (obesity, diabetes, lipids); tight **BP control** $\leq 130/80$ recommended by ESC (but NB the updated 2022 [NICE BP guideline](#) recommends we use the same BP targets for people with and without CVD i.e. $<140/90$ if <80 or $<150/90$ if ≥ 80); reducing **alcohol** can reduce AF recurrence; consider **sleep apnoea** and refer if suspected.
- **Refer:**
 - **Urgently** if acutely symptomatic and haemodynamic instability or decompensated heart failure.
 - **Routinely** if:
 - Suspected underlying structural or functional heart disease (e.g. heart failure, valve disease).
 - Rhythm control strategy being considered.
 - Ongoing symptoms despite 2nd line rate control drugs.



CHRONIC HEART FAILURE

Chronic heart failure (CHF) is an increasingly common condition for us to manage in primary care, the rising prevalence driven by an ageing population and improved survival from conditions that contribute to it. Heart failure now affects 10% of our patients aged over 75, yet diagnosis remains a challenge [BJGP 2023; 73 \(726\): 4-5](#). The big Hot Topic in CHF is *the new evidence of the role of SGLT-2 inhibitors in the management of heart failure with reduced ejection fraction (HFrEF), especially now that dapagliflozin and empagliflozin have been approved by NICE for this indication* [NICE 2021 TA679](#) and [NICE2022;TA773](#). On the back of this, we have new, ground-breaking evidence on the *use of SGLT2 inhibitors in heart failure with preserved ejection fraction (HFpEF)* [Lancet 2022; 400: 757-67](#). We also have new evidence supporting the use of IV iron for people symptomatic with HFrEF in the form of the [IRONMAN trial](#).

NICE last updated their heart failure guideline in 2018 [NICE 2018, NG 106](#), so a recent evidence review on the management of HFrEF in the BJGP is welcome [BJGP 2021;71;330](#), and more recently the European Society of Cardiology have updated its guidance on heart failure ([ESC Aug 2021](#)).

1 *Diagnosis of chronic heart failure and BNP testing* [BJGP 2023; 73 \(726\): e1-e8](#)

The introduction of the natriuretic blood tests, including NT-pro BNP, was a major step forward in heart failure diagnosis, but despite that the diagnosis of heart failure remains a challenge as discussed in a recent [BJGP editorial](#). One of the issues is the BNP test itself. As highlighted in the KISS below there are **substantial differences in the 'cut-off' value used for diagnosis/referral for echo between UK/NICE/SIGN (400ng/L) and the rest of Europe based on ESC guidance (125 ng/L)**. This was investigated in a diagnostic accuracy study comparing the two thresholds, recently reported in the BJGP:

- Data was collected from UK General Practice records including 229,580 patients who had NT-pro BNP testing, which was linked to secondary care records.
- Not surprisingly the **lower ESC threshold had better sensitivity** 94.6% (95% CI = 94.2-95.0) vs NICE/SIGN threshold sensitivity 81.7% (95% CI = 81.0-82.3).
- However, this was at a **cost of a substantially worse specificity** - ESC specificity 50.0% (95% CI = 49.7-50.3) vs NICE/SIGN specificity 80.3% (95% CI = 80.0-80.5).
- This translates to the following NPVs and PPVs:
 - NPVs: ESC = 98.9%, NICE/SIGN = 97.7%
 - PPVs: ESC = 16.4%, NICE/SIGN = 30.0%
- **Conclusion:** At the higher NICE/SIGN threshold about 1:5 cases of HF will initially be missed, but the lower ESC thresholds would require substantially more referrals/echos and assessments. *'The optimal threshold will depend on the priorities and capacity of the healthcare system'*.

Overall, the NICE/SIGN threshold still compares well - although it may miss 1:5 patients initially, a **NPV of 97.7% means it is still a good rule-out test at this threshold** and has much better diagnostic accuracy at that level. If we were to adopt the ESC threshold a specificity of 50% means only 1:2 of those referred will actually have HF on the back of substantially more referrals. This does not feel like a good use of resources and given the current pressures on NHS systems this data probably supports the current thresholds, but with the caveat that if we have an ongoing high clinical suspicion of HF but the NT-pro BNP levels are between 125-400 we should still consider specialist advice for consideration of echo.

► *Earlier diagnosis of heart failure* [BJGP 2023; 73 \(726\): 4-5](#)

The issue of HF diagnosis and barriers to diagnosis was explored more widely in a linked editorial in this BJGP. The **focus of this article was to highlight the ongoing challenges of HF diagnosis and promote earlier HF diagnosis**. The editorial bemoans the poor diagnosis rates in primary care - despite the advent of natriuretic peptide testing 80% of HF diagnoses are initially made following acute hospital admissions, and that much of the NHS annual £2 billion HF spend is on hospital care. The implication is that we are 'missing' these diagnoses in primary care, but the article does not point out that some of these people may have subclinical HF which is triggered into clinical HF from an acute event i.e. some of these primary hospital HF diagnoses will be triggered or caused by an acute event that we can neither predict nor prevent.



Be that as it may, the **editorial does explore the barriers to early diagnosis in primary care**. Many patients with HF will have other LTCs and be on multiple medications, and may well have presenting symptoms that could be caused by a number of other diagnoses. It is well known that the term 'heart failure' is frightening to patients possibly leading them to delay presentation, especially if they perceive their symptoms (typically SOB and fatigue) are simply a fact of them 'getting older'. BNP testing is helping diagnosis, but is by no means universally accessible in primary care, and then there is the thorny issues of access to echocardiography and cardiology specialist services.

The editorial concludes by stating that **HF should be seen as a 'malignant condition'** given it's prognosis, but that with earlier diagnosis the increasingly wide range of treatment options can be employed to improve outcomes. They **propose a new acronym**, similar to the 'FAST' campaign for stroke, to promote earlier presentation by patients and earlier investigation in primary care - **BEAT-HF (Breathless, Exhausted, Ankle swelling, Time to get a HF test)** is hoped to be promoted more widely in the future. However, as the editorial points out, a public health campaign like this will only bear fruit if the wider patient pathway is improved, including universal access to BNP testing, better access to echocardiography and better access to specialist HF clinics.

KISS: Diagnosis of Chronic Heart Failure

[NICE 2018, NG 106](#) [BMJ 2018;362:k3646](#) [BJGP 2019; 69 \(682\)](#) [ESC 2021 SIGN 147](#)

Clinical assessment:

- Consider risk factors e.g. MI, coronary revascularization, hypertension, AF, and diabetes.
- Symptoms/signs include SOB (exertional and orthopnoea), fluid retention, decreased exercise tolerance, fatigue/tiredness; enlarged liver, basal creps, pleural effusions and leg/sacral oedema.

Investigations:

- Check NT-pro BNP in anyone with suspected heart failure.
- ECG.
- Blood tests - U&E, thyroid function, LFT, lipids, Hba1c, FBC.
- Consider CXR +/- spirometry if alternative pulmonary diagnoses are being considered.

Referral:

- Routine referral (within 6 weeks) for specialist assessment and echo if NT-pro BNP 400-2000 ng/L (47 to 236 pmol/l).
- Urgent referral (within 2 weeks) for specialist assessment and echo if NT-pro BNP >2000 ng/L (> 236 pmol/l) due to poor prognosis in this group.
- NT-pro BNP <400 ng/L (< 47 pmol/l) makes a diagnosis of heart failure less likely in an untreated patient but there are a **number of caveats** to be aware of:
 - **'Cut off' values:**
 - NICE/SIGN both use much higher values for referral threshold for echo at ≥400 ng/L, compared to the ESC guideline which recommends echo for levels ≥125 pg/ml (same as ng/L).
 - NICE carefully say a level <400 ng/L means heart failure is 'less likely' but this **cut-off does not adequately rule out heart failure**; patients with levels <400 ng/L may still have heart failure and if we think this is the likely diagnosis we should still get specialist advice.
 - **NT-pro BNP levels may be falsely negative or positive:**
 - **False negatives may occur with:** obesity, African or African-Caribbean family origin, patients already taking diuretics, ACEI, ARB, beta-blockers, mineralocorticoid receptor antagonists.
 - **False positives may occur with:** LVH, tachycardia, hypoxaemia, CKD, COPD, cirrhosis, diabetes, old age...

2 *Classification of chronic heart failure* [ESC 2021](#) [BJGP2019;69;313](#) [NICE NG 106, 2018](#)

Whilst our primary role as GPs in the assessment of heart failure is to recognise the condition, do a BNP and refer, it is important we have an understanding of the classification of the different types of heart failure. This will obviously be specialist-led, and whilst this level of detail risks going red on the 'nerdometer' scale, it is important to understand and interpret the emerging research discussed below and the fact that **the evidence base is very different depending on the type of heart failure present.**

The historical picture of heart failure was one of poorly contracting ventricles, often as a result of myocardial infarction (or occasionally cardiomyopathy), leading to LV systolic impairment/reduced ejection fraction. Most of the evidence on the management of heart failure refers to this group of people with heart failure and reduced ejection fraction (HFrEF).

However, **heart failure with *preserved* ejection fraction** (HFpEF - previously known as diastolic heart failure or diastolic dysfunction) is an important and increasingly common condition. Depending on the cut-off value of ejection fraction used, **HFpEF accounts for anywhere between 40-70% of all patients with heart failure** and HFpEF is increasing by 10% per decade relative to HFrEF, driven by an ageing population with a cardiometabolic disease (diabetes, hypertension, obesity). It is a type of heart failure characterised by a stiff ventricle, that is more common in older people with these co-morbidities. HFpEF presents similarly to HFrEF with breathlessness, fatigue, fluid retention and raised BNP. Typically **echo will show preserved ejection fraction, with impaired LV relaxation rather than LV contraction, and evidence of diastolic dysfunction.** Common risk factors are older women, hypertension, diabetes, AF and ischaemic CHD.

A problem has arisen over time with **how the 'cut-off' values for HFrEF and HFpEF are defined**, which has slightly muddled the water with the evidence base. For quite some time it has generally been accepted that an ejection fraction of <40% clearly defines HFrEF, but there has been inconsistency in the ejection fraction cut-off for HFpEF, with the NICE guideline not giving a clear answer. The **ESC have now given clear guidance** on this, in the hope that future research/guidance will be more standardised:

HF type	HFrEF	HFmrEF*	HFpEF
Clinical features	symptoms +/- signs and raised BNP	symptoms +/- signs and raised BNP	symptoms +/- signs and raised BNP
Left Ventricular Ejection Fraction/Echo features	LVEF ≤40%	LVEF 41-49%	LVEF ≥50% <i>plus</i> structural/functional signs consistent with LV diastolic dysfunction

- *HFmrEF = heart failure with *mildly reduced* ejection fraction. This clarifies the 'grey' area between those with clearly preserved or reduced ejection fraction (previously known as heart failure with mid-range ejection fraction). Although this may seem like nit-picking semantics it is important when interpreting the heart failure evidence base, as this group have variously been used in studies of people with both 'preserved' and 'reduced' ejection fraction. We discuss this issue below in the HFpEF section.

3 *Heart failure with reduced ejection fraction (HFrEF)*

► **SGLT2 inhibitors for HFrEF**

SGLT2 inhibitors are now well-established oral hypoglycaemic drugs for diabetes, and have become the second line after metformin for many patients due to their proven cardiovascular benefits (see diabetes section for more on this). An unexpected finding in the key diabetes trials of these drugs such as [EMPA-REG](#), [CANVAS](#) and [DECLARE-TIMI](#) was that the improved CVD outcomes were largely driven by improvements in heart failure outcomes. These heart failure benefits were seen to be immediate suggesting a direct circulatory or cardiac effect and to be independent of glucose-lowering. How SGLT2i improve outcomes in heart failure is not established, but it is thought that it is independent of any glucose-lowering effect and may be due to direct cardioprotective and nephroprotective effects related to actions on sodium balance and natriuresis.

Recent studies have shown that SGLT2 inhibitors improve outcomes in both chronic HFrEF and also chronic kidney disease *even in patients without diabetes which is leading to a change in guidance on the use of these drugs*. Some of the key trials:

EMPEROR-reduced Trial [NEJM 2020;383:1413](#)

- This was a double-blind RCT in 3,730 patients with heart failure and a reduced ejection fraction of <40%
- They were randomised to empagliflozin 10mg daily or placebo in addition to recommended therapy
- The primary outcome was cardiovascular death or hospitalisation for worsening heart failure
- *Over a median of 16 months, the primary outcome was reduced from 25% to 20%, a RRR of 25% with an ARR of 5% NNT = 20*
 - There was a 30% RRR in heart failure hospitalisations
- This benefit was seen regardless of whether or not the patient had diabetes
- **Conclusion:** *In patients with heart failure with reduced ejection fraction taking recommended therapy, the addition of empagliflozin 10mg daily reduced the risk of cardiovascular death and hospitalisation for worsening heart failure, regardless of whether or not they had diabetes*

DAPA-HF Trial [NEJM 2019;381:1995](#)

- This was an RCT of 4,744 patients with heart failure with *reduced* ejection fraction <40%, randomised to dapagliflozin or placebo in addition to their standard CHF therapy
- The primary outcome was a composite of worsening heart failure or cardiovascular death
- Over a median of 18 months, the primary outcome occurred in 16% of the dapagliflozin group and 21% of the placebo group (NNT = 21), and there was no difference between the two groups in adverse events
- The magnitude of benefit seen was similar in different groups of patients both with and without diabetes
- A number of secondary CV outcomes were also positive, including recurrent hospitalisations, all-cause mortality and improved quality of life
- **Conclusion:** *Dapagliflozin improves outcomes in heart failure with reduced ejection fraction, regardless of the absence or presence of diabetes*

SGLT-2 inhibitors in patients with Heart failure and reduced EF [Lancet 2020;396:819](#)

- This was a meta-analysis of the EMPEROR-Reduced and DAPA-HF trials
- Neither of the two trials was sufficiently powered to look at overall mortality, or to characterise outcomes in clinically important sub-groups hence the need for this meta-analysis
- The combined results show a 13% RRR in all-cause mortality, a 25% RRR in hospitalisations for worsening heart failure and improved renal outcomes
- **Conclusion:** *the effects of empagliflozin and dapagliflozin on hospitalisations for heart failure are consistent in both trials; the data suggest that these drugs also improve renal outcomes and reduce all-cause and cardiovascular mortality*

So, should SGLT2i become part of standard therapy for heart failure with reduced ejection fraction?

The authors of the Lancet meta-analysis [Lancet 2020;396:819](#) point out how consistent the results are in the EMPEROR-Reduced and DAPA-HF trials suggesting that this is a class effect. They state *'the evidence supports the use of the SGLT2 inhibitors empagliflozin or dapagliflozin as an integral part of a comprehensive therapy that improves the event-free survival of patients with HFrEF'*. Furthermore, the benefits were seen whether or not patients had diabetes, in men and women, in younger and older patients and also whether they were receiving neprilysin inhibitors (sacubitril) or not. They conclude that their meta-analysis *'establishes a solid evidence base'* confirming an important role for these drugs in patients with HFrEF. For patients with type 2 diabetes and HFrEF, the evidence seems strong for an SGLT-2i and this *is* recommended in the joint [US and European diabetes guidance 2018/19](#). *We need to be aware of the risk of rare but serious side effects (e.g. DKA) with SGLT2i and to mitigate this risk and warn patients appropriately (see Safe Use of SGLT2i in the prescribing chapter)*. For patients on hypoglycaemic drugs such as sulfonylureas or insulin, the dose of these may need to be reduced and dapagliflozin should **not** be added in type 1 diabetes ([MHRA Dec 2021](#)).

For patients with HFrEF and without diabetes, SGLT2i do not appear to cause DKA. Empagliflozin and dapagliflozin now have a class 1 recommendation for HFrEF in the updated [ESC guidance 2021](#), and BOTH are now recommended as an option by NICE (see box below):

KISS: SGLT2i for Chronic Heart Failure with Reduced Ejection Fraction

[NICE 2021, TA 679](#) and [NICE2022, TA773](#)

- Dapagliflozin or empagliflozin is recommended as an option for treating symptomatic CHF with reduced EF in adults only if used as an add-on to optimised standard care with:
 - ACEi or ARB plus beta-blocker and if tolerated mineralocorticoid receptor antagonist (MRA)
 - Sacubitril valsartan (*Entresto*®) with BB and if tolerated MRA.
 - Note that optimised standard care means that patients have been titrated up to the maximal tolerated target dose of the standard medications.
 - *NB The evidence from the trials relates to patients with an EF < 40%.*
- Start treatment on advice of heart failure specialist.

► IV iron therapy for iron deficiency in HFrEF - the IRONMAN study [Lancet 2022; 400: 2199–209](#)

One of the recommendations in the updated 2021 ESC guidance is to consider IV iron therapy for people with symptomatic HF and EF <45% if iron deficient (defined as ferritin < 100 ng/ml or transferrin saturation <20%). There has been some evidence to support this, mostly for people who have had recent hospital admissions, although this doesn't seem to be routinely implemented in the UK. So a new, quite large, UK-based RCT the IRONMAN study has helped the evidence base by looking **more widely at IV iron for people with chronic heart failure and iron deficiency**:

- IRONMAN was a prospective RCT across 70 UK hospitals which recruited 1137 people with HFrEF and EF ≤45% and iron deficiency (ferritin <100 mcg/L or transferrin saturation <20%) irrespective of whether they were anaemic or not.
- They were randomly assigned to either usual care or IV iron treatment (ferric derisomaltose) and the **primary outcome was recurrent hospitalisation for HF or CV death**, with median follow-up 2.7 years.
- There were 18% fewer events in the IV iron group (RR 0.82, 95% CI = 0.66-1.02) with an absolute reduction of ~5 events per 100 patient-years. This improvement was driven by reduced hospital admissions with no differences in CV death rates.
- There were fewer serious cardiac events in the IV iron group as well.
- A pre-specified COVID-19 analysis also showed a reduction in primary endpoints (RR 0.76, 95% CI = 0.58-1.00).
- **Conclusion: For a broad range of patients with HFrEF and iron deficiency, IV iron replacement reduces the risk of hospital admission.**

An [associated editorial](#) discussed the results. It's important to note that the **results in the main analysis were not statistically significant** with wide CIs crossing the threshold, although the pre-specified COVID-19 analysis did reach statistical significance (but only just!). However as pointed out in the editorial this study was an open-label design and 17% of the usual care group moved across to the IV iron group during the study, which is likely to have attenuated the effects seen in the results. However, **this important study suggests that IV iron is likely to reduce hospital admissions for a broader range of people with HFrEF and iron deficiency**, and adds to the growing evidence base that IV for this group is beneficial for both hard clinical endpoints (hospital admissions) as well as important patient-related outcome measures (quality of life and exercise capacity). **These results and those from other studies have implications for us in primary care** - if people with HFrEF develop worsening symptoms we should screen for anaemia and iron deficiency and refer for consideration of IV iron if the ferritin is <100 mcg/L or transferrin saturation is <20%. It will be interesting to see whether similar results can be achieved with oral iron therapy, or whether patients with HFpEF may gain similar benefits.

► *Management of Chronic HFrEF* [BJGP2021;71;330](#) [ESC Heart Failure Guidance 2021](#)

This recent BJGP review is an excellent summary of current evidence, given that the NICE guideline dates back to 2018, and very much mirrors the recently published updated ESC guidance.

The review discusses that we now have **'four pillars' of pharmacological therapy for chronic heart failure with reduced ejection fraction (HFrEF)**: cardioselective beta-blockers, ACEi or ARBs (or the angiotensin-neprilysin inhibitor (ARNI) sacubitril valsartan), mineralocorticoid receptor antagonists (MRA) and now the SGLT2i. The ESC concur with this and give all these drugs class 1 recommendations.

The **mainstay of pharmacological treatment** continues to be **ACEi or ARBs** with a **beta-blocker (BB)** licensed for heart failure, both of which are well established to reduce mortality and improve outcomes in heart failure alongside flexibly used diuretics.

NICE (and ESC) guidance is to start with an ACEi (as greater evidence base) and substitute with an ARB if the ACEi is not tolerated. *The aspiration should be to slowly titrate to maximally tolerated doses (see table below).* As the BJGP points out, however, in some patients hypotension, bradycardia or hyperkalaemia will make it impossible to reach target doses and in such cases, most specialists would advise using all key groups of drugs but at lower tolerated doses. See also our 'troubleshooting' KISS below.

Mineralocorticoid-receptor antagonists (MRA, e.g. spironolactone) should be added after ACEi and BB if patients remain symptomatic. The use of MRAs has been less widespread in primary care, but it is recommended by NICE as a first line option in patients who remain symptomatic despite ACEi and BB. As for BB and ACEi there is level 1a evidence from [systematic reviews](#) that MRAs improve mortality and outcomes in HFrEF. Selective MRAs (eplerenone) and non-selective (spironolactone) are equally effective in improving outcomes, but the selective MRA eplerenone is less likely to cause gynaecomastia or sexual dysfunction in men. *Careful monitoring of U and E when adding a MRA to look for hyperkalaemia is essential.*

NICE guidance is to measure electrolytes and renal function, and BP:

- Before and after starting.
- After each dose increment.
- Once on target or maximal tolerated dose to monitor electrolytes monthly for 3 months, then at least 6 monthly and whenever the person has an intercurrent illness.

For patients who remain symptomatic despite BB, ACEi/ARB and MRA and who have an ejection fraction < 35%, the next step is referral for the specialist consideration of the **angiotensin-neprilysin inhibitor (ARNI)** sacubitril valsartan [NICE TA388](#), as a replacement for the ACEi/ARB. This recommendation was principally based on the [PARADIGM-HF](#) trial which showed reduced hospitalisations for heart failure and reduced CV and all-cause mortality for the ARNI vs enalapril. However, a recent study published in 2021 ([PARADISE-MI](#)) in people with HFrEF and recent MI did not show any superiority of the ARNI vs ramipril.

*However, we now also have the option of adding the **SGLT2i** dapagliflozin at this step [NICE 2021, TA 679](#) in patients with an ejection fraction <40%.*

So, which is better at this step? Adding an SGLT2i or ARNI? We have no direct head-to-head trials to answer this question, but a recent indirect comparison of outcomes comparing SGLT2i and ARNI in a network meta-analysis [ESC Heart Failure 2021](#) suggest that SGLT2i and ARNI had similar effects on improving cardiovascular outcomes, and that combination achieved a better prognosis compared to ARNI monotherapy.

For consideration of ARNI, we need specialist referral and then the ACEi/ARB need to be stopped before the ARNI can be initiated (minimum 48-hour washout period), and thus the BJGP review states *'the relative ease'* of prescribing dapagliflozin in primary care *may* make it preferable at this step. It does not need dose titration and ironically, starting dapagliflozin for heart failure is more straightforward in patients without diabetes as an SGLT2i does not lower blood glucose in people who are normoglycemic and there is not the risk of DKA. If a patient does have type 2 diabetes, their other medication may need modifying (if they are on insulin or a sulfonylurea then their dose may need to be reduced to reduce the risk of hypoglycaemia). **See the chapter in the prescribing section on the Safe Use of SGLT2i.**

What about SGLT2i and renal failure as many HFrEF patients have renal impairment? The BJGP summaries this nicely: *'Although previous advice was to avoid initiating SGLT2is in patients with eGFR <60 ml/min/m² and to discontinue if it fell to <45 ml/min/m², this reflected the diminishing glucose-lowering impact as renal tubular function declines. When prescribing dapagliflozin for HFrEF the marketing authorisation and NICE simply cautioned that there is limited experience with dapagliflozin in HFrEF when eGFR is <30 ml/min/m². It is now widely accepted that SGLT2is are not only safe when eGFR is >20 ml/min/m² but SGLT2is also reduce the rate of decline in renal function in HFrEF.'* This is backed up by [BNF guidance](#) (as at Jan 2022) that dapagliflozin can be initiated as long as eGFR >15.

Although there are many practical reasons why starting an SGLT2i for HFrEF may be easier than an ARNI, there is still debate amongst specialists as to which (ARNI or SGLT2i) is preferable, and indeed the choice may depend on patient-specific characteristics - and the current NICE TA679 does state that dapagliflozin should only be started *'on the advice of a heart failure specialist'*.

The paper concludes that pharmacotherapy with the 'four pillars' is of course though only one aspect of good management. Working closely with community heart failure specialist nurses, appropriate referral for cardiac rehabilitation, iron infusions and device implantation and considering palliative care when appropriate are all important.

KISS: The 'Four Pillars' of Pharmacotherapy for HFrEF BJGP2021;71;330		
Drug	Starting dose example	Target dose if tolerated
1. Beta-blocker	Bisoprolol 2.5mg od	Bisoprolol 10mg od
2. ACEi or ARB (option to replace with ARNI sacubitril valsartan on specialist advice if symptomatic and EF < 35%)	Ramipril 2.5mg od Losartan 12.5mg od	Ramipril 10mg od Losartan 150mg od
3. MRA	Spironolactone 25mg od	Spironolactone 50mg od
4. SGLT2i (if EF < 40%)	Dapagliflozin 10mg	Dapagliflozin 10mg

KISS: Management of Chronic Heart Failure with reduced ejection fraction

[BJGP2021;71:330](#); [NICE 2021, TA 679](#); [NICE NG 106, 2018](#) ; [BJGP2019;69:313](#) [ESC 2021](#)

General for all patients:

- *Community heart failure nurse input is key.*
- Smoking cessation, reduce alcohol, lose weight if obese.
- All patients should be offered exercise-based cardiac rehabilitation once stable.

Heart Failure with reduced Ejection Fraction (HFrEF)

- **Diuretics as required for symptoms:** Individualise dose to treat symptoms and signs of fluid overload, titrate dose up and down; record dry weight and encourage daily weighing.
- Check FBC and iron. If recently hospitalised and ferritin <100 ng/ml refer for i.v. iron infusion ([ESC](#)).
- AVOID verapamil, diltiazem and short-acting DHP calcium channel blockers.
- **FIRST LINE Beta-blockers and angiotensin drugs and titrate to maximal tolerated doses (see also KISS below on troubleshooting problems):**
 - ACEI (or ARB if intolerant) start at a low dose and titrate every ~2 weeks, checking U&E and BP 1-2 weeks after each dose increase.
 - **Beta-blockers** 'start low, go slow' - do not titrate any quicker than every 2 weeks, and many will need slower titration; check HR and BP after each dose increase.
 - Bisoprolol, carvedilol or nebivolol are the first choice.
 - BBs are NOT contraindicated in COPD or peripheral arterial disease.
- **If persistent symptoms:**
 - If ongoing symptoms add **mineralocorticoid receptor antagonist (MRA)**; monitor U&E carefully, before and after starting, after each dose change and once on target dose monthly for 3 months and then 6 monthly.
- **If symptoms remain discuss with the specialist team the option of adding:**
 - SGLT2i e.g. dapagliflozin if LVEF <40%.
 - There is also evidence for empagliflozin, NICE guidance expected in 2022.
 - Sacubitril-valsartan if LVEF < 35% (replacing existing ACEi or ARB).
- If still, ongoing symptoms **refer for specialist advice**; options may include:
 - Ivabradine, if sinus rhythm resting HR >75 bpm & LVEF<35%.
 - Digoxin.
 - Device therapy. Implantable cardiac defibrillators (ICD) improve life expectancy if reasonable prognosis and may be combined with cardiac resynchronisation pacing (CRT).

Palliative care:

- Be sensitive to the time when goals of treatment may change from disease modification to symptomatic treatment and quality of life.
- Reduce drugs that cause hypotension and may increase falls risk.
- Optimise diuretic therapy for symptoms of congestion.
- Consider low-dose morphine for pain, breathlessness or anxiety.

KISS: Troubleshooting drug management in HFrEF

[ESC Heart Failure Guideline Aug 2021](#) (NB in supplementary data section)

Background/General points:

- As discussed above ACEI/ARBs and B-blockers remain the mainstay of treatment for HFrEF and confer both symptomatic and survival benefits - important to highlight to patients.
- In many cases, these drugs are stopped or doses reduced unnecessarily due to concerns about side effects or complications.
- It is rarely necessary to stop these drugs altogether - remember **SOME** dose is better than **NO** dose.

ACEI (or ARB):

- **Titration** - Not less than 2 weekly intervals; check U&E 1-2 weeks after initiation and 1-2 weeks after final dose titration.
- It is **rarely necessary to stop altogether and clinical deterioration is likely if treatment is withdrawn** - ideally seek specialist advice before discontinuing treatment.
- **Asymptomatic low BP** - usually requires no changes in therapy.
- **Symptomatic hypotension:**
 - Dizziness/lightheadedness - common and usually improves over time - reassure.
 - Reduce/stop other vasodilators if possible e.g. nitrates, CCBs.
 - If not congested consider reducing diuretic dose.
- **Cough:**
 - Can be a sign of pulmonary oedema - rule this out first.
 - Consider other causes of cough, especially in smokers/those with lung disease.
 - ACEI-induced cough does not always require treatment discontinuation.
 - If cough is troublesome and **proven due to ACEI** (recurs after withdrawal and re-challenge) switch to ARB.
- **Decline in renal function** (see KISS below for details), but in principle:
 - An increase in urea/creatinine and decline in eGFR is to be **expected**.
 - Increase of K to ≤ 5.5 mmol/L is acceptable.
 - Increase of serum creatinine of 30% above baseline is acceptable, and up to a 50% rise may be acceptable.

B-blockers:

- **Contra-indications** - COPD is not a contra-indication, but asthma is a relative contra-indication - cardio-selective B-blockers *may* be initiated under close specialist supervision.
- Verapamil/Diltiazem should NOT be used concurrently (high risk of AV block).
- **'Start low, go slow'** - titrate no quicker than every 2 weeks and many patients will need slower titration.
- **Worsening symptoms/signs** ($\uparrow$ SOB, weight gain, fatigue, oedema):
 - If increasing congestion $\uparrow$ dose of diuretic or consider halving BB dose if this doesn't work.
 - If marked fatigue or bradycardia (see below) halve BB dose and review in 1-2 weeks.
- **Bradycardia:**
 - If bradycardia <50 bpm *and* worsening symptoms halve BB dose.
 - If severe bradycardia stop BB (rarely necessary).
 - Do ECG to exclude heart block.
- **Low BP/hypotension:**
 - If asymptomatic rarely needs change in therapy.
 - If symptomatic reduce/stop other vasodilators if possible e.g. nitrates, CCBs and if not congested consider reducing diuretic dose.
- **Important message to patients** - symptomatic improvement with B blockers takes a **long time** (often at least 3-6 months) and short-term side effects are common, but be patient and try to persevere as long-term quality of life and symptom improvement is the norm.

4 *Heart Failure with preserved Ejection Fraction (HFpEF)*

As discussed above in the classification section HFpEF is an important and increasingly common condition and it is not a benign - data from the UK National HF Audit 2009–2013 showed that all-cause mortality from HFpEF was *higher* than for HFrEF. A recent qualitative study [BJGP2020;70:e880](#) found that there is a lack of understanding and awareness of the condition in primary care, which acts as a barrier to timely diagnosis and treatment. This may be due to the fact that **until recently there has been no convincing evidence that any of the drugs traditionally used for HFrEF confer any improvement in clinical outcomes or survival benefit in HFpEF** (as concluded in the latest [cochrane 2021](#)) and treatment is mainly focused on improving symptoms and cardiometabolic risk factors. But the landscape is changing quickly with the seemingly ever-present group of drugs the SGLT2 inhibitors leading the charge. First we had some encouraging, but by no means conclusive, evidence in the form of the [EMPEROR-Preserved](#) trial. Then more recently we have had much more definite evidence from the [DELIVER trial](#) and a subsequent [meta-analysis](#).

► **Empagliflozin in HFpEF - EMPEROR-Preserved trial** [N Engl J Med 2021; 385:1451-1461](#)

There has been much excitement about the role of SGLT2 inhibitors in HFrEF with convincing evidence supporting their use, as discussed above. The sister trial to EMPEROR-reduced was the EMPEROR-Preserved trial which also looked at empagliflozin, but this time in **people with heart failure and preserved ejection fraction**. What did it show?

- Double-blind RCT, 5988 patients with heart failure and preserved ejection fraction (EF >40%) and class II-IV heart failure.
- Randomly assigned to either empagliflozin 10mg OD or placebo, on top of usual therapy.
- The primary outcome was a composite of CVD death or hospitalisation for heart failure.
- Overall the primary outcome was seen in 13.8% of the empagliflozin group and 17.1% of the placebo group, giving a HR of 0.79, 95% CI = 0.69-0.90.
- As with the above SGLT2i trials the improvements were seen **irrespective of whether the patient had type 2 diabetes or not**.
- Uncomplicated genital and urinary tract infections and hypotension were more common with empagliflozin.
- **Conclusion:** Empagliflozin reduces the composite of CV death and hospitalisation with heart failure for people with HFpEF by about 20%.

So on the face of it **this is a potentially ground-breaking trial** - the first drug to show significant CV outcome benefit for people with HFpEF, so it has inevitably caused quite a stir. However, as discussed in [our blog](#), there is one substantial caveat to this trial, which revolves around the definition of HF with *preserved* ejection fraction. As mentioned above in the classification section there has been a grey area as to where the bar lies on this cut-off. The ESC now defines HFpEF as having an EF of ≥50%, whereas **this trial also included people with EF of 41-49%**, which the ESC define as HF with *mildly* reduced EF. **Broken down by EF status subgroup analysis of the primary endpoint shows:**

- EF >40% to <50% (mildly reduced) ~33% of participants = HR 0.71, (CI = 0.57-0.88).
- EF ≥50% to <60% ~33% of participants = HR 0.80, (CI = 0.64-0.99) i.e. barely statistically significant.
- EF ≥60% ~33% of participants = HR 0.87, (CI = 0.69-1.10) i.e. not significant.

So, as we can see once we get to the now ESC definition of HFpEF (EF ≥50%) the results are barely significant and certainly not for people with EF ≥60%, with the group with EF 41-49% essentially deriving most of the benefit. Given the ESC assessment that *'patients with mildly reduced ejection fraction (41-49%) have, on average, features that are more similar to HFrEF than HFpEF'* it is possibly not surprising that this group do derive the greatest benefit as was seen in the trials in people with HFrEF.

► **Dapagliflozin in HFpEF - the DELIVER trial** [N Engl J Med 2022; 387:1089-1098](#)

So the EMPEROR-PRESERVED trial gave some encouraging, but not definitive, evidence supporting the use of SGLT2i in HFpEF. However, the DELIVER trial, looking at dapagliflozin showed much more concrete results, as discussed in our [NB blog](#):

- RCT of 6263 patients with HFpEF (EF >40%) randomised to either **dapagliflozin 10mg OD or placebo**, in addition to usual treatments, over a median of 2.3 years.
- The primary outcome was a composite of worsening heart failure (defined as unplanned admission to hospital or urgent visit) and CV death.
- There was an **18% reduction in the primary outcome in the dapagliflozin group** (HR 0.82, 95% CI = 0.73-0.92).
- Subgroup analysis showed significant reductions in worsening heart failure (HR 0.79, 95% CI = 0.69-0.91) and a non-significant trend to reductions in CV death (HR 0.88, 95% CI = 0.74-1.05).
- As with previous studies results were similar for those with or without diabetes.
- **Most importantly this trial showed similar improvements in those with EF >60% compared to those with EF <60% (unlike the EMPEROR-PRESERVED trial above).**
- **Conclusion: Dapagliflozin reduces the combined risk of worsening heart failure or CV death for people with HFpEF with similar results across those with preserved or mildly reduced ejection fraction.**

► **Meta-analysis of SGLT2i in patients with heart failure** [Lancet 2022; 400: 757-67](#)

Following on from DELIVER a meta-analysis puts together everything we have with regard to SGLT2i and heart failure. This review looked at the combined results of DELIVER and EMPEROR-preserved (i.e. just those with HFpEF) as well as a combined analysis of trials involving those with HFrEF as well:

- For those **patients with HFpEF there was a reduction in the composite of first hospitalisation with HF and CV death of 20%** (HR 0.80, 95% CI = 0.73-0.87).
- There were **consistent reductions** in both components of the primary outcome with reduced first hospitalisation for HF (HR 0.74, 95% CI = 0.67-0.83) **AND CV death** (0.88, 95% CI = 0.77-1.00).
- Comparing the results for patients with both preserved and mildly reduced EF in the broader context of the wider meta-analysis the data showed that **treatment effects were similar across subgroups, including NYHA functional class, other HF therapies, diabetes status and (importantly) ejection fractions.**
- **Conclusion: 'SGLT2 inhibitors reduced the risk of cardiovascular death and hospitalisations for heart failure in a broad range of patients with heart failure, supporting their role as a foundational therapy for heart failure, irrespective of ejection fraction or care setting'.**

These results are **incredibly significant, as this is the first data to show a class of drug that conclusively improves CVD outcomes in people with HF and preserved ejection fraction** (not just those with mildly reduced EF). The more concrete result is the reduction in hospitalisations for HF, but this data also gives the first indication that **SGLT2i may be the first drug to reduce CVD mortality** as well for people with HFpEF - the results for reduction in CV death were right on the borderline of statistical significance but suggest a 12% reduction. The [accompanying editorial](#) suggests that given the benefits seen across EF subgroups **SGLT2i should now become a 'bedrock of therapy for heart failure'**. The main caveat to these results was the woefully low number of Black participants (3.4%), which given their higher HF burden is disappointing and means we need to be careful of generalising these results more widely. This data has yet to be incorporated into formal guidance but it would be a surprise if NICE do not ultimately recommend SGLT2i for those with HFpEF (irrespective of diabetes status) as they have done for those with HFrEF.

KISS: Management of chronic heart failure with preserved ejection fraction

[BJGP2019;69;313](#) [ESC 2021](#) [NICE NG 106, 2018](#)

- The focus of managing HFpEF is on reducing symptoms with diuretics and optimal management of comorbidities and cardiometabolic risk factors, such as hypertension, AF, IHD and diabetes.
- Lifestyle is crucial - stop smoking, reduce alcohol, weight loss if overweight or obese.
- Increase aerobic exercise and refer for cardiac rehabilitation once stable.
- **Hypertension:**
 - Consider **ACEI/ARB** if **coexistent hypertension** and HFpEF.
 - Consider adding a **beta-blocker** if still hypertensive, especially if **coexistent AF or IHD**.
- **Diabetes:**
 - Good diabetic control important; the target will depend on life expectancy, frailty and comorbidities but generally aim for 53 mmol/mol (7%).
 - For those with **type 2 diabetes** offer **SGLT2 inhibitor** - recommended in both [ADA/EASD guideline 2022](#) and [NICE NG28](#)
- **Fluid overload:**
 - Treat with loop diuretics; NICE recommend low/medium doses (e.g. $\leq 80\text{mg}$ daily furosemide).
 - Thiazides can be considered alternatively if agents needed for hypertension.
- If patients remain symptomatic refer, and specialist options used in HFrEF may be tried.

NB Learning Points & Quality Improvement Ideas: Chronic Heart Failure

- Lower your threshold for checking a BNP level in older people with non-specific symptoms such as fatigue and breathlessness; **CHF is easily missed in these patients**, and BNP is a sensitive 'rule out test' but may be falsely negative if patients are already on heart failure treatments, and be aware that NT-pro BNP levels between 125-400 may miss as many as 1:5 people with heart failure.
- We may see promotion of earlier heart failure diagnosis from public health campaigns to '**BEAT-HF**' = Breathless, Exhausted, Ankle swelling, Time for a HF blood Test'.
- For patients with heart failure with reduced ejection fraction, if they remain symptomatic after the addition of BB and ACEi or ARB and careful titration to maximum tolerated doses, consider adding a mineralocorticoid receptor antagonist with careful monitoring of electrolytes.
- There is now a **strong evidence base for SGLT2i in patients with HFrEF**, even if they do not have diabetes, with NICE recommending both dapagliflozin and empagliflozin as options for HFrEF.
- For patients with CHF with reduced ejection fraction < 40% who remain symptomatic, discuss with specialist team the option of adding SGLT2 inhibitors or refer for consideration of ARNI (sacubitril valsartan).
- For patients with HFrEF **consider has their care been optimised with the '4 pillars'** of pharmacotherapy known to improve mortality and morbidity: beta-blockers, angiotensin drugs (ACEi or ARB or ARNI), mineralocorticoid receptor antagonist (MRA) and SGLT2i. Good subject for an audit?
- Be aware of useful **new guidance on monitoring renal function in patients with heart failure** on RAAS drugs, we can tolerate a dip in creatinine level < 30% and a rise in K levels to < 5.5 mmol/L.
- It is no longer advised to routinely suspend RAAS inhibitors during intercurrent illness with 'sick day rules', but to monitor closely and only suspend if creatine rises > 30% or K > 6 mmol/L.
- **Half of all patients with CHF now have heart failure with preserved ejection fraction (HFpEF)** - consider the diagnosis especially in older patients with the co-morbidities of hypertension, AF and ischaemic CHD.
- The mainstay of **management of HFpEF** consists of optimising the control of common co-morbidities of hypertension, AF and diabetes, addressing lifestyle factors and reducing fluid overload with loop diuretics.
- However, we have new evidence in the form of 2 RCTs and a subsequent meta-analysis that **SGLT2i significantly reduce HF hospitalisations, and probably also reduce CV deaths**, irrespective of ejection fraction level; this is highly significant data as **SGLT2i are the first drug to have shown significant CVD improvements in people with HFpEF**.
- **Quality Improvement Ideas:**
 - For patients with HFrEF **consider has their care been optimised with the '4 pillars'** of pharmacotherapy known to improve mortality and morbidity: beta-blockers and angiotensin drugs (ACEi or ARB or ARNI) titrated to maximum tolerated doses, +/- mineralocorticoid receptor antagonist (MRA) and SGLT2i.
 - For **patients with heart failure (irrespective of ejection fraction or diabetes status)** who remain symptomatic despite usual care consider seeking advice from the heart failure team as to whether they would benefit from a trial of an SGLT2i.
 - We have **increasing evidence supporting the use of IV iron for people with symptomatic HFrEF** (EF <45%) and iron deficiency (ferritin <100 mcg/L or transferrin saturation <20%) - if patients with HFrEF become symptomatic check FBC and ferritin and if ferritin <100 mcg/L refer/take advice for consideration of IV iron therapy.

5 *Prescribing in heart failure and changes in renal function*

New national guidance jointly produced by the Renal Association and the British Association for Heart Failure has been published to help us navigate the difficult balance of prescribing in heart failure and how to navigate changes in renal function [Heart 2019;105:904](#), summarised in the KISS below:

KISS: Prescribing in Chronic Heart Failure & Declining Renal Function

Guidance by the Renal Association and British Association for Heart Failure [Heart 2019;105:904](#)

- During initiation and titration of RAAS inhibitors testing renal function is mandatory
- Efforts should be made to initiate, titrate and maintain patients with heart failure with reduced ejection fraction (HFrEF) on RAAS inhibitors whether during intercurrent illness or worsening heart failure
- There is a danger that misplaced concerns over renal function may prevent patients from receiving treatment that is beneficial to their long-term outcome
- A decline in renal function of $\leq 30\%$ is acceptable
- In patients with fluid overload, high-dose diuretics are needed, and a decline in renal function is not an indication to reduce diuretic dose
- Towards the end of life, consider stopping RAAS inhibitor
- **Change in renal function compared to baseline with RAAS inhibitor in HFrEF:**
 - Increase in serum creatinine by $< 30\%$: continue RAAS inhibitor unless symptomatic hypotension
 - Increase in serum creatinine 30% to 50% : consider reduce dose of RAAS inhibitor or suspension
 - Increase in serum creatinine $> 50\%$: temporarily stop RAAS inhibitor
 - Severe renal dysfunction, stop RAAS inhibitor and specialist advice
 - In heart failure with preserved ejection fraction, consider stop RAAS inhibitor if rise in creatinine by $< 30\%$ and stop if $> 30\%$
- **Change in renal function compared to baseline with RAAS inhibitor in HFrEF and intercurrent illness:**
 - Do not routinely suspend RAAS inhibitors during intercurrent illness but monitor renal function
 - Stop any other medication which may worsen heart failure
 - Rise in creatinine by $< 30\%$, continue RAAS inhibitor but monitor closely
 - if $\geq 30\%$ suspend RAAS inhibitor and resume once intercurrent illness is over
 - If $K \geq 5.5$ mmol/L monitor closely and consider suspending RAAS inhibitor
 - If $K \geq 6$ mmol/L STOP RAAS inhibitor
- **Changes in renal function in patients with diuretics:**
 - In patients with fluid overload high dose diuretics are needed, and a decline in renal function is not an indication to reduce diuretic dose
 - An increase in creatinine of $\leq 25\%$ or a fall in eGFR of 20% is acceptable [BMJ2020;369:m2099](#)
 - if the patient remains congested more diuretic is needed, if the patient is hypovolaemic the diuretics should be withheld until needed
- **Patients who develop hyperkalaemia:**
 - If $K > 5.4$ check for dehydration and stop all other drugs that may cause increased potassium, including NSAIDs, trimethoprim, non-selective beta-blockers, amiloride and check for digoxin toxicity
 - If a patient clinically unwell with sepsis, hypovolaemia or AKI or decompensated heart failure specialist advice and likely admission needed
 - **If a patient is clinically well with no AKI**
 - Mild hyperkalaemia 5.5 to 5.9 mmol/L: consider a reduced dose of RAAS inhibitor but do not stop and increase the frequency of monitoring
 - Moderate k $6 - 6.4$ mmol/L: STOP RAAS inhibitor and repeat, cautiously re-start at a lower dose once $K < 5.5$
 - Severe $k > 6.5$ mmol/L: STOP RAAS inhibitor and ADMIT for immediate K-lowering treatment

ISCHAEMIA WITH NON-OBSTRUCTED CORONARY ARTERIES (INOCA)

You've seen the patient with exertional chest pain, referred them to your local service confident that they have angina (they even ticked all the boxes on the referral form!), the results are in... " *no obstructive lesions on coronary angiography - discharged back to GP*". So, we wave them off reassured that the angiogram is clear, but with a sense of unease as we have no explanation to account for their symptoms or a treatment plan...

This is a common scenario, *as up to 50% of patients undergoing elective coronary angiography for investigation of cardiac-sounding chest pain do not have evidence of obstructive coronary artery disease*. Previously, these patients were either dismissed as having 'non-cardiac pain', or given a label such as cardiac syndrome X, microvascular angina, Prinzmetal angina or coronary artery spasm and discharged. Both GP and patient were then left with uncertainty about the nature of the diagnosis and with many unanswered questions such as: *Should this patient be on anti-angina treatment? Do they need aspirin and statins? Are they at increased risk of coronary death or MACE (major adverse coronary events)? What monitoring (if any) do they need?*

Fortunately, in the last few years, a **new diagnostic definition** has appeared that includes these patients, validates their symptoms and gives some pointers to management. It is **ischaemia with non-obstructed coronary arteries (INOCA)** and this has usefully been reviewed in the [BMJ 2021;375:e060602](#) in a 'state of the art' clinical literature review.

This new diagnostic label of INOCA will have **two significant implications for us**. Firstly, we shall now understand why so many of our patients whom we thought had angina but had been reassured by normal angiograms have ongoing symptoms. Secondly, these patients are at increased cardiac risk so if symptoms are difficult to control we should have no qualms about referring them back to cardiology for specialist input.

The paper answers these questions for us, and we have summarised diagnosis and management into a KISS below.

► *So, what causes INOCA?*

- As you would expect there are multiple causes, but research reveals impaired epicardial blood flow ('slow flow') in the context of increased myocardial oxygen demand e.g. in response to exercise, a tachycardia or stress
- Patients with INOCA commonly report an association between stress and chest pain. Pathophysiological studies have shown that emotional stress could provoke epicardial coronary artery constriction and microvascular dysfunction, in the absence of coronary artery disease (NB this is likely to have contributed in the past to many of these patients being 'dismissed' as having psychogenic pain)
- Patients with INOCA are well recognised to have fluctuations in their condition, with so-called hot phases when the angina is frequent and difficult to control and cold phases when it is quiescent

► *Are patients with INOCA at increased risk of MACE (major adverse cardiovascular events)?*

- YES! There is a small increased risk. The review states that whilst there are problems with the heterogeneous nature of the data, the evidence supports an increased risk of death and MACE. A comprehensive meta-analysis has reported an incidence of mortality and MACE in patients with INOCA of 0.98 per 100 patient-years, compared to 0.2 per 100 patient-years in controls (put another way, if you have INOCA a 1% chance per year of MACE compared to 0.2% in controls)
- Smaller prospective studies have shown higher risks of INOCA, with an annual incidence of 6% for vasospastic angina and 8% for microvascular angina reported

► *Should patients with INOCA, therefore, be treated with cardio-protective drugs?*

- The paper argues that given the increased risk of MACE, patients with INOCA warrant cardio-protective treatment. However, they acknowledge that no large prospective randomised clinical trials have been carried out to evaluate this. These trials looking at treatment with high intensity statins, ACEi and anti-platelets are on-going
- All patients will benefit from lifestyle advice and exercise training, and avoiding precipitants - see KISS below

► *Should patients with INOCA be treated with anti-angina drugs?*

- Not necessarily. If attacks are mild and intermittent, they can be treated with prn nitrates
- However, it is estimated that 50% of patients with INOCA have recurrent episodes of chest pain, which is comparable to people diagnosed with obstructive coronary artery disease
- Before initiating treatment, we should be confident of the clinical diagnosis. So, if there is residual uncertainty after a normal angiogram we should refer back for further assessment for functional imaging
- See the KISS table below for treatment recommendations. The review recommendations are based on secondary care management, however, if there is a significant delay or patients will not be seen again there are useful learning points for us in primary care e.g.
 - Beta-blockers may help symptoms in microvascular angina, however, in vasospastic angina, they can precipitate angina and should be avoided
 - In vasospastic angina, calcium channel blockers improve infarct-free survival and their cessation is associated with an increased risk of MACE

NB Learning Points & Quality Improvement Ideas

- Patients with typical symptoms of angina that you have referred and been found to have non-obstructed coronary arteries are now defined under a new diagnostic umbrella called INOCA (ischaemia with non-obstructed coronary arteries)
- There are **two significant implications for us**. Firstly, we now understand why so many of our patients whom we thought had angina but had been reassured by normal angiograms have ongoing symptoms. Secondly, these patients are at increased cardiac risk so if symptoms are difficult to control REFER BACK to cardiology for specialist input
- There are two broad categories of INOCA, these are macrovascular dysfunction (vasospastic angina or coronary artery spasm) and microvascular dysfunction and these distinctions are important for management
 - Vasospastic angina or spasm will be diagnosed if it is visualised at angiography; microvascular dysfunction will not be diagnosed at angiography, it may be seen with functional perfusion scanning but a clinical diagnosis can be made on the basis of typical symptoms and non-obstructed arteries
- Patients with vasospastic angina should receive CCB for cardioprotection as well as symptoms, not be prescribed beta-blockers and SSRIs used with caution as may precipitate attacks
- Patients with microvascular dysfunction need individualised decisions on management based on the level of risk and frequency and severity of symptoms - see the KISS table below
- There is a strong link with stress triggering pain in patients with INOCA
- **Quality improvement idea:** *Are all patients with known coronary artery spasm on a CCB? and do they have an alert on their notes not to stop it, as this may trigger MACE?*

KISS: INOCA Ischaemia with non-obstructed coronary arteries Based on [BMJ 2021;375:e060602](#)

Definition

- **Patients who have typical symptoms and signs suggestive of angina but found to have no obstructive coronary arteries on angiography (i.e. no coronary lesions on angiography > 50%) should be diagnosed with INOCA**
 - non-cardiac causes of chest pain need to be excluded (these can be excluded on history alone, but may need further investigation if there is uncertainty)
- **There are two sub-types of INOCA.**
 - **Macrovascular dysfunction:** patients with coronary artery spasm, or vasospastic angina, have non-obstructed arteries but spasm is visualised at angiography; they have 'coronary macrovascular dysfunction'
 - **Microvascular dysfunction:** patients without spasm are considered to have 'coronary microvascular dysfunction'
- The diagnosis is supported by functional imaging from cardiology e.g. perfusion scanning may show impaired myocardial perfusion on stress imaging, or stress-induced regional wall motion abnormality in patients with INOCA. However, a diagnosis of INOCA can be made on the basis of typical symptoms and no obstructive lesions on angiography

Clinical implications

- Patients with INOCA are at increased risk of MACE (major adverse coronary events) and death, and ongoing symptoms have a significant negative impact on quality of life
 - Cardio-protective treatments (e.g. statins, ACEi) should be considered, but there is a lack of evidence with prospective trials on-going
 - All patients with vasospastic angina should receive CCB as these are cardio-protective in this group
- There is wide variation in severity and frequency of symptoms, and patients will often go through 'hot and cold' periods
- There is a strong association with stress-triggering symptoms
- Individualised clinical decisions need to be made on treatment based on severity, frequency and INOCA subtype - see table below
- **Refer back to cardiology if diagnostic uncertainty or continuing symptoms**

Management	Coronary macrovascular dysfunction (vasospastic angina)	Coronary microvascular dysfunction
Avoid precipitants	Smoking, beta-blockers, sympathomimetics SSRIs with caution	Stress
Cardioprotective treatment	Calcium channel blockers (CCB) Unknown if anti-platelets improve outcomes	Consider statin +/- ACEi Unknown if anti-platelets improve outcomes
Anti-ischaemic treatment	Exercise training or cardiac rehabilitation If infrequent mild symptoms, regular CCB + prn short-acting nitrates If more disabling or frequent pain consider long-acting nitrates and nicorandil	Exercise training, relaxation training and cardiac rehabilitation If infrequent mild symptoms, for prn short-acting nitrates If more disabling or frequent pain consider anti-anginals meds with individualised choice based on co-morbidities e.g. beta-blocker, CCB, nitrate etc

STABLE ANGINA

The management of **stable angina** has recently been updated in the [CKS 2022](#) and also by [SIGN 151 in 2018](#). NICE produced their **chest pain** guideline in 2010, updated in 2016 [NICE CG95](#), which was summarised and discussed in [BJGP2018;68:20](#). Evidence from the SCOT-HEART Trial [NEJM2018;379:924](#) confirms that CT coronary angiography should be the secondary care first-line investigation, in line with recent national guidelines.

Although by definition, patients with stable angina are considered 'stable', their risk of cardiac death remains significant. *It is estimated that those with stable angina have a CV death rate of 1.4–6.5% per year and a non-fatal MI rate of 0.3–5.5% per year.* This highlights the importance of good secondary prevention. Although most patients with stable angina will need a referral to secondary care for confirmation of diagnosis and risk stratification, primary care has a vital long-term role as most patients will ultimately be managed in General Practice.

► *When is angina considered stable or unstable?* [SIGN 151 2018](#) [BJGP2018;68:202](#)

Stable refers to angina which is classically retrosternal, triggered by exercise and relieved by rest. Symptoms occurring at rest or persisting once the activity stops, are **unstable** and need urgent referral to assess for acute coronary syndrome.

► *How should we assess suspected angina?* [CKS 2022](#) [BJGP2018;68:20](#) [SIGN 151 2018](#)

Despite the complicated plethora of modern diagnostic tests, all these guidelines agree that it remains the combination of **clinical history and risk factors that is the key component in evaluating possible angina**. The diagnosis can be made on the basis of clinical history alone. NICE recommend characterising presentations into typical, atypical and non-anginal pain based on the following **three features of the pain**:

- Precipitated by physical exertion
- Constricting discomfort in the front of the chest, in the neck, shoulders, jaw, or arms
- Relieved by rest or GTN within about 5 minutes

- **Typical angina** = 3/3 features
- **Atypical angina** = 2/3 features
- **Non-anginal pain** = 1/3 features

The likelihood of angina will increase with increasing numbers of CVD risk factors but of course, not all patients will present with typical symptoms e.g. no pain but breathlessness on exertion. If other risk factors are present (e.g. smoking, diabetes, FH, hypertension, dyslipidaemia etc) referral should be considered based on clinical judgement even in the absence of the typical features.

- Patients with **suspected stable angina should have the following assessed**:
 - Examination including BMI or waist circumference and heart sounds
 - Blood: FBC, renal function, lipid profile, fasting blood glucose or HbA1c, thyroid function
 - Resting ECG (being aware that a normal ECG does not rule out angina but that an abnormal ECG does increase the likelihood of angina)

SIGN and CKS also highlight a **key area for us in General Practice is to make a positive diagnosis of non-cardiac chest pain** which can reduce anxiety/distress and reduce unnecessary hospital admissions and consultations. They suggest those low-risk patients should be assessed in primary care wherever possible.

- **Features that make angina less likely include**:
 - pain that is continuous or prolonged or not related to activity.
 - Pain that is sharp or stabbing in nature.
 - Pain influenced by respiration, eased with antacids and simple analgesia.
 - Pain associated with anxiety symptoms - dizziness, palpitations, tingling or difficulty swallowing.

► *Who should we refer?*

Unstable patients (resting chest pain, features consistent with an acute coronary syndrome or rapidly progressive symptoms) obviously need urgent referral. In stable patients, NICE & SIGN advise patients with typical angina or atypical + risk factors or ECG changes consistent with coronary disease should be referred for further investigation.

► *What tests will I have doc?*

Although this is the remit of secondary care, patients may well ask us what tests they may have done after referral. If investigations are needed both SIGN & NICE ***propose a significant change in practice with non-invasive CT coronary angiography (CTCA) now the investigation of choice.***

One of the objectives of these new guidelines is to reduce the use of invasive angiography as the first line. CTCA is non-invasive and has very high diagnostic sensitivity making it an excellent rule-out test. With CTCA multiple 'slice' images of the heart are taken in rotation to provide detailed 3D images. Fast heart rates reduce quality so patients are often prescribed beta-blockers pre-testing. Exercise tolerance testing ETT should no longer be used as a first line diagnostic tool due to its variable diagnostic accuracy. Decisions on further investigation will vary depending on local pathways but options may include functional tests (e.g. ETT, stress echocardiography, stress perfusion MRI, myocardial perfusion scintigraphy). Only in the event of refractory symptoms despite maximal medical therapy or evidence of proximal or extensive disease on CTCA or functional imaging should invasive angiography now be offered.

A recent study from Scotland, the SCOT-HEART Trial [NEJM2018;379:924](#) randomised patients with stable angina referred to chest pain clinics to usual care +/- CTCA. The group who had investigation using CTCA (with consequent changes in treatment) had a significantly lower rate of death or MI (40% RRR) than the usual care group at 5 years, without resulting in a significantly increased risk of coronary angiography or revascularization.

How should we medically manage stable angina? See KISS below

► *What about revascularization with percutaneous coronary intervention (PCI) or CABG?*

Clearly, a specialist-led decision but SIGN recommends that patients with stable angina who remain symptomatic on optimal medical therapy should be considered for revascularization by CABG or PCI.

NB Learning Points: Stable Angina

- Make a positive diagnosis of non-cardiac chest pain and manage low-risk patients in GP.
- All patients with suspected stable angina should be referred to cardiology/chest pain services for confirmation of diagnosis +/- risk stratification.
- Beta-blockers remain the first line for symptom control (and may be cardio-protective) including patients with COPD; aim for a resting heart rate of 60 (shows adequate b-blockade).
- Patients with stable angina remain at significant risk of coronary death (1.4% to 6.4% per year) so vigorous lifestyle advice and secondary prevention is key. In addition to aspirin + statins, the routine use of ACEi is more controversial, but SIGN advise them in all patients with stable coronary disease for additional cardioprotection.

NB Quality Improvement Ideas: Stable Angina

- Are all your patients with stable angina on aspirin and a statin (ideally atorvastatin 80mg)?
- Review if your patients with stable angina should be on ACEi, especially those with CKD, previous MI and heart failure.

KISS: Stable Angina

Based on [CKS 2022](#) [NICE 2019, TA607](#) & [SIGN 151 2018](#) & [BJGP2018;68:202](#)

- **Assessment of possible stable angina (clinical history is the key component):**
 - Typical features of stable angina include 1) pain precipitated by physical exertion 2) constricting discomfort in the front of the chest, in the neck, shoulders, jaw, or arms 3) pain relieved by rest or GTN within about 5 minutes.
 - If ≥ 2 features above refer for confirmation of diagnosis +/- risk stratification.
 - More CVD risk factors increase the risk of angina; be aware some patients will present with atypical symptoms e.g. GI discomfort, exertional SOB with no pain.
 - All patients with suspected stable angina should be referred for confirmation of diagnosis; if symptoms are unstable (e.g. rest pain, rapidly progressive or features of acute coronary syndrome) refer immediately.
- **Secondary care investigation:**
 - non-invasive CT coronary angiography should be the investigation of choice.
- **Cardiovascular risk reduction is the most important part of management:**
 - **Lifestyle change** and lifestyle risk factor modification +++
 - **Aspirin** 75mgs daily; use clopidogrel if PAD, previous stroke, or aspirin not tolerated Dual antiplatelet therapy is only recommended in stable angina after PCI max 1 year.
 - Rivaroxaban 2.5mg plus low dose aspirin, recommended by [NICE 2019, TA607](#) as an option in patients with stable coronary artery disease at high risk of recurrence and low risk of bleeding.
 - **Statin** - atorvastatin 80mg daily and aim for reduction in baseline non-HDL-C of $>40\%$ (see lipid chapter for detail and other options if not at target based on updated [NHSE guidance](#)).
 - **Consider ACEI** if stable angina and diabetes, previous MI, CKD or LV systolic dysfunction. Recent research suggests they do not improve outcomes [BMJ2017;356:j4](#) in patients with stable CAD but without heart failure but is recommended by SIGN.
 - **Blood pressure** - NICE confirmed in [2022 update to NG136](#) to use the same BP targets for people with and without cardiovascular disease i.e. for aged <80 BP $<140/90$ and for aged ≥ 80 BP $<150/90$.
- **Symptomatic drug treatment:**
 - **GTN prn** - If develop chest pain, stop the activity and take immediately; repeat the dose at 5 minutes, if the pain persists at 5 minutes call 999; use before activities are known to bring on angina
 - **First line for symptoms**
 - **Beta-blockers, aim for resting heart rate <60 bpm** (Atenolol 100mgs, or metoprolol 50-100mg bd, or bisoprolol 5-20mg od).
 - BB can be used in the common co-morbidity of angina + COPD.
 - If beta-blockers not tolerated or contraindicated, use **rate-limiting CCB** e.g. diltiazem as monotherapy (but note C/I in heart failure or heart block).
 - If both BB and CCB C/I consider alternative monotherapy (consider cardiology advice):
 - Long-acting nitrate; Nicorandil (note restrictions/ulceration risk [MHRA 2016](#)); Ivabradine (note need to monitor for bradycardia [MHRA 2014](#)); Ranolazine (not recommended by SIGN).
 - **Adding second drugs if still symptomatic on monotherapy:**
 - CCB plus BB is the preferred combination. Preferably use amlodipine or felodipine + BB; diltiazem + BB can be used but with caution as may cause bradycardia & hypotension; AVOID BB + verapamil.
 - Other options are BB or CCB plus one of long-acting nitrate or nicorandil.
 - If symptoms become unstable, refer urgently.
 - If symptomatic on maximal doses of two drugs, refer for consideration of PCI. Do not try to control symptoms with 3 drugs in primary care.



ACUTE CORONARY SYNDROMES & SECONDARY PREVENTION

We don't tend to think of acute coronary syndromes as a diagnostic challenge, but an interesting recent BMJ 'Easily Missed' paper [BMJ2022;377:e069591](https://doi.org/10.1136/bmj-2022-069591) reminds us how it can present in vague and protean ways and diagnostic and referral delays may easily occur.

► **Recognising acute coronary syndrome** [BMJ2022;377:e069591](https://doi.org/10.1136/bmj-2022-069591)

- Approximately half of all patients experiencing an acute coronary syndrome (ACS) do not directly call emergency services, as they do not perceive their symptoms to be due to a major health problem leading to delays in diagnosis. In clinical practice, almost a third of cases are initially misdiagnosed.
- These delays are more common in women, older patients and those with multimorbidity.
- Women represent the majority of patients with ACS aged over 75, yet they are less likely than men to have the diagnosis considered.
- Patients often do not present with the classic symptoms of a sense of pressure in the central chest and left arm, leading patient and clinician to not suspect the diagnosis.
- **As well as classic chest and left arm pain, ACS can present with:**
 - Dyspnoea.
 - Isolated jaw or arm pain, bilateral arm pain.
 - Nausea and vomiting without pain.
 - Palpitations.
 - Pre-syncope and syncope.
- Missed diagnoses have significant clinical consequences e.g. in missed cases the 12-month mortality is 21%, compared to 6% and 8% for STEMI and NSTEMI diagnosed appropriately
- Physical examination is often normal, abnormal findings such as hypotension, tachycardia and lung creps are important when found.
 - When chest pain can be reproduced by palpation, this greatly reduces the chance of ACS.

So, what should we do if we clinically suspect ACS? And is an ECG useful?

- The paper is clear: *'when a GP suspects ACS based on clinical judgement the appropriate action is to refer the patient immediately to the emergency department'*
- A pre-hospital ECG helps to detect ischaemic changes and arrhythmia and can potentially affect the destination hospital, however, *'a normal ECG alone cannot be used to rule out ACS as up to 6% of patients with ACS have a normal initial ECG'*
- Whilst waiting for ambulance transfer we should assess patient's vital status and give a loading dose of aspirin (300mg), and to relieve symptoms GTN should be given unless the patient has hypotension
- The paper concludes we need to consider the possibility of ACS in any patient with new or worsening chest discomfort, dyspnoea or localised discomfort outside the chest such as arm, throat or jaw occurring while at rest and have a low threshold for emergency department referral of patients with ongoing symptoms

► **Acute coronary syndromes** [NICE NG185, 2020](https://www.nice.org.uk/guidance/ng185)

In November 2020 NICE updated their guidelines for Acute Coronary Syndromes NICE 185, with updated guidance on secondary prevention. Acute coronary syndromes comprise ST-elevation myocardial infarction (STEMI), non-ST elevation MI and unstable angina (NSTEMI). The guidelines relate mostly to secondary care management, our roles being early identification and hospital transfer and then secondary prevention. This new NICE guideline was summarised in the [BMJ 2021;372:m4760](https://doi.org/10.1136/bmj-2021-04760) complete with their usual infographics. ***For both STEMI and NSTEMI patients should be offered 300mg loading dose of aspirin as soon as possible and arrange immediate hospital transfer.*** We should NOT offer dual antiplatelet therapy to patients with chest pain before hospital transfer.

Early management of ACS is then reperfusion therapy, either via primary percutaneous coronary intervention (PCI) or if not possible with fibrinolysis. If interested in the pathway our cardiology colleagues follow, NICE have produced flow charts on the early management of [STEMI](#) and [NSTEMI](#).

KISS: Coronary Artery Disease, Secondary Prevention

Based on [NICE 2020, NG185](#) & [NICE 2019, TA607](#)

■ Cardiac Rehabilitation:

- For all patients, start before discharge and first appointment within 10 days of discharge.
- Programme based around physical activity, lifestyle advice and stress management.

■ Lifestyle:

- Mediterranean diet; stop smoking; weight loss to healthy BMI; moderate alcohol <14U weekly; regular exercise 20-30 mins per day and increase with fitness.

■ Drug therapy for secondary prevention:

- **ACEi** and continue indefinitely (ARB if intolerant); titrate to maximal tolerated dose within 6 weeks.
- **Dual antiplatelet therapy (DAPT)** with aspirin + another anti-platelet for up to 12 months, and then continue aspirin unless an indication for anticoagulation.
- **Beta-blockers** titrate to max tolerated dose; continue indefinitely if reduced LV ejection fraction otherwise can stop after 12 months; consider rate-limiting CCB (verapamil or diltiazem) if BB contraindicated and normal LV function.
- Aldosterone antagonist if heart failure with reduced ejection fraction after MI; start after ACEi and with careful monitoring of renal function & potassium.
 - If new HFrEF post-MI for eplerenone; if already on spironolactone for HFrEF continue.

■ Antiplatelets & dual antiplatelet therapy (DAPT):

- Aspirin should be continued lifelong, or clopidogrel 75mg if they are aspirin intolerant.
- DAPT is indicated in all patients following STEMI and NSTEMI:
 - Drug choices for DAPT (aspirin + clopidogrel, ticagrelor or prasugrel) will be secondary care led
 - But see also chapter 'which antiplatelet when' for more detail on general recommendations on DAPT after ACS depending on whether medical management or PCI acute vs stable.
- ◦ Some high-risk patients may be offered aspirin +/- clopidogrel + low dose rivaroxaban 2.5mg bd [NICE TA335](#)
- Usual recommendations for gastroprotection with a proton pump inhibitor.
- Low dose rivaroxaban 2.5mg plus aspirin, recommended by [NICE 2019, TA607](#) as an option in patients with stable coronary artery disease at high risk of recurrence and low risk of bleeding.
- **Anticoagulants?** Specialist-led decision, but...
 - If patient IS already on an anticoagulant, continue and add clopidogrel (if had PCI) or aspirin (if no PCI) for 12 months if not high bleeding risk.
 - For a new indication for anticoagulant, offer oral anticoagulant and clopidogrel or aspirin for up to 12 months.
 - Usual recommendations for gastroprotection with a proton pump inhibitor.

■ Statins (see Lipids chapter for more detail): [Click here](#) for NHSE guideline

- Intensive treatment with **atorvastatin 80mg** indefinitely [NICE 2016](#)
- **Target** - Non-HDL-C of >40% reduction from baseline or non-HDL-C < 2.5mmol/L or LDL-C <1.8 mmol/L (if baseline non-HDL-C not available).
- If not at target add ezetimibe 10mg OD.
- If still not at target, inclisiran or PCSK9 inhibitors may be options (see NHSE guideline for detail).

■ Hypertension:

- NICE confirmed in [2022 update to NG136](#) to **use the same BP targets for people with and without cardiovascular disease** i.e. for aged <80 BP <140/90 and for aged ≥80 BP <150/90

STROKE AND TIA

Stroke and TIA have become a really Hot Topic over the last 12 months, with some **landmark primary research and some significant guideline changes**, some of which may be practice altering immediately, and some of which may be considered aspirational currently.

In 2020 the *Treat Stroke to Target Trial* has published in the [NEJM2020;382;9](#) showing improved outcomes for secondary prevention after ischaemic stroke treating to a lower target of LDL cholesterol. We also have new primary research on stroke outcomes in the UK ([BMJ 2019;365:11778](#)) which gives cause for optimism by showing improved survival rates after stroke but also gives cause for concern by confirming increasing stroke rates in middle-aged people.

In 2018 the [POINT trial](#) published data suggesting **dual antiplatelet therapy (DAPT) after high-risk TIA or minor stroke was superior to aspirin alone**, which led to a practice-changing guideline recommendation from the BMJ later in the year ([BMJ 2018;363:k5130](#)). In May 2019, came the **updated NICE guideline (NG128)** on the initial management of stroke and TIA, which is notable for two things - one that IS in the guideline and one that ISN'T! And finally, we have some excellent research in the form of the [RESTART trial](#) asking the crucial question of whether we should, or should not, restart antiplatelets in people who have had intracerebral haemorrhage but who have vascular disease.

► *Stroke mortality and event rates in England* [BMJ 2019;365:11778](#)

This observational study published in the BMJ in May 2019, which was [widely reported in the media](#), highlights cause for optimism in stroke management, yet simultaneously throws up concerns and challenges for us to work on in primary care. First, the good news: this large population study, based on data in England from 2001-2010, showed that stroke mortality over this period reduced by 55%; although stroke event rates reduced by 20% over this period, the largest factor in reducing mortality was improved survival from stroke (71% of the improvement in mortality was attributed to improved survival). The study could not determine which factors were the most important in this improvement, but it is likely to be a combination of better acute stroke management and improved vascular risk factor control.

However, this study also highlighted a significant concern, namely a significant increase in stroke event rate in younger adults; stroke event rates in people aged 35-54 increased by 2% per year on average, in contrast to the downward trend seen in the older age groups. This concern was apparent in [PHE data published in 2018](#) and was discussed in an [editorial in the Lancet](#) which highlighted that over a third of first times strokes (38%) now occur in people aged 40-69.

As the BMJ editorial linked to the study above discusses ([BMJ 2019;365:12150](#)), whilst this is encouraging data by showing reduced stroke mortality, the major reduction in mortality was due to improved survival, not stroke prevention, conversely meaning a higher burden of people living with stroke. Whilst good news it reinforces the importance of strategies to prevent stroke in the first place, especially in younger adults, which *'requires a concerted global effort to improve stroke prevention, care and surveillance'*.

► *Treat Stroke to Target Trial, a comparison of LDL targets after ischaemic stroke* [NEJM2020;382;9](#)

The use of intensive lipid-lowering after an ischaemic stroke or TIA is established and recommended in guidelines. However, the optimal 'target level' of LDL cholesterol that should be aimed for has not been well studied, and guidelines often recommend intensive treatment but do not give a specific target we should aim for. The *Treat Stroke to Targets Trial* was a French study, funded by the French government, to answer the question: *is a lower LDL better for preventing recurrence?*

- This was a large prospective RCT in which patients who had suffered an ischaemic stroke in the previous 3 months or a TIA in the previous 14 days were randomised to an LDL target of < 1.8 mmol/L (70 mg/dl) or a higher target group of 2.3 - 2.8 mmol/L (90 to 110mg/dl)
- n = 3,000 patients followed for a median of 3.5 years



- They were treated with statins or ezetimibe or both to reach target levels; the lower target was achieved by adjustment of statin dose and with the addition of ezetimibe in 34% of patients
- The primary endpoint was major adverse cardiovascular events (MACE) or death from CVD
- **Results:**
 - there was a 22% RRR in MACE and death in the lower target group
 - In absolute terms, the risk was reduced from 10.9% to 8.5%
 - There was no significant difference between the two groups in the incidence of intracranial haemorrhage and newly diagnosed diabetes
- **Conclusion:**
 - *After an ischaemic stroke or TIA, treating to a lower target of LDL of < 1.8 mmol/L reduces the risk of recurrent CV events*

An associated editorial [NEJM2020;382;81](#) discusses the results and the 'complicated' relationship between statins and stroke. Previous studies have suggested that aggressive cholesterol reduction may increase the risk of haemorrhagic stroke and cause systemic side effects. Whilst this trial was very reassuring that intracranial haemorrhage or diabetes was not increased, 30% of patients in both groups discontinued the trial and other adverse events were not recorded. Unfortunately, the study was stopped earlier than planned due to a lack of funding. Despite these limitations, they conclude that these results '*could guide clinicians in the use of statins in patients with CVD who have had strokes*'.

► ***Effects of antiplatelet therapy after intracerebral stroke - RESTART trial*** [Lancet 2019;393:2613](#)

What should we do if we have patients on antiplatelets for secondary vascular prevention but they sustain an intracerebral haemorrhage? Patients surviving an intracranial haemorrhage are at increased subsequent risk of both haemorrhagic and ischaemic cardiovascular events. Do we restart the antiplatelet or not?

This was the question posed by the REstart or STop Antithrombotics Randomised Trial (RESTART), the first randomised trial to compare restarting vs avoiding antiplatelet therapy after intracerebral haemorrhage. It was funded by the British Heart Foundation.

- The trial was based in 122 UK hospitals and randomised 537 participants who suffered an intracerebral haemorrhage, whilst they were taking antiplatelets or anticoagulants for the prevention of occlusive vascular disease
- They were randomised to either 're-start' or 'avoid' an antiplatelet:
 - Patients were followed up for up to 5 years (median 2 years)
 - The antiplatelets (clopidogrel or aspirin monotherapy in nearly all cases) were re-started at a median of 76 days after the intracerebral haemorrhage
- Patients allocated to restart antiplatelet therapy experienced proportionally fewer recurrences of intracerebral haemorrhage, fewer major haemorrhagic events and similar numbers of occlusive vascular events
 - Recurrent intracerebral haemorrhage was seen in 4% of the 're-start' group vs 9% of the 'avoid' group, but this difference was not statistically significant (HR 0.51, 95% CI: 0.25-1.03)
- **Conclusion:** restarting antiplatelet therapy after an intracerebral haemorrhage that occurred while taking antithrombotic therapy is safe, and any risk of recurrent haemorrhage is probably too small to exceed the established benefits of antiplatelets for secondary prevention.

The finding that restarting antiplatelets may even *reduce* the risk of subsequent intracerebral haemorrhage (this was not a statistically significant finding but the trend was in this direction) was surprising but is not as counter-intuitive as it seems as some haemorrhagic strokes are due to underlying ischaemic disease. Whilst the study is reassuring on restarting antiplatelets, what it doesn't answer is when the optimum time is to restart the antiplatelet. The timing of restarting antiplatelets in this trial was very variable, with a median of 76 days, so as always further research is needed...

► *Stroke and TIA: diagnosis and initial management* [NICE guideline NG 128, May 2019](#)

There were a couple of significant changes to this updated NICE guideline which will have raised the eyebrows of both clinicians and commissioners alike. There is no doubt that some NICE guidelines are published with recommendations on what can be considered best practice, but which given current service provision and funding can only be considered aspirational in many areas of the country. This is one of them. First a **quick review of stroke management**, which hasn't really got any changes from a primary care perspective but there are some interesting changes which may happen in stroke units:

- If there is **suspected stroke in the community** (i.e. ongoing neurological symptoms at the point of assessment):
 - Use the FAST (Face, Arm, Speech, Time to call 999) screening tool
 - Exclude hypoglycaemia and admit immediately to the hospital
 - Do NOT give antiplatelet therapy
- Thrombolysis is still recommended within 4.5 hours of symptoms starting if a patient is suitable and has had haemorrhage ruled out
- The major change is that **thrombectomy** is now also recommended as an option (+/- thrombolysis) for people with an **occlusive proximal disease** if there is potential to salvage brain tissue
- Aspirin 300mg daily is still recommended until discharge (or max 2 weeks) at which point 'long term antithrombotic treatment' should be commenced; this is not specified but will be clopidogrel in most cases (+/- initial dual antiplatelet therapy with clopidogrel and aspirin in some cases)

On the flip side, there were some **significant changes for us with regard to TIA** assessment and management:

- **Refer immediately people with suspected TIA for specialist assessment within 24 hours**
- **Do NOT use the ABCD2 scoring system to assess the risk of subsequent stroke or inform urgency of referral**
- Although not directly relevant to us it is interesting to note that NICE now recommends that CT scans should not be done routinely in TIA clinics as CT rarely picks up alternative diagnoses, but that if imaging is considered necessary MRI should be the preferred modality

So, why have NICE made these recommendations and will it impact us in practice?

There are two reasons for these recommendations. Although we have become very comfortable with the use of the ABCD2 score for risk-assessing TIA, the **evidence suggests these risk-scoring tools are actually poor discriminators at determining the low or high risk of subsequent stroke after TIA**. The other important factor is that TIA diagnosis in the community is actually quite difficult, and the evidence suggests that between **30-50% of people referred to TIA clinics actually do *not* have TIA**, but an alternative diagnosis. So NICE has made these recommendations to a) improve diagnostic accuracy and b) reduce the risk of subsequent stroke in people that have been traditionally thought of as 'low risk' on the scoring systems. However, in many areas of the country, these will currently be aspirational objectives, with the obvious elephant in the room that many areas simply have neither staff nor resources to set up daily TIA clinics. NICE acknowledges that setting up daily TIA clinics would *'require significant additional resource and this may result in a substantial resource impact for the NHS'*.

The one notable area that was **not mentioned at all in the NICE guidance**, which came as a surprise to many, was the significant emerging evidence on dual antiplatelet therapy after minor stroke and high-risk TIA...

► *Dual antiplatelet therapy after a minor ischaemic stroke & high-risk TIA*

The POINT Trial, [NEJM2018;379:215](#) (1b)

After acute coronary syndromes or intervention, dual antiplatelet therapy (DAPT) has become standard practice. The picture after TIA or acute ischaemic stroke, however, has been less clear. Around 1 in 10 people will have a recurrent stroke within 90 days after a high-risk TIA or acute minor stroke. Aspirin reduces this by about 20% so it is an important research question as to whether or not additional gain can be made by early use of dual antiplatelets. The [POINT Trial, NEJM2018;379:215](#) sets out to compare aspirin alone with aspirin + clopidogrel in a large, international population.

- Patients with minor acute ischaemic stroke or high-risk TIA (ABCD2 score ≥ 4) were included
- They were randomized to dual antiplatelet therapy DAPT (clopidogrel 600mg loading dose then 75mg daily + aspirin) or aspirin alone for 90 days (aspirin dose varied from 75-325mg daily)
- **Results:** At 90 days, the DAPT group had fewer major ischaemic events compared to aspirin alone (5% vs. 6.5%) but at a cost of more major bleeds (0.4% vs. 0.9%)
 - In absolute terms, treating 1000 patients with DAPT rather than aspirin alone would prevent 15 major ischaemic events but cause 5 major haemorrhages
 - Most major ischaemic events were prevented with DAPT in the first week

► *Clopidogrel + Aspirin vs. Aspirin alone, Systematic review* [BMJ2018;363:k5108](#) (1a)

This new systematic review published in December 2018 was triggered by the findings of the POINT trial, and was the basis for the BMJ guideline recommendation below:

- They included data from 3 eligible trials, including POINT, n= 10,500 patients
- Patients were included following a high-risk TIA (ABCD2 score ≥ 4) or acute minor stroke
 - Patients were excluded if they had an indication for anticoagulation e.g. AF
 - Minor stroke was defined as a score of ≤ 3 on the [NIHSS Scale](#) and no residual neurological deficit
- They compared aspirin alone to aspirin + clopidogrel started within 24 hours of symptom onset
- Compared with aspirin alone, DAPT reduced the risk of recurrent stroke (RRR 30%, ARR 1.9%) with a small increase (absolute risk increase 0.2%) in moderate or severe bleeding
- DAPT also reduces moderate or severe functional disability by 1.4%
- There was no significant reduction in all-cause mortality
- Most recurrent stroke events occurred within the first 10 days and 'any benefit beyond 21 days is extremely unlikely'; this was also the conclusion drawn in editorials after the POINT trial
- **Conclusion:** This is 'high-quality evidence' that DAPT with clopidogrel + aspirin given with 24 hours of high-risk TIA or acute minor stroke will reduce subsequent stroke risk by about 20 in 1,000 people, with a possible increase in severe bleeding in 2 per 1,000 people, compared to aspirin alone

The research does have some limitations. Different studies used variable loading doses of clopidogrel (either 300mg or 600mg), and there have been no studies comparing aspirin + clopidogrel with clopidogrel alone which could be argued to be the optimal comparator.

► *Rapid Recommendation Clinical Practice Guideline Dec 2018* [BMJ 2018;363:k5130](#)

Following this research, this new 'rapid recommendation' clinical practice guideline was published with **a strong recommendation for DAPT** after high-risk TIA (ABCD2 score ≥ 4) and acute minor stroke.

- DAPT when started within the first 24 hours and continued for the first 10 to 21 days:
 - Reduces the rate of recurrent stroke by 1.9%
 - Reduces the rate of moderate or severe functional disability by 1.4%
 - Increases the risk of bleeding by 0.2%

■ *When should patients begin DAPT, and at what dose?*

- If brain imaging is done within 24 hours then DAPT should commence as soon as scanning has excluded haemorrhage or other cause
- If it is suspected there will be a delay of ≥ 24 hours in imaging, then primary care clinicians should commence DAPT
- Trials have used different dosages but based on the evidence they recommend
 - Clopidogrel 300mg loading dose and 75mg thereafter
 - Aspirin a dose between 75 mg and 345mg 'seems reasonable'

► *In summary, thoughts on referral and management of TIA in the community*

So for many of us, it is likely to be business as usual by using the (albeit imperfect) ABCD2 score and our current referral pathways, as the NICE guidance that all TIA patients be reviewed within 24 hours is likely to be more aspirational than practical currently. If patients with TIA won't get a specialist assessment within 24 hours the ABCD2 score still has an important role in helping to determine which patients may benefit from DAPT based on the BMJ guidance above. However, we do need to be aware of the high numbers of patients we see in the community with possible TIA that will *not* have had a TIA (may be as high as 30-50%). Thus, if there is any doubt on the diagnosis of TIA we should think carefully before exposing patients to the potential increased bleeding risk with DAPT and consider discussing such cases with the on-call stroke team before commencing DAPT. What we certainly will see in the near future (and probably already are) is patients coming out of TIA clinics on DAPT for 10-21 days.

NB Learning Points: Stroke and TIA

- If a patient has had a **suspected stroke** do NOT give antiplatelet drugs but arrange urgent admission for immediate scanning and specialist treatment:
 - Thrombolysis is still recommended within 4.5 hours of symptoms starting if a patient is suitable and has had haemorrhage ruled out
 - Thrombectomy is now recommended as an option for occlusive proximal disease
- If a patient has had a **suspected TIA**:
 - NICE now recommend all patients be assessed by a specialist within 24 hours, and we don't risk stratify using the ABCD2 score; however, in most areas, this will not be possible, in which case...
 - Calculate their [ABCD2 score](#)
 - If ABCD score ≥ 4 and urgent review within 24 hours is not possible to consider starting DAPT with clopidogrel + aspirin
 - However, be aware that current data suggests 30-50% of people referred to TIA clinics do NOT have TIAs so we need to be clear on the diagnosis of TIA if considering DAPT
- **Mortality rates from stroke have dropped by >50% between 2001-2010 in England**, however, this is mostly due to improved survival from stroke rather than stroke prevention
- In contrast to older adults, stroke rates in younger adults are increasing by ~2% per year, and now **over a third of first-time strokes occur in middle-aged (age 40 to 69) adults**
- The RESTART trial suggests **re-starting antiplatelets in those with vascular disease who have intracerebral haemorrhage is safe**, and may even reduce the risk of subsequent intracerebral haemorrhage, although it is unclear when the optimal time is to re-start antiplatelets

KISS: Ischaemic Stroke and TIA

[NICE May 2019](#) [NEJM2020;382;9](#) [BMJ 2018;363:k5130](#) [CKS 2018](#)

Acute management

- **Is it a stroke?**
 - Use a validated tool outside a hospital setting, e.g. [FAST test](#).
 - Could it be a posterior circulation event? e.g. visual disturbance, ataxia, vertigo etc.
 - **If stroke suspected**, exclude hypoglycaemia, do not give antiplatelets and arrange **emergency admission**.
 - Be aware patients may have ongoing neurological deficits despite a negative FAST test - if so manage as acute stroke.
- **If TIA suspected:**
 - *Initiate aspirin 300mg daily and refer ALL patients with TIA for specialist assessment within 24 hours, without doing an ABCD2 score* [NICE 2019](#).
 - If specialist assessment within 24 hours is not possible, assess using [ABCD2 score](#):
 - If ABCD2 score ≥ 4 consider starting dual antiplatelet therapy (DAPT - clopidogrel loading dose 300mg and 75mg daily thereafter, with aspirin 75-300mg daily); however, *be aware 30-50% of people with suspected TIA in the community do NOT have TIA* so if diagnostic doubt consider discussing with on-call stroke team before commencing DAPT.
 - If ABCD2 ≤ 3 , start aspirin 300mg daily and refer for specialist assessment ASAP.
 - If a patient is on an anticoagulant, arrange urgent admission.
 - If ≥ 1 TIA discuss the need for admission with a stroke specialist.

Secondary prevention

- **Antiplatelet therapy:**
 - **DAPT may be recommended for first 10 to 21 days** after ischaemic stroke or high risk TIA (ABCD2 ≥ 4), thereafter:
 - Clopidogrel 75mgs daily first line.
 - If clopidogrel not tolerated, dipyridamole M/R 200mg bd + low dose aspirin.
 - If both clopidogrel & dipyridamole contraindicated or not tolerated, aspirin alone.
 - If patient already has coronary artery disease or peripheral vascular disease low dose rivaroxaban 2.5mg BD plus aspirin 75mg OD is an option ([NICE TA607 2019](#))
 - If in AF for anticoagulation (see AF chapter).
 - **Statins (see Lipids chapter for more detail):** [Click here](#) for NHSE guideline
 - Intensive treatment with **atorvastatin 80mg** indefinitely.
 - **Target** - Non-HDL-C of $>40\%$ reduction from baseline or non-HDL-C $< 2.5\text{mmol/L}$ or LDL-C $< 1.8\text{mmol/L}$ (if baseline non-HDL-C not available).
 - If not at target add ezetimibe 10mg OD.
 - If still not at target, inclisiran or PCSK9 inhibitors may be options (see NHSE guideline for detail).
- **Hypertension/Blood Pressure:**
 - Some guidelines have advocated lower BP targets (e.g. $<130/80$) in the context of stroke/TIA.
 - However recent NICE updates to the Hypertension guideline [NG136 March 2022](#) have reviewed the evidence and felt the evidence was not strong enough to advocate lower targets than for the general population so recommend we **use the same blood pressure targets** for people with and without cardiovascular disease - if **aged <80 years old target $<140/90$** and if **≥ 80 target $<150/90$** .
- **Carotid endarterectomy?** Those considered as candidates should have rapid carotid imaging and offered surgery if a stenosis of $>50\%$.

WHICH ANTIPLATELETS WHEN?

KISS: Antiplatelet Treatment NICE CKS, Antiplatelet Treatment 2022	
Indication	Preferred antiplatelet regimen
Primary prevention of CVD	Do not routinely prescribe; <i>consider</i> aspirin in people at high risk of MI or stroke bearing in mind it is not licensed for primary prevention, there is limited evidence of benefit and risk of harm
Stroke or TIA	- Clopidogrel 75mg daily preferred, if unsuitable: - Aspirin 75mg OD + dipyridamole M/R 200mg bd - Evidence BMJ2018;363:k5108 supports dual antiplatelet therapy (DAPT) with aspirin + clopidogrel for the first 10 to 21 days after minor ischaemic stroke or high-risk TIA
Peripheral arterial disease (PAD) or multi-vascular disease	- Clopidogrel 75mg daily - Aspirin 75mg if clopidogrel unsuitable
Symptomatic PAD or coronary artery disease at high risk of ischaemic events	- Consider aspirin 75mg plus low dose rivaroxaban 2.5mg BD (NICE TA607), if considered high risk: - Aged >65 years old, or - atherosclerosis in ≥2 territories, or ≥2 risk factors (smoking, diabetes, eGFR 15-59, heart failure)
Angina	- Aspirin 75mg daily indefinitely - Clopidogrel 75mg daily if aspirin unsuitable
Acute coronary syndrome ACS (unstable angina, STEMI or NSTEMI) medically managed Initiated in secondary care	- DAPT with Aspirin 75mg daily + ticagrelor 90mg bd for 12 months, then continue aspirin 75mg - If high bleeding risk DAPT only for one month. If low bleeding risk, DAPT may be continued after a year with aspirin + ticagrelor 60mg bd or clopidogrel for up to 36/12
ACS, managed with a percutaneous coronary intervention PCI Initiated in secondary care	- DAPT with aspirin 75mg daily + ticagrelor 90mg bd or prasugrel 5mg or 10mg daily usually for 12/12 then continue aspirin 75mg - DAPT with aspirin + clopidogrel if ticagrelor/prasugrel unsuitable - If high bleeding risk DAPT for the shorter time period; if low-risk DAPT may be continued with aspirin + ticagrelor 60mg bd or clopidogrel for up to 36/12
Stable coronary artery disease undergoing PCI (Initiated in secondary care) or coronary bypass grafting	- DAPT with aspirin + clopidogrel usually for 6/12 - DAPT with ticagrelor or prasugrel may be advised for shorter or longer periods up to 36/12 based on individual risk profile. - People who undergo CABG should continue DAPT when safe to do so after surgery; specialist recommendation on length of treatment

HEART VALVE DISEASE

Towards the end of 2021, NICE produced new guidance ([NG 208, Nov 2021](#)) on investigating and managing heart valve disease in adults. How this fits into Primary Care and how much we will get involved as GPs will be variable depending on local resources, referral pathways and access to echocardiography. However, it does help give us some guidance on what to do if we pick up new murmurs and thresholds for referral for intervention. Not all valve disease is the same, and when (and when not) to intervene very much depends on the type of valve disease present, severity of the disease (which should be graded by the echo technician into mild/moderate/severe) and severity of symptoms. But **NICE are clear that not all people with valve disease need specialist referral** - although they go short of saying we should not/never refer people with mild valve disease, they stress *'that patients and healthcare professionals should be aware that mild valve disease is very common in people over 70, it seldom causes symptoms and does not progress in most cases.'* Given the data, they quote that **in the >65s asymptomatic heart disease may be present in >50%, but clinically significant disease is only present in 11%**, the rationale for streamlining referrals is clear. The **exceptions/caveats** being **bicuspid aortic valve disease** (which should always be referred) and **mild aortic and mitral stenosis** that should at least have follow-up echos every 3-5 years due to the risk of progression.

KISS: Heart Valve Disease (in adults)

NICE NG 208 November 2021

- **Consider referral for echo** if murmur and no symptoms/signs, depending on the nature of the murmur, family history, age (esp if >75) and medical history (e.g. AF). (**NB comment** - this wording reflects the uncertainty that many asymptomatic murmurs will not turn out to be valve disease but NICE accept that differentiating innocent murmurs from pathological ones clinically is not easy so referral for echo should remain an option due to this uncertainty).
- **Offer referral for echo if:**
 - Symptoms (e.g. angina, SOB) or signs (e.g. oedema) or abnormal ECG, or
 - Ejection systolic murmur with reduced second heart sound even if asymptomatic (increased risk of significant aortic stenosis (AS) and more likely to need intervention).
- **Refer for urgent (within 2 weeks) specialist assessment (or urgent echo):**
 - Systolic murmur with exertional syncope (if caused by AS high risk of poor outcome), or
 - Murmur with severe symptoms (angina, SOB at rest/minimal exertion).
- **Routine referral for specialist assessment:**
 - Moderate or severe valve disease of any type, or
 - Bicuspid aortic valve disease of any severity.
 - Note **mild valve disease is common and rarely progresses to become clinically significant.**
- **Refer for consideration of intervention** (may be relevant if access to direct echo - note these parameters are based on common echo result parameters but for more detailed parameters/referral guidance see the full guideline):
 - Aortic stenosis - if LVEF <55% or BNP/NT-pro-BNP >2x upper limit of normal.
 - Aortic regurgitation - if LVEF <55%.
 - Mitral regurgitation - if LVEF <60%.
- **Monitoring:**
 - Patients with more severe disease should be under specialist review every 6-12 months.
 - Mild aortic or mitral stenosis should have echo every 3-5 years (more at risk of developing severe disease needing intervention).
- **Anticoagulation/antiplatelet after intervention for valve disease** (should be directed by secondary care):
 - Anticoagulation (DOAC or warfarin) is *not* needed after surgical biological valve replacement unless needed for another indication.
 - Single antiplatelet therapy (aspirin or clopidogrel if not tolerated) should be considered after TAVI (transcatheter aortic valve implantation).

VENOUS THROMBOEMBOLISM

Much has changed in the world of venous thromboembolism (VTE) since the previous iteration of NICE guidance dating back to 2012...First, we had NOACs, then they became less novel so they morphed into DOACs, suspected pulmonary emboli (PEs) were very much the realm of inpatient treatment in hospitals and there was concern that unprovoked deep vein thrombosis (DVTs) could be associated with a 20% risk of occult cancer with subsequent recommendations for multiple further investigations. But the evidence, and medicine, marches on giving us a **substantially different guideline** ([NICE NG158 March 2020](#)) on **diagnosis and management of VTE** with new recommendations on DOACs, tweaking to the assessment pathway, more emphasis on outpatient treatment of low-risk PEs, and a more selective investigation framework for those with unprovoked DVTs. This was reviewed in the BMJ ([BMJ 2020;369:m1565](#)) with the **usual BMJ infographic** ([click here](#)) and has also been covered in the [BJGP BJGP2021;71;379](#).

As is increasingly the 'norm' with NICE guidance now, there are recommendations we can put into practice now (or consolidate what we're already doing) e.g. use of DOACs, and some that will be aspirational for many e.g. introduction of point of care testing for D-dimer.

For a summary of the assessment, investigation and management of DVT see our KISS summary below, but first a **bit more detail of what's new, and why, in this guidance** ([NICE NG158 March 2020](#)):

► D-dimer testing for suspected DVT or PE

- Consider a **point-of-care D-dimer test** if lab facilities are not immediately available.
- When using D-dimer tests (point of care, or lab) consider using an **age-adjusted threshold for patients over the age of 50**.

We are slowly (many would say far too slowly!) starting to see the introduction of various point of care (POC) tests in a primary care setting. D-dimer POC testing is the next one that is likely to get introduced over time as ultimately this can be a cost-effective intervention, reducing the numbers of people needing interim anticoagulation while awaiting lab tests. NICE recommend that lab testing is still the 'gold standard' due to better quality assurance, but recognise that getting results back within 4 hours is not realistic in most cases and if so POC testing can be used. NICE also recognised the high number of false positives with D-dimer testing in the >50 age group, and although they admit the 'evidence was not plentiful' they felt there was enough to suggest that adjusting the thresholds in this age group did not seem to affect the accuracy of the test, but did reduce the numbers having unnecessary further testing and treatment.

► High risk for DVT (Wells score ≥ 2) with positive D-dimer, but negative scan

- For people with Wells score ≥ 2 with an initial negative leg vein ultrasound but a positive D-dimer offer a repeat scan 6-8 days later.

Many of us may have been faced with this scenario - clinically we think there may be a DVT, D-dimer is positive but the scan is negative...what do we do? Well, there is a recognition that venous ultrasound can miss some small or early DVTs, so NICE reconcile this by recommending, in this scenario, we repeat a scan 6-8 days later. Use of the word 'offer' implies we should do this in most situations, which will have an impact on service provision but lowers the risk of us missing that early/small and evolving DVT.

► Which anticoagulant?

It came as no surprise that for the **majority of patients a DOAC is recommended as a first line treatment**:

- Offer apixaban or rivaroxaban to people with confirmed or suspected DVT or PE UNLESS:
- Both are unsuitable, in which case use LMWH for ≥ 5 days followed by dabigatran or edoxaban or LMHW concurrently with warfarin until INR ≥ 2.0 .

However, there are some groups we need to be aware of, in whom a **DOAC may not be suitable or needs careful consideration**:

- **People at extremes of body weight (<50kg or > 120kg)** - consider anticoagulation treatment with regular monitoring of therapeutic levels to ensure effective anticoagulation (but note updated guidance from ISTH below).
- **Renal failure** (remember we need to use CrCl for dosing, NOT eGFR - [click here](#) for CrCl calculator):
 - The same recommendations essentially apply for first/second-line treatments above as long as CrCl ≥ 15 ml/min, with the caveat that if dabigatran is considered the CrCl needs to be ≥ 30 ml/min.
 - If CrCl < 15 ml/min then the option is LMWH either alone or concurrently for ≥ 5 days with warfarin until INR ≥ 2.0 followed by warfarin alone.
- **Active cancer** - DOACs now CAN be considered as an option, taking into account bleeding risk with the underlying tumour and chemotherapy drug interactions, or alternatively LMWH either alone or concurrently for ≥ 5 days with warfarin until INR ≥ 2.0 followed by warfarin alone.

The **extremes of body weight group is one, in particular, we need to be aware of** - firstly we need to make sure we use CrCl for dosing otherwise, we risk overdosing in the underweight group or under-dosing in the overweight group. NICE highlight the limited evidence of the effectiveness of different treatments at the extremes of weight, and that an International consensus statement recommended (at the time of NICE publication) that DOACs not be used in people ≥ 120 kg. They thus recommend that anticoagulation with monitoring is used - they have not specified this be warfarin, as some hospitals may have access to DOAC level monitoring, in which case this can be considered, but for the majority, this is not available so warfarin and INR monitoring will be needed - **NICE recommend that we take specialist advice in this group**. However, since the NICE guidance, the **ISTH have changed their recommendations ([ISTH guideline on DOACs in patients with obesity July 2021](#))** for people with **weight >120kg or BMI >40**, based on accumulating evidence. They **now recommend we can use rivaroxaban or apixaban at standard doses in this group** (without the need for DOAC blood level monitoring), and comment the evidence is best for rivaroxaban. They recommend dabigatran and edoxaban should NOT be used in people with weight >120kg or BMI >40.

The other big change here is that **DOACs CAN be considered for those with active cancer**. Again NICE highlight treatment in this group has not been studied sufficiently to make definitive conclusions. There was some limited direct and indirect evidence that DOACs may be as effective as LMWH, but probably with higher bleeding risks, particularly those with GI or GU malignancies. But as we know LMWH is obviously a pain to use and expensive, hence the change of stance that DOACs can at least be considered, depending on tumour type and chemotherapy agents etc. Although not specified by NICE the decision on what anticoagulant to use, and for how long (usually 3-6 months) should realistically be specialist-led.

► Investigation for cancer after unprovoked DVT/PE

- Review the medical history, and baseline blood tests including FBC, renal and hepatic function and do an examination.
- Do NOT offer investigations for cancer with unprovoked DVT/PE unless there are relevant clinical symptoms or signs.

This is a **significant change from the first iteration of this NICE guideline**. At that time there were concerns from observational data that as many as 10-20% of people with unprovoked VTE may have an underlying malignancy. NICE thus recommended considering a number of further investigations including CT scans. However, further data has questioned those numbers, and more importantly, we now have **RCT evidence that routine screening tests (including CT and PET scanning) for cancer do not improve outcomes**. NICE accept that unprovoked VTE is associated with cancer in some people, but that screening did not improve outcomes and risks costly and time-consuming investigations with subsequent risks of overdiagnosis and overtreatment; they thus recommend further investigation is not routinely recommended unless there are concerning symptoms or signs.

► Signs or symptoms of PE

- If clinical suspicion of PE is low, consider using the pulmonary embolism rule-out criteria (PERC) to help determine whether any further investigations for PE are needed.

So what is the PERC rule and how might it apply in General Practice? The **PERC rule is a 'rule-out' test that has been studied in low-risk populations** (those in whom the likelihood of PE is <15%) and has been shown to be able to accurately rule out PE, as long as none of the criteria is met. It, therefore, sits nicely with the low-risk presentations we may see in General Practice. NICE highlight that the original data has been studied in emergency departments but could see *'no reason why its use should be limited to this setting or why the diagnostic accuracy of PERC would differ in other settings'* and hope its implementation will lead to a reduction in unnecessary D-dimer testing and anticoagulation treatment. PERC criteria:

- Age ≥50 years old
- HR ≥100
- O2 sats (room air) <95%
- Unilateral leg swelling
- Haemoptysis
- Recent surgery or trauma (within 4 weeks requiring GA)
- Prior PE or DVT
- Hormone use (oral contraceptives, HRT or oestrogenic hormones in men or women)
- If **ALL the above are negative risk of PE is <2% so can safely be ruled out (in LOW-RISK populations)**

Whether we can safely assess patients with a higher risk of PE in primary care is not clarified in the NICE guidance - when discussing assessment and management of people with suspected PE, the guidance talks about ambulatory care unit management and 'outpatient treatment' which implies secondary care-based management. However, given the advice above to start using POC D-dimer testing (if available) it is feasible that primary care could consider doing a **Two-level PE Wells score** in these situations:

Clinical Feature	Points
Signs/symptoms of DVT (minimum of leg swelling and tenderness of deep veins)	3
Alternative diagnosis is less likely than PE	3
Heart rate >100 bpm	1.5
Immobilisation for >3 days or surgery within last 4 weeks	1.5
Previous DVT/PE	1.5
Haemoptysis	1
Malignancy (on treatment of treated in last 6 months, or palliative)	1

- Score >4 = high risk of PE admit/refer for CTPA.
- Score ≤4 = lower risk of PE - do D-dimer for results within 4 hours (consider POC test if available):
 - If D-dimer -ve consider alternative diagnoses.
 - If D-dimer +ve admit/refer for CTPA.

So as is often the case, how we manage patients presenting at higher risk of PE will depend on our clinical judgement and facilities for testing, as well as local pathways.

KISS: DVT

[NICE NG 158 March 2020](#) [BMJ 2020;369:m1565](#) [BJGP BJGP2021;71;379](#)

Diagnosis of DVT

- If VTE is clinically suspected use the modified [Wells score](#) for DVT
 - The Wells score is used to determine which subsequent investigation is needed, the Wells score alone is not intended to be a rule-out test
- DVT likely (score ≥ 2):
 - Refer for an urgent ultrasound - Ideally within 4 hours; if unable to scan in 4 hours:
 - Check D-dimer and give interim anticoagulation and scan ASAP.
 - If scan -ve check D-dimer if not already done:
 - If scan and D-dimer both -ve stop anticoagulation and look for alternative cause.
 - If scan -ve but D-dimer +ve, stop anticoagulation and repeat scan in 6-8 days.
- DVT unlikely (score ≤ 1):
 - Check D-dimer test ideally with result within 4 hours - if results >4 hours give interim anticoagulation whilst awaiting result:
 - If D-dimer +ve refer for urgent ultrasound and continue anticoagulation till result is back.
 - If D-dimer -ve stop anticoagulation and look for alternative cause.
 - **D-dimer testing:** Lab results within 4 hours considered the gold standard, but **consider point of care D-dimer test** if access to test and lab results not available within 4 hours. Age-adjusted D-dimer results recommended for patients >50 years old (lab dependent).

Management of DVT

- DOACs now considered first-line treatment for most people:
 - Remember to dose ALL patients using CrCl NOT eGFR - [click here](#) for CrCl calculator.
 - Rivaroxaban and apixaban first line options.
 - If both unsuitable use LMWH for ≥ 5 days followed by dabigatran or edoxaban or LMWH concurrently with warfarin until INR ≥ 2.0 .
- Caution in the following groups (Specialist advice recommended):
 - **People at extremes of body weight (<50kg or > 120kg):** Anticoagulation with regular monitoring of therapeutic levels required - for most this will be warfarin (after 5 days of LMWH) unless you have access to DOAC level monitoring; however, more recent [ISTH guidance July 2021](#) now recommends we *can* use rivaroxaban (best evidence) or apixaban at standard doses (without DOAC blood level monitoring) for people with weight >120kg or BMI >40.
 - **Renal failure with CrCl ≤ 15 ml/min:** In most cases likely to be LMWH for ≥ 5 days concurrently with warfarin until INR ≥ 2.0 ; remember dabigatran can only be used if CrCl ≥ 30 ml/min.
 - **Active cancer:** DOACs can now be considered depending on tumour type, chemotherapy and overall bleeding risk, otherwise LMWH for ≥ 5 days and warfarin.
- **Duration of therapy:** ≥ 3 months in all but may be extended depending on risk factors. If longer-term anticoagulation is recommended but declined NICE say aspirin 75-150mg daily can be considered.

Further investigations after unprovoked DVT

- **Cancer?:** Review the medical history, and baseline blood tests including FBC, renal and hepatic function and do an examination; Do NOT offer further investigations for cancer with unprovoked DVT/PE unless there are relevant clinical symptoms or signs.
- **Thrombophilia testing?** On specialist guidance; will depend on presentation and family history.



KISS: Assessment for PE

[NICE NG 158 March 2020](#) [BMJ 2020;369:m1565](#) [BJGP BJGP2021;71;379](#)

Assessment of people at LOW risk of PE (PE risk <15%)

- Consider the PERC criteria:
 - Age ≥ 50 years old
 - HR ≥ 100
 - O₂ sats (room air) <95%
 - Unilateral leg swelling
 - Haemoptysis
 - Recent surgery or trauma (within 4 weeks requiring GA)
 - Prior PE or DVT
 - Hormone use (oral contraceptives, HRT or oestrogenic hormones in men or women).
- If **ALL** the above are negative risk of PE is <2% so can safely be ruled out.

NB Learning Points & Quality Improvement Ideas: VTE

- Rivaroxaban and apixaban are now the preferred treatment for most people with suspected or confirmed VTE.
- But we need to be aware of **situations with DOACs should be used with caution** (and consider specialist advice):
 - Extremes of body weight <50kg or > 120kg.
 - Renal failure (remember to use [eCrCl](#), NOT eGFR when dosing DOACs).
 - Some cancers, especially those at higher bleeding risk e.g. GI or GU cancers.
- Recent [ISTH updated guidance](#) now advises we can use rivaroxaban or apixaban for people with weight >120kg or BMI >40.
- **Further investigation for cancer is now not routinely recommended in people with unprovoked VTE** unless clinical features or baseline blood abnormalities suggest this possibility.
- **For people at low risk of PE, consider using the PERC criteria** - a simple-to-use clinical decision system which can safely rule out PE in low-risk populations without the need for further investigation, as long as all the criteria are met.

PERIPHERAL ARTERIAL DISEASE

NICE updated their guideline on peripheral arterial disease (PAD) in 2020 [NICE CG 147, Dec 2020](#). They have also made a recommendation that people with *symptomatic peripheral arterial disease be considered for low-dose rivaroxaban plus aspirin for the prevention of major adverse cardiovascular events* [NICE 2019, TA607](#).

Peripheral arterial disease (PAD) was recently reviewed in an excellent BMJ review article ([BMJ 2018;360:j5842](#)), and was accompanied by a useful infographic ([click here](#)). It reinforces the key role we have in GP in managing patients with PAD, in particular reducing their overall cardiovascular risk, and subsequently, we've had an interesting piece of primary research asking whether low-level exercise therapy is as effective as the high-intensity exercise therapy that is currently recommended [JAMA. 2021;325\(13\):1266-1276](#) (spoiler it isn't!). There has also been a great paper on the 'easily missed' chronic limb-threatening ischaemia [BMJ 2018;360:j5460](#).

► Rivaroxaban for secondary prevention in people with symptomatic peripheral arterial disease

In 2019 NICE published new technology appraisal guidance [NICE 2019, TA607](#) on the role of low-dose rivaroxaban in combination with aspirin for secondary prevention in **people with coronary artery disease or symptomatic peripheral artery disease at high risk of ischaemic events**. They recommend:

- Low dose rivaroxaban 2.5mg twice daily in combination with low dose aspirin in patients with coronary artery disease, or *symptomatic peripheral arterial disease at high risk of ischaemic events*
- For people at high risk, this is defined as:
 - Aged 65 or over, or
 - Atherosclerosis in ≥2 territories e.g. coronary, cerebral or peripheral arterial, or
 - ≥ 2 of the following risk factors: current smoker, diabetes, CKD with eGFR between 15 and 60 ml/min, heart failure or previous ischaemic stroke
- Assess a person's bleeding risk before starting rivaroxaban

NICE made these recommendations following the results of the peripheral arterial disease branch of the [COMPASS Trial Lancet 2017](#).

- This was a large, placebo-controlled prospective RCT of patients with stable coronary artery disease or peripheral arterial disease (see coronary artery disease chapter for the CAD component)
- 7,500 patients with stable CAD were randomised to rivaroxaban 2.5mg bd plus low-dose aspirin, rivaroxaban 5mg bd alone or aspirin alone
- The mean study duration was two years
- The combination of rivaroxaban 2.5mg bd plus low dose aspirin reduced the primary outcome of major adverse cardiovascular events with a RRR of 28%, in absolute terms reducing it from 7% to 5% and reduced major adverse limb events including major amputation from 2% to 1%
- The combination of rivaroxaban with aspirin increased the risk of major bleeds from 2% to 3%, but there was no significant increase in cerebral or fatal bleeding
- By contrast treatment with rivaroxaban alone did not improve outcomes compared to low-dose aspirin
- **Conclusion:** *In patients with PAD, the combination of rivaroxaban and low-dose aspirin lowered major adverse cardiovascular events and total mortality. Although the risk of major bleeds was increased, the risk of fatal or cerebral bleeds was not.*

The authors conclude that this combination is now an 'important advance' for patients with peripheral arterial disease and could replace aspirin alone as the standard of care in patients who are not at high risk of bleeding. NICE now recommend this as a treatment option in people with symptomatic peripheral arterial disease at high risk of CV events fulfilling these criteria, after taking an individual's bleeding risk into account and an informed discussion about potential risks and benefits. They conclude this will be cost-effective and that *'aspirin plus rivaroxaban is therefore recommended as a treatment option in people at high risk of having atherothrombotic events who do not have an increased risk of bleeding.'*

► Peripheral arterial disease, evidence update [BMJ 2018;360:j5842](#)

- PAD is often part of a picture of more widespread systemic vascular disease - ~60% will have IHD and ~30% will have cerebrovascular disease
- Smoking is the biggest risk factor and is thought to account for half of all PAD, with diabetes being the other major risk factor
- Symptoms are a poor marker of disease severity; **most patients will be asymptomatic**
- Prognosis is quite variable - most patients with claudication have stable or improved symptoms at 5 years after diagnosis with <5% progressing to critical limb ischaemia; however if critical limb ischaemia develops there is a 30% risk of limb amputation at 1 year, and a 50% all-cause mortality at 5 years

Assessment and diagnosis

- **Symptoms:** most patients are asymptomatic, but if claudication develops it presents as aching or burning pain in leg muscles (buttock, thigh or calf depending on the level of occlusion), which is reliably reproduced at a set walking distance and is relieved within minutes of rest
- **Differential diagnoses** include spinal stenosis (pain relieved by position, not rest), OA (pain not quickly relieved by rest) and lumbar nerve root irritation (positive straight leg raise)
- Acute limb ischaemia is rare and presents with 1 or more of the '6 Ps' (Pain, pallor, pulseless, paraesthesia, paralysis, perishingly cold); chronic limb-threatening ischaemia can be difficult to pick up (see separate section below)
- Do the **ankle-brachial pressure index (ABPI)**:
 - ≤ 0.9 is diagnostic of PAD
 - > 1.2 suggests a falsely elevated reading, often due to calcified arteries (may be seen in diabetes and renal failure)
 - < 0.5 suggestive of severe PAD
 - In a 2018 update of the [NICE PAD Guideline](#) they state we should NOT exclude PAD in patients with diabetes even if the ABPI is normal or raised; this is because people with type 2 DM and PAD *may* have a normal or raised ABPI due to hardened arteries

Management in primary care

- One of the key management aims is to **reduce cardiovascular disease complications**:
 - All lifestyle factors are important to address but **smoking is absolutely crucial** - ongoing smoking with PAD increases the risk of amputation and 5-year survival is halved compared to non-smokers
 - Good diabetic control with HbA1c < 48 ; this decreases the risk of microvascular complications but does not reduce amputation risk
 - Blood pressure $< 140/90$ ($< 150/90$ if over 80 years old); this improves cardiovascular outcomes; ramipril is recommended first-line and ACEI may improve claudication distance
 - Clopidogrel 75mg OD (or aspirin 75mg OD if not tolerated)
 - Atorvastatin 80mg OD; there is evidence that statins reduce all-cause mortality and the incidence of stroke, but may not reduce the risk of limb amputation
- **Symptom management**:
 - Refer for **supervised exercise programmes** (if available) - this improves walking distance (even if no improvement in ABPI) and has similar benefits to angioplasty in terms of walking ability and quality of life; ideally, patients should walk for > 30 mins ≥ 3 x/week to near-maximal pain levels for 6 months
 - **Vasoactive drugs** - Naftidrofuryl oxalate and cilostazol both improve walking distance; NICE recommends a trial of naftidrofuryl for 3-6 months as an option if there is no improvement after supervised exercise and the patient does not want a referral to secondary care

■ Referral:

- **Routine referral** if worsening symptoms or failure of supervised exercise despite risk factor management +/- vasoactive drugs
- **Urgent referral** if critical limb ischaemia, rest pain, tissue loss or ABPI < 0.5
- **Immediate referral** if there is concern over limb-threatening ischaemia (acute or chronic)
- Increasingly surgical management is done via endovascular techniques and newer techniques (e.g. drug-eluting stents and endovascular removal of atheroma) are currently under assessment

► **Low vs high intensity exercise therapy for peripheral arterial disease** [JAMA. 2021;325\(13\):1266-1276](#)

As discussed above one of the cornerstones of treatment for symptom improvement is supervised exercise programmes. There is evidence that pushing patients to near maximal (ischaemic) pain levels improves walking distance, but adherence can inevitably be poor. So the question posed by this RCT recently in JAMA was useful - **Does lower intensity exercise work as well as the more painful higher intensity exercise?** If so it may help adherence to exercise programmes. Findings:

- 305 patients with confirmed lower limb PAD were randomised to one of 3 arms - high-intensity exercise enough to induce ischaemic symptoms, low-intensity exercise without inducing ischaemic symptoms and a non-exercise control group.
- Both exercise groups exercised in an unsupervised setting 5 times per week for up to 50 minutes per session, and accelerometer data was fed back to a coach who gave telephone support to make sure they were exercising at the right intensity.
- **Results were based on changes in the 6-minute walk distance at 12 months.**
- 250 patients completed the trial:
 - There was **no significant change in the 6-minute walk distance in either the low-intensity exercise group or non-exercise control group.**
 - There was however a **significant increase in the 6-minute walk distance in the high-intensity exercise group** with an increase of 34.5m (95% CI = 20.1-48.9) which was a (probably clinically significant) increase of >10% from baseline.

So although walking is obviously a good form of exercise and may have other CVD benefits for people with PAD, this study sadly suggest that low-level intensity exercise will not improve their PAD symptoms. This study **supports the recommendation that patients need to exercise to near maximal pain levels if they want to improve PAD symptoms, and the old adage of 'no pain, no gain' still stands!**

► Chronic limb-threatening ischaemia ([BMJ 2018;360:j5460](#))

As GPs we will be well aware of the symptoms of acute limb ischaemia (the 'six Ps': pain, pallor, pulseless, paraesthesia, paralysis, perishing cold) with a recognition that this is a surgical emergency. However, what we may not be so aware of is the risk of **chronic limb-threatening ischaemia (CLTI)**, and that this also needs urgent vascular review. This condition was the subject of an article in January 2018 in the BMJ 'easily missed' section ([BMJ 2018;360:j5460](#)). It highlights the difficulties of picking up CLTI, but that **prompt surgical intervention can significantly reduce the risk of amputation**.

Why is it easy to miss this condition?

- It can have a slow and insidious onset
- There is often no history of intermittent claudication
- It is relatively rare; even in patients with intermittent claudication only 1-2% will go on to develop CLTI (a full-time GP may expect to see 1 case of CLTI per year)
- Symptoms can easily be confused with other causes of foot pain e.g. cellulitis, gout, arthritis
- The foot often looks pink or red with apparently normal capillary refill time when sitting with the foot down, due to hyperaemia

What features should we look out for/how should we assess possible CLTI?

- Persistent pain in the forefoot and toes, typically worse at night and relieved by hanging legs down; typically patients are woken in the early hours with pain which is eased by hanging their foot down, and they may take to sleeping in a chair
- Non-healing foot wound +/- infection (be aware patients with peripheral neuropathy may present with CLTI with NO pain due to the neuropathy)
- Foot pulses should be palpated but be mindful that there is a substantial false positive and negative rate even by experienced vascular surgeons
- Perform **Buerger's test**: lay the patient down, elevate both legs to 45 degrees for 1-2 minutes and look for pallor; then hang legs down at 90 degrees and look for reactive hyperaemia
- **ABPI is unreliable in suspected CLTI**; do not use ABPI in primary care to determine referral in patients in whom CLTI is suspected
- Have a high index of suspicion in patients presenting with foot symptoms if they have risk factors for PAD - IHD, stroke, smoking, diabetes (especially if peripheral neuropathy as symptoms will be masked)

Management?: All cases of *suspected CLTI* should be discussed immediately with a vascular surgeon

NB Learning points: chronic limb-threatening ischaemia (CLTI)

- CLTI can be easily missed - it typically presents insidiously with pain in the forefoot or toes, worse overnight and eased by hanging the foot down, or with a non-healing foot wound
- Elevate the limb to look for pallor (Buerger's test) as the foot often looks pink/red in the sitting position with an apparently normal capillary refill time
- ABPI is unreliable in assessing CLTI and should not be used to determine referral
- Refer all cases of suspected CLTI immediately to a vascular surgeon

KISS: Peripheral Arterial Disease (PAD)

Based on [NICE 2019, TA607](#), [BMJ 2018;360:j5842](#) and [NICE CG147, 2020](#)

■ Diagnosis:

- Typical **claudication symptoms** present as burning or aching pain in leg muscles brought on at a reliable walking distance and is relieved within minutes of rest
- However **many patients with PAD are asymptomatic** so have a low threshold for considering PAD and doing ABPI in those at risk (especially smokers):
 - Suggestive symptoms or unexplained leg pain in at-risk patients
 - Diabetes, non-healing leg wounds or need to have compression hosiery
- **Acute limb ischaemia** is rare but presents with 1 or more of the '6 Ps' - pain, pallor, pulseless, paraesthesia, paralysis, perishing cold
- **Chronic limb-threatening ischaemia** is more difficult to pick up:
 - Usually presents with pain in forefoot/toes, worse at night, eased by hanging foot out of the bed
 - ABPI is unreliable in this situation - do Buerger's Test to aid assessment (lay patient down, elevate both legs to 45 degrees for 1-2 minutes and look for pallor; then hang legs down at 90 degrees and look for reactive hyperaemia)

■ ABPI = ankle-brachial pressure index:

- ABPI 0.91 to 1.0 = borderline PAD
- **ABPI $\leq$ 0.9 = PAD**
- ABPI $\leq$ 0.5 = severe PAD (ulceration risk, low threshold for referral)
- ABPI $>$ 1.2 suggests a falsely elevated reading due to calcified arteries
- NICE update 2018 specifically says NOT to exclude PAD in diabetics based on a normal or raised ABPI alone

■ Management:

- **Cardiovascular risk reduction**
 - **Lifestyle advice** +++ especially smoking
 - **Antiplatelets**: clopidogrel 75mg OD (aspirin 75mg OD if not tolerated)
 - Addition of **low-dose rivaroxaban 2.5mg bd to low-dose aspirin** is an option following the COMPASS trial [Lancet 2018;391:219](#), and recommended by NICE [NICE 2019, TA607](#) in patients with **symptomatic peripheral arterial disease at high risk of ischaemic events**:
 - Aged 65 or over, or
 - Atherosclerosis in ≥ 2 territories e.g. coronary, cerebral or peripheral arterial, or
 - ≥ 2 of the following risk factors: current smoker, diabetes, CKD with eGFR between 15 and 60 ml/min, heart failure or previous ischaemic stroke
 - **High-intensity statins** e.g. atorvastatin 80mg aiming for target 40% reduction in non-HDL-C from baseline (see lipid chapter for more guidance and other options based on [NHSE update](#))
 - **BP** $<$ 140 /90 ($<$ 150/90 if over 80 years old) with ACEI first line
- **Functional improvement in walking time**
 - Refer for supervised exercise programmes if intermittent claudication
 - Consider vaso-dilators if remain symptomatic, naftidrofuryl 200mg tds (NICE recommendation)

■ Referral:

- **Routine referral** if worsening symptoms or failure of supervised exercise despite risk factor management +/- vasoactive drugs
- **Urgent referral** if critical limb ischaemia, rest pain, tissue loss or ABPI $<$ 0.5
- **Immediate referral** if there is concern over limb-threatening ischaemia (acute or chronic)

FAMILIAL HYPERCHOLESTEROLAEMIA (FH)

The [NICE guideline on familial hypercholesterolaemia \(FH\)](#) was updated in October 2019, and it has some significant implications for General Practice. The original guideline was published in 2008 with the aim of identifying individuals and families with FH in order to get them appropriately treated and to get family members tested for the FH gene. With improvements in diagnostic testing, further data on high-intensity statin therapy, and more safety data on the use of statins in children and young people, NICE have updated their guideline with a 'push' to identify more people with FH. **We have a crucial role to play in helping to identify potential FH cases**, making appropriate referrals, and checking these patients are treated with high-intensity statins. The recent [European lipid guideline 2019](#) also has implications for people with FH, by suggesting much more aggressive lipid targets, with increasing use of the PCSK9 inhibitors if required (see also our lipid and statin chapter).

► *What is FH?*

FH is an inherited genetic defect causing high cholesterol levels and early atherosclerotic disease, leading to premature coronary artery disease. It is an autosomal dominant gene and is **more common than we think** estimated to have a prevalence of between 1:250 and 1:500, equating to 130,000-260,000 cases in the UK, many of which are undiagnosed. Heterozygous FH leads to a greater than 50% risk of coronary heart disease in men by the age of 50 years and at least 30% in women by the age of 60 years. Homozygous disease is rare and often presents in childhood.

► *Why is FH important for General Practice?*

Is this not just for specialists? Well yes, and no. Yes, the testing for FH and most of the management is likely to be undertaken by specialists, but **we are the crucial cog in the pathway to identify these patients and families and refer them** as highlighted in the updated NICE guidance. Based on [primary care data from Medway](#), a practice of 10,000 patients would expect to have ~30 patients with FH, many of whom are undiagnosed.

► *The good news*

Whilst the outcomes of untreated FH are very poor, the good news is that treatment with statins can give patients with FH dramatic improvements. A landmark study published in 2008 ([BMJ 2008;337:a2423](#)) found that treating FH, even with modest statin doses, in patients >55 **improved life expectancy to near normal population levels**.

► *So what has changed in this guideline?*

Whereas much of the guidance on testing and treatment is similar, the key change for us in General Practice is to start systematically identifying individuals and families with possible FH and get them referred for genetic cascade testing. It is recommended 1st, 2nd, and where possible 3rd-degree relatives should be offered cascade testing. The guideline recommends that we:

- **Systematically search primary care records for people:**
 - <30 years old with total cholesterol >7.5mmol/l and ≥ 30 years old with total cholesterol >9.0 mmol/l
 - With a family history of premature coronary heart disease (event <60 years old in 1st-degree relative) whose total cholesterol is unknown, and offer to measure their cholesterol
- We then need to use the **Simon Broome criteria or Dutch Lipid Clinical Network criteria** to identify potential cases and refer (see below KISS)
- Importantly **we should NOT use standard CVD risk assessment tools** e.g. QRISK as these **SIGNIFICANTLY underestimate risk** in people with suspected FH

► *What treatment options are available?* [NICE CG71 Oct 2019](#) [SIGN 149 2017](#) [European guideline 2019](#)

Whilst the treatment for people with FH may be delivered by specialists (NB note updated [NHS lipid guideline 2020](#) advises all patients with clinical diagnosis of FH should be referred to a specialist lipid clinic) it's worth being aware of the treatment options available, as in some areas once the diagnosis of FH is made initial prescribing may fall on the GP. The range of options has increased over the last few years.

- **High-dose and high-intensity statins should remain the mainstay of treatment for people with FH** - we have good evidence they both reduce lipid levels and improve CVD outcomes (see evidence above); however, we know some may not tolerate statins, or may not reach their target lipid levels on statins, so others options include:
 - **Ezetimibe, ideally as an add-on to statin treatment** (but can be used as monotherapy if statins are not tolerated or contraindicated) is recommended as the next step of treatment by both [NICE](#), [SIGN](#) and the [European guidelines](#).
 - If patients **do not reach their target (see below) with statins +/- ezetimibe we should refer them** (if not already done so) to a specialist with expertise with FH.
 - **PCSK9 inhibitors** are recommended in both the European guidelines, and by SIGN and NICE if patients with FH have not reached their lipid targets with statins +/- ezetimibe, but they will need to be prescribed by specialists. **PCSK9 inhibitors are monoclonal antibodies** that inhibits PCSK9, an enzyme involved in down-regulation of LDL receptors. Thus they increase receptor density and lowers low-density lipoprotein cholesterol (LDL-C). They are given by sub-cut injections every 2 or 4 weeks and are extremely expensive. The 2 recommended by NICE are:
 - Evolocumab - [TA394](#)
 - Alirocumab - [TA393](#)
 - **Bempedoic acid** has been recently approved by NICE in April 2021 ([TA694](#)), and although it is unlikely to be on our primary care prescribing formularies it will be available for specialist initiation. Bempedoic acid is now an option for people with familial or non-familial hypercholesterolaemia or mixed dyslipidaemia, in combination with ezetimibe (as separate tablets or fixed-dose combination) for those in whom statins are contraindicated or not tolerated AND whom don't reach LDL target with stand-alone ezetimibe. There is no direct comparison data between bempedoic acid and the PCSK9 inhibitors, but indirect comparisons suggest it is not as effective as either alirocumab or evolocumab, and although not cheap at ~£55/month it is considerably cheaper than the eye-wateringly expensive PCSK9 inhibitors (~£340/month to >£1000/month).

► *Treatment targets?*

- NICE recommend aiming for a reduction in baseline LDL of >50%.
- The new combined [NHS guidance on lipid management](#) also recommends we refer for consideration of PCSK9 inhibitors if:
 - LDL > 5.0 mmol/L (primary prevention) or
 - LDL > 3.5 mmol/L if established CVD/secondary prevention.

► *What about people with severe hyperlipidaemia who don't have a family history?*

[NHSE lipid guideline 2020](#)

We may occasionally come across people who have significant hyperlipidaemia but no family history. This situation doesn't really fall in the FH remit but updated NHSE lipid guideline 2020 helps address this. This guideline recommends that patients who are very high risk based on their lipid profile, irrespective of family history, should be referred to a specialist lipid clinic.

- **Refer to a lipid clinic for assessment, irrespective of family history, anyone who has:**
 - Total cholesterol > 9.0 mmol/l, or
 - LDL > 6.5 mmol/l, or
 - non-HDL > 7.5 mmol/l

KISS: Familial Hypercholesterolaemia (FH) [NICE guideline 2019](#)

Case finding

- **Suspect FH in adults with:**
 - Total cholesterol > 7.5 mmol/l and/or
 - Personal or family history of premature CHD (event <60 in an index case or 1st-degree relative).
- **Systematically search primary care records for people:**
 - <30 with total cholesterol >7.5 mmol/l or ≥30 with total cholesterol >9.0 mmol/l.
 - With personal or family history of premature CHD (event <60 in an index case or 1st-degree relative) whose total cholesterol is unknown, and offer to measure it.

Diagnosis

- Exclude secondary causes of hypercholesterolaemia and do 2 LDL measurements ([expert opinion in CKS FH](#) suggests tests should be fasting but NICE does not specify).
- **Do NOT use standard CVD risk estimation tools** e.g. QRISK.
- Use either the Simon Broome criteria or Dutch Lipid Clinical Network (DLCN) criteria:
- **Simon Broome criteria** (and [click here for DLCN criteria](#)):
 - **Definite FH:**
 - A) **ADULT:** total chol >7.5 mmol/l or LDL > 4.9 mmol/l; **CHILD** <16: Total chol > 6.7 mmol/l or LDL > 4.0 mmol/l PLUS
 - B) Tendon xanthomas in patient, or in 1st-degree relative (parent, sibling, child), or in 2nd-degree relative (grandparent, uncle, aunt).
 - **Possible FH:**
 - As above for A) PLUS ONE OF C) OR D):
 - C) Family history of MI <50 in 2nd-degree relative or <60 in 1st-degree relative
 - D) Family history of raised cholesterol >7.5 mmol/l in adult 1st or 2nd-degree relative or > 6.7 mmol/l in child or sibling under 16.

Referral to specialist FH service for DNA testing

- If either possible or definite FH on Simon Broome criteria or DLCN score >5.
- Genetic cascade testing should be offered to 1st, 2nd and if possible 3rd-degree relatives.

Treatment

- Although drug therapy is crucial, relevant lifestyle changes are also vital.
- **High intensity statin with NICE recommending >50% reduction in baseline LDL +/- Ezetimibe** as an option if statins are contraindicated or not tolerated, or as an add-on to statins if target LDL not reached with a maximum tolerated statin dose.
- **Refer to a specialist with expertise in FH:**
 - If target LDL not reached with statin +/- ezetimibe consider referral for PCSK9 inhibitors (NICE [TA393](#), [TA394](#)) if:
 - LDL > 5 mmol/L (for primary prevention) or LDL > 3.5 mmol/L (for secondary prevention)
 - If established CHD, family history of premature CHD or ≥2 other CVD risk factors.
 - All children - they should be offered statins by age 10 or as soon after as possible.
 - NB [NHSE lipid guideline](#) recommends all patients with FH referred to specialist lipid clinic.
- **Patient information** can be found on the [British Heart Foundation website](#)



SAFE USE OF SGLT2 INHIBITORS

As the range of indications for SGLT2 inhibitors widens and with concerns about rare but serious side effects such as diabetic ketoacidosis (DKA), their safe use in primary care has become a real Hot Topic. We have *new evidence* published [Ann Int Med 2020;173:417](#) on the risk discussed in the [DTB 2021](#), there has been a review of DKA with SGLT2i in the [BMJ2020;371:m4147](#) and the *Association of British Clinical Diabetologists (ABCD) and Diabetes UK have published a joint position statement with recommendations for non-diabetes specialist prescribers ABCD SGLT2i Guidance 2021*. It is also important to note that the [MHRA Dec 2021](#) have said that *Dapagliflozin should no longer be used in type 1 diabetes*.

As discussed elsewhere in the relevant sections of this book (diabetes, heart failure and chronic kidney disease) SGLT2 inhibitors have been shown to be effective hypoglycaemic drugs in Type 2 diabetes with the added advantages of a low risk of hypoglycaemia and moderate weight loss. There is now also strong evidence that in people with T2DM they reduce major adverse cardiovascular events, improve outcomes in heart failure with reduced ejection fraction and also progression of kidney disease. This evidence base for heart failure with reduced ejection fraction and chronic kidney disease with proteinuria now extends to patients without diabetes.

Despite all of this positive data, however, many GPs remain reluctant to prescribe SGLT2i in primary care because of concerns relating to serious side effects such as diabetic ketoacidosis (DKA), Fournier's gangrene and lower limb amputation.

GPs are aware of the relatively common side effects of UTI and genital mycotic infections with SGLT2 inhibitors. These are generally mild, occur soon after initiation of the drug and are amenable to the usual treatments. However, the risk that concerns us most is the increased risk of diabetic ketoacidosis with these drugs ever since an [MHRA alert](#) on this risk back in 2016. Five years later and with these drugs now in much more widespread use, including in patients without diabetes, we have a number of questions that need answering:

- *Just how common is DKA with SGLT2 inhibitors?*
- *Can DKA occur in patients without diabetes on these drugs?*
- *How can DKA be avoided, and what should we look for?*
- *And what about other rare but serious side effects, such as the risk of amputations or Fournier's gangrene with these drugs?*

► *How common is DKA with SGLT2 inhibitors?*

Back in 2016, the MHRA estimated that DKA occurred between 1 in 1,000 and 1 in 10,000 patients prescribed these drugs. This question has been more definitively answered by new research [Ann Int Med 2020;173:417](#) discussed in the [DTB 2021](#). It was a large observational study in which 200,000 patients with Type 2 diabetes who were new users of SGLT2 inhibitors were compared to 200,000 matched controls who were prescribed DPP-4 inhibitors. The primary outcome was the risk of hospitalisation with DKA. **Results:**

- The relative risk of DKA in patients taking an SGLT2 inhibitor was almost 3 times higher compared to those taking a DPP-4 inhibitor (incidence rate 2.03 vs 0.75 per 1,000 patient-years)
- Molecule-specific analyses for dapagliflozin, empagliflozin and canagliflozin were conducted and suggest that it is a class effect
- The paper did not report how many of these cases had euglycaemic DKA (DKA with low or normal blood glucose levels)
- The risk of DKA is higher in patients with Type 2 diabetes on insulin or other glucose-lowering medication
- **Conclusion:** *compared to matched controls on DPP4 inhibitors, the relative risk of DKA in patients prescribed SGLT2 inhibitors is 3 times greater; however the absolute risk remains relatively small at 1 case per 500 patients per year.*

- The recent BMJ review ([BMJ 2020;371:m4147](#)) reviewed this important adverse effect of the SGLT2i:
 - A recent high-quality meta-analysis has concluded that SGLT2i does increase the risk of DKA compared to other drugs used in T2DM, with an absolute event rate of 3/1000 patient years.
 - The risk is highest shortly after starting the drug with >75% of cases happening within 6 months
 - Patients usually present with typical symptoms of DKA - polyuria, polydipsia, nausea/vomiting and abdominal pain, often with hyperglycaemia.
 - However, up to 1/3 of patients will present atypically with only mildly elevated blood sugars (<14 mmol/L), with less severe osmotic symptoms (euglycaemic DKA) which can delay diagnosis.
 - Factors that increase the risk include excessive alcohol or illicit drug use, low carbohydrate or ketogenic diets, vomiting or dehydrating illness, acute infections (see KISS below)

Whilst on the subject of DKA, there was an important update from the [MHRA December 2021](#) stating that **Dapagliflozin should no longer be used for people with type 1 diabetes**. Although we would never initiate an SGLT2i in type 1 diabetes in primary care, it had been an option that Dapagliflozin could be considered in some situations under careful guidance from specialists. Although the MHRA confirm there is no new safety data it is known that the risk of DKA is much higher if SGLT2i are used in type 1 vs type 2 diabetes and they obviously feel the risks outweigh benefits now in type 1 diabetes so have now withdrawn that indication.

► ***Can DKA occur in patients without diabetes who are prescribed SGLT2 inhibitors?***

As patients without diabetes are increasingly prescribed these drugs for other indications such as heart failure, this is an important question. The answer is NO. There have been no reported cases of DKA in the SGLT2i trials in patients without diabetes [Lancet 2020;396:819](#). This is very reassuring, however in theory it is possible some patients with undiagnosed Type 2 diabetes may be started on these drugs for other indications such as heart failure, so it is vital that in these patients their diabetes status is confirmed before starting.

► ***How can DKA be avoided in patients prescribed SGLT2 inhibitors?***

This was discussed in the recent UK guideline from the Association of British Clinical Diabetologists [ABCD SGLT2i Guidance 2021](#).

Essentially, this rare but very serious side effect is more likely in patients with a low beta-cell function reserve (e.g. type 1 diabetes, LADA or diabetes due to pancreatic disease or damage), conditions that lead to restricted food intake or dehydration (e.g. acute illness or starvation) or acute illness.

See the KISS summary below for when these drugs should be avoided, used with caution, how we should modify risk and detect and manage suspected cases.

► ***What about the risk of other rare side effects with SGLT2 inhibitors?***

Top of most doctors' list of 'conditions which I would least like to have' is **Fournier's gangrene**. This is necrotising fasciitis of the perineum or genitalia, and in 2019 the MHRA issued an alert after several reports of this life-threatening infection occurring with SGLT2i [MHRA Alert 2019](#). It is important to note that it is an association, that there is no direct evidence of causation and T2DM is of course the major risk factor for Fournier's gangrene itself. Fortunately, it is incredibly rare, estimated to occur in 1 in 100,000 patients per year.

All SGLT2i increase the risk of genital mycotic infections, and also of course UTI. Patients should be advised to practice good personal hygiene and to treat such infections early, and the SGLT2i can be suspended if necessary to aid recovery. If the genital infections are recurrent, SGLT2i may need to be stopped. *However, if patients experience severe pain, tenderness, erythema or swelling in the perineal or genital area accompanied by malaise or fever they should be urgently referred to secondary care.*

The other 'rare but scary' potential side effect was the increased risk of lower limb **amputation** reported with canagliflozin following the CANVAS trial, which led to an [MHRA alert 2017](#). Fortunately, this finding has not been seen with other SGLT2i or replicated in other studies such as the CREDENCE trial. However, there is enough of a concern that in the [updated 2019 ADA/EASD guideline](#) they recommend that:

- *Patients with foot ulcers or at high risk for amputation should only be treated with SGLT2 inhibitors after careful shared decision-making around risks and benefits with a comprehensive education on foot care and amputation prevention.*

The ABCD guideline also states that SGLT2i should be used with caution in the **frail elderly**, due to the osmotic diuresis causing volume depletion side effects such as thirst, postural hypotension and dehydration. They may also potentiate the effect of thiazide and loop diuretics increasing the risk of these side effects.

► *I think you've just put me off ever prescribing these drugs...*

Sorry! But, it is important to remember that the evidence base now for prescribing these drugs across these different indications is compelling, that these are VERY rare side effects and that the risk of them can be greatly modified by following the guidance in the KISS below. A recent BJGP paper [BJGP2021;71;330](#) states that in the landmark DAPA-HF trial for heart failure, the NNT was only 21 but the NNH (number needed to harm) relating to DKA was 790.

Furthermore, when SGLT2i first started being widely used in type 2 diabetes, there were concerns about increased **risks of urinary tract infection** and of possible serious UTI events such as pyelonephritis and urosepsis. More recent data is much more reassuring, and even questions whether SGLT2i do actually increase the risk of UTI compared to other second-line agents. A large population-based cohort study [Ann Int Med 2019](#) compared patients with T2DM who had SGLT2i added to their medication compared to those who added a DPP4i or GLP-1 agonist. *The risk for severe and non-severe UTI events among those initiating SGLT2i therapy was similar to that among patients initiating treatment with other second-line agents.* This reassuring finding has been replicated in another new study [Therapeutic Adv Drug Safety 2021](#) looking at adults aged over 65.

► *Further practical guide for when/when not to prescribe SGLT2 inhibitors* [Diabetes March 2022](#)

In March 2022 A Steering Group published an updated 'traffic light' guideline on the practicalities of prescribing SGLT2 inhibitors to summarise when we should prescribe SGLT2i (Green), when we should not prescribe them (Red) and when it can be considered but the benefit/risks need to be weighed up (Amber). This guidance and summary table very much mirrors the ABCD guidance discussed above

SGLT2i should be offered	SGLT2i can be considered	SGLT2i should not be prescribed
Fist line (metformin intolerant)	Frail/elderly/cognitive impairment	Acute illness
2nd line to metformin	History of PAD	DKA or previous DKA
Established CVD inc previous stroke	History of foot ulceration	Excess alcohol intake
Heart failure	Previous lower limb amputation	Eating disorder
Overweight/obesity	Current diabetic foot ulcers	Rapid progression to insulin <1 yr
Risk of hypoglycaemia	Ketogenic/low CHO diet	Multiple risks for Fournier's
CKD	BMI <25	Pregnancy/breast feeding
No history of lower limb amputation	High HbA1c (>86 mmol/mol)	Recent major surgery
No PAD	Recurrent UTI	
On loop diuretics	Recurrent genital mycotic infections	
Osteoporosis or previous fractures	On systemic steroids	

KISS: SGLT2 inhibitors and diabetic ketoacidosis (DKA)

[ABCD Diabetes UK, SGLT2i Guidance 2021](#)

■ **Background**

- DKA is estimated to affect 2 per 1,000 patients per year with type 2 diabetes prescribed SGLT2i
- It may present with normal or near-normal glucose levels, so-called euglycaemic DKA

■ **Avoid SGLT2i in the following situations**

- Past history of DKA
- Eating disorder or very low carb (ketogenic) diet
- eGFR lower than allowed in the current licensing of the SGLT2 inhibitor
- Person with excessive alcohol consumption or IVDU
- Patients who are acutely unwell
- Any diagnosis or suspicion of any other type of diabetes other than Type 2. *Avoid in Type 1 diabetes (dapagliflozin has recently had its recommendation as an option in type 1 diabetes withdrawn by the MHRA Dec 2021), LADA, other genetic causes of diabetes, diabetes due to pancreatic disease or injury or in people with Type 2 DM who rapidly progressed to needing insulin within one year of diagnosis*
- Pregnant, breastfeeding or female of child-rearing age and sexually active without contraception

■ **Use SGLT2i with caution in the following situations**

- Body mass index < 25 kg/m² and < 23 in South Asian patients, or patients with recent weight loss
- People considered at high risk of acute hyperglycaemia e.g. due to non-adherence to medication
- People with very elevated levels of HbA_{1c} > 86 mol/mol (> 10% in old units)
- People with or at risk of frailty, or with cognitive impairment
- On chronic oral steroids
- Patients with long duration of diabetes e.g. over 10 years from diagnosis

■ **Reducing the risk of DKA when prescribing SGLT2i**

- When SGLT2i are started review other medications which may cause hypoglycaemia (e.g. sulfonylureas and insulin) and *consider reducing the dose when the SGLT2i is started*, particularly if the HbA_{1c} is at target when the drug is started, to reduce the risk of hypoglycaemia. Dose reductions should be personalised, but typically 20% to 25%.
- Drink plenty of fluids to avoid dehydration
- Suspend SGLT2i in acutely unwell patients
- Suspend SGLT2i with admission for elective surgery or procedure requiring starvation
- **Sick day guidance:** *if unwell with diarrhoea, vomiting or fever STOP SGLT2i and do not restart until feeling better and eating and drinking normally again for at least 24 hours, and drink plenty of fluid*

■ **Assessing possible DKA in patients taking SGLT2i**

- Be aware that glucose levels may be normal despite ketoacidosis
- Suspect DKA in patients with nausea, vomiting, abdominal pain, stupor, fatigue and difficulty breathing
- Urine ketone levels are not reliable, test capillary ketones using a ketone meter (if not available admit for ketone testing). If ketones are > 1.5, refer to secondary care for further testing (bicarbonate and venous pH) for acidosis.

Excellent table with traffic light summary for when to prescribe SGLT2i, when not and when to use with caution, summarising/expanding ABCD guidance above - [click here](#)

Patient information leaflets

[TREND UK, SGLT2i and DKA](#)

DIRECT ORAL ANTICOAGULANTS (DOACs)

The DOACs (direct oral anticoagulants - the new terminology for NOACs as they are not so 'novel' anymore...) are now very much part of the landscape, and their indications are growing. They may be more simple for patients than warfarin, but they are far from simple drugs and have a broad range of indications with different doses. The change to using DOACs instead of warfarin has been accelerated due to the COVID pandemic, with a push towards using drugs that require less monitoring. Recently DOACs are back in the headlines due to changes in the [Impact and Investment Fund \(IIF\) component of the Network Contract DES 2022/23](#). One of the focuses in the IIF is DOAC use in AF, and a **push to use edoxaban as the first line DOAC for AF** after the NHS made a deal with the drug manufacturer of edoxaban.

But as highlighted in a recent DTB article ([DTB July 2020](#)) DOACs are by no means benign drugs, with significant bleeding risks, need dose adjustments in a number of situations and do require blood monitoring, albeit generally less frequently than for warfarin. We will cover some important reviews on the DOACs, the key Drug Safety Updates pertaining to the DOACs, but first 'lets talk about edoxaban'..

► [Why am I being asked to prescribe edoxaban for patients with AF? IIF 2022/23](#)

Back in January 2022 when the UK health system was somewhat busy with a massive surge in COVID cases, not to mention a backlog of general work, a [commissioning statement](#) was published that probably missed most people's inboxes. NHS England had done a deal with DOAC manufacturers which was [announced in November 2021](#) recommending that:

- Edoxaban should be used first line *'where this is clinically appropriate'*.
- Or if not clinically appropriate *'clinicians should then consider rivaroxaban first, then apixaban or dabigatran'*.
- The recommendations went one stage further though to recommend *'For patients already prescribed a DOAC for the treatment of AF: subject to the criteria specified in the relevant NICE technology appraisal guidance, commissioners may wish to consider developing local policy to review patients currently prescribed apixaban, rivaroxaban or dabigatran, where clinically appropriate'*.

The aim of this deal is to **widen access to more affordable DOACs**, and the money saved from these deals will get put back into Detect, Protect and Perfect schemes to promote better detection and treatment of AF, ultimately in the hope it will lead to fewer strokes.

Subsequently, these recommendations were incorporated into the [2022/23 Impact and Investment Fund \(IIF\)](#), which forms part of the Network Contract DES, and is a financial incentive scheme that focuses on supporting PCNs to deliver health improvements. The NHS Long Term Plan has boldly committed to preventing 150,000 strokes, heart attacks and dementia cases by 2029 through earlier detection of CVD. One of the domains in the IIF is CVD prevention focusing on the 'ABC' of AF, Blood pressure and Cholesterol. Within the AF section **PCNs are incentivised for getting more people with AF onto DOACs (based on appropriate CHA2DS2Vasc scores) but also for using a higher percentage of DOACs as edoxaban.**

A procurement process happened in 2021 when NHSE gave all DOAC suppliers an opportunity to effectively bid for lower tariffs to be recommended as a preferred NHS drug and Daiichi Sankyo offered the most significant discount for their drug edoxaban. One can see the reason why NHSE is so keen to try to incentivise a more cost-effective DOAC - **DOACs are the highest spend drug in primary care accounting for >£500 million/yr** and if no action is taken DOAC expenditure is expected to double. But the increasing number of deals done by the NHS with drug companies is drawing criticism as the nature of the deals (and therefore cost-effectiveness) is hidden behind commercially sensitive agreements.

The IIF will incentivise PCNs to prescribe more edoxaban, as a % of the DOACs used for new AF but also supports the (potential) switch to edoxaban for people already with AF and on another DOAC. The IIF does note that *'It is important that switching patients who are currently prescribed a different brand of DOAC to Edoxaban is done in a clinically appropriate way and as the result of a shared decision-making conversation.'* It's worth noting that the NICE AF guidance which was only published last year, highlighted that we have no head-to-head studies between the different DOACs, so we don't really know which is the most clinically effective option yet, but that *'When all these outcomes were combined in the cost-effectiveness analysis, apixaban was the most clinically effective and cost-effective anticoagulant based on UK drug tariff prices at the time'*. So if locally DOACs other than edoxaban have been used first-line e.g. apixaban, making these thresholds/switches could be difficult. Also, there is a risk that future data may show edoxaban to be inferior to other DOACs, and if so further headaches may be in store...

► **DOACs therapeutics review** [Lancet November 2020](#)

This useful therapeutics review in the Lancet gave a good summary of where we are, and where we might be going, with the DOACs. A few useful points and reminders:

- DOACs bind to the active sites of either thrombin (factor IIa) or factor Xa.
- They are all at least partially renally cleared (with dabigatran being 80% really cleared), hence the need for **careful dose adjustments in renal impairment** (see below).
- They all have a short half-life of ~12 hours in people with normal renal function.
- They can interact with drugs that induce or inhibit either P-glycoprotein or CYP3A4 (e.g. macrolides).
- The rationale for DOAC use over warfarin is not so much a significant reduction in events (which in most RCTs is small, with high NNTs), but a better safety profile, with lower serious bleeding risks, and greater convenience.
- The **range of indications is ever-expanding**. These include:
 - Prevention of VTE after hip or knee surgery.
 - Treatment of VTE (now extended to include those with active cancer, although the bleeding risk is increased vs LMWH in some cancers e.g. GI cancers).
 - Stroke prevention in AF.
 - Low-dose rivaroxaban in conjunction with antiplatelets after ACS.
 - Low-dose rivaroxaban with aspirin for secondary prevention of vascular events in patients with coronary artery disease or peripheral arterial disease at high risk of further ischaemic events.
- **Uncertainties and challenges:**
 - **Renal dysfunction** - we will deal with this below, but this review points out that although some of the DOACs are licensed to be used down to estimated creatinine clearance (eCrCl) of 15 ml/min, this is based on pharmacokinetic data, not trial data, so the true bleeding risk in patients with renal dysfunction is still somewhat uncertain.
 - **Severe obesity** - patients with BMI >40 are significantly under-represented in DOAC trials and because of the high volume of distribution and increased clearance in obesity, there is a risk of under-dosing in this group. Although not mentioned in this review, the reverse is true, namely a risk of over-dosing in underweight people, hence the crucial importance of using eCrCl, not eGFR for dosing DOACs.

► **DOACs: how do we modify and reduce bleeding risk?**

The recent NICE guidelines on Atrial Fibrillation [NICE NG196, 2021](#) state that we should assess bleeding risk using the ORBIT tool (see AF chapter for more on this). If this identifies patients at higher risk of bleeding we should not deny them anticoagulation, but seek to modify that risk. *How should we do that?*

Simple steps include looking at other medications which increase bleeding risk (e.g. aspirin, NSAIDs, SSRI etc), controlling hypertension to < 130/80, and making sure we have the correct DOAC dose for the patient's creatinine clearance (see below). A recent network meta-analysis [Eur J Gastro 2021](#) has found that apixaban has the lowest rate of major GI bleeding compared to other DOACs and also warfarin.



But should we add a PPI to provide additional gastro-protection? This is an important clinical question that is not addressed in national guidelines. Recent evidence [Heart 2021;0:1-6](#) suggests that PPI cover may be protective, especially in elderly patients with AF

- This was a large observational cohort study of patients initiated on a DOAC for atrial fibrillation (n=164,000) and patients who were co-prescribed PPI were compared to matched controls who were not
- PPI use was associated with a reduced risk of major upper GI bleeding
- Overall the effect was modest but found to be highest in patients aged over 75 and with higher bleeding risk identified on the HAS-BLED score
- In absolute terms, the authors estimate a 5-year NNT of 158 for adults aged over 75, 134 aged over 85 and 76 if identified as a higher bleeding risk (HAS-BLED score >2)
- **Conclusion:** *PPIs can be useful to reduce the risk of major upper GI bleeds in DOAC-treated patients with a high risk of bleeds due to age or co-morbidities*

► DOACs and antiphospholipid syndrome [Drug Safety Update June 2019](#)

The [Drug Safety Update June 2019](#) highlighted the recent [TRAPS Study, Blood 2018](#) which compared warfarin to rivaroxaban in patients with high-risk antiphospholipid syndrome. The trial was terminated early after 12% of patients in the rivaroxaban group had VTE events compared to none on warfarin. The European Medicines Agency and the MHRA have therefore concluded that **DOACs are NOT recommended in patients with anti-phospholipid syndrome.**

► DOAC dose adjustment in renal impairment [DTB June 2020](#)

A common cause of confusion is that we use eGFR to assess renal function, but for some drugs (including the DOACs), it is advised in the BNF and their product literature to adjust dosage according to creatinine clearance based on the Cockcroft Gault (CG) equation. As highlighted by the [MHRA Oct 2019](#), there is evidence and case reports of adverse outcomes when DOACs were dosed on eGFR rather than CrCl. A useful review from the [Specialist Pharmacy Service 2018](#) discusses DOAC dosing in renal impairment and caveats even when the CG equation is used. The equation is based on age, body weight and serum creatinine. Serum creatinine is derived from muscle turnover, so the weight element should be based on ideal body weight e.g. CG calculation may not be accurate in oedematous patients, those with muscle wasting or limb amputation. There is a complex formula, but it is easy to calculate from calculators in GP computer systems or **online** calculators e.g. [MDCalc](#)

The issue of **DOAC dose adjustment in renal impairment** was highlighted again recently in the [DTB June 2020](#). A study from the US analysed changes in renal function over time in people with AF, and assessed whether appropriate dose adjustments for DOACs had been made:

- Between 2013-2016 >6500 patients were enrolled, median age 72 years old.
- 83% were treated with DOACs.
- During the study, 22% had worsening renal function (decrease in CrCl of >20% from baseline).
- Of the patients on standard DOAC dose at baseline 3.7% had CrCl levels that met the threshold for dose adjustment, but of that group, **only 20% actually had the appropriate dose adjustment.**
- Of the group that met the threshold for dose adjustments, all of the major bleeding episodes occurred in those that had not had their DOAC dose-adjusted; there were no major bleeds in the group that did have appropriate DOAC dose adjustments (admittedly small numbers).

Whilst only a single study, and one based in the US, this study does highlight the importance of appropriate dose adjustment in people with renal impairment, and the increased risk of serious bleeds if not done. The DTB remind us that **people with, or at risk of, renal impairment need regular reviews of their DOAC dose.** See below a KISS with a summary of dose adjustments and frequency of monitoring for those with renal impairment. The DTB give a useful 'rule of thumb' recommended by the [European Heart Rhythm Association 2018](#), that the **frequency of renal function testing, once CrCl has fallen below 60, can be guided by dividing the CrCl by 10, to give the frequency of monitoring in months** (e.g. CrCl = 50; divide 50/10 = 5, so do renal function every ~5 months).



KISS: DOACs in Reduced Renal Function

CKS Oral Anticoagulation 2022

	CrCl > 50 ml/min	CrCl 30-49 ml/min	CrCl 15-29 ml/min	CrCl <15 ml/min
Apixaban: AF	5mg BD. Reduce dose to 2.5mg BD if 2 of: - Age ≥ 80 - Weight ≤ 60 kg - Serum Cr ≥ 133	5mg BD. Reduce dose to 2.5mg BD if 2 of: - Age ≥ 80 - Weight ≤ 60 kg - Serum Cr ≥ 133	Reduce dose 2.5mg BD.	Do not use.
Apixaban: Treatment of VTE*	Initially 10mg BD for first 7 days, then 5mg BD.	Initially 10mg BD for first 7 days, then 5mg BD.	Use with caution.	Do not use.
Rivaroxaban: AF	20mg OD.	Reduce dose 15mg OD.	Reduce dose 15mg OD.	Do not use.
Rivaroxaban: Treatment of VTE*	15mg BD for 3 weeks then 20mg OD.	15mg BD for 3 weeks, then consider reduced dose 15mg OD.	15mg BD for 3 weeks, then consider reduced dose 15mg OD.	Do not use.
Edoxaban: AF	60mg OD. Reduce dose 30mg OD if: - Weight ≤ 60kg - On P-gp inhibitors**	Reduce dose 30mg OD.	Reduce dose 30mg OD.	Do not use.
Edoxaban: Treatment of VTE*	Dose as above for AF (after 5 days of LMWH).	Reduce dose 30mg OD. (after 5 days of LMWH).	Reduce dose 30mg OD. (after 5 days of LMWH).	Do not use.
Dabigatran: AF	150mg BD. Reduce dose 110mg BD if: - Age > 80 - High bleeding risk - On verapamil	Consider Reduce dose 110mg BD depending on thrombo-embolic/bleeding risk.	Do not use.	Do not use.
Dabigatran: Treatment of VTE*	Dose as above for AF (after 5 days of LMWH).	Dose as above (after 5 days of LMWH).	Do not use.	Do not use.
Frequency of U&E monitoring	Minimum yearly. Every 6 months if age > 75 or frail.	At least every 6 months.	At least every 3 months.	

*These refer to standard treatment doses for DVT and PE, usually for minimum 3 months, but length of treatment should be specialist-led depending on VTE recurrence risk.

*For **prophylaxis of recurrent DVT/PE**, lower doses may be recommended after minimum 6 months of full treatment dose, but should be **specialist-led and will be determined by individual risk of VTE recurrence and bleeding risk** (e.g. apixaban 2.5mg BD; rivaroxaban 10-20mg OD;)

**P-gp inhibitors = ciclosporin, dronedarone, erythromycin, ketoconazole



RISKS OF LONG TERM NITROFURANTOIN AND AMIODARONE

We are well used to monitoring 'high risk' drugs in primary care but how many of us would put nitrofurantoin in that category? The **risks of long-term nitrofurantoin use and the lack of awareness of these risks** (at both primary care and secondary care level) have been highlighted by a DTB review article ([Drug and Therapeutics Bulletin 2022;60:59](#)) and a study in the BJGP Open ([BJGP Open 2021; 5 \(6\)](#)), and would suggest that many of us are unaware of the potential risks of this drug. Our recent [NB Blog](#) on this subject also discusses this subject. So why are many of us unaware of the risks with nitrofurantoin, what even are the risks, and what monitoring should we be doing?

The focus of both the original BJGP study and subsequent discussion in the DTB is a relatively rare, but serious complication of long-term nitrofurantoin, namely **Nitrofurantoin induced Interstitial Lung Disease (or NFILD)**, although the risk of hepatic-toxicity was also discussed. So what did this study show?

- This study assessed prescribing and monitoring activity in both primary and secondary care by use of questionnaires, review of GP databases and case note reviews in secondary care MDTs.
- 242 GPs (86%) and 41 urologists (32%) responded, 125 of whom (with similar GP/urologist split) prescribed long-term nitrofurantoin.
- **Most long-term prescriptions for nitrofurantoin were for recurrent UTI (66%).**
- **Many prescribers were unaware of the potential for hepatotoxicity (42%) or lung toxicity (28%)** and more than half (53%) did not measure baseline lung or liver function.
- The most common practice was to 'never' monitor liver (41%) or lung (53%) function.
- When prescribers did monitor patients, it was usually performed every 12 months (32%).
- The analysis of monitoring in practices found that baseline monitoring was absent in 74% of the 265 patients who received long-term nitrofurantoin within the previous 2 years.
- In a subset of patients prescribed nitrofurantoin for 6–24 months, **many received no monitoring beyond 6 months (45%)** and just 21% of these patients received both liver and lung function tests after 6 months of treatment.

The secondary care MDT case notes review of those that developed NFILD also highlighted some useful information, although one has to be careful about interpretation as the case numbers were small (46 cases):

- Median age 72 years old.
- ~20% of cases developed NFILD within 6-12 months of drug exposure, and ~40% developed NFILD after ≥12 months of drug exposure (the rest duration was unknown).
- The majority ~60% had partial or complete resolution of disease once nitrofurantoin was stopped but **some did have onward disease progression (~16%)** despite stopping the drug, with some ending up on oxygen therapy.

Nitrofurantoin can cause both chronic lung reactions, including fibrosis as well as hepatitis, and monitoring of both lung and liver function is recommended by the BNF. The risks of pulmonary complications are more common in older patients, as highlighted by this study, and can occur after 6 months of treatment but may occur many months or years later. This study and the DTB identified 3 main barriers to effective monitoring for these potential long-term complications:

1. **Lack of awareness.**
2. **Lack of guidance** - As discussed in the BJGP study *Current adult BNF guidelines state: 'On long-term therapy, monitor liver function and monitor for pulmonary symptoms, especially in the elderly (discontinue if deterioration in lung function).'* However, neither a method nor frequency of hepatic or pulmonary monitoring, nor a definition for 'long-term' is suggested. And the NICE guideline on long-term antibiotics for recurrent UTI simply states to return for 'review within 6 months'. Indeed over a third of responders to the survey stated that **current guidelines were not at all sufficient/clear.**
3. **Confusion over responsibility** for both baseline and ongoing monitoring - and it's worth noting that nearly all GPs considered it to be the prescriber's responsibility to do the monitoring whilst only ~50% of urologists believed themselves to be responsible for this...

As always one does need to put this in perspective as **these complications are rare** - controlled trials suggest severe adverse events occur in ~0.2% of people, with observational data suggesting the event rate to be 0.02-1.5 per 1000 nitrofurantoin users. However, as highlighted above, in **small numbers this can cause progressive disease despite cessation of therapy**, and it's also worth noting that the [Medical Protection Society](#) has dealt with a number of cases of inadequate monitoring of long-term nitrofurantoin and they recommend as a minimum 6 monthly reviews of LFTs and respiratory symptoms.

So what can we do as Primary Care teams to mitigate these risks and address the barriers above?

1. **Awareness** - hopefully, better now you've read this! Share this with your teams and read our [NB Blog](#).
2. **Guidance**. The BJGP calls for more explicit guidance to help us, but in the meantime give some useful pragmatic suggestions to guide monitoring:
 - Consent patients and make them aware of the long-term effects of nitrofurantoin and to **stop immediately if they develop new respiratory symptoms**.
 - **Baseline LFT, U&E and respiratory assessment** (respiratory symptoms and exam, MRC dyspnoea score, oxygen sats and CXR).
 - **Monitoring** (ideally every 3 months) - LFTs, respiratory sx/exam, MRC dyspnoea score, O2 sats.
3. **Responsibility**. This inevitably draws on an age-old frustration for us in General Practice regarding drug recommendations from secondary care and the lack of responsibility taken by them, as highlighted by the survey results. The vast majority of these patients will have had prophylactic antibiotics recommended by secondary care and the [BMA guidance on drugs recommended by hospital outpatients states](#) *'The doctor recommending a prescription should ensure that the prescription is appropriate, including carrying out any tests required to ensure safety.'* From a personal perspective, this would imply to me that baseline testing should be done by the initiating/recommending doctor, whilst ongoing monitoring would probably fall under primary care's remit. But the key message is **we need to communicate clearly with our secondary care colleagues and be absolutely clear about whose responsibility it is for both baseline and ongoing monitoring testing**. Ideally, this would be done under shared care agreements but it is likely this is not in place in many areas.

► **Amiodarone - reminder on respiratory, liver and thyroid risks and monitoring** [MHRA March 2022](#)

Whilst on the subject of high risk drugs, a reminder on the risks of this drug, and monitoring requirements, was published in the March 2022 MHRA Drug safety update. There are no new safety concerns, but a recent coroner's review highlighted a case of multi-organ failure associated with lung disease and long-term amiodarone therapy. The coroner raised concerns that no routine lung imaging is required whilst patients are on amiodarone. An MHRA review confirmed that routine lung imaging is not necessary but they highlighted some reminders on amiodarone safety/monitoring (see Learning Points box below):

Learning Points and Quality Improvement Ideas: Long Term Nitrofurantoin and Amiodarone

- People on **long-term nitrofurantoin need monitoring for both lung and liver complications**. There are no clear national guidelines on this but a recent review article suggests (**consider audit**):
 - **Baseline LFT, U&E and respiratory assessment** (respiratory symptoms and exam, MRC dyspnoea score, oxygen sats and CXR).
 - **Monitoring** (every 3-6 months) - LFTs, respiratory symptoms/exam, MRC dyspnoea score, O2 sats.
 - Warn patients to stop nitrofurantoin immediately if they develop any new respiratory symptoms, particularly a new cough or SOB.
- **Amiodarone** has a long half-life (~50 days), so adverse events may occur even weeks after cessation.
 - Liver function and thyroid function should be checked before treatment and every 6 months.
 - Amiodarone can also cause pneumonitis which can progress to fibrosis but is usually reversible on cessation of therapy.
 - Routine lung imaging is not required, but patients should **stop amiodarone immediately if any new symptoms develop** (generally cough, SOB +/- systemic features e.g. fatigue, weight loss, fever). If **pulmonary toxicity suspected refer for CT** (may require respiratory referral).

MONTELUKAST, NEUROPSYCHIATRIC REACTIONS

Since its introduction in the late 1990s, montelukast has proved to be a useful drug in the management of atopic asthma and allergic rhinitis. In more recent years, its use has increased as it has become established in the main asthma guidelines as a preventative drug for both adults and children. It is a particularly attractive option in children due to the predictable oral dosing when inhalers can be difficult. There has been awareness that the drug may be associated with neuropsychiatric adverse reactions for some years, but in the last few years we have had strengthened guidance from the [European Medicines Agency](#), the [MHRA](#) and there has been a new review of the evidence published in the [BMJ 2022;376:e067554](#).

► Neuropsychiatric reactions associated with montelukast

Neuropsychiatric reactions with montelukast have been reported in all ages. They range significantly in both phenotype and severity (see box below). At the mild end of the spectrum (e.g. sleep disturbance, memory disturbance, anxiety or low mood) they may not be attributed by the patient, parent or clinician to the drug. At the severe end of the spectrum suicidal thoughts, OCD symptoms or hallucinations may occur. The European review highlighted cases where there was a delay in neuropsychiatric reactions being associated with the medication. Fortunately, severe cases are very rare, however, reactions such as behavioural change, mood change and sleep disturbance may occur in up to 1 in 100 patients.

Neuropsychiatric reactions associated with Montelukast [BMJ 2022;376:e067554](#)

Uncommon (may affect up to 1 in 100 people taking montelukast)

- Agitation, including aggressive behaviour
- Sleep disturbance, insomnia, vivid dreams, sleepwalking
- Depression
- Anxiety
- Restlessness and irritability

Rare (may affect up to 1 in 1,000 people taking montelukast)

- Memory and attention problems
- Tremor or shakiness

Very rare (may affect up to 1 in 10,000 people taking montelukast)

- Obsessive-compulsive symptoms
- Hallucinations
- Stammering
- Suicidal thoughts and actions
- Disorientation and confusion

NB Learning Points & Quality Improvement Ideas:

- Be alert for neuropsychiatric reactions including mood and behaviour change when we prescribe montelukast, and advise patients and parents to read the [updated patient information leaflet](#) and to seek medical advice if any of these reactions occur.
- We should review the drug's effectiveness and tolerability at 4 to 6 weeks, proactively ask about noticeable changes in sleep, mood or behaviour. If adverse reactions are noted, or the drug is ineffective, it should be withdrawn.
- **QI idea:** do you have any patients on regular montelukast coded as having these problems?

SEROTONIN SYNDROME

The DTB has recently reviewed serotonin syndrome [DTB June 2022](#)

KISS: Serotonin Syndrome

Based on [DTB June 2022](#)

- **Definition:** The serotonin syndrome is a life-threatening adverse drug reaction resulting from excess serotonergic agonism due to interactions between multiple drugs, poisoning (e.g. overdose) or less commonly due to the action of a single drug
- **Clinical features:** The central triad of features of the serotonin syndrome are:
 - **Altered mental state:**
 - 'Positive' features include insomnia and anxiety leading to agitation, confusion, hallucinations and delirium; negative features include drowsiness and reduced consciousness
 - **Autonomic hyperactivity**
 - Hyperthermia, sweating and flushing, shivering, tachycardia, nausea and diarrhoea
 - With fever and tachycardia, it may be misdiagnosed as sepsis
 - **Neuromuscular excitation**
 - Hyperreflexia, clonus, tremor, rigidity; and there may be signs of dilated pupils
 - **Onset is rapid**, typically within 6 hours of after initial administration or change to medication and may progress rapidly
- **Potential causative agents**
 - **Antidepressants**
 - Monoamine oxidase inhibitors e.g. phenelzine, moclobemide
 - SSRIs and SNRIs e.g. venlafaxine, duloxetine
 - TCAs, clomipramine and imipramine (these are much more serotonergic than other TCAs)
 - **Opioid analgesics**
 - Tramadol, pethidine, fentanyl and tapentadol
 - **Miscellaneous**
 - Lithium, buspirone, St John's Wort
 - ADHD meds e.g. amphetamines and methylphenidate
 - Synthetic stimulants (e.g. 'ecstasy') and amphetamines
 - Triptans and metoclopramide are **not** considered as likely to cause serotonin syndrome
- **Initial management**
 - Prompt clinical identification is the most important and crucial step
 - ***Severe serotonin syndrome is a medical emergency, and it is marked by rapidly rising temperature and rigidity and a temperature > 38.5 necessitates urgent cooling***
 - The [NEWS2 score](#) is not validated in serotonin syndrome, but an elevated score will trigger an urgent secondary care response
 - Mild toxicity may present with tremor, hyperreflexia and insomnia/anxiety with nausea and diarrhoea. Mild to moderate cases usually resolves in one to three days after stopping the serotonergic drugs [BMJ2014;348:g1626](#)
- **Avoiding serotonin syndrome**
 - Beware the most common and severe culprits which are SSRI + MAOI, and SSRI + opioid
 - Avoid co-administration of serotonergic agents e.g. SSRI for depression and tramadol for pain
 - Reassure patients that serotonin syndrome is rare, but make them aware of the risks with dose and medication changes and to seek urgent advice if they experience symptoms
 - Patients should be made aware of the risks of taking other drugs and illicit substances
 - When changing medications, allow suitable 'wash out' periods of serotonergic drugs and follow [NICE CKS 2022 Guidance on switching antidepressants](#)

OPIATES AND BENZODIAZEPINES

There have been considerable concerns about the risks of adverse outcomes if opiates and benzodiazepines (BZPs) or Z-drugs are used concurrently. Whilst we have done a very good job in primary care over the last decade or so by reducing the amount of BZPs and Z-drugs prescribed, opiates are still prescribed commonly for acute and chronic pain so the risks of concomitant opiate/BZP prescribing is still a very real one - the patient on long term BZPs for severe anxiety gets acute back pain and has some tramadol or codeine, or the patient on long term opiates for their chronic back pain gets acute spasm and asks for some diazepam - scenarios that we see frequently. And we all know about the theoretical risks of respiratory depression and death if these two classes of drug are used together as the MHRA warned us about in [March 2020](#), but I suspect many of us question the absolute risks of this and see it as a rare complication. However new data from Canada, which was reviewed in the DTB recently ([Drug and Therapeutics Bulletin 2021;59:116](#)), has really highlighted the substantial risks of concurrent opiate and BZP prescribing, and should seriously make us sit up and think.

- This was a large cohort study from Canada with >1 million people who had received an opioid prescription and had been hospitalised or attended ED.
- ED attendance, hospitalisations and death rates were examined and compared between opiate-only use and opiate plus BZP concurrent use.
- **Risks of ED attendance and hospitalisation were significantly higher if opiates were concurrently used with BZP/Z-drugs vs opiate use alone with OR = 1.13, 95% CI 1.10-1.17.**
 - The risks were **especially apparent if aged >65** (OR = 1.50, 95% CI 1.39-1.61)
 - Importantly the **risk was highest in those who had used both drugs for <1 month** (OR = 2.47, 95% CI 2.26-2.70).
 - Inevitably risks were higher with morphine and oxycodone, but there was also significantly increased risk if tramadol was used and even with codeine.
- There was also a **substantially increased mortality risk** if the 2 classes were used concurrently (OR = 1.90, 95% CI 1.76-2.05)
 - The mortality risk was higher if higher doses of opiate were used.
 - As with the hospitalisation/ED attendance risk, **mortality risk was substantially higher in the first month of concurrent use** (OR = 4.93, 95% CI 4.29-5.66)
- **Conclusion:** concurrent use of opiates and BZPs substantially increases the risk of ED attendance, hospitalisation and death, with the risks being highest in the first month of concurrent use and in the over 65s, with the risks being dose-dependent.

As always we need to bear in mind the usual caveat with observational studies that there is always a risk of confounding, and that much of the excess risk was in people on the stronger opiates at higher doses but this study confirms that the **risks of concurrent opiate and BZP use (even if we are using codeine or tramadol) are real and potentially substantial**. To put in context the 90% increased overall relative risk of death of concurrent use equates to a **frighteningly low number needed to harm (in this case death) of only ~17**. The implication for us in primary care is simple - we need to be very wary of adding in an opiate to people on BZPs or adding in BZPs to people on opiates, especially if they are on high-strength opiates such as morphine or oxycodone, and especially those aged over 65.

NB Learning Points & Quality Improvement Ideas:

- New research reveals that the co-prescription of benzodiazepines and opioids (including codeine and tramadol) substantially increases the risk of hospitalisation or death, especially in the acute phase in older adults.
- **Quality Improvement Idea:** Search for patients aged >65 who are concurrently being prescribed benzodiazepines and opiates (particularly potent opiates e.g. morphine and oxycodone)
 - Book a medication review (?could your clinical pharmacist do this?)
 - Discuss the increased risks of this combination, and can doses be weaned, or ideally stop one or other of the drug classes?

MEDICINES ASSOCIATED WITH DEPENDENCE OR WITHDRAWAL

In 2022 NICE produced a new guideline on **Medicines associated with dependence or withdrawal symptoms** [NICE 2022, NG215](#). A *report from Public Health England in 2019 on prescribed medicines* [PHE Prescribed Medicines review 2019](#) revealed the scale of the problem, that a *staggering 1 in 4 people in England are taking prescription medicines with potential for dependence and withdrawal*. Prescribing rates are strongly linked with areas of socio-economic deprivation.

► Medicines associated with dependence or withdrawal symptoms [NICE 2022, NG215](#)

The guideline covers opioids, benzodiazepines (BZPs), gabapentinoids, Z- drugs and antidepressants. It does not refer to gabapentinoids for epilepsy, nor opioids for acute or cancer pain, palliative care for the management of illicit drug dependence. Antidepressants (AD) are not considered as dependence-forming medications but can of course cause withdrawal symptoms when they are stopped. The recommendations are very general and will not surprise you, but the *guideline has significant implications for prescribers in terms of being explicit with people about the risk of dependency of prescribed drugs before prescribing and has significant workload implications requiring detailed medication reviews for patients to consider their options and offering supported withdrawal*. All this should be done following the principles of [shared decision making](#).

■ Key recommendations

- **Before starting or continuing treatment** with one of these drugs, ensure all non-pharmacological options have been discussed and offered to that person
- When making prescribing decisions, determine if there are any factors that might increase the potential for problems associated with dependence e.g. co-morbid mental health problem, a history of drug or alcohol misuse, not having a clearly defined diagnosis to support the prescription and taking an opioid together with a BZP
- During the first discussion about prescribing give the person information to help them balance the potential benefit of the drug against long-term consequences including potential for dependence
- **Take steps to reduce the risk of dependence** e.g. start with lowest possible dose, consider avoiding prolonged-release opioids, regular review to establish lowest effective dose and avoid dose increase if the therapeutic response is not sustained
- **Offer regular medication reviews** during which discuss the benefits and risks of continuing the current dose, adjusting the dose or stopping the medication
 - Look for signs that person is developing problems with dependence e.g. running out of a medicine early, frequent requests for dose increases, reporting that the medicine is no longer working
- **Discuss supported withdrawal** if the medication is no longer benefitting the person, problems associated with dependence have developed, the person wants to stop or the harms are outweighing the benefit
- **Make individualised shared decisions on withdrawal**, and make an individualised withdrawal plan
 - Drugs should not be stopped abruptly
 - If withdrawing from an opioid, BZP or Z-drug suggest slow stepwise reduction proportionate to the existing dose so that the decrements become smaller as the dose is lowered
 - If withdrawing from a gabapentinoid, reduce the dose by a fixed amount at each decrement
 - If withdrawing from a BZP with a short half-life, consider switching to a BZP with a longer half life
- Complete withdrawal may not be the initial goal, or for people who find complete withdrawal too difficult a more realistic aim may be dose reduction with ongoing reviews

NB Learning Points & Quality Improvement Ideas

- We need to discuss with patients before prescribing opioids, benzodiazepines (BZPs), gabapentinoids, Z- drugs and antidepressants the risk of dependence and withdrawal reactions
- Look for signs that the person is developing problems with dependence e.g. running out of a medicine early, frequent requests for dose increases, reporting that the medicine is no longer working
- Offer patients guided and supported withdrawal according to an individualised withdrawal plan

PRESCRIBING IN RENAL IMPAIRMENT

In October 2019 we had a very useful review in the monthly [MHRA Drug Safety Updates](#) on prescribing in renal impairment. The review/advice was undertaken due to a number of reports and queries to the MHRA around the choice of renal function estimate when prescribing in renal impairment. We are getting more used to using creatinine clearance (CrCl) due to the advent of DOACs, but *in which other situations should we use CrCl as opposed to eGFR?*

For most drugs and most adult patients of average height and weight, eGFR is an appropriate measure of renal function to use to adjust drug doses. It is simple and easy to use as it is now routinely reported from the labs with renal function parameters. However, **eGFR is based on an average body surface area of 1.73m²** so those at the extremes of weight may get false renal function estimates - in overweight people, eGFR can underestimate true renal function, but possibly more importantly (from a prescribing perspective) in underweight people eGFR can overestimate true renal function running the risk of overdosing on drugs. Also, drugs that are predominantly renally excreted or have a narrow therapeutic window need more accurate renal function adjusted dosing based on CrCl, as do the elderly, and some drugs (e.g. DOACs) that have been licensed based on drug dosing using CrCl, not eGFR.

The MHRA have had a **number of reports of overdosing of drugs when eGFR has been used instead of CrCl**, and a cross-sectional study in the BJGP in 2018 ([British Journal of General Practice 2018; 68 \(670\)](#)) particularly highlighted the risks of overprescribing in the elderly; the study revealed that eGFR overestimated kidney function in 3-28% of those ≥ 65 years old, and in 13-58% for those ≥ 85 years old. The upshot is we should be **using CrCl instead of eGFR more widely, in particular, those aged ≥ 75 years old**. Tools for doing this are integrated into GP software systems, but it can quickly be calculated using the [Cockcroft-Gault formula via MDCalc](#)

KISS: Prescribing in Renal Impairment

[MHRA Drug Safety Update October 2019](#)

- For most drugs and for most adult patients of average height and weight eGFR can be used to adjust drug doses in renal impairment.
- However, **eGFR may overestimate (or underestimate) true renal function in a number of circumstances** leading to overdosing (or underdosing) of medications.
- In these situations, **Creatinine Clearance (CrCl)** should be used as a more accurate predictor of true renal function using the **Cockcroft-Gault formula**
- This can be calculated using an integrated tool in GP software, or alternatively, [click here for a link to MDCalc](#)
- Use CrCl as opposed to eGFR in the following circumstances when prescribing in renal impairment:
 - When prescribing DOACs
 - Elderly patients ≥ 75 years old
 - Patients at extremes of muscle mass (BMI <18kg /m² or > 40kg/m²)
 - Drugs with narrow therapeutic index e.g. digoxin, sotalol

NB Learning Points & Quality Improvement Ideas

- **eGFR can overestimate true renal function (and therefore risk overdosing of drugs) in the elderly**, use the Cockcroft-Gault formula to estimate Creatinine Clearance (CrCl) when prescribing drugs for patients ≥ 75 years old.
- **Quality Improvement Idea:** Review all patients ≥75 years old on DOACs - have they had their CrCl estimated to guide drug dose?



MEDICAL CANNABIS

Medical cannabis has been a major media story of the past year, following a number of high-profile cases, particularly in [children with severe epilepsy](#), where cannabis oil dramatically improved their symptoms but could not be prescribed in the UK nor imported into the country. High-quality RCTs have proven efficacy for CBDs compared to placebo in these rare childhood syndromes with severe epilepsy [NEJM 2018](#) and [NEJM 2017](#). Following this and a huge amount of public pressure, the UK government has acknowledged that cannabis has medicinal value and relaxed the rules around cannabis products so that throughout the [UK doctors will be able to prescribe](#) them. The editor of the BMJ calls this “[an important step](#)”, but the reality for clinicians is that the rules are still [highly restrictive and patients will struggle to obtain cannabis medications](#). **More recently we had a systematic review in the BMJ ([BMJ 2021;374:n1034](#)) looking at medical cannabis for chronic pain, with a somewhat surprising and certainly controversial Rapid Recommendation ([BMJ 2021;374:n2040](#)) based on this evidence.**

► *What is medical cannabis?*

Cannabis comes from the cannabis plant and, in its illicit form typically comes as dried leaves which are smoked. The first reported use is documented as ~440 BC, with widespread use throughout the western world over the past two thousand years. It has been used in many populations as a 'medicine' often to relieve pain, tetanus and seizures, often as a tincture. Interestingly, a BMJ [essay from David Nutt](#) explains the 'demise' of cannabis stems from the USA in the 1930s where the government were looking to find work for those policing alcohol prohibition which was rescinded during this time.

Cannabis contains >100 components including cannabinoids, terpenes and flavonoids. The most studied active chemicals are tetrahydrocannabinol (THC) and cannabidiol (CBD). The former is associated with recreational highs, whereas the latter is not. The ratio between the two may be important as CBD may act as an antagonist to some of the more harmful effects of THC. As well as being found naturally in the cannabis plant, these compounds can be synthesised, the resulting products being either chemically identical or mimicking the effects of cannabis. Cannabinoid receptors are found throughout the body and are termed the “endocannabinoid system”. It is involved in a range of functions: metabolic, endocrine, neural and more. Cannabinoids interact with this system and other neural transmission systems.

► *What are the risks of medical cannabis?*

The short-term effects, particularly with THC predominant products, that are the same as illicit cannabis: dizziness, euphoria, drowsiness, dry mouth, confusion, disorientation, somnolence, balance problems and fatigue. More of a concern is around dependency (~9% of users) and neuropsychiatric effects. Data suggests a likely causal link between cannabis use and psychosis, particularly in those with a personal or family history. There are also concerns over other mental health issues such as depression and anxiety disorders and [possible teratogenicity](#), including with paternal use.

► *What evidence is there that cannabis products may be beneficial and what might be the risks?*

And here is the issue. Data is limited and of low quality. Different assessments often draw different conclusions. The [BMJ 2018;362:k3230](#) featured an essay summarising what we know, [Dame Sally Davies provided an evidence review](#) to government supporting a recommendation to alter the “scheduling” (level of legal restrictions on a drug) and [a recent systematic review from the Australian Department of Health](#) provides a further summary of the literature.

NICE have now produced a new guideline [NICE Cannabis-based medical products 2019](#) which advises on the very limited medical roles it sees for cannabis. Meanwhile, it has also published two technology appraisals on cannabis products in the management of [Lennox-Gastaut](#) and [Dravet syndrome](#). Since the publications above a systematic review and meta-analysis [BMJ 2021;374:n1034](#) attempted to review the evidence and make recommendations on the use of medical cannabis and cannabinoids for chronic pain.

1 Cannabis-based Medical Products [NICE 2019 NG144](#), [TA614](#), [TA615](#)

Inevitably, patients and families who were hoping for easier access to cannabis-based rugs (CBDs) are likely to be disappointed by this guideline, as it reflects the lack of high-quality data for most indications, the lack of long-term safety data and also accounts for cost-effectiveness. The key recommendations are:

- GPs should be able to take on prescribing of CBDs, once it has been initiated by specialists in expert centres as part of shared care agreements
- **Chronic pain**
 - Do not offer nabilone, dronabinol, THC or CBD/THC products to manage chronic pain
 - Do not offer CBD unless part of a clinical trial
 - Those already on cannabis-placed medicinal products prior to the guideline publication can continue until they and their clinician think it appropriate to stop.
- **Nausea and vomiting**
 - "Consider" nabilone as an add-on treatment for adults with chemotherapy-induced nausea and vomiting failing to respond to conventional licensed treatments.
 - *Nabilone* is the only licensed cannabis-based drug that NICE recommends; the 'limited evidence' are mostly old, low quality and used poor comparators. Nabilone was also associated with more adverse events.
- **Spasticity**
 - Offer a 4-week trial of THC:CBD spray to treat moderate to severe spasticity in adults with MS if 1) other treatments are ineffective and 2) the company provides it via its pay-for-responders scheme
 - Specialist initiation and supervision is required.
 - *Sativex* is the only licensed product in the UK.
- **Severe epilepsy**
 - The main guideline makes research recommendations for the management of severe epilepsy. Two technology appraisals provide further recommends for specific rare syndromes.
 - **Dravet ([TA614](#)) and Lennox-Gastaut ([TA615](#)) syndromes:**
 - Cannabidiol with clobazam is recommended as an option for treating seizures associated with Lennox-Gastaut or Dravet syndrome in people 2y+ if: 1) the frequency of drop seizures is checked every 6/12 and cannabidiol is stopped if the frequency has not fallen by at least 30% vs the 6/12 pre-treatment, and 2) the company provides cannabidiol via its commercial arrangement with the NHS.

Perhaps the most surprising recommendation, and disappointing for many members of the public, was the recommendation against using cannabis-based products for chronic pain. In [Dame Sally Davies report](#) she concluded there was "conclusive or substantial evidence" that cannabis (rather than THC or CBD) was effective for managing chronic pain.

NICE acknowledged that there was some evidence of benefit with CBD in chronic pain but the effect was 'modest' (it quotes an improvement of 0.4 on a scale of 0-10). It concluded that *given so many people have chronic pain and the medical version of treatment would be very expensive, once the limited benefit and ongoing costs are considered it was not a good use of NHS resources.*

The conclusion from NICE marries neatly with a recent systematic review in Pain journal ([2018;159:1932](#)) which found in data on almost 10k patients chronic non-cancer pain cannabinoids do reduce pain compared to placebo albeit only modestly (NNT = 24) but also increase adverse events (NNH = 6), concluding the balance appears to not favour the active treatment.

► **Systematic review, meta-analysis and Rapid Recommendation of medical cannabis or cannabinoids for chronic non-cancer and cancer pain** [BMJ 2021;374:n1034](#) [BMJ 2021;374:n2040](#)

Given the uncertainties and varying interpretations on the benefits of cannabis for chronic pain as discussed above, and the insistence of many that it does help, this meta-analysis and Rapid Recommendation was a valiant attempt to review the state of play and make some pragmatic recommendations. **In summary:**

- This study reviewed 32 trials, comprising just over 5000 patients taking medical cannabis or cannabinoids for chronic pain (both cancer and non-cancer related) against placebo with follow-up 1-5.5 months.
- **Compared with placebo non-inhaled medical cannabis probably results in a small increase in the proportion of patients with a reduction in pain levels** (deemed as 1cm on a 10cm pain scale).
- Oral medical cannabis resulted in a very small improvement in physical function and sleep but did not improve emotional or social functioning.
- There was moderate certainty evidence that medical cannabis results in a small increased risk of transient cognitive impairment, vomiting, nausea, drowsiness and impaired attention.

The evidence above led to the BMJ producing a [Rapid Recommendation](#) guideline with a **weak recommendation that non-inhaled medical cannabis or cannabinoids should be offered to people with chronic pain when standard care is not sufficient**.

To say that this Rapid Recommendation slightly took us by surprise would be a massive understatement. There are more holes in this evidence than a Swiss cheese and **seems to ignore many of the wider implications and risks of making a recommendation such as this**, as highlighted in the associated editorial ([BMJ 2021;374:n1942](#)):

1. First the **quality of the underlying evidence is frankly a bit of a mess**. Follow-up periods of less than 6 months is very short for chronic pain conditions. Sample sizes were small in the individual studies. Different outcome measures were often used and finally, industry sponsorship and funding was apparent in some of the data. However, you want to rearrange the deck chairs, if the deck chairs are missing legs, backs and are of various sizes it's never going to look great whatever you do.
2. Second is the **0.5-1cm improvement on a 10cm pain scale really clinically relevant?** Most studies consider 1.5cm to be more clinically relevant so the recommendations seem somewhat questionable. However, analysis of a sub-set of the trials showed a 7% increase in the proportion of people reporting a >30% reduction in pain with medical cannabis, so it may be that there are a proportion of people that get more substantial benefits, with others (probably the majority) getting no benefit at all.
3. **Self-medication by patients may lead to a higher risk-benefit ratio if they use higher THC concentrations** than those used in the studies, **especially if they start inhaling** in the hope of getting quicker results (this data specifically excluded inhalation) as we know there are definite risks both if inhaled in cigarette or vape form.
4. There is also a **substantial risk that recommendations such as this 'blur the lines' between recreational and medical cannabis**, with the risk that people will consider recreational cannabis 'medically recommended' and therefore safe, despite the known risks of that. This risk could be higher for younger populations whose brains are still developing, or for vulnerable populations at risk of other neuropsychiatric complications or substance abuse.
5. And **what about the long-term effects?** These studies were <6 months in duration and can only highlight the short-term side effects. Long-term complications may take years to develop, and **what are the risks of dependency and abuse?** We just need to remember the problems we now face with first opiates, and then gabapentinoids for the treatment of chronic pain, to know we've been down this route before.
6. And finally **the most fundamental and practical question of all - Access**. We can't prescribe these drugs and it would be a big leap of faith to recommend patients self-medicate due to the risks above, the variability of the products that can be bought and the fact that these products are unregulated.

All in all, **it does not seem that this current evidence or Rapid Recommendation should change the current NICE recommendation that cannabis products should not be recommended for use in chronic pain**. What we really need is some high-quality evidence to see if we can unpick those patients who may get significant benefit from medical cannabis and those who won't. The call for urgent high-quality research was echoed by a BMJ Editorial ([2019;367:l5871](#)) which set out how trials could be designed to include different arms encompassing different combinations of cannabis products, the role of N=1 trials to identify treatment responders who merit continuing prescription, and the need for industry co-operation to find products which could ultimately benefit patients.



2 Cannabinoids for the treatment of mental health disorders [Lancet Psych 2019;6:995](#)

Along with chronic pain management, within the general population, the other common use of cannabis products is to attempt to alleviate mental health issues. Most of us might consider that highly likely to be counter-productive, but what does the data show?

This new systematic review and meta-analysis ([Lancet Psych 2019;6:995](#)) searched for data on using cannabinoids for the treatment of a variety of mental health disorders in adults, including depression, anxiety, ADHD, Tourette syndrome, PTSD and psychosis.

The authors concluded that despite 83 eligible studies (42 for depression, 31 anxiety, 8 Tourette syndrome, 12 PTSD, 11 psychosis), in which most used THC +/- CBD vs placebo, ***there was "scant evidence" to suggest that cannabinoids improve any mental health disorder***, with the exception of anxiety, for which there was "very low-quality evidence" that THC +/- CBD led to small improvements in symptoms of anxiety among individuals with other medical conditions.

Data did demonstrate double the risk of adverse events and almost triple the risk of withdrawal effect compared to placebo across of group of conditions. There was little research into pharmaceutical CBD or 'medicinal cannabis'.

► Conclusions

Ultimately cannabis-based products appear to be a promising area with potential benefit in a number of important conditions including treatment-resistant epilepsy, spasticity and (possibly) chronic pain. There is a relatively strong evidence base for rare forms of severe childhood epilepsy which are elected in the relevant NICE TA guidelines. The limitation is a lack of good data on efficacy for other indications and on short and long-term adverse events, particularly with concerns around the teratogenic effect and mental health. This lack of evidence and concern over cost-effectiveness has resulted in the recommendations from the NICE guideline 2019 that the only cannabis-based products recommended are nabilone (for chemotherapy-induced nausea/vomiting not responding to usual treatments) and Sativex (for moderate/severe spasticity in MS, under specialist initiation/supervision).

Good quality RCTs are required to explore further not just whether they work but which components are most effective and which have the least side effects. With legal restrictions loosened there is hope that we may start to see this happen.

NB Learning Points: Medical Cannabis

- There are very limited medical cannabis products licensed in the UK.
- NICE published guidance in 2019, which recommends limited use of nabilone and Sativex in very specific situations, initiated by a specialist with the potential to continue prescribing in primary care under shared care agreements.
- A recent BMJ Rapid Recommendation, based on a meta-analysis, makes a weak recommendation that non-inhaled medical cannabis or cannabinoids should be offered to people with chronic pain when standard care is not sufficient. This is a questionable recommendation given the quality of the underlying evidence and multiple wider risks and goes against the clear NICE recommendation that cannabinoid products should not be offered to manage chronic pain.
- Cannabis products appear largely ineffective for the treatment of mental health conditions.
- For clinical guidance from NHS England on the use of cannabis-based products [click here](#)
- For NHS England FAQs [click here](#)
- For patient information, the website [End Our Pain](#) has useful info and further links.

DISEASE MODIFYING DRUGS (DMARDs)

Although initiation of DMARDs will be done by secondary care, the vast majority of prescribing and monitoring of these drugs is done in primary care. These drugs have revolutionised the management of inflammatory diseases but keeping tabs on the monitoring requirements can be bewildering at times. So it was helpful that the British Society of Rheumatology gave updated guidelines on the prescribing and monitoring of rheumatological DMARDs in June 2017 ([Rheumatology, Volume 56, Issue 6, 1 June 2017, Pages 865–868](#)), which is supported by [CKS Dec 2021](#). However, this all adds up to a significant amount of testing and monitoring, and recent data ([BJGP 2022; 72 \(720\): e528-e537](#)) suggests we are over-testing significant numbers of people on two of the commonest DMARDs - methotrexate and azathioprine.

► Prescription and monitoring of non-biological disease-modifying anti-rheumatic drugs

[BSR guideline June 2017](#) and [CKS Dec 2021](#)

General principles and recommendations:

- Initiation of DMARDs should be done by a specialist in secondary care.
- Once stabilized on treatment, prescribing and monitoring can be taken over in primary care but should **always be done as part of a shared-care protocol**.
- **Baseline assessment** should include height, weight, BP and bloods for FBC, U&E (inc eGFR), ALT +/- AST and albumin.
- **Vaccinations** for pneumococcus and influenza are recommended.
- Ensure patients on DMARDs have been given suitable **patient information**. Good info at:
 - [Arthritis care](#); [Arthritis Research UK](#); [National Rheumatoid Arthritis Society](#)

Prescribing with co-morbidities:

Although this decision will be made by secondary care it is useful to be aware of the following:

- Pre-existing lung disease is not a specific contraindication to DMARDs but caution needed if using drugs associated with pneumonitis in patients with poor respiratory reserve.
- In patients with deranged LFTs caution is needed if using hepatotoxic drugs.
- In patients with liver cirrhosis, extreme caution is needed with DMARDs.
- Use DMARDs with caution in CKD - appropriate dose reduction and monitoring advised.
- Cardiovascular disease and malignancy are not contraindications to DMARDs.

Drug-specific recommendations:

- **Methotrexate** - co-prescribe folic acid at a minimum dose of 5mg weekly.
- **Azathioprine** - Patients should have baseline thiopurine methyltransferase (TPMT) status (blood test for levels of TPMT enzyme which is important for metabolising azathioprine).

Standard monitoring schedule and troubleshooting for DMARDs:

- See KISS below.

► Persistently normal blood tests in patients taking methotrexate for RA and azathioprine for IBD

[BJGP 2022; 72 \(720\): e528-e537](#)

But these monitoring schedules lead to a huge amount of work for clinicians, labs and admin staff, as well as significant burdens for our patients. Many of us I'm sure will have questioned whether all this monitoring is really needed, and it became clear during COVID that you could safely reduce monitoring for a number of patients. As this BJGP study points out there is **no clear evidence that monitoring actually reduces harm**. This study aimed to determine the frequency and associations of persistently normal blood tests in patients with rheumatoid arthritis on methotrexate, and patients with IBD on azathioprine. What did it find?

- This was a 2-year retrospective cohort study covering >1.4 million people in the UK.
- 3001 patients with rheumatoid arthritis (RA) being prescribed methotrexate, and 1162 patients with IBD being prescribed azathioprine were identified.
- Blood test results were then reviewed against the normality thresholds defined by NICE (see KISS below for defined thresholds) and associated patient factors were cross-referenced.
- **In patients with RA on methotrexate:**
 - 1585 (52.8%) had persistently normal blood tests.
 - Those with abnormalities were due to eGFR (19.5%), ALT (7.3%), WBC (7.2%), neutrophils (6.7%).
 - Not surprisingly **those with persistently normal results were younger and without co-morbidities**, especially CKD and heart failure; among people aged under 60, >60% had persistently normal results vs <30% of those aged ≥80.
- **In patients with IBD on azathioprine:**
 - 657 (56.5%) had persistently normal blood tests.
 - Those with abnormalities were due to WBC (11.9%), eGFR (7.6%), ALT (6.2%), neutrophils (5.6%).
 - Similar patterns of persistently normal results were seen in younger people without co-morbidities.
- **Conclusion: Persistent normality is common with DMARD monitoring in the cohorts examined, and more stratified monitoring approaches should be considered.**

The authors point out the significant workload, and burdens on patients, with the current recommended monitoring regime - the vast majority of patients had ≥9 blood tests in a 2-year period. Even those with abnormal results (dominated by eGFR abnormalities) may have abnormal but stable parameters that don't need action. However, they do rightly point out that some abnormalities may take longer to develop than the 2-year study period e.g. methotrexate-induced liver disease.

However, overall this data does suggest that there are significant numbers of people, particularly the younger group without co-morbidities, that could safely have their monitoring reduced. As things currently stand we will still have to continue the nationally recommended monitoring regime, but we can hope that, as the authors call for, over time we can use a more patient-centred stratified approach to monitoring, which will reduce the significant burdens on our patients while reducing unnecessary workload for primary care.

NB Learning Point: Monitoring of DMARDs

- **During a serious infection** methotrexate, leflunomide, sulfasalazine, azathioprine, mycophenolate, ciclosporin and tacrolimus should be temporarily discontinued; we should seek specialist advice if considering this

NB Quality Improvement Ideas: Monitoring of DMARDs

- Patients on hydroxychloroquine > 5 years - are they having annual eye assessments?
- Patients on leflunomide - are they having their BP checked at each monitoring?
- Patients stable on sulfasalazine > 1 year - are they still having routine lab monitoring? The BSR guidance recommends they do not need to continue this

KISS: Monitoring of Disease Modifying Drugs (DMARDS)

[BSR guideline June 2017](#) and [CKS Dec 2021](#)

■ Standard monitoring schedule:

- FBC, U&E (inc. eGFR), ALT +/- AST and albumin every 2 weeks after initiation or dose increase until on stable dose for 6 weeks
- Then same blood tests monthly for 3 months
- Then same blood tests every 3 months

Drug	Lab monitoring	Other monitoring
Apremilast	No routine lab monitoring needed	None
Azathioprine (AZA)	Standard monitoring schedule	None
Ciclosporin	Extend monthly monitoring long-term*	BP & glucose at each monitoring
Gold	Standard monitoring schedule	Urine for blood/protein prior to each dose
Hydroxychloroquine	No routine lab monitoring needed	If on treatment >5 years annual eye assessment needed
Leflunomide (LEF)	Standard monitoring schedule	BP & weight at each monitoring
Mepacrine	No routine lab monitoring needed	None
Methotrexate (MTX)	Standard monitoring schedule	None
MTX and LEF combined	Extend monthly monitoring long-term*	None
Mycophenolate	Standard monitoring schedule	None
Sulfasalazine (SSZ)	Standard monitoring schedule for 12 months then no routine monitoring needed	None
Tacrolimus	Extend monthly monitoring long-term*	BP & glucose at each monitoring
*	If stable for 12 months reduced the frequency of monitoring can be considered on an individual patient basis	

■ Consider stopping treatment if any of the following develop (contact rheumatology urgently for advice):

- WCC $<3.5 \times 10^9/l$
 - MCV >105 fL
 - Neutrophils $<1.6 \times 10^9/l$
 - Platelets $<140 \times 10^9/l$
 - Unexplained eosinophilia $>0.5 \times 10^9/l$
 - Creatinine increase $>30\%$ over 12 months and/or eGFR <60 ml/min
 - ALT and/or AST $>100U/l$
 - Unexplained reduction in albumin <30 g/l
- During a **serious infection** MTX, LEF, SSZ, AZA, apremilast, mycophenolate, ciclosporin and tacrolimus should be discontinued - seek specialist advice

BIOLOGIC AGENTS

Biologic agents in inflammatory arthritis [Br J Gen Pract 2018; 68 \(669\)](#)

Biologic agents are now widely used in our patients with inflammatory arthritis. Whilst they are initiated and monitored in secondary care, primary care is often the first port of call for patients on biologic agents when they get acutely unwell, so having an understanding of our role is important.

► *Just remind me what biologic agents and biosimilars are?*

They are immune-modulating proteins derived from human genes that are genetically engineered to inhibit precise components of the immune system. They suppress specific immune pathways and thus are more targeted in their approach than conventional DMARDs. They are options for managing inflammatory arthritis in those who do not achieve adequate control on conventional DMARDs, and if tolerated they are often used in combination with conventional DMARDs such as methotrexate to enhance efficacy.

Biosimilars are the biological medicines equivalent of generics. They are almost identical in structure to biologics produced at lower cost once the original is beyond patent, and are now often recommended ahead of biologics e.g. the biologic etanercept is Enbrel but has a biosimilar benepali. Studies have shown that biosimilars have a similar mechanism of action and safety profile as biologics e.g. for infliximab short-term data suggests the biosimilar is as effective and it reduces the cost from £10,000 to £5,000 per year [DTB2017](#) In the UK it is advised that biosimilars are prescribed by brand name

KISS: Guide to different biologics & biosimilars licensed for inflammatory arthritis				
Drug name	Brand names B/S = Biosimilar	Mechanism of action	Route of admin	Specific cautions
Infliximab	Remicade Inflectra (B/S) Remsima (B/S)	Anti-TNF	IV	FBC & LFT monitoring Risk of TB reactivation
Etanercept	Enbrel Benepali (B/S)	Anti-TNF	S/C	FBC & LFT monitoring Risk of TB reactivation
Adalimumab	Humira	Anti-TNF	S/C	FBC & LFT monitoring Risk of TB reactivation
Certolizumab Pergol	Cimzia	Anti-TNF	S/C	FBC & LFT monitoring Risk of TB reactivation
Golimumab	Simponi	Anti-TNF	S/C	FBC & LFT monitoring Risk of TB reactivation
Tocilizumab	RoActemra	Anti-IL-6	IV or S/C	Lipid monitoring Reduced CRP response to infection
Abatacept	Orencia	Anti-CD28	IV or SC	
Rituximab	Mabthera	Anti-CD20	IV	Rarely causes PML; alert specialist if any neurological symptoms

► *So what are the key points we, and our patients on biologic agents, need to be aware of?*

- These are potent immune-modulating drugs, and **the main risk is infection** due to immune suppression; this includes common bacterial infections as well as fungal and TB
- **Prior to prescription** patients will be screened for a history of recurrent infection or signs or symptoms of malignancy, and specifically be tested for TB and virology (e.g. HIV, hepatitis etc) as well as standard screening tests such as FBC, LFT and lipids
 - Avoid contact with those suffering from chickenpox or shingles

- **Vaccination:** all patients should be up to date with varicella, hepatitis b and MMR vaccines prior to their use
 - Regular influenza and pneumococcal immunisations are advised
 - Avoid live vaccinations; if clinically indicated, specialist advice should be sought
 - **Side-effects:**
 - The most common s/e is local irritation at the injection site
 - Flu-like illness, GI upset and headache may occur
 - If the patient develops symptoms e.g. flu-like illness or gastro-intestinal upset following biologic dosing discuss with a specialist team
 - Patients should be advised to seek medical attention urgently if **symptoms of infection** develop, and they should stop their biologic treatment until medical advice is sought
 - With regards to **travel**, patients must ensure they are up to date with appropriate vaccinations, and caution must be taken regarding refrigeration and storage of biologic drugs
 - Long-term use of certain biologic agents can increase **malignancy risk**, such as lymphoma and skin cancers, so we should have a low threshold for referral
 - **Smoking** cessation advice and support is advised; studies have shown improved efficacy of biologic agents following smoking cessation
 - Be aware that as the prescribing is done in secondary care, these drugs are often not listed on the primary care clinical record drug list, so we need to actively enquire of any patients with inflammatory arthritis if they are on biologic agents.
- **So, what should we do if patients on biologic agents present with infection?**
- If a patient has evidence of active infection requiring antibiotics or antifungal therapy (regardless of severity), we should withhold the biologic agent and treat the infection urgently; seek advice from the specialist rheumatology team regarding when the biologic agent can be restarted
 - Patients on biologic agents may not respond to the standard short course of antibiotics and may need longer courses (refer to local antibiotic guidelines for antibiotic choice and duration)
 - If there are symptoms of viral illnesses, consider holding the biologic agent if a dose is due imminently and seek advice from the rheumatology specialist team

NB Learning Points & Quality Improvement Ideas

- If a patient on a biologic or biosimilar develops an active infection requiring antibiotic therapy, withhold the biologic and treat the infection urgently; longer courses of antibiotics may be needed.
- **Quality Improvement Idea** - Ensure all biologics are recorded on the primary care drugs record (under the 'hospital prescription only' or 'not to prescribe in primary care' section or similar) and consider adding an alert message if they contact us with symptoms of infection.

ASPIRIN, BLEEDING RISK & PPI

'Low dose aspirin causes 3,000 preventable deaths per year in older people' was the message that came out loud and clear in the popular media in Summer 2017, following the latest publication from the OXVASC study [Lancet 2017;390;490](#). These headlines may well have led to many people stopping aspirin when they really need it for secondary prevention. However, *the research is high quality and has important implications for clinical practice given that 40-66% of adults aged 75 or over, take low-dose aspirin. PPIs are under-used in patients on antiplatelet therapy, despite the fact that they reduce upper GI bleeding risk by 70% to 90% in this group.*

More latterly we have had the results from the **ASPREE trial** ([Drug and Therapeutics Bulletin 2021;59:117](#)) which is a high-quality RCT and confirms the findings of the observational OXVASC trial that there is a substantial GI bleed risk in elderly people taking aspirin, and being a big RCT it gives quite robust data on absolute risks. First the results of the OXVASC trial:

► Age-specific risks with long-term antiplatelet therapy [Lancet 2017;390;490](#)

This important paper from Peter Rothwell and co was the latest publication from the long-going OXVASC study, that has already yielded important clinical findings that have changed practice (e.g. the seminal [EXPRESS Study](#), showing that early active intervention after TIA reduces recurrent stroke risk by 80%).

What did they do?

This was a prospective cohort study in patients with a first TIA, ischaemic stroke or MI treated with anti-platelets (mainly aspirin without routine PPI use) in the OXVASC study and followed up for 10 years. They estimated bleeding risks and age-specific numbers needed to treat (NNT) to prevent upper gastrointestinal (GI) bleeding using proton pump inhibitors (PPIs). They gathered data on first bleeds in over 3,000 patients with over half of those aged over 75. Most previous studies have only had two to four years of follow-up, so the long follow-up here is significant. Patients were not routinely prescribed PPIs, in line with existing guidance, but approximately 30% of the cohort were taking them enabling them to estimate the efficacy of PPIs for preventing bleeds.

What did they find?

The principal finding is that the risk of major and fatal bleeding in patients receiving antiplatelets without PPI cover is significantly higher in older patients than younger patients. This is not surprising, what was surprising was that *the risk was significantly higher and more sustained than shown in previous trials* with a 'substantial risk of disabling or fatal upper GI bleeding' in those aged 75 or over.

- Risk of *major bleeding increases steeply with age* especially aged over 75 (≥ 75 years, hazard ratio (HR) for a major GI bleed is 4 and for fatal or disabling bleeds HR is 10)
- The absolute risk aged ≥ 75 years for a major or fatal bleed is 9 per 1,000 patients pa (NNH = 111)
 - The absolute annual risk of a major bleed is around 1% aged under 75, rising to 2% aged over 75 and reaches 4% in those aged over 85
- In adults aged ≥ 75 years, the risk of having an upper GI bleed has previously been under-estimated; 62% of these bleeds were fatal or disabling
- Proton pump inhibitors have previously been shown to reduce the relative risk of a major bleed with antiplatelets by about 75%. This study estimated the NNT over 5 years for preventing a major bleed with a PPI was 75 in patients aged 65 to 74, but only 23 in those aged 75 to 84 and 21 in those aged 85 or older. For fatal and disabling bleeds the NNT fell from 338 in adults aged under 65 to just 25 in those aged over 85
- **Conclusion:** The risk of major bleeding with low-dose aspirin and antiplatelets in older adults is higher and more sustained than previously thought, these bleeds when they occur are highly dangerous and the estimated NNTs to prevent a GI bleed with a PPI in this group is low. Co-prescription of PPIs with anti-platelets in older adults 'should be encouraged'.

What are the implications of this research?

The clear implication is preventing major GI bleeds in elderly patients, discussed in an associated editorial [Lancet 2017;390:435](#). They state:

- We should reassess the risk: benefit ratio of antiplatelets in older adults every 3 to 5 years (NB and more research is needed into how to best identify the overall balance of risks and benefits in older people).
- We should routinely use PPIs for patients on antiplatelet therapy aged over 75.

► **Aspirin and GI bleed risk in older people** [Drug and Therapeutics Bulletin 2021;59:117](#)

The ASPREE trial was a primary prevention RCT (in contrast to OXVASC which was a secondary prevention trial) which published its first set of results in 2018 [N Engl J Med 2018; 379:1509-1518](#) and confirmed that aspirin for primary prevention did not reduce the risk of CVD events, but that it did increase haemorrhagic risk. This **second publication looked more specifically at GI bleed risks and associated risk factors**. What was the trial and what did it show?

- Large RCT (almost 20,000 participants) based in USA and Australia taking aspirin as a primary preventative measure for CVD events, randomised to 100mg daily enteric-coated aspirin and placebo.
- Median age 74 years, 44% men, median follow-up 4.7 years, and participants were not considered to be at high risk of GI bleed and had no history of CVD, dementia and not expected to die within 5 years.
- The **primary outcome was significant GI bleed (requiring hospital admission, transfusion, prolonged hospital stay, surgery or death)**.
- Absolute risks of the primary endpoint were 1.7% in the aspirin group and 1.1% in the placebo group
 - Giving relative risks of HR = 1.61, 95% CI 1.26-2.08.
 - This gives a **number needed to harm (NNH) of 156 over 4.7 years**.
- There was a relatively higher risk of aspirin causing upper rather than lower GI bleed risks, but interestingly the lower GI bleed risks were also significant.
- As with the OXVASC study **ASPREE confirmed the GI bleed risk goes up with age**:
 - Absolute 5-year risk of GI bleed in a healthy 70-year-old = 0.25% without aspirin and 0.40% with aspirin.
 - These risks go up for an 80-year-old to 0.60% without aspirin and 0.96% with aspirin.
- **ASPREE also showed that GI bleed risks went up significantly with other risk factors** (smoking, hypertension, overweight, CKD); if any of these present:
 - 5-year absolute GI bleed risk in 70-year-old 1.42% without aspirin and 2.26% with aspirin.
 - 5-year absolute GI bleed risk in 80-year-old 3.22% without aspirin and 5.03% with aspirin.
- **Conclusion:** Aspirin for primary prevention in older people causes excess GI bleed risk without improving CVD outcomes, but the risks increase with both age and other comorbidities.

What are the implications for us with this data?

First the authors and DTB highlight the importance of us **reviewing the need for aspirin as people get older** - as they get older the benefit/risk ratio will change especially if they accumulate other comorbidities, and if they need to continue with their aspirin we need to address other risk factors that increase their bleeding risk (e.g. hypertension). To put this in context the **5-year risk of a serious GI bleed in a 70-year-old without risk factors is 1 in 250, but for an 80-year-old with risk factors it is as high as 1 in 20**. Second, they highlight the **importance of adding a PPI for people at higher GI bleed risk** - in this study only 25% of people were taking a PPI and so there was not enough data to extrapolate risk reduction with PPI, but given the OXVASC data above we should generally have a very low threshold for adding a PPI for people over age 70 on aspirin.

Finally, this data confirms the current guidelines - that **GI bleed risks of aspirin for primary prevention of CVD outweigh any CVD benefits and should not be routinely prescribed**. However, the one factor that still is yet to be answered fully is whether aspirin may give benefits for the primary prevention of cancer (specifically GI cancers)...we have observational data that suggests it may do, but further RCTs in progress will give us more data, notably the [ADD-ASPIRIN](#) trial which is looking at primary and secondary preventative benefits of aspirin.

NB Learning Points and Quality Improvement Ideas: Aspirin, bleeding risk and PPI

- The benefit of low-dose aspirin for **secondary prevention** after MI, TIA and stroke is unquestioned, however, this research highlights that the risk of serious and fatal GI bleeding with aspirin is higher and more sustained than previously thought in those aged ≥ 75 years and raises questions on the long-term risk: benefit ratio if a PPI is not co-prescribed.
- In people aged under 75, the benefits of secondary prevention clearly outweigh the small risk of bleeding and these patients need not worry.
- In people aged over 75, the risk of bleeding is significantly higher but can be substantially prevented by taking a proton pump inhibitor (NNT only 23 over 5 years to prevent a major bleed in adults aged over 75).
- For **primary prevention** of CVD the results of the 2 ASPREE publications clearly show that the GI bleed risks of aspirin outweigh any CVD benefits, with an overall NNH of ~ 150 over ~ 5 years.
- The ASPREE data also confirms the results of the OXVASC trial that GI bleed risks increase with age, but also that other co-morbidities substantially increase GI bleed risk - as with OXVASC the data suggests a low GI bleed risk with aspirin - in an otherwise healthy 70-year-old on aspirin the 5-year risk is ~ 1 in 250, but for an 80-year-old with other risk factors (smoking, hypertension, overweight, CKD) the risks are substantially higher at ~ 1 in 20.
- These trials suggested that elderly patients should have the risk: benefit ratio re-assessed every 3 to 5 years, and that we should try to mitigate any other bleeding risk factors (e.g. smoking, hypertension).
- NICE advise licensed dose for gastroprotection e.g. omeprazole 20mg daily or lansoprazole 15mg daily.
- **NB Quality Improvement Ideas:**
 - **Audit of patients aged over 75 on antiplatelet therapy:**
 - Do they still need it and are they on a PPI?
 - Any patients on aspirin for primary prevention are highly unlikely to gain any CVD benefit and should probably stop their aspirin.
 - Are there other risk factors that can be mitigated? Hypertension, smoking, overweight?

TYPE 2 DIABETES - PHARMACOLOGICAL MANAGEMENT

'The management of hyperglycaemia in type 2 diabetes has become extraordinarily complex with the number of glucose-lowering medications now available'; thus concluded the ADA/EASD guideline on hyperglycaemic management in Type 2 Diabetes from 2018 ([Diabetes Care 2018 Dec; 41\(12\): 2669-2701](#)), and I'm sure this is a sentiment we all agree on! Given the complexities of diabetes management, **we now have a bespoke course on Diabetes in Primary Care** addressing this hugely important area for us, which can be accessed together with all our other course material through an NB Plus subscription ([click here](#)).

In this chapter, we summarise the latest updates and studies in the pharmacological management of type 2 diabetes, particularly around the CVD and renal benefits of GLP1 agonists and SGLT2 inhibitors, as well as the (complicated!) licensing and guideline recommendations of the different SGLT2i depending in eGFR thresholds. SGLT2 inhibitors and GLP1 agonists are now firmly established treatments and in the case of SGLT2i are now recommended alongside metformin as a dual therapy for most people with T2DM in the recently updated [NICE guideline NG28](#) (see guidelines chapter for more detail). Given that SGLT2i are now also used in some patients without diabetes, we have a separate section on their rare but important side effects (including the risk of DKA, and how we should mitigate this risk) in the prescribing chapter. But first, we discuss some data flagging concerns about incretin-based drugs and biliary disease:

► Association of GLP1a and DPP4i with biliary disease

We have had two recent systematic reviews and meta-analyses investigating uncertainty regarding the use of GLP1 agonists and DPP4 inhibitors and biliary disease. The first [JAMA Intern Med. 2022;182\(5\):513-519](#) looked at GLP1a alone:

- 76 RCTs comparing use of GLP1a drugs with placebo or other non-GLP1a drugs, covering >100,000 people, mean age ~60 years.
- **GLP1a were associated with significant increases in gallbladder or biliary disease** (RR 1.37, 95% CI = 1.23-1.52), **cholecystitis** (RR 1.36, 95% CI = 1.14-1.62) and **cholecystectomy** (RR 1.70, 95% CI = 1.25-2.32) but not biliary cancers.
- The risks of gallbladder or biliary disease were **higher in trials for weight loss** (RR 2.29, 95% CI = 1.64-3.18) but were **still present in trials for T2DM** (RR 1.27, 95% CI = 1.14-1.43).
- Risks were dose and exposure dependent.

This study suggests there are increases in clinically meaningful biliary disease outcomes associated with the use of GLP1a drugs. But as always there are some caveats; first, weight loss in people with obesity has been associated with the development of gallbladder disease so it is possible that some of the effect seen is due to the weight loss rather than the drug, and second although the relative risks are significant the absolute increased risks are relatively small - just 27 extra cases per 10,000 persons treated years. However, there is a biologically plausible mechanism for this effect so it may need to be borne in mind as a discussion point when considering GLP1a, especially in people with obesity and if higher doses are being considered.

The second [BMJ 2022;377:e068882](#) was a systematic review and network meta-analysis which focussed more on DPP4i:

- The initial meta-analysis reviewed 82 RCTs with just over 100,000 people and compared the risks of biliary disease of DPP4i with placebo or non-incretin drugs.
- There was a **significant increased risk of cholecystitis with DPP4i compared to placebo/non-incretin drugs** (OR 1.43, 95% CI = 1.14-1.79), which equates to ~15 more events per 10,000 person-years, with the **associations stronger for a longer duration of use**.
- There was also an increased risk of a composite of gallbladder and biliary disease.
- A further network meta-analysis comparing DPP4i, GLP1a, SGLT2i and control drugs supported the findings of the study above with **significant increased risks of cholecystitis (and composite gallbladder/biliary disease) for GLP1a vs both SGLT2i and control drugs, with similar findings for the DPP4i vs both SGLT2i and control drugs**.

In summary, these studies suggest a **potential increased risk of biliary disease and cholecystitis for both GLP1a and DPP4i drugs**, although further studies will be needed to confirm/refute this. If true there are implications for both class of drug - GLP1a use is on the rise significantly for both diabetes and weight loss, and whereas DPP4i are used less these days, they are still often used in frailer elderly people with T2DM due to the low risk of hypoglycaemia and other side effects - in this group biliary disease/cholecystitis may be more difficult to pick up. However, the **balance of benefits/risks is still favourable for the vast majority of people** - the overwhelming data showing improved CVD outcomes for GLP1a strongly supports their use for people with or at risk of CVD, and the still generally good side effect profile for DPP4i makes them a good option for older/frailer people.

► **Meta-analysis of CVD and renal outcomes with SGLT2i and GLP1a** [BMJ 2021;372:m4573](#)

The relative merits of both SGLT2i and GLP1a in people with type 2 diabetes were further cemented in this very recent systematic review and meta-analysis in the BMJ. Importantly they reviewed patients across different CVD & renal risk groups and then gave estimates of the absolute benefits, and risks, of these two groups, the results of which can be seen through an interactive decision support tool ([click here](#)). Summary:

- 764 trials including 421,346 patients with type 2 diabetes, comparing SGLT2i and GLP1a treatment with each other, other diabetes treatment, or placebo.
- **Both SGLT2i and GLP1a, when added to current diabetes treatment, lowered all-cause mortality, cardiovascular mortality, non-fatal MI and kidney failure (high certainty).**
- SGLT2i decreased death from any cause and hospitalisation for heart failure compared to GLP1a.
- GLP1a decreased non-fatal stroke risk compared to SGLT2i.
- The study also showed that the rate of genital infections was increased with SGLT2i when compared to GLP1a, but that severe GI events may be increased in those taking GLP1a compared to SGLT2i.
- The data showed that the **absolute benefits of both drugs varied substantially depending on baseline CV or renal risks** (e.g. SGLT2i resulted in 5 fewer deaths in 1000 patients over 5 years in those with no CVD risk factors, but 48 fewer deaths in 1000 patients over 5 years in those with known CVD and CKD).
- Although there was considerable heterogeneity in the trials the **consistency of results was notable**.
- **Conclusion:** This study further confirms that both SGLT2i and GLP1a reduce both CV and renal adverse outcomes, but absolute benefits vary considerably with **substantially more benefit seen in those with higher baseline CVD or renal risk**.

► **Observational data supporting SGLT2i in reducing MACE in the real world** [BMJ 2020;370:m3342](#)

We have had numerous RCTs showing the SGLT2 inhibitors (SGLT2i) reduce major adverse cardiovascular events (MACE), however as we all know these trials often have quite strict inclusion criteria, and may not always be representative of our wider primary care population. So this observational study is important as it probably better reflects our usual populations:

- Retrospective cohort study comparing MACE (composite of MI, ischaemic stroke, CV death) in new users of SGLT2i and matched users of DPP-4i across Canada and the UK.
- 209,867 patients in each group were followed up for a mean of 0.9 years.
- The population reflected a wide variety of people in keeping with usual primary care populations - almost 60% had had diabetes for >10 years, ~43% were > age 65 (with 11% being > age 75), and most (>85%) were already on metformin.
- There was a **significant decrease in MACE in the SGLT2i group** (HR 0.76, CI = 0.69-0.84); secondary outcome data also revealed significant decreases in some of the individual components of MACE - MI, cardiovascular death, as well as significant decreases in heart failure and all-cause mortality.
- **Similar benefits in MACE were seen in all SGLT2i** (canagliflozin, dapagliflozin, empagliflozin).
- **Conclusion:** Although there is always the risk of confounding in such studies, this study provides real-world data in more usual primary care populations supporting the CVD benefits of SGLT2i vs DPP-4 inhibitors, in keeping with the results from the more selective RCTs.



► **Diabetes, renal disease and the SGLT2 inhibitors - summary of key evidence:**

- **The CREDENCE trial** [NEJM June 2019](#) (1b)
 - Double-blind RCT evaluating patients with T2DM and albuminuric CKD (eGFR 30-90 with raised ACR), also on renin-angiotensin drugs, randomised to canagliflozin 100mg daily or placebo.
 - The primary outcome was a composite of end-stage kidney disease (dialysis, transplant, sustained eGFR <15), doubling of serum creatinine or death from renal or CVD causes.
 - The **relative risk was 30% lower in the canagliflozin group** (HR = 0.70; 95% CI, 0.59-0.82).
- **Scandinavian cohort study** [BMJ 2020;369:m1186](#)
 - Cohort study of almost 30,000 new users of SGLT2i matched to a similar number using active comparator (DPP4i). 19% had CVD but only 3% had CKD.
 - The primary outcome measure was a composite of renal replacement therapy, death from renal cause or hospital admission for renal events.
 - Use of **SGLT2i vs DPP4i was associated with a 58% reduced risk of the primary outcome** (2.6 vs. 6.2 events/1000 person-years) giving a hazard ratio of 0.42 (95% CI = 0.34-0.53) and an **absolute risk reduction of 3.6 events/1000 person-years**.
- **Meta-analysis of CV and renal outcomes with SGLT2 inhibitors** [Lancet Nov 2018](#) (1a)
 - This systematic review and meta-analysis analysed progression of renal disease, as well as CVD outcomes, in people with type 2 diabetes taking SGLT2i.
 - From a renal perspective, **SGLT2i reduced the risk of progression of renal disease by 45%** (HR = 0.55, CI 0.48-0.64), with a **similar benefit in those with and without atherosclerotic CVD**
- **DAPA-CKD** [N Engl J Med 2020; 383:1436-1446](#) (see CKD section for more detail)
 - This RCT has gone one stage further by showing that **SGLT2i (dapagliflozin in this case) may improve renal outcomes in albuminuric CKD in people with and without type 2 diabetes**.
 - We discuss this trial in our CKD section but it further adds weight to the renal benefits of SGLT2i, possibly even in people without established type 2 diabetes.
- **EMPA-KIDNEY** [N Engl J Med 2023; 388:117-127](#) (see CKD section for more detail)
 - This subsequent RCT supports the DAPA-CKD study by **confirming a class effect for renal benefits for the SGLT2i for those with and without diabetes**.
 - This study looked at people with CKD (either eGFR 20-45 or eGFR >45 with raised ACR).
 - There was a reduction in the composite outcome (progression of renal disease and death from renal or CVD cause) in the empagliflozin group which was similar for those with or without diabetes.

► **Licensing and guidelines for SGLT2i in renal disease and eGFR thresholds**

The evidence for the use of SGLT2i in people with type 2 diabetes to improve renal outcomes is now strong and led to the ADA/EASD (US/European Associations) [updating their recommendations in 2019](#) that **SGLT2i are recommended to prevent the progression of CKD in people with type 2 diabetes and CKD, and in 2022 NICE (NG28) have followed suit**, although the thresholds between the guidelines do differ (See the guidelines chapter for more detail on the ADA/EASD and NICE updated guidance).

One of the issues we have had in the UK, until recently, has been the guidance that SGLT2i should not be initiated if eGFR was < 60ml/min (which comprised the majority of people with diabetic CKD). It's important to point out this licensing restriction was not due to any safety concerns of SGLT2i at lower eGFR, but more that initial data suggested these drugs were less effective at lower eGFRs for glycemic control. However, **we now have strong evidence this class of drug offers both renal and CVD benefits at much lower eGFRs**, which has led to changes in licensing and guidance although the two do not always tally up! It is important to remember that **once eGFR falls below ~45 the glycaemic impact of SGLT2i wanes** so we may need to consider additional glycemic lowering drugs if the eGFR falls, even if they remain on the drugs for cardio-renal benefits.

NICE have recently published their final CKD guidance ([NG203, Aug 2021](#)) which was then updated again in November 2021 to tally with the updated type 2 diabetes guidance ([NG28 Nov 2021](#)) on CKD (see following chapter on diabetes - guidelines). For people with **diabetes and CKD they recommend (once ACEI/ARB has been titrated to highest tolerated dose)**:

- We *OFFER* SGLT2i for T2DM with CKD *and* ACR ≥ 30 mg/mmol.
- We *CONSIDER* SGLT2i for T2DM with CKD *and* ACR 3-30 mg/mol.
- As long as they meet the criteria in the marketing authorisation (including relevant eGFR thresholds). In [March 2022 NICE TA775](#) was published recommending eGFR should be ≥ 25 for use of dapagliflozin in the context of CKD.

Due to the rapidly changing evidence base **licensing for the SGLT2i** has changed a lot over the past couple of years. The EMC licensing recommendations can be [found here](#), but as of June 2022:

SGLT2i	Indications	Licensing eGFR thresholds
Dapagliflozin	T2DM, CKD, HFrEF	>15 (but note NICE TA775 above that for CKD NICE recommend eGFR >25)
Canagliflozin	T2DM, diabetic kidney disease	>30 (can continue down to ESDR if ACR >30) 300mg dose only if eGFR >60
Empagliflozin	T2DM, HFrEF	>60 for T2DM (but can continue to eGFR 45 if eGFR then falls) >30 if T2DM and established CVD >20 for HFrEF 25mg dose only if eGFR >60
Ertugliflozin	T2DM	>60 (can continue down to 45 if eGFR then falls)

► What about metformin and the sulphonylureas?

The pharmacological management of type 2 diabetes used to be straightforward! Metformin, +/- sulphonylurea (SU), consider insulin as a 3rd line. Now the range of drugs can be somewhat overwhelming, and amongst all that, it can be easy to forget about our old friends metformin and the SUs.

Sulphonylureas:

- The debate on the CVD safety of SUs has been going on for years but in 2019 the results of the only cardiovascular outcome trial to look at an SU was published. [CAROLINA](#) compared glimepiride to linagliptin in a population of 6000 patients with type 2 diabetes of whom 42% had pre-existing CV disease and they were followed for over 6 years. The CV safety of linagliptin had already been demonstrated in its own CV outcome trial in 2018, [CARMELINA](#). CAROLINA showed no significant difference in the primary endpoint of major adverse cardiovascular events between the DPP4 inhibitor and the sulphonylurea. Although, as you would expect, considerably more hypoglycaemia in the SU group, with a 73% relative risk increase of hypoglycaemia at all doses.
- The study certainly supports CV safety for SUs (or at least for glimepiride) although the debate is likely to roll on and the argument that there are better options for people at high CV risk still remains.
- SUs remain an important (and often forgotten drug) due to their rapid glucose-lowering effect, and are still **recommended as a key 'rescue' drug by NICE for symptomatic hyperglycaemia**; they also have a **crucial role in managing steroid-induced diabetes** (see separate chapter).

Metformin:

- But what about metformin? For many years it has been the bedrock and first-line treatment for type 2 diabetes, but this is starting to be questioned given the increasingly positive results of the SGLT2i and GLP1a drugs. The debate was further ignited by the [ESC recommendation in 2019](#) that SGLT2i or GLP1a be considered in people at high CVD risk whether or not they are on metformin.

- This was reviewed in a recent [Lancet editorial Nov 2020](#) questioning whether it is too early to abandon metformin as a first-line treatment. The editorial points out that although the CVD benefits of SGLT2i and GLP1a are now without question, in these CVD outcome trials most participants were on at least one other glucose-lowering drug at baseline. Thus we really do not have enough evidence to know if the CVD benefits of SGLT2i and GLP1a are similar in treatment-naive patients.
- The editorial concludes that the current evidence contributes to knowledge on the optimal first-line drug in a limited way only and that further RCTs comparing metformin with an SGLT2i or GLP1a as first-line agents are required.
- The [updated NICE guideline](#) still recommend metformin as the first-line treatment unless contraindicated, even in those with established, or at high risk of, CVD or CKD, however the updated version of the [ADA/EASD guideline 2022](#) has changed its stance on metformin with an increasing focus on the CVD benefits of GLP1 agonists and SGLT2 inhibitors, **with or without metformin** and conclude that that *'these agents should be considered in people with established or high risk of CVD, HF, or CKD, independent of metformin use'*.
- **Metformin and B12 deficiency:** [MHRA drug safety update June 2022](#)
 - Whilst on the subject of metformin....the concern over the association between metformin and B12 deficiency has reared its ugly head again in the form of a recent MHRA drug Safety Update.
 - The actual prevalence of this issue, and indeed the clinical impact/relevance is still debated.
 - The MHRA reviewed the evidence including a secondary analysis of a US trial which suggested that the prevalence of B12 deficiency was about double those in the metformin group vs placebo group; however, at 5 years the absolute rates of B12 deficiency was only 4.3% in the metformin group vs 2.3% in the placebo group (and the definition of B12 deficiency used <203 pg/ml in a screened population could be questioned). The evidence also suggests the risk of B12 deficiency increases with both higher doses, longer duration of treatment and in patients with risk factors.
 - But the **MHRA now recommend the following:**
 - **Test B12 levels if deficiency is suspected** e.g. new onset neuropathy, megaloblastic anaemia, glossitis, recurrent mouth ulcers etc (no controversy there...)
 - **Consider periodic monitoring for people with risk factors for B12 deficiency** e.g. elderly people, partial gastrectomy, IBD, vegan and some vegetarian diets, PPI or colchicine use. (This recommendation is much more difficult to reconcile and comes with a large risk of over-testing, incidental low/normal results, cascade testing and significant work-load implications. Careful clinical judgement will be needed here to balance the known risks of over-testing and over-treatment.)

NB Learning Points & Quality Improvement Ideas: Pharmacological Management of T2DM

- We now have strong evidence that the SGLT2 inhibitors improve renal outcomes in people with T2DM and CKD, and their use is now reflected in all international guidelines.
- The **updated NICE T2DM/CKD guideline** now states we can *offer* SGLT2i for T2DM with CKD and ACR ≥ 30 mg/mmol and *consider* SGLT2i for T2DM with CKD and ACR 3-30 mg/mmol.
- We have recent meta-analysis data which confirms the CVD and renal benefits of both the GLP1a and SGLT2i, and gives useful absolute relative benefits; this data shows that the absolute benefits vary substantially depending on baseline CVD or renal risk.
- Recent meta-analyses suggest there may be an increase in significant biliary disease/complications associated with both GLP1 agonists and the DPP4 inhibitors, although the benefits/risks of both these drugs will be favourable for most people.
- **Quality Improvement Ideas:**
 - It can be complicated to remember all the different renal thresholds for SGLT2i use! Use the table above which summarises **when to use which SGLT2i and at what eGFR threshold**.
 - **Audit patients with T2DM and CKD with ACR ≥ 30 mg/mmol** - are they on a SGLT2i and if not consider starting one.

DIABETES - GUIDELINES

There has been an absolute explosion in the evidence base for the pharmacological management of type 2 diabetes over the past few years leading to substantial changes to international guidelines. Somewhat late to the party (some would argue) **NICE in March 2022 updated their guideline and made substantial changes (NICE NG28)**. One of the main themes in the guideline is that the newer medications available (particularly the SGLT2 inhibitors) should be seen as **treatments that are for cardio-renal protection, not just glycemic control**. This requires a change of mindset for both us and our patients that we may add a SGLT2i even when their glycemic control is good, to protect their kidneys and heart.

Later in 2022 the [ADA/EASD](#) made further updates to its guideline, including the importance of **24-hour behavioural change involving the '5 Ss'** which we discuss below. Finally, hot off the press in January 2023 NICE have announced [draft guidance](#) recommending use of **hybrid closed-loop systems for some people with type 1 diabetes** who aren't managing to achieve adequate HbA1c targets.

1 NICE Guideline on Type 2 Diabetes [NG28 March 2022](#)

The long-awaited NICE guideline update happened in early 2022. The update focused almost entirely on prescribing in type 2 diabetes, although the guideline includes all aspects of care. There is a **useful 5-page visual summary** ([click here](#)) which includes 2 prescribing algorithms, one for choosing first-line medications ([click here](#)) and another for further medications ([click here](#)). See KISS below for a summary.

► Summary of the main changes in the updated guidelines:

- **Further promotion of individualised care.** They have produced a useful **patient decision aid** ([click here](#)) to help to jointly set a target HbA1c. They ask us to take into account personal preferences, co-morbidities, polypharmacy, likelihood of benefiting from a long-term intervention and remind us that it is especially **important to consider relaxing HbA1c targets in the presence of multimorbidity** (See also our chapter on diabetes in the older person which is relevant to this).
- **General principles of HbA1c targets** remain the same:
 - Aim < 48 mmol/mol (6.5%) on monotherapy or < 53 mmol/mol (7.0%) if on SU.
 - Add therapy once >58 mmol/mol (7.5%).
 - Aim < 53 mmol/mol (7.0%) on multiple therapies.
- **Use of continuous glucose monitoring (CGM):**
 - In the updated guidance NICE have recommended use of CGM, in particular intermittently scanned CGM (isCGM, AKA 'Flash monitoring' e.g. Freestyle Libre®) for people with **T2DM who are on insulin in certain circumstances** (e.g. recurrent hypos, impaired hypo awareness, disabilities, multiple testing required ≥8x/day) - [see here for details](#) 1.6.17-1.6.26
 - It's important to note these people are likely to be under specialist-led care and NICE recommend *'CGM should be provided by a team with expertise in its use'*.
- **Choosing first-line medications:**
 - In patients with type 2 diabetes and **heart failure or established atherosclerotic cardiovascular disease (ASCVD), start metformin and offer an SGLT2i from the outset**. NICE advises starting these sequentially but adding the SGLT2i as soon as metformin tolerability is confirmed.
 - In patients with type 2 diabetes and **high CV risk** (Qrisk2 > 10% if aged over 40, or a high lifetime risk (1 or more CV risk factor in a person under 40 years), **start metformin and consider adding an SGLT2i from the outset**, again once metformin tolerability confirmed.
 - In patients not at high CV risk, metformin monotherapy.
 - If metformin not tolerated and est. ASCVD – use SGLT2i alone.
 - If metformin not tolerated and high CV risk – consider SGLT2i alone.
 - If metformin not tolerated and not high CV risk, consider pioglitazone/SU first line, or an SGLT2i first line (rather than DPP4i).
 - If symptomatic hyperglycaemia, consider SU/insulin initially as rescue therapy.

■ Choosing further medications:

- If **CV risk or CV status changes** ie the person develops CVD, HF or their Qrisk increases to > 10% and they are not already on an SGLT2i, start or consider one (either switching or adding, depending on individual circumstances).
- If **HbA1c remains above individual target**, switch or add treatments using dual/triple therapy. Consider:
 - DPP4i
 - Pioglitazone
 - SU
 - SGLT2i
 - Insulin (if dual oral therapy not effective)
- **Consider switching one drug to a GLP1 if triple oral therapy is not effective or tolerated when BMI > 35 (or adjusted acc. to Ethnic group)**, or at lower BMI where wt loss will be helpful / insulin problematic).
 - Only continue the GLP1 if there is a reduction in HbA1c of 11mmol/mol and weight loss of 3% in 6 months.
- NICE have relaxed the combination of therapies we can use together, where previously it needed to be metformin/sulphonylurea/GLP1 we can now use **different combinations, including GLP1 alongside SGLT2i's**.

► What does this mean?

Essentially NICE are advising dual therapy with metformin and an SGLT2i from the outset for the majority of our patients with type 2 diabetes. This is based on the clinical and cost-effectiveness analyses of this combination, and here we are **adding the SGLT2i primarily for organ protection (cardiorenal)** rather than glycaemic control - so we add the SGLT2i even if the HbA1c is to target with metformin - a very different approach from we are used to. It takes some time to get our head around using medication in type 2 diabetes in this way and we will have to have different conversations with our patients, talking to them about the **medication 'protecting your heart and kidneys'** - a bit like we do when adding a statin.

► How soon do we start the SGLT2i?

NICE do not specify, but state we should add *"without delay to ensure people do not remain on metformin monotherapy for prolonged periods"*. Our NB Diabetes Lead Dr Sarah Davies advises *'My approach is that in the highest risk groups with ASCVD or HF, I am advising they start it after 4 weeks, if they are ok with the metformin. In the high CV risk group, I am adding it at around 12 weeks, when we are often reviewing anyway.'*

► Which SGLT2i?

NICE state we should use an **SGLT2i with proven CV benefits**. Essentially in their analysis, all had some benefits but it was most certain for **dapagliflozin, empagliflozin and canagliflozin** (not ertugliflozin). Dapagliflozin had the most consistent evidence for cost-effectiveness in NICE analysis for outset combination therapy.

► Why are GLP1s still so far down the treatment options and so restricted when other new guidelines have them near the top?

This is the most controversial part of the new NICE guideline. As we know there has been evidence of CV benefits for this class also. However, NICE grouped them all together as one class (the class is much more heterogeneous than the SGLT2i's) and found the cost-effectiveness benefits in terms of CV disease not to be clear enough to recommend them. They did not consider the additional effectiveness of glucose and weight lowering and this has been an area for much discussion, as have the continued use of BMI cut-offs and stop criteria.

► Do I Switch or add medications?

NICE give us a useful reminder about whether to switch or add, which as always needs to be an individualised decision. They remind us that we should **consider stopping medications that have had no impact on glycaemic control and weight, *unless there is an additional clinical benefit such as CV or renal protection***, from continued treatment. This is a useful reminder to **continue an SGLT2i for cardiorenal benefit, even if there is little impact on HBA1c**. For this reason, it can be useful to record the indication for starting the treatment, to make sure that it is not inappropriately stopped by a well-meaning person later on.

► What else is new in the guideline - SGLT2i and CKD

See chapters on CKD for more detail, as well as pharmacological management of T2DM which includes changes to licensing in renal disease for SGLT2i. The advice on SGLT2i use in diabetes and kidney disease was updated in the new [NICE CKD guideline](#) which was published in Nov 2021. This is from strong evidence of reducing progression of renal disease and CV events. It is included in the diabetes guideline update also, although in the small print rather than the prescribing algorithms. Remember, at lower eGFRs (< 45), there will be little impact of HBA1c, another reason to record the initial indication for starting the medication.

- For people with T2DM and CKD (once ACEI/ARB has been titrated to the highest tolerated dose):
 - **OFFER** SGLT2i for T2DM with CKD and ACR ≥30 mg/mmol.
 - **CONSIDER** SGLT2i for T2DM with CKD and ACR 3-30 mg/mol.
 - And they meet the criteria in the marketing authorisation (including relevant eGFR thresholds) - In [March 2022 NICE TA775](#) was published recommending eGFR should be ≥25 for use of dapagliflozin in the context of CKD

NB it's worth noting the **license for dapagliflozin in CKD is down to a lower eGFR of 15** ([click here](#) for EMC licensing website), but NICE went for a eGFR threshold of ≥25 based on the [DAPA-CKD study](#) - realistically most patients in this group are likely to be under renal care but it's worth noting the discrepancy.

2 [ADA/EASD Consensus Statement 2022](#)

This colourful, very visual algorithm was updated in 2022 jointly between the European and American Diabetes Associations. The statement focuses not just on medication but also on a key decision cycle, emphasising the importance of assessing key patient characteristics, using a shared decision-making process, and avoidance of clinical inertia. There is a real emphasis on diet and lifestyle interventions, education and self-management as well as a focus on obesity, addressing the evidence for lifestyle, medication and surgical interventions to target weight loss. Although most of us are likely to be using the NICE guideline now it has been updated, there are some very useful themes and concepts in this guideline that are worth mentioning. One of the overriding themes is the further move away from a glucose or glycemic-centric approach to management, towards a focus on the wider determinants of health that are part of diabetes management. Some of the new/key themes includes:

- More **detailed guidance on lifestyle and healthy behaviours**, with a real positive focus on the benefits of 24-hour behaviour modifications based on **the five S's -Sitting/breaking up prolonged Sitting, Sweating, Stepping, Strengthening and Sleep**. [Click here for summary](#). There are some great messages here we can share with patients eg a 5-6 min brisk intensity walk per day equates to around 4 years greater life expectancy.
- An **increased focus on weight reduction** including lifestyle, drugs and surgery options - maintaining weight management goals is now given the same status as glucose management in the algorithms.
- The **medication algorithm** ([click here](#)) allows individualised choice according to a number of patient characteristics but firmly focuses on the evidence from cardiovascular outcome trials on the benefits seen in the newer agents, with a **strong focus on cardio-renal benefits rather than just glycemic control**. This inevitably means management is increasingly focussed around the GLP1 agonists and SGLT2 inhibitors, **with or without metformin** and that *'these agents should be considered in people with established or high risk of CVD, HF, or CKD, independent of metformin use'*.

3 Glycemic and BP targets [NICE Hypertension guideline NG136 August 2019](#) & [Type 1 diabetes NG17](#)

► Hba1c targets

Whilst all the key international guidelines now stress the importance of individualised Hba1c targets, for those using QoF it's worth a quick reminder on the contractual targets, with the caveat that if the QoF target is inappropriate for a given individual we can use an appropriate exception code to reflect their individualised target. The change to take into account frailty that was initially made back in 2018/19 remains current in the [2022/23 contract](#):

- Hba1c target = 58 mmol/mol (for those WITHOUT moderate/severe frailty).
- Hba1c target = 75 mol/mol (for those WITH moderate/severe frailty).

► BP targets

One of the surprises of the new NICE Hypertension guideline was a **relaxing of the previous BP targets for people with Type 2 diabetes**. Previously targets for people with type 2 diabetes were 140/80 or a lower target of 130/80 in the presence of end-organ damage (renal, eye or CVD). In the updated NICE guideline the committee reviewed all the evidence and concluded there was *'insufficient evidence to recommend a different blood pressure target for this subgroup'* (those with type 2 diabetes); they, therefore, recommend the **BP targets for those with type 2 diabetes are the same as those without type 2 diabetes**:

- < 80 years old target <140/90 and ensure that it is maintained below that level.
- ≥ 80 years old target <150/90 and ensure that it is maintained below that level.

This is very much 'against the flow of traffic' with most other International guidelines taking a different view of the evidence and recommending tighter BP control. The European Guidelines recommend (albeit not differentiating between type 1 or type 2 diabetes) targets of <130 /80 for those <65 years old and <140/80 for those ≥65 years old. Navigating these variations in guidance is tricky, but we should continue to set individualised targets based on co-morbidities and preferences. Younger patients with type 2 diabetes or those at greater CVD risk with reasonable life expectancy may well still benefit from lower targets.

However, it is worth noting that in **March 2022 NICE have slightly altered (and strengthened) their wording** in the [Hypertension guideline NG136](#) on targets to include the line that we should reduce BP below the target level and *'ensure that it is maintained below that level'* i.e. **aim to get below the BP target and not tolerate BP levels at or slightly above the target level** - their re-wording is aimed to ensure *'neither patients nor practitioners will be accepting of clinic blood pressure values' above target levels*.

For those with **T2 diabetes and CKD** we go by the CKD guideline from 2021 [NG203](#)

- Target <140/90
- Unless ACR ≥70 mg/mmol in which case target <130/80

Finally, in August 2022 the [Type 1 diabetes guideline NG17](#) was updated to align with BP targets across other guidelines and co-morbidities. For **people with type 1 diabetes**:

- BP Target <140/90
- Unless ACR ≥70 mg/mmol, in which case target <130/80 (i.e. same as CKD guideline)
- If aged ≥80 BP target <150/90 irrespective of ACR level

That leaves the only outstanding discrepancy with BP targets in QoF which as of the [2022/23 contract](#) remains with the BP target for diabetes of <140/80 (unless moderate/severe frailty).

4 NICE Type 1 diabetes guidelines [NG17 updated March 2022 DRAFT closed loop systems for T1DM](#)

Although generally the remit of specialist-led clinics there were some important updates to the NICE Type 1 diabetes guideline in March 2022:

- **Diagnosis** - whilst the diagnosis of T1DM is generally thought of as straightforward, it has become clear over the years that this is not always the case. **Overlap or confusion with other types of diabetes** including type 2 diabetes, other autoimmune diabetes (e.g. LADA) and genetic-driven diabetes (e.g. MODY) is much more recognised. Making the correct diagnosis is crucial as treatments (and risks) can be substantially different. In the updated guidance NICE highlight these issues and make **recommendations on when further testing** with diabetes-specific autoantibodies (e.g. GAD antibodies) or C-Peptide levels should (and should not) be used.
- **CGM - continuous glucose monitoring is now recommended for all people with T1DM**, either as real-time CGM (rtCGM) or intermittently scanned CGM (isCGM AKA 'flash' monitoring e.g. Freestyle Libre®).

Then hot off the press, in January 2023 in a [blaze of publicity](#), [NICE announced](#) new [DRAFT guidance](#) recommending **closed loop systems as an option for some people with type 1 diabetes**. These systems, often thought of as the closest (current) technology to an 'artificial pancreas', comprise of a continuous glucose monitor sensor attached to the body which transmits data to a body-worn insulin pump. This uses the data to run a mathematical calculation to work out how much insulin needs to be delivered via the pump. Clinical trial and real-world evidence suggests these systems are superior to usual care, especially for people running higher HbA1c levels >64 mmol/mol (8.0%). The **draft recommendations are that:**

- Hybrid closed loop systems are an option for managing blood glucose levels in type 1 diabetes for people who are having difficulty managing their condition and have an average HbA1c of around 64 mmol/mol (8.0%) or more, despite optimal management with at least 1 of the following:
 - continuous subcutaneous insulin infusion
 - real-time continuous glucose monitoring
 - intermittently scanned continuous glucose monitoring.
- They are also recommended for people who are pregnant or planning pregnancy.
- One crucial caveat to all this is that delivery of these systems will **only be recommended if the companies and NHS England agree a cost-effective price for the systems**.

NB Learning Points & Quality Improvement Ideas: Type 2 diabetes guidelines

- Current guidelines strongly encourage an individualised approach to diabetes care with shared decision-making; when discussing treatment goals take into account the complexity of regimens and make sure goals are achievable and realistic and choose regimens that optimise adherence.
- NICE have made substantial changes in it's updated guideline **recommending SGLT2i for most people alongside metformin** once metformin tolerability is confirmed:
 - If **heart failure or established CVD**, **start metformin and offer an SGLT2i from the outset**.
 - If **high CV risk** (Qrisk2 > 10%) **start metformin and consider adding SGLT2i from the outset**.
- NICE have also updated the CKD recommendations - they recommend we **OFFER** an SGLT2i (if CKD with ACR ≥30 mg/mmol) and **CONSIDER** an SGLT2i (if CKD with ACR 3-30 mg/mol) once ACEI/ARB have been titrated to max tolerated dose.
- The ADA/EASD guideline was also updated in 2022 with some interesting new additions including the benefits of **24 hour behaviour modifications based on the five S's** (Sitting/breaking up prolonged Sitting, Sweating, Stepping, Strengthening and Sleep) and this update also now promotes the use of GLP1 agonists and SGLT2 inhibitors for cardio-renal benefits, **independent of metformin**
- **Quality Improvement Ideas:**
 - Audit all patients with T2DM and either heart failure or CVD - if not on SGLT2i consider starting one.
 - Audit all patients with T2DM and CKD with ACR ≥30 - check they are titrated to maximum ACEI/ARB dose and if not on a SGLT2i consider starting one; dapagliflozin is now recommended in [TA775](#) down to an EGRF of 25.

KISS: Type 2 Diabetes NICE guidance - General Care, Targets and Glycemic ControlNICE NG28 2022Individualised care, Hba1c targets and non-pharmacological management:

- *'Adopt an individualised approach to diabetes care that is tailored to the needs and circumstances of adults with type 2 diabetes, taking into account their personal preferences, comorbidities and risks from polypharmacy, and their likelihood of benefiting from long-term interventions. Such an approach is especially important in the context of multimorbidity.'*
- **Hba1c targets should be individualised depending on the person's individual circumstances** - [click here for good PDA](#) - but NICE make the following guideline recommendations:
 - If managed by diet/lifestyle +/- a single drug not associated with hypoglycaemia (e.g. metformin) aim for 48 mmol/mol (6.5%) or if hypoglycaemic drug (e.g. SU) aim for 53 mmol/mol (7.0%).
 - If further drug treatment is required aim for target 53-58 mmol/mol (7-7.5%).
 - **Consider relaxing targets:**
 - Especially if frail/limited life expectancy.
 - If tight blood glucose control will increase risks (e.g. falls, impaired hypo awareness, drive/operating machinery as part of their job).
 - If intensive management is not appropriate e.g. significant comorbidities.
- All people should be offered structured education.
- **Dietary and physical activity advice should be tailored to the individual's needs.**
- **Do not forget to consider referral to Tier 3 weight loss services for those with obesity** - bariatric surgery (via Tier 3 services) should be considered and offered if BMI ≥ 35 .

Drug treatment for glycemic control ([click here](#) for a visual summary):

- **1st line:** standard release metformin gradually increased over several weeks - switch to modified-release if GI side effects not tolerated.
- If **established CVD or heart failure** *offer* dual therapy with SGLT2i in addition to metformin - introduce SGLT2i as soon as metformin tolerability is confirmed.
- If **QRISK2 $\geq 10\%$** *consider* dual therapy with SGLT2i plus metformin - add sequentially as above.
- If metformin is contraindicated or not tolerated:
 - If established CVD or heart failure offer SGLT2i, or if QRISK2 $\geq 10\%$ consider SGLT2i.
 - Otherwise, options include DPP-4 inhibitor, pioglitazone, sulphonylurea (SU), SGLT2i.
- If **further intensification is required:**
 - Consider adding one of (depending on which drugs already started) DPP-4 inhibitor, pioglitazone, SU or SGLT2i, or insulin.
- If on **triple oral therapy and still not adequate control** consider triple therapy by substituting one drug for GLP-1 agonist* if:
 - BMI ≥ 35 and specific psychological or other problems associated with obesity, or
 - BMI < 35 if black, Asian, other minority ethnic group, or if insulin would have occupational implications, or weight loss would benefit obesity-related comorbidities.
 - Continue GLP1 agonist if Hba1c drops by ≥ 11 mmol/mol (1.0%) and weight loss of $\geq 3\%$.
- **Before starting SGLT2i be aware risks of DKA and if present SGLT2i may not be appropriate:**
 - Previous episode DKA or intercurrent illness.
 - If on (or may start in the future) ketogenic or very low carb diet do not use SGLT2i.
 - [Click here](#) for excellent PIL from TREND and [here](#) for 'traffic light' guidance on SGLT2i prescribing.

*NB comment: This recommendation has proved controversial and is very much at odds with both the underlying evidence base and other international guidelines which recommend GLP1 agonists higher up the list, especially for those with established CVD or those with obesity; however NICE make recommendations not just on the evidence base, but on cost-effectiveness analysis and their assessment is that currently, the GLP1 agonists do not meet their cost-effectiveness thresholds for most people.

KISS: Type 2 Diabetes NICE Guidance - Cardio-renal protection [NICE NG28 2022](#)SGLT2i and CKD:

- If ACR ≥ 3 mg/mmol titrate ACEI to the highest tolerated dose.
- In addition to ACEI:
 - Offer SGLT2i if ACR ≥ 30 mg/mmol.
 - Consider SGLT2i if ACR 3-30 mg/mmol.
- Licensing and use of SGLT2i in CKD ([click here](#) for latest licensing recommendations):
 - The original higher eGFR thresholds were not set due to increased risks at lower eGFRs but due to the fact that they have less glycaemic lowering efficacy (minimal glycaemic effect at eGFR < 45).
 - However, the **cardiovascular and renal protection is still effective at lower eGFRs**. So, if we are adding an SGLT2i for CV/renal protection where the eGFR is low, we may need to consider an additional agent if further glycaemic control is needed.
 - **Dapagliflozin**: Licensed for use if eGFR ≥ 15 (consider lower 5mg dose in severe renal impairment), but [NICE TA775](#) recommends use only if eGFR ≥ 25 .
 - **Canagliflozin**: Do not initiate if eGFR < 30 (use lower dose 100mg OD if eGFR < 60); can be used at lower eGFRs if initially eGFR ≥ 30 then drops below this threshold.
 - **Empagliflozin** (use lower 10mg OD dose if eGFR < 60):
 - Do not initiate if eGFR < 60 , but can be used down to eGFR of ≥ 45 if initially ≥ 60 then drops.
 - Can be used down to eGFR ≥ 20 if heart failure with reduced ejection fraction.
 - Can be used down to eGFR ≥ 30 if established CVD based on latest licensing.

Cardiovascular risk reduction:

- The **greatest benefits on mortality in type 2 diabetes are seen in CVD risk reduction, not glycaemic control**, although the SGLT2i and GLP1 agonists can improve both HbA1c and CVD risk.
- **Antiplatelets** only for secondary prevention.
- **Blood pressure**:
 - Target $< 140/90$ if < 80 years old
 - Target $< 150/90$ if ≥ 80 years old
 - Target $< 130/80$ if ACR ≥ 70 mg/mmol (NICE updated CKD guidance [NG203 Nov 2021](#))
 - Note these targets are higher (at least in the context of younger fitter people without frailty) than in many other guidelines and low targets of 130-140/80 may be appropriate in younger fitter adults.
- **Statins/lipids** (see recent combined [NHSE guidance here](#)) and (NB non-HDL-C = total chol - HDL):
 - **Primary prevention if QRISK $\geq 10\%$** :
 - Offer atorvastatin 20mg OD.
 - Target reduction of non-HDL-C $> 40\%$ from baseline.
 - If not at target increase to max 80mg atorvastatin.
 - If max tolerated dose does not achieve target/intolerant consider switch to alternative statins.
 - If still not at target add ezetimibe 10mg OD.
 - If statin is contraindicated or intolerant consider ezetimibe 10mg OD and switch to ezetimibe 10mg/bempedoic acid 180mg combination if target not achieved.
 - **Secondary prevention**:
 - Atorvastatin 80mg for all (unless significant CKD, see below).
 - Target reduction of non-HDL-C $> 40\%$ from baseline - if this value not available target non-HDL-C < 2.5 mmol/L (equivalent to LDL-C < 1.8 mmol/L).
 - If not at target add ezetimibe 10mg OD.
 - If statin C/I or not tolerated use ezetimibe monotherapy or combo with bempedoic acid (as above).
 - If non-HDL-C remains > 2.5 mmol/L consider inclisiran or PCSK9i if LDL > 4.0 mmol/L or > 3.5 mmol/L if recurrent CVD events.
 - In the context of **CKD with eGFR < 60** use atorvastatin 20mg OD for primary and secondary prevention and consider higher doses if not at target and eGFR ≥ 30 ; seek renal advice if eGFR < 30 .

KISS: SIGN Pharmacological Management of Glycaemic Control in Type 2 Diabetes

[SIGN 154 Nov2017](#) and [Click here for summary algorithm](#)

- **Intensive lifestyle advice for all patients at all stages**
- **HbA1c targets for glycaemic control:**
 - 53 mmol/mol (7%) is reasonable to reduce the risk of micro & macrovascular disease
 - 48 mmol/mol (6.5%) may be appropriate at diagnosis
 - Individualise targets to balance benefits with harms, in particular, hypoglycaemia and weight gain
 - Continue medication at each stage if EITHER target achieved OR HbA1c falls more than 0.5% (5.5 mmol/mol) in 3–6 months. Discontinue if medication ineffective
- **Drug choice**
 - Make individualised choices based on the risk of weight gain and hypoglycaemia
 - Consider using SGLT-2 or liraglutide in patients with cardiovascular disease
- **Metformin:**
 - Metformin should remain the first-line oral treatment option for all patients
 - If intolerant of metformin, then sulfonylurea
- **Second line, in addition to lifestyle measures, ADD one of:**
 - **Sulfonylurea (SU)**
 - Potent; risk of hypo & weight gain; no CV benefit
 - **SGLT2 inhibitor,**
 - Moderate potency; low risk of hypo; causes weight loss; proven CV benefit; main side effect is genital mycosis or UTI; warn re DKA; do not use in CKD with eGFR<60 or peripheral arterial or diabetic foot disease
 - **DPP-4 inhibitor**
 - Moderate potency; low hypo risk; neutral effect on weight; no CV benefit; few side effects
 - **Pioglitazone**
 - Moderate potency; low hypo risk; weight gain; probable CV benefit; side effects of fluid retention, oedema and fractures
- **Third line, in addition to lifestyle measures**
 - Add either an additional oral agent from a different class OR
 - An injectable agent
 - if BMI >30 GLP 1 agonist (stop DPP-4, consider reducing su, can continue metformin, pioglitazone and/or SGLT-2)
 - if BMI <30 basal insulin (stop SU, continue other oral drugs)

STEROID-INDUCED DIABETES AND HYPERGLYCAEMIA

Steroid-induced diabetes and hyperglycaemia is a reasonably common phenomenon that we see either 'de novo' in primary care, or on discharge from hospital. The Joint British Diabetes Society (JBDS) updated its guidance in this area in May 2021, and although the focus of the guideline is very much in-patient care, there are a number of very useful points that may help guide management in Primary Care, and it highlights the importance of our old (and often forgotten) friend gliclazide...

KISS: Steroid Induced Diabetes and Hyperglycaemia

JBDS guideline 2021

Background:

- Although most steroid use is short-term, in >20% of cases it is for >6 months.
- The risk of hyperglycaemia is dependent on the dose, length of duration and potency of the steroid.
- **Steroid-induced diabetes** = A rise in glucose related to steroids in someone without known diabetes.
- **Steroid-induced hyperglycaemia** = rise in glucose in a person with known diabetes (almost inevitable).

Assessment of hyperglycaemia in people taking steroids (NB CBG = capillary blood glucose):

- **No previous diagnosis of diabetes:**
 - Assess if high risk e.g. obesity, FH diabetes, previous gestational diabetes, PCOS, ethnic minorities, known pre-diabetes, previous hyperglycaemia on steroids (or consider [risk calculator](#)).
 - If high risk or any diabetic symptoms check HbA1c and commence CBG once daily pre or post-lunch or evening meal.
 - If CBG <12 mmol/L consider them to be low risk and do CBG daily post-breakfast or post-lunch.
 - If CBG consistently <10 mmol/L consider stopping CBG.
 - If CBGs >12 mmol/L increase CBG to 4x daily.
 - If CBG >12 mmol/L consistently (e.g. ≥2 occasions in 24 hours) start treatment below.
- **Known diabetes:**
 - Set CBG target e.g. 6-10mmol/L, 6-12mmol/L acceptable but less tight control indicated for those at higher risks of hypos/complications from hypos e.g. frail, cognitive impairment, falls risk etc.
 - Check CBG 4x daily.
 - If Type 1 diabetes check for ketones daily if CBG >12 mmol/L.

Treatment:

- **No previous diabetes (i.e. steroid-induced diabetes):**
 - Titrate until maximum dose of gliclazide 240mg mane or glycemic targets reached. (The guideline says titration can occur daily but this is in the setting of hospital care so judgement will need to be taken on how quickly to titrate in primary care.)
 - Seek specialist help if no improvement in control despite 160mg gliclazide.
 - If steroids reduced/discontinued consider dose reductions but continue CBG monitoring if >12.
- **Known Type 2 diabetes:**
 - If **not on SU or insulin** and no hypo symptoms titrate until maximum dose of gliclazide 240mg mane or glycemic targets reached.
 - If **already on max mane SU** consider evening dose of SU (max total daily dose 320mg i.e. 240mg mane, 80mg evening) or add morning NPH insulin (e.g. insulatard).
 - If on **once-daily night insulin** switch to morning and titrate in increments of 10-20%; consider adding gliclazide up to max dose; if not at target consider BD dose or basal-bolus regime.
 - If multiple daily doses of insulin are needed **involve the community diabetes team**.
- **Type 1 diabetes** - titrate insulin doses in ≥2 unit increments every 24-48 hours but **diabetes specialist nurses or community diabetes teams should be involved**.
- Glucose lowering treatment doses will need to be reduced in tandem with reductions in steroid doses.

DIABETES IN THE OLDER PERSON

How should we be approaching the management of diabetes in the older person, and how do we go about that all-important aspect of de-prescribing where appropriate? Of course, older people vary widely in terms of functional level, number of comorbidities, polypharmacy etc. For the older person, diabetes care should be individualised taking into account functional status, co-morbidities, the presence of established CVD, risk of hypoglycaemia, and the presence of microvascular complications. The risk-benefit balance must be considered when setting HbA1c targets. One of the major problems we have is that studies on frail older adults are lacking, with this group commonly excluded from most diabetes trials.

However, we have recently had new primary evidence from Canada reviewing community-dwelling older patients with diabetes, which highlights the risks of overtreatment in this group, and helps inform the debate on treatment in the elderly ([Diabetologia 2021 May;64\(5\):1093-1102](#)). On the back of this, we are increasingly seeing elderly people being diagnosed with pre-diabetes. But is this really helpful in the elderly population? Will an early 'diagnosis' improve outcomes or is it yet another example of 'too much medicine' and us chasing numbers that will not improve outcomes or quality of life? Recent primary research ([JAMA Intern Med. 2021;181\(4\):511-519](#)) suggests the push for 'diagnosis' of pre-diabetes in the elderly is highly questionable.

► Background evidence on diabetes in the elderly:

- **Treatment burdens** are important to consider in all people with diabetes, especially for older people. A study ([JAMA Intern Med 2014;174\(8\):1227-1234](#)) investigated how treatment burden affects the benefits of intensive versus moderate glycaemic control in patients with T2DM. The main outcomes were quality-adjusted life-years (QALYs) and a reduction in the risk of microvascular and cardiovascular complications. This study was discussed in the BMJ ([BMJ 2014;349:g4335](#)). They concluded that the benefits of glycemic treatments declined with age, and by age 75 the harms of most treatments were likely to outweigh any potential benefits.
- Following on from this, a study ([JAMA Intern Med 2015;175\(3\):356-362](#)) looked at glycaemic control among older adults and the prevalence of **potential overtreatment of diabetes**. A third of included patients had significant impairments of their ADLs as well as at least 3 co-morbidities. Nearly two-thirds of those included had an HbA1c below 53mmol/mol. The authors concluded that the harms of intensive treatment likely exceeded the benefits in these more frail patients, and the majority were over-treated.
- It is now recognised that **frailty and muscle loss** (sarcopenia) are complications of diabetes itself and major risk factors for poor outcomes. [Diabetic Medicine 2018;35:838-845](#)
- **Hypoglycaemia** is a particular concern for older people with diabetes; it carries a worse prognosis and higher mortality risk with age. Symptoms of hypoglycaemia in older people tend to be non-specific - including fatigue, irritability, confusion, drowsiness, unsteadiness and light-headedness and therefore more likely to go unrecognised. Hypoglycaemia was associated with 40-70% increased risk of a fall-related fracture in people > 65 years. [Diabet ObesMetab 2012;14:634-43](#)

► New evidence on the risks of overtreatment in older adults [Diabetologia 2021 May;64\(5\):1093-1102](#)

- This was a retrospective cohort study of 108,620 community-dwelling adults in Canada aged ≥75 with diabetes on at least one anti-hyperglycaemic agent.
- They were categorised as having intensive (HbA1c < 53 mmol/mol, <7%) or conservative control (HbA1c 54-69 mmol/mol, 7.1-8.5%).
- They were also categorised as being on high risk (insulin and sulphonylureas) or low-risk (other) agents.
- The study analysed the **composite risk of emergency department visit, hospitalisation or death**.
- Overall 61% were treated to intensive targets and 21.6% to conservative targets.
- There was **almost a 50% increased risk of the composite outcome** in those with intensive control on high-risk agents, compared to those with conservative control on low-risk agents (RR 1.49, 95% CI = 1.08-2.05).
- The authors inevitably conclude that we need to *'re-consider the use of anti-hyperglycaemic medications that may lead to hypoglycaemia, especially in setting of intensive glycaemic control.'*

► Specific issues to consider for older people with diabetes

Hypoglycaemia

- Hypoglycaemia is a particular concern for older people with diabetes; it carries a worse prognosis and higher mortality risk with age.
- **Symptoms of hypoglycaemia may be non-specific**, older people lose the typical autonomic response to hypos such as palpitations and sweating.
- Symptoms may include fatigue, irritability, confusion, drowsiness, unsteadiness and light-headedness and therefore more likely to go unrecognised.
- **Recurrent hypoglycaemia is associated with significant morbidities** leading to physical and cognitive dysfunction and eventually frailty and disability.
- An individual at an advanced age (e.g., ≥75 years), identified as being moderately or severely frail, with an HbA1c value <53 mmol/mol (<7.0%) would suggest a high risk of occurrence of complications from hypoglycaemia.
- **In older people, a higher HbA1c is associated with a lower average blood glucose reading** than in younger people:
 - Older people have lower red cell turnover. Their cell membranes are more friable and prone to glycosylation. Therefore the same average glucose is associated with a higher HbA1C.
 - We need to take this into account when individualising HbA1c targets.

Loss of insulin resistance

- As frailty ensues, older adults lose adipose tissue, which, in turn, reduces underlying insulin resistance.
- As a result, much smaller doses of insulin are often sufficient to provide adequate and reproducible glycaemic control
- Often during reviews of frail older adults, individuals will be identified with a low HbA1c (<48 mmol/mol; <6.5%). It should be **considered that insulin resistance may have subsided sufficiently that no therapies are required**. In this scenario, it is reasonable to discontinue all anti-hyperglycaemic agents and review control after 3 months, against an individualised HbA1c goal.

Diabetes and cognitive impairment

- The relationship between diabetes and dementia is complex and multifactorial. However, people with diabetes appear to be at an increased risk of dementia, both Alzheimer's and Vascular dementia, estimated at an overall increased risk of 2-3 fold.
- There is also some evidence suggesting that people with diabetes have a faster rate of overall cognitive decline.
- Proposed mechanisms include microvascular disease, macrovascular disease, insulin resistance, inflammation, depression and recurrent hypoglycaemia.

Renal impairment increases hypoglycaemia risk and restricts treatment options

- Renal impairment is very common in older people with diabetes.
- As **eGFR falls, the risk of hypoglycaemia increases** as excretion on medication takes longer. The risks of using medications such as gliclazide in this group are much higher and insulin doses should be reviewed and monitored very carefully.

In people over the age of 80, low body weight is a negative prognostic indicator

- It is now recognised that frailty and muscle loss (sarcopenia) are complications of diabetes itself and major risk factors for poor outcomes. [Diabetic Medicine 2018;35:838-845](#)
- We need to therefore protect against this and take it into consideration when prescribing in diabetes.

► **Older adults with pre-diabetes - risk of progressing to diabetes** [JAMA Intern Med. 2021;181\(4\):511-519](#)

But what about older adults with pre-diabetes? Is this even a significant diagnostic category and what is their risk of progressing to diabetes? Given the increased number of older adults being flagged as 'pre-diabetic' this is an important question, and this research suggests we are generating a lot of work for minimal benefit:

- This prospective cohort study of 2497 older adults (mean age ~75 years old) was followed for 6.5 years and reviewed for progression to diabetes.
- 44% had pre-diabetes based on HbA1c levels (39-47 mmol/mol, 5.7-6.4%). NB important to note the lower HbA1c level used in this study is lower than the standard UK definition of 42 mmol/mol, 6% so the total numbers with pre-diabetes is higher than we would accept.
- 59% had pre-diabetes based on fasting plasma glucose/FPG (100-125 mg/dL, 5.6-6.9 mmol/L).
- This means a **staggering 73% met the criteria for pre-diabetes based on either HbA1c levels or FPG**, with 29% meeting both criteria.
- Of those with **pre-diabetes based on HbA1c**:
 - 9% progressed to diabetes.
 - 13% regressed to normoglycaemia.
 - 19% died.
- Of those with **pre-diabetes based on FPG levels**:
 - 8% progressed to diabetes.
 - 44% regressed to normoglycemia.
 - 16% died.
- The authors conclude that *'regression to normoglycemia or death was more frequent than progression to diabetes. These findings suggest that prediabetes may not be a robust diagnostic entity in older age.'*

There is concern generally that identifying 'pre' or 'sub-clinical' states in the elderly often derives no benefit for the individual and has substantial workload implications for us in general practice, at a time when we can ill-afford unnecessary work. Parallels can be drawn with subclinical hypothyroidism in the elderly and even CKD, where unnecessary testing can lead to treatment burdens or potentially harmful treatment. The issue of pre-diabetes in the elderly has been exacerbated by the new [QoF rules from 2020/21](#) recommending that all adults with pre-diabetes should have an annual HbA1c or fasting glucose level done. This **data suggests a substantial proportion of our elderly population will fulfil the criteria for pre-diabetes, and more of them will revert to normoglycaemia or die than develop diabetes**. And of the 8-9% that may develop diabetes, how many of those will actually require treatment? Given the concerns of overtreatment in the elderly, it's likely most will need no treatment at all. Too much medicine? Highly likely.

► **Guidance to help manage older people with diabetes** [Diabetic Medicine 2018; 35 \(7\): 838–845](#)

Given the complexities of dealing with older people with diabetes and the lack of specific guidance in national guidelines, this excellent national collaborative statement is extremely welcome. As eluded to above, **we need to be careful about thinking of advancing age and frailty as synonymous**. The two can go hand in hand, but not necessarily. Some elderly people will be fit and expect to have life expectancies beyond 10 years, so more intensive treatment may be appropriate to reduce micro-vascular or macro-vascular complications. But in the frail older adult these treatments are unlikely to give improvements in quality of life, and may well be harmful. As this guidance points out, **traditional macro-vascular and microvascular diabetes complications are thought to account for less than half of the diabetes-related disability seen in older people**. The guidance stresses the importance of regularly assessing for frailty in older adults, as this will help **guide de-escalation of treatment, and reduce overprescribing and treatment burdens**. This statement is explicit that *'It is important that establishing less aggressive targets is not regarded as a decision 'not to treat'; indeed, it could be argued that failure to establish individualized targets represents a dereliction of appropriate care.'*

KISS: Treatment targets and de-escalation thresholds in older adults with Diabetes[Diabetic Medicine 2018; 35 \(7\): 838–845](#)

	De-escalation threshold	De-escalation interventions	Treatment target	Treatment interventions
Fit older adult	53 mmol/mol (7.0%)	- Evaluate long-acting SUs and insulins that may cause hypoglycaemia. - Consider appropriate dose in setting of renal function.	58 mmol/mol (7.5%)	Avoid initiating new agents that may cause hypoglycaemia or weight loss.
Moderate-severe frailty	58 mmol/mol (7.5%)	- Discontinue SUs. - Avoid TZDs (risk heart failure). - Cautious use of metformin and insulin, be mindful of renal function.	64 mmol/mol (8.0%)	- DPP4i and longer-acting insulins have demonstrated safety. - SGLT2i may provide benefit in people with heart failure but can exacerbate diabetic symptoms.
Very severe frailty	64 mmol/mol (8.0%)	- Withdraw SUs and short-acting insulins. - Review timings and suitability of NPH insulins.	70 mmol/mol (8.5%)	- DPP4i. - Consider OD morning NPH insulin or analogue alternatives if symptomatic nocturnal hyperglycaemia. - Educate carers on hypo risks.

- SU = sulphonylurea
- TZD = thiazolidinediones (e.g. pioglitazone)
- DPP4i = DPP4 inhibitors (e.g. sitagliptin, linagliptin)
- SGLT2i - SGLT2 inhibitors (e.g. canagliflozin, empagliflozin)
- NPH insulin = Neutral Protamine Hagedorn insulin AKA 'intermediate insulin' or 'isophane insulin'

NB Learning Points & Quality Improvement Ideas: Diabetes in the Older Person

- Priorities for glycaemic control may shift in older populations to favour safety and avoidance of hypoglycaemia, rather than achieving HbA1c targets.
- Older people are at higher risk of hypoglycaemia, with a poorer outcome in terms of both physical and mental health from recurrent hypoglycaemic episodes.
- Recent evidence suggests treating elderly people aged ≥75 to intensive targets (HbA1c < 53 mmol/mol, <7%) with insulin or sulphonylureas significantly increases the risk of hospital attendance/admission and death.
- **Deprescribing** is a vital part of managing diabetes in our frailer older patients.
- We have new evidence questioning the benefit of 'diagnosing' elderly people with pre-diabetes, with significantly more reverting to normoglycemia or dying, than progressing to diabetes.
- **Quality Improvement idea:**
 - Search for patients over 75 or ideally those with a moderate/high frailty score with an HbA1c < 58 mmol/mol to review for potential deprescribing, prioritising those on SUs or insulin.



PRE-DIABETES

Pre-diabetes, an opportunity or an example of overdiagnosis? This was the theme of a recent interesting editorial [BJGP2018;68:172](#). Since the term pre-diabetes came in over the last 5 or 6 years it has become controversial, particularly so in the elderly population with new evidence questioning this 'diagnostic' label ([JAMA Intern Med. 2021;181\(4\):511-519](#)). But it seems the term is here to stay, especially since it's embedded in the [QoF rules from 2020/21](#) recommending that all adults with non-diabetic hyperglycaemia should have an annual HbA1c or fasting blood glucose. However, we have new primary care-based evidence from the UK ([JAMA Nov 2020](#)) showing supported lifestyle interventions in people with pre-diabetes reduces the risk of developing type 2 diabetes, thus giving some indirect support for the England-based Diabetes Prevention Programme. First a review of the BJGP editorial:

- In the UK pre-diabetes is diagnosed on the basis of an HbA1c level of 42-47 mmol/mol (6% to 6.4%).
- It is estimated that up to 1 in 3 of the adult population may fall into this range.
- 5% to 10% of these people will develop diabetes each year, with a similar percentage reverting to normoglycemia.
- Those with risk factors such as family history, higher-risk ethnic groups or polycystic ovarian disease or previous gestational diabetes are at higher risk.
- In England, the NHS 'Healthier You' [Diabetes Prevention Programme](#) has been rolled out to give intensive personalised lifestyle advice to people with pre-diabetes. Those referred will be offered tailored personalised lifestyle behaviour change over at least 13 face-to-face sessions over 9 months. Although an appealing idea, the evidence for real-world efficacy of such programmes is described as 'sparse'.
- Is this **overdiagnosis**? Expanding the definition of an abnormal blood sugar result and attaching a new label of pre-diabetes has implications for individuals and society, and may have unintended consequences (e.g. health anxiety and stigma). It is similar to other conditions (e.g. CKD) where the distinction becomes blurred between what is a risk factor, a biochemical abnormality or a disease.
- **Conclusion:** the term pre-diabetes appears to be here to stay and the nationwide rollout of the diabetes prevention programme means that GPs, nurses and HCAs will be having frequent conversations about it; we need to ensure we maximise the benefits and minimise the harms of these conversations by encouraging the patient to take ownership of the label as a motivator for a lifestyle change.

► [Role of diet in Type 2 diabetes incidence](#) [BMJ 2019;366:l2368](#)

The role of dietary factors in preventing, or increasing, the risks of diabetes is a hot topic, both medically and politically. Whilst not specific to pre-diabetes this recently published study in the BMJ does help give some more useable evidence on what foods patients at high risk of diabetes, including those with pre-diabetes, should avoid and which ones may be protective (given the usual caveats of potential confounding with observational studies):

- This was a systematic overview of meta-analyses of observational studies looking at dietary factors and incidence of type 2 diabetes - evidence quality was high for 75% of the data and moderate for 23%.
- The high-quality evidence suggests (summary hazard ratios in brackets):
 - **Reduced diabetes incidence** with increased intake of **cereal fibre** (0.75) and **whole grains** (0.87).
 - **Increased diabetes incidence** with increased intake of **red meat** (1.17), **sugar-sweetened drinks** (1.26), **processed meat** (1.37) and **bacon** (2.07).

► [Lifestyle interventions for pre-diabetes](#) [JAMA November 2020](#)

So is there any evidence for interventions that can reduce the risk of progression from pre-diabetes to type 2 diabetes? **Very early data from the England Diabetes Prevention Programme** ([Diabetes Care 2019 Nov; dc191425](#)) shows that, of the people who attended at least one intervention session mean weight loss was 2.3kg with an average HbA1c reduction of 1.26 mmol/mol. This is obviously early data, and it is not clear if this translates to clinically meaningful improvements, so only time will tell if this programme is effective (or indeed cost-effective).

There is data from elsewhere in the world to support lifestyle interventions to reduce the risk of progression from pre-diabetes to overt type 2 diabetes, but this **UK-based randomised clinical trial** is important for the obvious reason that it was both UK and primary care-based. Results:

- 1028 Patients with pre-diabetes (fasting plasma glucose 6.1 to < 7.0 mmol/L or HbA1c $\geq$ 6.0% to < 6.5%) were recruited from General Practices in the East of England (Norfolk).
- Mean age 65 years old, mean BMI 31, mean follow-up 2 years.
- Randomised to 3 arms: Control arm with usual care and 2 intervention arms - theory-based lifestyle intervention (group-based 6 core sessions and up to 15 maintenance sessions) with or without support from diabetes prevention mentors.
- There was no difference between the 2 interventions arms but **both intervention arms significantly reduced the risk of developing type 2 diabetes by ~40%** (combined OR 0.57, 95% CI = 0.38 - 0.87), giving an **NNT of 11**.
- The effect size was similar across age and social deprivation groups and was low cost (£122 per person).
- The **other notable finding was that in the control arm 22.8% had developed type 2 diabetes by 2 years**, supporting current recommendations that we should be screening this group yearly for potential progression to type 2 diabetes.
- **Conclusion:** Supported lifestyle interventions can significantly reduce the risk of progression from pre-diabetes to type 2 diabetes at low cost, but further support from diabetes mentors did not further reduce diabetes risk.

► **Liraglutide for pre-diabetes** [NICE TA664 Dec 2020](#)

Whilst on the subject of interventions there has been a recent NICE TA supporting the use of liraglutide. Although we should not be prescribing this ourselves in primary care, it's worth us being aware:

- Liraglutide can be considered (alongside lifestyle interventions) for people with pre-diabetes (HbA1c 42-47 mmol/mol or fasting plasma glucose 5.5-6.9 mmol/L) and obesity (BMI $\geq$ 35 or $\geq$ 32.5 for those in ethnic groups at higher risk).
- This should only be prescribed in secondary care through multidisciplinary tier 3 weight management programmes.

► **Older adults with pre-diabetes - risk of diabetes progression** [JAMA Intern Med. 2021;181\(4\):511-519](#)

But what about older adults with pre-diabetes? Is this even a significant diagnostic category and what is their risk of progressing to diabetes? Given the increased number of older adults being flagged as 'pre-diabetic' this is an important question, and this research suggests we are generating a lot of work for minimal benefit:

- This prospective cohort study of 2497 older adults (mean age ~75 years old) was followed for 6.5 years and reviewed for progression to diabetes.
- 44% had pre-diabetes based on HbA1c levels (39-47 mmol/mol, 5.7-6.4%). NB important to note the lower HbA1c level used in this study is lower than the standard UK definition of 42 mmol/mol, 6% so the total numbers with pre-diabetes is higher than we would accept.
- 59% had pre-diabetes based on fasting plasma glucose/FPG (100-125 mg/dL, 5.6-6.9 mmol/L).
- This means a **staggering 73% met the criteria for pre-diabetes based on either HbA1c levels or FPG**, with 29% meeting both criteria.
- Of those with **pre-diabetes based on HbA1c**:
 - 9% progressed to diabetes.
 - 13% regressed to normoglycaemia.
 - 19% died.
- Of those with **pre-diabetes based on FPG levels**:
 - 8% progressed to diabetes.
 - 44% regressed to normoglycemia.
 - 16% died.

- The authors conclude that *'regression to normoglycemia or death was more frequent than progression to diabetes. These findings suggest that prediabetes may not be a robust diagnostic entity in older age.'*

There is concern generally that identifying 'pre' or 'sub-clinical' states in the elderly often derives no benefit for the individual and has substantial workload implications for us in general practice, at a time when we can ill-afford unnecessary work. Parallels can be drawn with subclinical hypothyroidism in the elderly and even CKD, where unnecessary testing can lead to treatment burdens or potentially harmful treatment. The issue of pre-diabetes in the elderly has been exacerbated by the new [QoF rules from 2020/21](#) recommending that all adults with pre-diabetes should have an annual HbA1c or fasting glucose level done. This **data suggests a substantial proportion of our elderly population will fulfil the criteria for pre-diabetes, and more of them will revert to normoglycaemia or die than develop diabetes**. And of the 8-9% that may develop diabetes, how many of those will actually require treatment? Given the concerns of overtreatment in the elderly, it's likely most will need no treatment at all. Too much medicine? Highly likely.

NB Learning Points: Pre-diabetes

- In the UK pre-diabetes is diagnosed on the basis of an HbA1c level of 42-47 mmol/mol (6% to 6.4%).
- 5% to 10% of people with pre-diabetes will develop diabetes each year, higher risk if additional risk factors and a similar percentage will revert to normoglycaemia.
- The label has become very controversial as up to 1 in 3 of adults in the UK may fit into this category.
- The label is particularly controversial in the elderly with recent research suggesting that older adults are far more likely to revert to normoglycemia or die, than develop diabetes, with the authors concluding that their findings *suggest that prediabetes may not be a robust diagnostic entity in older age.'*
- New [QoF rules from 2020/21](#) recommend that any person with non-diabetic hyperglycaemia (defined as an HbA1c 42-47mmol/mol or a fasting plasma glucose (FPG) of 5.5-6.9mmol/l) should have an annual HbA1c or fasting blood glucose level.
- Across England, the [Diabetes Prevention Programme](#) is being rolled out across the NHS for patients with pre-diabetes to receive intensive individualised face-to-face lifestyle advice, with early data suggesting small improvements in weight loss and HbA1c reduction, although it is not yet clear if those reductions are clinically meaningful.
- Recent evidence suggests cereal fibre and whole grains reduce diabetes incidence, whilst red meat, processed meat, bacon and sugar-sweetened drinks increase diabetes incidence.
- We have UK-based primary care data showing that group-based supported lifestyle interventions can significantly reduce the risk of progression to diabetes by as much as 40%.
- Patient information on preventing diabetes from [Diabetes UK](#).

REMISSION OF TYPE 2 DIABETES

The argument for promoting the concept of remission of type 2 diabetes was made in an analysis in the BMJ in 2017 [BMJ 2017;358:j4030](#). Following this, the DiRECT study was published in 2018 to much fanfare, showing very low-calorie diets (VLCD) may be a strategy for inducing diabetes remission. Ever since the idea of diabetes remission has become a red hot topic, with the [announcement by NHSE of a 5000 patient pilot](#) to mirror the DiRECT design; this was [widely publicised in the press](#). And summarising all this we have had a recent review article discussing the nutritional basis of remission in type 2 diabetes and how both theory and evidence has changed in this interesting area over the last decade ([BMJ 2021;374:n1449](#)). First a review of the DiRECT study:

► *Primary Care weight management for remission of T2DM (DiRECT)* [Lancet 2018;391;541](#)

- This was a UK-based RCT conducted in 49 primary care practices in Scotland and Northern England, led by Prof Roy Taylor and his team in Newcastle, funded by Diabetes UK.
- It recruited 300 patients, aged 20-65 with T2DM diagnosed within the last 6 years and BMI 27-45kg/m².
- Patients were randomised then to either an intensive weight loss management programme (Counter Weight Plus) or best practice by guidelines (control).
 - The intervention group had a low-energy formula diet (825-853 kcal/day in shakes or soups) for 3-5 months then structured food reintroduction and support.
 - All anti-diabetic drugs and antihypertensives were discontinued on day 1 of the programme in the intervention group.
 - The programme was delivered in a primary care setting by a practice nurse or dietician who had received a day.
- At 12 months weight loss ≥15kg was recorded in 24% of the intervention group and 0% in controls - of those that lost ≥15kg 86% achieved diabetic remission (HbA1c <6.5% /48 mol/mol).
- At 12 months there was diabetes remission in 46% of the intervention group and 4% of the controls.
- Of those that lost 5-10kg, 34% still went into remission.
- At 12 months 74% of the intervention group were taking no anti-diabetic medication.
- **Conclusion:** at 12 months almost half of the patients in the intervention group achieved remission to a non-diabetic state off anti-diabetic drugs; remission of T2DM is a practical target for primary care.

In March 2019 the 2-year follow-up of the DiRECT trial was published ([Lancet March 2019](#)):

- ≥15kg weight loss was sustained in 11% of the intervention group and 2% of the control group.
- There was sustained diabetes remission in 36% of the intervention group and 3% of controls.
- **Conclusion:** Although there was substantial attrition in those who had achieved ≥15kg weight loss in the intervention group between 1 year (24%) and 2 years (11%), **over a third of patients were still in diabetic remission by year 2**, which was strongly linked to sustained weight loss.

How should we interpret the DiRECT Study?

The study has been discussed in the [Drug and Therapeutics Bulletin 2018](#) and in a BMJ analysis [BMJ2018;360:k1122](#)

- Whilst the results are very positive and encouraging it is clearly not an easy intervention and will only appeal to our most committed patients. Only 30% of patients invited into the study agreed to take part, however of those that did 80% completed the programme.
- 40% of the patients were from the two most socio-economically deprived groups and 40% had been on 6 different medications or more.
- The [DTB 2018](#) conclude that the evidence is impressive and that 'these results should prompt greater interest in strategies for weight management in primary care'.
- It was somewhat inevitable that attrition would happen in those with large degrees of weight loss, with half of those who managed ≥15kg weight loss at 1 year not being able to sustain this; however, that over a third of people in the intervention group were still in diabetes remission at 2 years, is encouraging.

► ***Sustained diabetes remission - cohort data from ADDITION-Cambridge study*** [BMJ 2020;368:l6775](#)

So, we know VLCDs are difficult to adhere to, and the concern all along with the DiRECT study is that weight loss would quickly wane and benefits, in the longer term, would not be sustained. So is there any data to suggest that diabetes remission can be sustained in the longer term? Well yes, in the form of a cohort study derived from the ADDITION-Cambridge RCT, which was picked up through the **NIHR research signals** and published in the [BMJ 2020;368:l6775](#) (see [Diabetic Medicine September 2019](#) for full data):

- The ADDITION-Cambridge study was an RCT based in GP Practices in east England which randomised newly diagnosed type 2 diabetic patients to either intervention (additional support and follow-up from nurses and GPs) and control (usual care).
- Importantly this did not include VLCDs and was a **longer-term study with results at 5 years**.
- The data from the 2 arms were pooled to give a prospective cohort encompassing 867 people, of whom 730 completed the 5-year follow-up and data requirements.
- **257 (30%) had achieved diabetes remission at 5 years** (defined as HbA1c <48 mmol/mol/6.5% on no medications).
- **People who achieved ≥10% weight loss at 1 year had a much higher chance of achieving remission at 5 years** (RR 1.77; CI = 1.32-2.38);
 - Similar trends were seen in those achieving 5-10% weight loss at 1 year but were not statistically significant.
- Sustained weight loss at 5 years significantly increased the chances of diabetes remission:
 - Weight loss ≥10% RR 2.43; CI = 1.78-3.31
 - Weight loss 5-10% RR 1.43; CI = 1.03-1.98
- There were some caveats, as always - the group were screen-detected diabetics (throwing up the possibility that they may be more motivated than the general population presenting symptomatically) and the population was mostly white.
- **Conclusion:** People newly diagnosed with type 2 diabetes should be encouraged that relatively achievable weight loss in the first year after diagnosis or within five years can lead to disease remission.

► ***Meta-analysis of low carbohydrate diets for type 2 diabetes remission*** [BMJ 2021;372:m4743](#)

Recently a systematic review attempted to quantify the benefits (and risks) of low (and very low) carbohydrate diets in terms of diabetes remission, with only limited success to be honest:

- The authors identified 23 studies and on the basis of moderate to low certainty, the evidence concluded that **low carbohydrate diets (LCDs) achieved a 32% higher remission rate at 6 months** compared to other diets with **NNT = 20**.
- LCDs also caused large and clinically significant weight loss, reduction in insulin resistance, as well as reduced medication use, at 6 months.
- In general, most of their benefits had diminished by 12 months.
- There was very limited data on very low carbohydrate diets.

The authors cited problems with significant heterogeneity in studies and many of the studies using surrogate endpoints. The other major obstacle they faced was **variations in the definition of diabetes remission**, a problem we shall now touch on further.

► *How do we define remission of diabetes and how should patients be followed up?*

One of the problems, as mentioned above, is that there has been no consensus on the criteria for remission in diabetes and definitions continually seem to shift.

The **American Diabetes Association (ADA)** have proposed that remission (no longer having diabetes) should be defined as both HbA1c < 6.5% (48 mmol/mol) and fasting blood glucose 5.6-6.9 mmol/L without antidiabetic drugs, maintained for 1 year.

In March 2019 The **Primary Care Diabetes Society (PCDS)** and **Association of British Clinical Diabetologists(ABCD)** published a [position statement](#) for the UK to define remission.

“ Remission of type 2 diabetes can be diagnosed when a person with confirmed type 2 diabetes has achieved all three of the following criteria: (i) Weight loss; (ii) Fasting plasma glucose or HbA1c below the WHO diagnostic threshold (<7mmol /l or <48mmol/mol, respectively) on two occasions separated by at least 6 months; (iii) the attainment of these glycaemic parameters following the complete cessation of all glucose-lowering therapies. ”

Then most recently in **August 2021 Diabetes UK** have also published a [position statement](#) with yet different criteria! Their key definition is:

- Diabetes UK defines type 2 diabetes remission as a sustained metabolic improvement where HbA1c levels return to below 6.5% (48mmol/mol) and which is sustained for at least 3 months in the absence of glucose-lowering medications.

They do helpfully give guidance on **what follow-up is expected**:

- As there is insufficient evidence on the impact of remission on specific diabetes complications, **everyone who is in remission should continue to receive regular monitoring at least annually including retinal screening**, and ongoing support for self-management, tailored to their specific needs.

► *Read Coding*

- Read code **C10P** (“diabetes in remission”) should be used. Using this code means that patients remain on annual recalls for HbA1c and also the retinal screening programme. These patients will need annual HbA1c for life. The code “Diabetes resolved” is NOT advised to be used.

► *Nutritional basis of type 2 diabetes remission* [BMJ 2021;374:n1449](#)

The concept of remission in type 2 diabetes has quickly become part of the language of type 2 diabetes management. However, it has not always been thus. As discussed in this interesting review article by Roy Taylor, who has been heavily involved in type 2 diabetes remission and was one of the DiRECT study authors, turn back the clock just over 10 years and the assumption was that type 2 diabetes was an irreversible and progressive disease. However back in 2011, the Counterpoint study showed that with restricted energy intake, both beta-cell function and hepatic insulin sensitivity could be restored.

So how does this all work? Type 2 diabetes is characterised by the accumulation of more fat in the liver and pancreas than the individual can cope with, but that threshold is different in different people, explaining why some people with type 2 diabetes can have normal BMI. The excess fat in the liver causes insulin resistance, which can resolve if liver fat returns to low-normal levels. Insulin can then act normally again and reduce the glucose excretion from the liver into the bloodstream, normalising blood glucose levels. The reduction in liver fat also leads to lower circulating triglycerides, thus lowering fat deposition in the pancreas which helps beta-cell function return to normal.

Which is the 'best' diet to achieve lower calorie intake seems to be a never-ending discussion. In the context of type 2 diabetes remission, the simple answer is we don't really have enough evidence to say whether one diet is better than another, with evidence supporting both low carbohydrate and low calorie (often low fat) diets. One of the main problems, as is often the case with diet studies, is the significant heterogeneity of definition of the different diets. There is also systematic review evidence supporting both sustained portion control and intermittent fasting (e.g. 5:2 diet). But **probably the key component of any diet is whether an individual can adhere to it or not** and as this article points out, there are other key factors determining the success or not of diets, with the support of family and friends being key. Inevitably there can be significant attrition with these very restrictive diets, with weight gain occurring over time. But encouraging data from the US national registry suggests that for people aiming for substantial weight loss over 10 years although weight regain was quite high in the first few years of follow-up, there was decreasing rates of weight regain over the subsequent five years, suggesting that over time maintenance may become easier.

However, the article highlights that we are still in the early days and only time will tell if maintained weight loss, and sustained remission in type 2 diabetes is possible. More data is obviously needed but **observational evidence suggest that sustained weight loss requires sustained dietary change, regular physical activity and frequent self-weighing, together with strong support from family and friends, as well as health care professionals.** The other question yet to be answered is whether longer-term remission of type 2 diabetes will translate into wider health benefits including lower rates of cardiovascular disease and lower cancer rates. Watch this space.

NB Learning Points & Quality Improvement Ideas: Remission of Type 2 Diabetes

- The DiRECT trial is a landmark UK-based primary care trial demonstrating the potential for remission of type 2 diabetes with weight loss, especially in the first few years after diagnosis.
- The ADDITION-CAMBRIDGE study suggests that relatively achievable weight loss in the first year after diagnosis if maintained can potentially result in remission in the longer term at 5 years.
- The global **definition of remission is in debate and frustratingly there is no consensus yet:**
 - In the UK the **PCDS/ABCD consensus statement 2019** recommends that it should be defined as 2 x HbA1Cs < 48 mmol/mol separated by 6 months, off all diabetes medications.
 - More recently in **2021 the Diabetes UK position statement** recommends it be defined as 'HbA1c levels return to below 6.5% (48mmol/mol) and which is sustained for at least 3 months in the absence of glucose-lowering medications.'
- Remission should be **read coded as "Diabetes in Remission"** not "Diabetes resolved".
- Patients who have achieved diabetes remission should continue to have (as a minimum) annual HbA1c levels and annual retinal screening.
- The significance for us in primary care is the change in conversation at diagnosis from "you have type 2 diabetes" to "you can do something about this".
- **Quality Improvement Ideas:**
 - Consider changing the narrative for people with newly diagnosed type 2 diabetes who are overweight to reflect the positive message that this is a disease which is potentially in their own hands and that with weight loss diabetes remission is an achievable goal, with all the health and social benefits that will come with reversing the diagnosis.
 - Direct them to good further information from **Diabetes UK** - [click here](#)

HYPOTHYROIDISM

Hypothyroidism is a relatively common problem, affecting ~2% of the UK population, with a much higher incidence in women than men (5-10 times more common in women). **NICE have recently produced a comprehensive guideline on assessing and managing thyroid disease NICE NG 145 2019.** One of the key areas reviewed is primary hypothyroidism (i.e. conditions arising from the thyroid gland rather than those arising from the pituitary gland i.e. secondary hypothyroidism). Whereas we have new evidence questioning the need to treat, or even monitor, subclinical hypothyroidism (see separate chapter), diagnosing and managing overt hypothyroidism (usually caused by autoimmune Hashimoto's thyroiditis) is important as it is a risk factor for hyperlipidaemia and cardiovascular disease.

The KISS below gives a summary of the NICE guidance, but first a few background points when considering thyroid disease. It is interesting to note that they recommend to consider thyroid testing in people with depression and unexplained anxiety. There is little evidence for this, hence the guidance is to *consider* rather than to *offer blood tests*. However, given how often we see patients with depression and anxiety in whom we very rarely do blood tests, it is an interesting idea.

Background Points: Thyroid Disease

[NICE NG 145 2019](#)

■ Symptoms:

- As we know there are a plethora of potential symptoms associated with thyroid disease (especially hypothyroidism), many of which are non-specific, and often come on slowly and insidiously, however, NICE do point out that single symptoms are less likely to be indicative of thyroid disease.
- Day-to-day variations in unexplained symptoms are unlikely to be related to thyroid disease.
- Be aware that in menopausal women symptoms of thyroid disease can be mistakenly attributed to the menopause.

■ Testing:

- Do NOT test for thyroid dysfunction in acute illnesses (unless the acute illness is suspected to be due to thyroid disease) as acute illnesses can affect thyroid function tests.
- Offer thyroid function tests for those with new-onset atrial fibrillation, Type 1 diabetes and other autoimmune diseases
- Consider thyroid function tests for those with depression or unexplained anxiety.
- Consider testing for thyroid dysfunction for children/young people with abnormal growth, or unexplained change in behaviour or school performance.
- Anyone with a diagnosis of autoimmune thyroid disease should have a coeliac screen done at the time of diagnosis.

■ Treatment:

- Warn patients that symptom improvement may lag behind treatment changes by several weeks or even months.
- Even when thyroid function results are within the reference range, changes to treatment may improve symptoms for some people

KISS: Primary Hypothyroidism (Adults)

[NICE NG 145 November 2019](#) [NICE CKS 2020](#)

Diagnosis/Testing:

- Do not test for thyroid disease in acute illness, unless you suspect the acute illness may be caused by thyroid disease, because the acute illness will affect the test results
- The diagnosis of primary (overt) hypothyroidism is based on clinical features & biochemical evidence of elevated TSH >10mU/L & low thyroxine (freeT4 = FT4).
- Once primary hypothyroidism confirmed, consider measuring thyroid peroxidase antibodies (TPOAbs) as a one-off test (does not need repeating).

Treating Primary Hypothyroidism:

- Commence levothyroxine (LT4):
 - Starting dose **1.6mcg/kg/day** (rounded to nearest 25mcg) for adults <65 years old and no CVD.
 - If >65 years old or CVD, commence smaller starting dose **25-50mcg** & titrate cautiously.
- Do NOT offer liothyronine (LT3) or 'natural thyroid extracts'; there is no evidence to support treatment with LT3 either alone or in combination with LT4, or for natural extracts

Monitoring Treatment:

- **Aim for TSH levels within the reference range**; if symptoms persist consider adjusting dose further but avoid TSH suppression (risk of osteoporosis and AF).
- Check TSH every 3 months until levels have stabilised then yearly.
- Consider measuring FT4 as well as TSH if patients remain symptomatic, but warn patients symptom improvement may lag behind treatment changes by several weeks or even months.

In those with persistently abnormal TSH levels:

- Check thyroxine dose is appropriate for body weight.
- Check & encourage compliance & correct administration; with water on an empty stomach at least 30 minutes before breakfast or at bedtime.
- Several foods can impair LT4 absorption - milk, coffee, soya products and papaya.
- Several medications impair LT4 absorption or increase LT4 requirements e.g. antacids, iron, PPIs & H2RAs, anti-epileptics (carbamazepine, phenytoin).
- Exclude coeliac disease, pernicious anaemia, gastritis & malabsorption which all affect LT4 absorption.

In those with persistent symptoms despite normal TSH consider:

- Other endocrine or autoimmune conditions e.g. diabetes, Addison's, hypopituitarism.
- Nutritional deficiency e.g. iron, vitamin B12, folate.
- Other causes of symptoms e.g. sleep apnoea, fibromyalgia, depression, anxiety, alcohol/drug misuse, other drug side effects (e.g. beta-blockers, opiates).

Pregnancy: Untreated hypothyroidism in pregnancy is associated with adverse maternal & fetal outcomes: adequate pre-conception counselling and monitoring and the dose increased when pregnancy confirmed is essential (see the separate chapter on Thyroid, Fertility & Pregnancy).

SUBCLINICAL HYPOTHYROIDISM

Important new research has been published [JAMA2018;320;1349](#) on subclinical hypothyroidism (SCH) which challenges established practice and has led to a new '*rapid recommendations*' guideline from the BMJ [BMJ2019;365;I2006](#). Further new research has also led to a new recommendation to avoid repeat thyroid testing in healthy older adults [BMJ2019;364;1804](#). In [November 2019 NICE](#) published guidance on thyroid disease which unfortunately gives slightly conflicting advice on managing SCH in adults <65 years old.

► Subclinical Hypothyroidism (SCH): Background Points [BMJ2019;365;I2006](#)

- SCH (raised TSH with normal thyroxine levels) is very common in primary care, affecting 4% to 20% of the population and is more common in women, people of white ethnicity and older adults.
- 90% of people with SCH have a TSH level between 4 -10 mIU/L; 60% of these normalise within 5 years.
- TSH levels increase with age and a slight increase in TSH is normal in older people.
- TSH levels show wide biological variation and levels may rise in response to stress or disease.
- Up to 25% of people with SCH report one or two symptoms typical of overt hypothyroidism (e.g. fatigue, cold sensitivity, skin changes, constipation etc.) but the relation between TSH and symptoms is unclear.
- The risk of progression to overt hypothyroidism ranges from 2% to 5% per year; this risk is increased with higher levels of TSH and if the patient has positive thyroid peroxidase antibodies.
- Observational data suggest that SCH is associated with an increased risk of CVD and CV mortality, particularly in those with a TSH > 10 mIU/L but we lack evidence that treatment improves CV outcomes.

Key Evidence on Subclinical Hypothyroidism (SCH)

► Systematic review questions treatment of SCH [JAMA2018;320;1349 \(1a\)](#)

This new study was a systematic review and meta-analysis of trials looking at the treatment of people with SCH with thyroxine, looking at outcomes in thyroid-related symptoms or quality of life.

- 21 trials, n=2192, comparing thyroxine to placebo in non-pregnant adults with SCH.
- The outcomes were quality of life and thyroid-related symptoms, with follow-up from 3 to 18 months.
- The quality of evidence was moderate to high with a low risk of bias in the studies.
- The systematic review included the [TRUST trial NEJM2017](#), a high-quality RCT that found no benefit of treating adults older than 65 years with thyroxine compared to placebo.
- **Results:**
 - There was **no benefit in a wide range of thyroid-related outcomes** such as quality of life, fatigue, depression, cognitive function, muscle strength, body weight etc.
 - There was **no benefit seen in cardiovascular outcomes or mortality with treatment**; however, only the TRUST trial included this as an outcome and only a few events were observed so that 'definitive evidence is lacking' for or against treatment of SCH with regard to CV outcomes.
- **Conclusion:** *Treatment does not improve symptoms or quality of life in non-pregnant adults with SCH and these findings do not support the routine use of thyroid hormones in these patients.*

► The Birmingham Elderly Thyroid Study [BJGP 2018;68;e718](#)

This study aimed to establish the stability of thyroid tests over time in adults > 65 in UK primary care:

- In 2,700 patients with normal thyroid function at baseline, at 5 years 96% still had normal thyroid function, only 0.5% developed overt thyroid disease and the remaining 3.5% developed SCH.
- 143 patients had SCH at baseline; at 5 years 58% stayed the same, 40% normalised and 2% developed overt hypothyroidism; higher baseline TSH was associated with higher risk of overt hypothyroidism.
- **Conclusion:** *In older adults, there is a high degree of thyroid stability over time which should discourage repeat testing and save costs.*

► **Guidelines and Recommendations****1** *Thyroid treatment for SCH: a clinical practice guideline 2019* [BMJ2019;365:l2006](#)

Following the JAMA systematic review above, the BMJ published a '**rapid recommendations**' **clinical practice guideline for SCH**. This is incorporated in the KISS box below. They state that the evidence consistently demonstrates no benefit for treating SCH with regard to quality of life or thyroid-related symptoms and that thyroid hormones may have little or no effect on CV events or mortality. The guideline group were concerned about the possibility of harm in older adults who were treated.

They conclude with a strong recommendation against treatment with thyroid hormones for adults with SCH

This recommendation does NOT apply to women who are pregnant, wish to become so or are at risk of unplanned pregnancy (see the separate chapter on TSH, fertility and pregnancy), or those with very high TSH levels > 20 mIU/L (who were not included in the studies). The recommendation may not apply to the following groups who were under-represented in the trials: those with very severe symptoms, young adults aged under 30 and patients already taking thyroid hormones.

2 *NICE guideline on Thyroid disease* [NG 145 November 2019](#)

NICE have taken a slightly different viewpoint to the BMJ guideline recommendation above with regard to SCH. NICE have taken a more conservative approach, having felt there was not enough evidence to make definitive guidance on not treating SCH in younger adults; they point out that most of the evidence for treating (or not) SCH comes from studies with older adults >65 (as discussed above the JAMA meta-analysis was quite heavily influenced by the TRUST trial which only included adults >65 years old). **NICE still have concerns about the observational data suggesting an association between SCH and CVD, especially if TSH >10mIU/L.** However NICE concur with the BMJ that the risks and benefits of treating most adults >65 with SCH favours no treatment. **Key points from the NICE guidance:**

- Consider a (single, not to be repeated) test for thyroid peroxidase (TPO) antibodies.
- Consider levothyroxine for adults with SCH and TSH $\geq$ 10mIU/L on 2 separate occasions 3 months apart.
- Consider a 6-month trial of levothyroxine for adults < 65 years old with SCH and TSH < 10mIU/L on 2 separate occasions 3 months apart AND symptoms of hypothyroidism:
 - **If symptoms persist when TSH normalises consider stopping the levothyroxine.**

So although there are differences in the two guidelines, they both agree that **adults >65 with SCH and TSH < 10mIU/L do not need treatment.**

3 *Monitoring of adults with SCH*

The recent [NICE guidance](#) has essentially stuck with the status quo when it comes to monitoring untreated SCH in adults and gives the following recommendations:

- Measure TSH and FT4 yearly if raised TPO antibodies or previous thyroid surgery/radioactive iodine.
- Measure TSH and FT4 every 2-3 years for all other adults.

The new research from the [Birmingham Elderly Thyroid Study](#) questions this recommendation in older people. Thyroid tests are expensive and widely requested, so this study has important implications. Around 30% of older adults have thyroid function tests done every year, and the total annual cost of thyroid testing in the UK alone is £30M [BMJ2019;364:1805](#). Following this study, the National Institute of Health Research has published a recommendation [NIHR signal 2019](#) discussed in [BMJ2019;364:1804](#) that **repeat thyroid testing for healthy older adults with normal or subclinical thyroid dysfunction is not needed unless patients have other risk factors or develop symptoms suggestive of overt disease.** They conclude that '*avoiding unnecessary testing will reduce the burden for patients and save NHS resources*'.

NB Learning Points & Quality Improvement Ideas: Subclinical Hypothyroidism (SCH)

- SCH is a common clinical finding in primary care and new research has questioned the benefits of treating with thyroxine, suggesting no improvement in symptoms or quality of life; a BMJ clinical practice guideline makes a strong recommendation against treating SCH with thyroxine.
- However NICE still suggest to consider treating adults with SCH and repeated TSH levels $\geq 10 \text{ mIU/L}$, or considering a 6-month trial of treatment for < 65 s if repeated TSH $< 10 \text{ mIU/L}$ AND symptomatic.
- Treatment is more likely to be beneficial for young adults < 30 years old or those with higher TSH $> 20 \text{ mIU/L}$, but is unlikely to be beneficial for older adults > 65 years old.
- New evidence suggests thyroid function remains stable in adults > 65 and that repeat testing in adults > 65 with SCH is not needed unless there is a clinical change.
- **Quality Improvement Idea:** Audit repeat thyroid testing in adults > 65 years old with SCH.

KISS: Subclinical Hypothyroidism (SCH) in adults

[BMJ2019;365;l2006](#) & [JAMA2018;320;1349](#) & [NICE NG 145](#)

- **Background:**
 - **Definition:** a raised TSH but normal free thyroxine levels.
 - Occurs in up to 20% of the population & in many cases is transient.
 - TSH levels vary with stress, transient disease and age with TSH increases in older adults normal.
 - 2%-5% progress to overt hypothyroidism annually (more likely if thyroid peroxidase (TPO) antibodies are positive, or in those with higher TSH levels $> 20 \text{ mIU/L}$).
- **Management:**
 - Confirm SCH in adults < 65 : Repeat TSH, FT4 and check TPO antibodies in 3 months.
 - **NICE recommend the following:**
 - *Consider* levothyroxine for adults with SCH and TSH $\geq 10 \text{ mIU/L}$.
 - *Consider* a 6-month trial of levothyroxine for adults < 65 years old with SCH and TSH $< 10 \text{ mIU/L}$ AND symptoms of hypothyroidism BUT if **symptoms persist when TSH normalises consider stopping the levothyroxine**.
 - When considering treatment take into account factors associated with increased risk of progressing to overt hypothyroidism i.e. raised TPO antibodies, previous thyroid surgery or radioactive iodine.
 - However, be aware that based on new research [JAMA2018;320;1349](#), there is a new **STRONG guideline recommendation** [BMJ2019;365;l2006](#) **NOT to offer thyroxine treatment to adults with SCH even if they are symptomatic** - evidence most strong for adults > 65 :
 - This recommendation does NOT apply to the following groups: Pregnancy, planning pregnancy or risk of unplanned pregnancy (see separate chapter Thyroid, pregnancy and fertility), patients with very severe symptoms, a TSH $> 20 \text{ mIU/L}$ or young adults aged under 30.
- **Monitoring:**
 - **NICE recommend the following:**
 - Measure TSH/FT4 yearly if raised TPO antibodies or previous thyroid surgery/radioactive iodine.
 - Measure TSH/FT4 every 2-3 years for all other adults.
 - However, be aware [NIHR recommendation](#) that in **healthy adults aged over 65, repeat thyroid testing is not needed** unless patients have risk factors for, or develop signs or symptoms of overt hypothyroidism [BMJ2019;364;1804](#)

CLINICAL & SUBCLINICAL HYPERTHYROIDISM

NICE have recently produced a new guideline on the assessment and management of thyroid disease in November 2019 [NICE 2019, NG145](#). The guideline covers clinical and subclinical hyperthyroidism but the detail around these two areas was quite thin. Most clinical hyperthyroidism is likely to be managed in secondary care and the NICE guidance on this is summarised in the KISS below. However, we see a surprising amount of subclinical hyperthyroidism in primary care so it was useful that in 2018 a new paper from the NEJM summarised the evidence on subclinical hyperthyroidism [NEJM2018;378:2411](#).

► Subclinical Hyperthyroidism [NEJM2018;378:2411](#)

What is subclinical hyperthyroidism?

In subclinical hyperthyroidism the TSH is low but patients have normal levels of free thyroxine, FT3 or FT4. Most people have mild subclinical hyperthyroidism with a TSH of between 0.1 to 0.4 mU/L, but 25% have more severe subclinical hyperthyroidism with a fully suppressed TSH of < 0.1 mU/L. Prevalence figures vary between studies ranging from 0.5-10% of the population.

What causes it?

- Subnormal TSH levels are often transient, and 50% will normalise especially when only mildly lowered.
- We commonly see **exogenous** subclinical hyperthyroidism when patients are taking too much thyroxine, and surreptitious ingestion of thyroid products can also be a cause.
- The causes of **endogenous** subclinical hyperthyroidism are the same as the causes of overt hyperthyroidism e.g. multinodular goitre or Grave's disease. Other things can cause low TSH levels including non-thyroidal illness, drugs (e.g. steroids) or pituitary or hypothalamic disease.

Is subclinical hyperthyroidism clinically significant?

Yes, is the clear answer from the paper. It may be called subclinical hyperthyroidism but it is clinically significant with possible progression to overt hyperthyroidism or cardiovascular or bone complications:

- **Progression to overt hyperthyroidism:**
 - Patients with mildly reduced TSH levels frequently normalise with follow-up. NICE recommend checking TFTs every 6 months in this group.
 - Patients with severe subclinical hyperthyroidism with a fully suppressed TSH of < 0.1 mU/L usually have persistent disease or progress to overt hyperthyroidism.
- **Cardiovascular conditions:**
 - Severe subclinical hyperthyroidism is associated with sinus tachycardia, ectopics and diastolic heart failure. Prospective observational studies have shown a significantly increased risk of developing atrial fibrillation, heart failure and major cardiovascular events and cardiac death compared to controls. The risk seems to increase with the degree of TSH suppression.
 - Increased risk of cardiovascular disease and mortality is also seen in patients with iatrogenic exogenous severe subclinical hyperthyroidism.
- **Bone loss and fractures:**
 - Severe subclinical hyperthyroidism is associated with an increased risk of osteoporotic fractures and some studies have shown an increased risk in patients with mild subclinical hyperthyroidism.
 - This has also been seen in patients with iatrogenic exogenous subclinical hyperthyroidism.
- **Dementia:**
 - Subclinical hyperthyroidism has been associated with an increased risk of dementia. Prospective [research](#) in older adults aged over 70 shows that people with severe subclinical hyperthyroidism have an increased risk of developing dementia and a larger cognitive decline.

How should we evaluate patients with subclinical hyperthyroidism?

See the KISS summary below. Assess for signs (e.g. goitre, neck lump or eye signs of Grave's disease) or symptoms (e.g. tremor or tachycardia) of thyroid disease, exclude other causes of a low TSH and then repeat the tests a few months later.

How is it treated?

Clearly, patients with exogenous subclinical hyperthyroidism need to have their dose of thyroxine reduced. Specialist treatment is usually offered to people with severe subclinical endogenous hyperthyroidism if they are symptomatic or if they are aged ≥ 65 or with risk factors for osteoporosis or cardiac disease, however, we lack evidence to know if treating these people improves outcomes. Younger or non-treated patients will need regular follow-up and observation, with NICE recommending TFTs every 6 months for those not treated.

What about patients with thyroid cancer who need to have TSH fully suppressed?

In these patients who need full TSH suppression, beta-blockers may be considered if they are symptomatic. If they have low-risk thyroid cancer with no measurable disease, the paper advises reducing levothyroxine dosage but this should obviously be a specialist decision.

What about older adults with subclinical hyperthyroidism?

Because these patients are at greater risk of developing osteoporosis or atrial fibrillation, the NEJM recommends a lower threshold for treatment for adults with subclinical hyperthyroidism, however other new research [BMJ2019;364;1804](#) has shown that older adults have great stability of thyroid tests over time, and only 4% of adults aged over 65 (albeit small numbers) with subclinical hyperthyroidism developed overt hyperthyroidism over 5 years, leading to a recommendation [NIHR signal 2019](#) that repeat thyroid testing in healthy older adults is not needed unless there is clinical change or other risk factors that warrant it.

KISS: Subclinical Hyperthyroidism

[NEJM2018;378:2411](#) & [NICE CKS 2020](#) & [NICE NG 145 Nov 2019](#)

Definition:

- Mild: low TSH 0.1 to 0.4 mIU/L and free thyroxine in the normal range.
- Severe: TSH < 0.1 mIU/L and free thyroxine in a normal range.

Management:

- Clinically assess the patient for any symptoms or signs of thyroid disease (e.g. goitre or lump; urgent referral of lump if malignancy is suspected).
- **Exclude other causes of a low TSH**, such as:
 - Recent or current non-thyroidal illness.
 - Drugs that suppress TSH (e.g. corticosteroids, dopaminergic anti-Parkinsonian medication or octreotide) or drugs that alter TSH secretion (e.g. lithium or amiodarone), or external thyroxine.
- If other causes of a low TSH have been excluded, **repeat thyroid tests after 3 to 6 months**:
 - If both TSH readings < 0.1 mIU/L and evidence of thyroid disease (e.g. goitre, +ve thyroid antibodies) refer for specialist advice (NEJM would recommend referring all with repeat TSH < 0.1 mIU/L).
 - If mild (TSH 0.1-0.4 mIU/L) repeat TFTs every 6 months.
 - Stop monitoring if 2 TSH readings within reference range within a 3-6 month period.
- Treatment may be considered for people with severe subclinical hyperthyroidism if they are symptomatic or if they are aged ≥ 65 , or with risk factors for osteoporosis or cardiac disease, however, there is currently no evidence treatment improves outcomes (e.g. AF, cardiac disease, dementia).

KISS: Clinical Hyperthyroidism

[NICE NG 145 November 2019](#)

Background:

- Caused by either increased production/secretion of thyroid hormones (hyperthyroidism) or release of thyroid stores (thyroiditis).
- Prevalence of ~2% in UK populations.
- Commonest cause is autoimmune hyperthyroidism (Graves' disease) - accounts for 60-80% of cases.

Testing:

- If thyrotoxicosis suspected check TSH and if below normal range FT3 and FT4 as well.
- If thyrotoxicosis confirmed measure TSH receptor antibodies (TRAbs) to confirm Graves' disease.

Primary Care Management:

- If transient thyrotoxicosis present without hyperthyroidism (i.e. thyroiditis) usually supportive treatment only is needed (e.g. beta-blockers).
- If hyperthyroidism present refer to secondary care for further assessment and management.
- Whilst awaiting specialist assessment consider starting antithyroid drugs, as well as supportive care:
 - Carbimazole is the usual first line drug starting dose **15mg daily** - [click here for BNF](#)
- Secondary care management will depend on underlying cause but options include radioactive iodine, surgery or antithyroid drugs.

Monitoring of antithyroid drugs:

- Measure TSH, FT3 and FT4 every 6 weeks until TSH in normal range then TSH +/- FT3/FT4 every 3m.
- Do not monitor FBC and LFTs routinely whilst on antithyroid drugs unless clinical suspicion of agranulocytosis or liver dysfunction.
- **Safety information on carbimazole** ([BNF](#) and [MHRA Feb 2019](#)):
 - Carbimazole can cause bone marrow suppression and should be stopped promptly if clinical or lab suspicion of neutropenia.
 - Patients should report symptoms of infection, esp. sore throat - if infection present check WCC.
 - Carbimazole is associated with increased risk of congenital malformation so women of childbearing age need to be on effective contraception.
 - Carbimazole is associated with an increased risk of acute pancreatitis - avoid in anyone with a history of acute pancreatitis - stop immediately and permanently if acute pancreatitis occurs.

NB Learning points & Quality Improvement Ideas: Clinical and Subclinical Hyperthyroidism

- Subclinical hyperthyroidism is a common finding in primary care - if repeated TSH levels < 0.1 mIU/L and signs of thyroid disease refer for specialist assessment; for all others monitor every 6 months.
- Subclinical hyperthyroidism is associated with a number of poor outcome measures including AF, cardiac disease, cardiac death and osteoporosis/fractures; however, we lack evidence to know if treating the subclinical hyperthyroidism improves these outcomes.
- The same risks are associated with exogenous subclinical hyperthyroidism, most often iatrogenic due to overtreatment with levothyroxine - **aim to keep TSH in normal range**.
- In clinical hyperthyroidism refer for specialist advice, but consider starting antithyroid drugs (Carbimazole 15mg daily) whilst awaiting review; also give supportive care with beta-blockers.
- **Quality improvement idea:** Audit of patients on thyroxine with fully suppressed TSH and reduce the dose (except in patients with thyroid cancer).

THYROID, PREGNANCY & FERTILITY

There is increasing controversy about thyroid hormone replacement in the preconception period and pregnancy, especially of borderline thyroid dysfunction and its potential impact on fertility and pregnancy outcomes. GPs are increasingly being asked by specialists and patients to 'treat' seemingly normal levels of TSH both pre-conception and during pregnancy, and there is a debate about current recommendations to always treat subclinical hypothyroidism in pregnancy. Recent evidence has added to this debate, and there has been a recent BJGP editorial [BJGP 2019;69;282](#) and also a [DTB 2019;57;22](#) review on the topic for women with both overt and also subclinical hypothyroidism. Then most recently we had a DTB review ([Drug and Therapeutics Bulletin 2022;60:101](#)) of an RCT which showed that thyroxine treatment for euthyroid women with miscarriage or delayed fertility and +ve TPO antibodies did not improve pregnancy outcomes.

► *Thyroid hormone replacement for women with established hypothyroidism in pregnancy and pre-conception* [BJGP 2019;69;282](#)

This editorial discussed the important issue of the optimal care of women with established overt hypothyroidism in pregnancy. Although this has often been considered a specialist topic and specialist referral is recommended for all cases, as the editorial points out it is a vital primary care issue given that pregnancies are often unplanned, present late or women of reproductive age on thyroxine are unaware of the need for pre-pregnancy planning.

Sub-optimal thyroid function in pregnancy carries significant risks of poor pregnancy outcomes, including increased pregnancy loss and impaired neurodevelopment. Whilst the devastating neurological consequences of severe iodine deficiency and congenital hypothyroidism are now fortunately rare, landmark studies in the 1990s showed that children born to women with uncorrected hypothyroidism in pregnancy had a 7-point IQ defect compared to babies born to euthyroid mothers. Since then numerous observational studies have shown an increased risk of pregnancy loss and child neurological-intellectual impairment in women with hypothyroidism who are sub-optimally treated. A developing fetus cannot synthesise thyroid hormone until the second trimester, thus during the crucial first trimester of neurodevelopment, it relies exclusively on maternal thyroxine hence the recommendation for an increased dose of thyroxine in the first trimester.

Women with hypothyroidism require a 25% to 50% increase in thyroxine dosage, but dose adjustments are often inadequate or late. Furthermore, early pregnancy may be accompanied by vomiting or the use of antacids or iron preparations which reduce thyroid bioavailability. Only 50% of pregnancies are planned and are often not recognised until late in the first trimester or after, and although guidelines advise specialist referral for women, access to specialist services is often delayed. An adequate pre-conception counselling strategy is thus essential for primary care and to make sure that women are aware of this. ***So, what should we advise women?***

■ Pre-conception counselling:

- clear, consistent and repeated information should be given to all women of reproductive age about the need to optimise thyroid replacement before conception
 - [British Thyroid Foundation Guidance for Patients 2019](#) can be downloaded and printed during the consultation or attached to their repeat prescription
- New UK NICE guidelines are in development, so the BJGP editorial recommends the recent [US guidelines](#) that women with hypothyroidism who are planning pregnancy should have their levothyroxine dosage adjusted to achieve a TSH < 2.5 mu/L at the time of conception
- **On conception**
 - women should be advised to increase the dose immediately (as below), book a blood test and arrange a review:
 - increase by 25mcg daily in women on a daily dose of ≤100mcg daily
 - increase by 50 mcg daily in women on a daily dose of > 100mcg daily
 - specialist endocrine advice should be sought as soon as possible when pregnancy is confirmed
 - to optimise the bioavailability, women also need clear guidance on taking thyroxine on an empty stomach with water at least 30 minutes before food, beverages or other medications

► ***Management of subclinical hypothyroidism (SCH) in pregnancy:*** [DTB 2019;57;22](#)

This is much more controversial...! The DTB state that in pregnancy a TSH of $< 4\text{mU/L}$ should be considered normal in pregnancy after 7 weeks of gestation, and that SCH is diagnosed in pregnancy if TSH is $4\text{--}10\text{ mU/L}$ with a normal T4 concentration.

The standard advice from guidelines (see KISS summary below) is to treat subclinical hypothyroidism in women who are pregnant or trying to become so. This has huge implications as SCH is estimated to affect up to 5% of all pregnancies. Observational studies have shown that SCH is associated with a range of adverse pregnancy-related outcomes, however, there is no consistent evidence that SCH causes harm in pregnancy or that treatment with thyroxine is beneficial.

We do have clear evidence that overt hypothyroidism is associated with impaired neuropsychiatric development, see above, but the evidence that this extends to subclinical disease is lacking. Furthermore in SCH by definition, maternal thyroxine levels are in the normal range, and it is hypothesised that maternal T4 levels rather than TSH are the crucial determinant for neurodevelopment.

Although guidelines recommend treatment, we lack evidence that treatment with thyroxine improves pregnancy-related outcomes such as increased pregnancy loss. Most clinical concerns relate to the increased risk of miscarriage but this is difficult to study and requires intervention at an earlier stage than has been done in most studies. No RCT has demonstrated that starting thyroxine treatment for SCH is associated with improved outcomes including gestational age, pre-eclampsia, pregnancy loss, birth weight, neonatal complications and childhood IQ and in [2016 a large systematic review](#) concluded that the value of treatment remains 'uncertain'.

The DTB review concludes that evidence that SCH causes harm to either pregnancy or the long-term development of the child is inconsistent and that there is insufficient evidence that treatment with thyroxine is beneficial. Healthy women do not need to be screened for SCH prior to pregnancy, and that if it is found in pregnancy uncertainty regarding the evidence for treatment should be shared.

► ***Does subclinical hypothyroidism contribute to infertility?***

Overt hypothyroidism is known to cause infertility, but we are increasingly being asked to prescribe levothyroxine to women seeking help for infertility with seemingly normal thyroid function. New research [J Clin Endo 2018](#) has published showing a link between unexplained infertility and a higher TSH. It was an observational study comparing TSH levels in women with unexplained infertility compared to controls.

- Women with unexplained infertility had significantly higher TSH levels than controls
- This finding remained consistent after controlling for age, body weight and smoking
- Nearly twice as many women with unexplained infertility had a TSH >2.5 compared to controls
- These findings, however, do not establish causality, and 'based on this data we cannot recommend thyroid replacement treatment until studies have been done to see if treatment improves pregnancy outcomes'
- **Conclusion:** *women with unexplained infertility have higher TSH levels compared to controls, studies are needed to determine whether treatment of high-normal TSH levels with levothyroxine reduces time to conception*

► *What about euthyroid women with +ve TPO antibodies?* [DTB 2022;60:101](#)

The concern over the link between subfertility and subclinical hypothyroidism has also been expressed in previous research that has suggested that **even euthyroid women with +ve thyroid peroxidase (TPO) antibodies may be at increased risk of miscarriage and sub fertility** (the theory being that if you have +ve TPO antibodies you are more likely to develop subclinical or overt hypothyroidism). However, the evidence reviewed by the DTB confirms treatment with thyroxine in these women does not improve outcomes, as there had been some conflicting evidence on this question in the past:

- Almost 1000 women across 49 UK hospitals, with a history of miscarriage or delayed fertility with +ve TPO antibodies but euthyroid on blood tests were recruited.
- This was a double-blind placebo-controlled trial and those women were randomised to 50mcg levothyroxine or placebo, with the **primary outcome being live birth at ≥ 34 weeks gestation**.
- **There was no difference between the 2 groups with live births at ≥ 34 weeks of $\sim 37\%$ in both groups.**
- There was also no difference in secondary outcomes (pregnancy at 7 weeks, ongoing pregnancy at 12 weeks, miscarriage or maternal/neonatal complications) between the 2 groups.
- **Conclusion: thyroxine treatment for euthyroid women with +ve TPO antibodies does not improve pregnancy rates.**

KISS: Hypothyroidism & Pregnancy

Based on [BJGP 2019;69;282](#) & [DTB 2019;57;22](#) & [NICE CKS 2020](#)

Overt hypothyroidism & pregnancy

- **pre-conception counselling**
 - clear, consistent and repeated information should be given to all women of reproductive age about the need to optimise thyroid replacement before conception
 - [British Thyroid Foundation Guidance for Patients 2019](#) can be downloaded and printed during the consultation or attached to their repeat prescription
 - women with hypothyroidism who are planning pregnancy should have their levothyroxine dosage adjusted to achieve a TSH < 2.5 mu/L at the time of conception
- **on conception**
 - on conception, women should be advised to increase the dose immediately, book a blood test and arrange a review
 - the dose should be increased
 - increase by 25mcg daily in women on a daily dose of ≤100mcg daily
 - increase by 50 mcg daily in women on a daily dose of > 100mcg daily
 - specialist endocrine advice should be sought as soon as possible when pregnancy is confirmed
 - to optimise the bioavailability, women also need clear guidance on taking thyroxine on an empty stomach with water at least 30 minutes before food, beverages or other medications
- **Recommended reference range** of TSH in pregnancy [BTA Guidelines](#)
 - First trimester: 0.4 - 2.5 mU/L
 - Second & third trimester: 0.4 - 3.0 mU/L
 - *'These reference ranges should be achieved where possible with appropriate doses of levothyroxine pre-conception & most importantly in the first trimester'*

Subclinical hypothyroidism & pregnancy

- Guidelines advise treatment of SCH pre-conception and with dose increases on conception as for overt hypothyroidism above
- Although guidelines routinely advise treating subclinical hypothyroidism in pregnancy, this is a more controversial recommendation and a recent paper concludes that *'there is no consistent evidence that SCH causes harm in pregnancy or that treatment with thyroxine is beneficial'* and that we should share this uncertainty with women [DTB 2019;57;22](#)

TSH & Fertility

- Overt hypothyroidism is associated with infertility
- There is limited evidence that subclinical hypothyroidism & euthyroid women with thyroid antibodies are associated with unexplained infertility in women
- Recommendations: [US Guidelines 2017](#)
 - Check TSH in all women seeking care for infertility
 - Treatment with LT4 is recommended for all women with overt hypothyroidism
 - Subclinical hypothyroidism
 - Insufficient evidence exists to determine if LT4 improves fertility in women with subclinical hypothyroidism in women attempting natural conception, but low dose treatment (25-50mcg daily) 'may be considered and carries minimal risk'
 - Treatment improves outcomes in women undergoing assisted reproduction (e.g. IVF) & these women should be offered treatment

ABDOMINAL PAIN AND CANCER

We have always been aware that new onset of abdominal pain in older people is an 'orange flag' for a new cancer diagnosis. However, it is difficult as abdominal pain in this group of patients due to constipation, diverticulitis or myriad other causes is common. *An excellent new research paper, a prospective cohort primary care study, has helped to explore what happens when people present to use with abdominal pain and to quantify their risk of developing cancer over the next year.* This research will help us to decide on investigation and referral decisions.

► **Intra-abdominal cancer risk with abdominal pain** [BJGP2022;72:e361](#)

- This was a primary care-based, prospective primary care cohort study by Prof Hamilton's team in Exeter, who have produced some fantastic research in recent years on cancer risk
- The aim of the study was to quantify the risk of intra-abdominal cancer in patients aged over 40 who present to us with newly reported abdominal pain
- The authors looked at any abdominal pain presentation, irrespective of other symptoms, signs or blood tests suggestive of cancer
- They looked at records of 126,000 patients aged over 40 presenting in primary care with new onset of abdominal pain and looked at the cumulative incidence of cancer over the next year
- **Results**
 - The risks of developing cancer over the next year were greater in men
 - *In men aged over 70 newly presenting with abdominal pain, 3.4% were diagnosed with cancer over the next year (expected annual incidence for this age group is 0.88%)*
 - This is greater than the 3% threshold generally set for urgent investigation
 - Having additional clinical features not surprisingly increased cancer risk
 - The most common cancer was, again not surprisingly, colorectal (1.9%) followed by pancreatic, oesophago-gastric and in women ovarian
 - In women aged over 70, the annual incidence of cancer was 2.3%
 - In men aged 60 to 70, their risk of cancer was only > 3% if they also had additional clinical features e.g. diarrhoea
 - Additional weight loss, nausea and/or vomiting increased risk with pancreatic more likely
- **Conclusions**
 - New onset of abdominal pain in men aged > 70 presenting in primary care, increases intra-abdominal cancer risk nearly fourfold and surpasses the 3% threshold warranting investigation
 - The threshold is only reached for those aged over 60 with additional clinical features
 - These results support appropriate testing and referral strategies for these patients, based on their individual clinical presentation and characteristics
 - The authors suggest non-invasive strategies first (such as FIT testing) before colonoscopy referral; CT scan or referral if pancreatic cancer or intra-abdominal lymphoma considered as possible; robust safety netting is needed and if ongoing concern and uncertainty, referral to one of the new Rapid Diagnostic Centres for evaluation is recommended

NB Learning Points & Quality Improvement Ideas

- New research reveals the risk in patients presenting aged over 70 with abdominal pain developing intra-abdominal cancer over the next year
- For men aged over 70 presenting with abdominal pain, their risk of cancer is 3.4% above the generally recommended 3% threshold for investigation
- Non-invasive investigations (such as blood tests and FIT test) are recommended first line, with robust safety netting
- Although colorectal cancer was most common, be wary of the risk of pancreatic cancer in patients with any associated clinical symptoms such as weight loss, nausea and of course new or recent onset of diabetes

FIT TESTS

The FIT test (faecal immunochemical test) for occult haemoglobin in the faeces has been around for a number of years now, both in the colorectal screening programme and at a lower threshold as a diagnostic test in symptomatic patients to help decide on referral decisions for possible colorectal cancer. ***However in the last year, it has become a real Hot Topic again with important new research, guideline changes and its role is evolving fast.*** NICE updated their guidance on FIT tests in their suspected cancer referral guideline [NICE NG 12 in 2021](#), and in 2022 announced a [review of the use of FIT testing](#) within the urgent/2-week wait referral pathway. They will be updating their guideline, due in November 2023. ***However, since then:***

- New guidelines have been published by the Association of Coloproctology of GB & Ireland and the British Society of Gastroenterology [ACPGBI & BSG FIT in patients with signs or symptoms of suspected colorectal cancer 2022](#)
- NHS England in October 2022 advised primary care to adopt the ACPGBI & BSG guidance above, NHSE [Using the FIT in the lower GI Pathway 2022](#)
- New guidance in Scotland, [FIT testing for patients with colorectal symptoms: primary care guidance 2022](#)
- There has been an editorial in the BJGP in August 2022 [FIT Test: challenges and opportunities for cancer diagnosis in primary care BJGP2022;72:366](#)

These introduce significant changes for how we should use the FIT test, summarised in the KISS box below. Important points raised in the BJGP editorial [BJGP2022;72:366](#) include:

- All bowel symptoms have poor sensitivity and specificity for colorectal cancer
- Until now we have always relied on 'red flag' symptoms or signs to prompt urgent referral; whilst these remain crucially important, 50% of newly diagnosed cases of colorectal cancer do not present with any red flag symptoms or signs at all
- ***A positive FIT test has greater sensitivity for the detection of colorectal cancer than any single or combination of symptoms, so we should 'adopt with confidence' this technology in primary care***
 - Recent research shows that 67% of primary care positive FIT tests were diagnosed with early-stage colorectal cancer, illustrating the potential of the test to improve outcomes
- ***A significant change, endorsed in the new guidelines above, and a significant change from previous NICE NG12 guidelines, is that FIT testing can also be used in patients with rectal bleeding***
 - Research from the [NICE FIT Study](#) shows that FIT testing performs equally well for cancer detection in patients with rectal bleeding as in those without; it shows that patients with rectal bleeding that are FIT negative are more likely to have a benign anorectal source of bleeding, and these patients can safely be investigated by flexible sigmoidoscopy
 - Patients with rectal bleeding who are FIT positive, suggests a higher source of the bleeding and these patients are more likely to have colorectal cancer and warrant urgent referral for colonoscopy
- ***However, we also need to be aware of the limitations of the test***
 - Remember that the usual FIT threshold for a symptomatic test of 10mcg of Hb/g of faeces is much lower than the threshold in the screening programmes (typically around 120 mcg/g) which is why we should always repeat the test in a patient with symptoms even if they have had a recent negative screening test
 - At the current threshold of 10 mcg/g for a symptomatic test, FIT has a sensitivity of 89% for colorectal cancer
 - This means that 11% of cancers may have a negative FIT test at this threshold and be missed
 - Clinical assessment remains important, and FIT testing should be adjunct to (and not a replacement for) clinical history and examination
 - FIT testing is less sensitive for other significant bowel disease, such as inflammatory bowel disease, so we still need access for non-urgent but timely specialist investigation in patients with symptoms who are FIT negative
 - For patients who test FIT negative but have continuing symptoms, robust safety netting needs to be put in place; the [Scottish Guideline 2022](#) and also [NHSE 2022](#) recommend ***a second repeat FIT test if there is ongoing clinical concern in patients who have had an initial negative FIT***

KISS: FIT testing for symptomatic patients in primary care

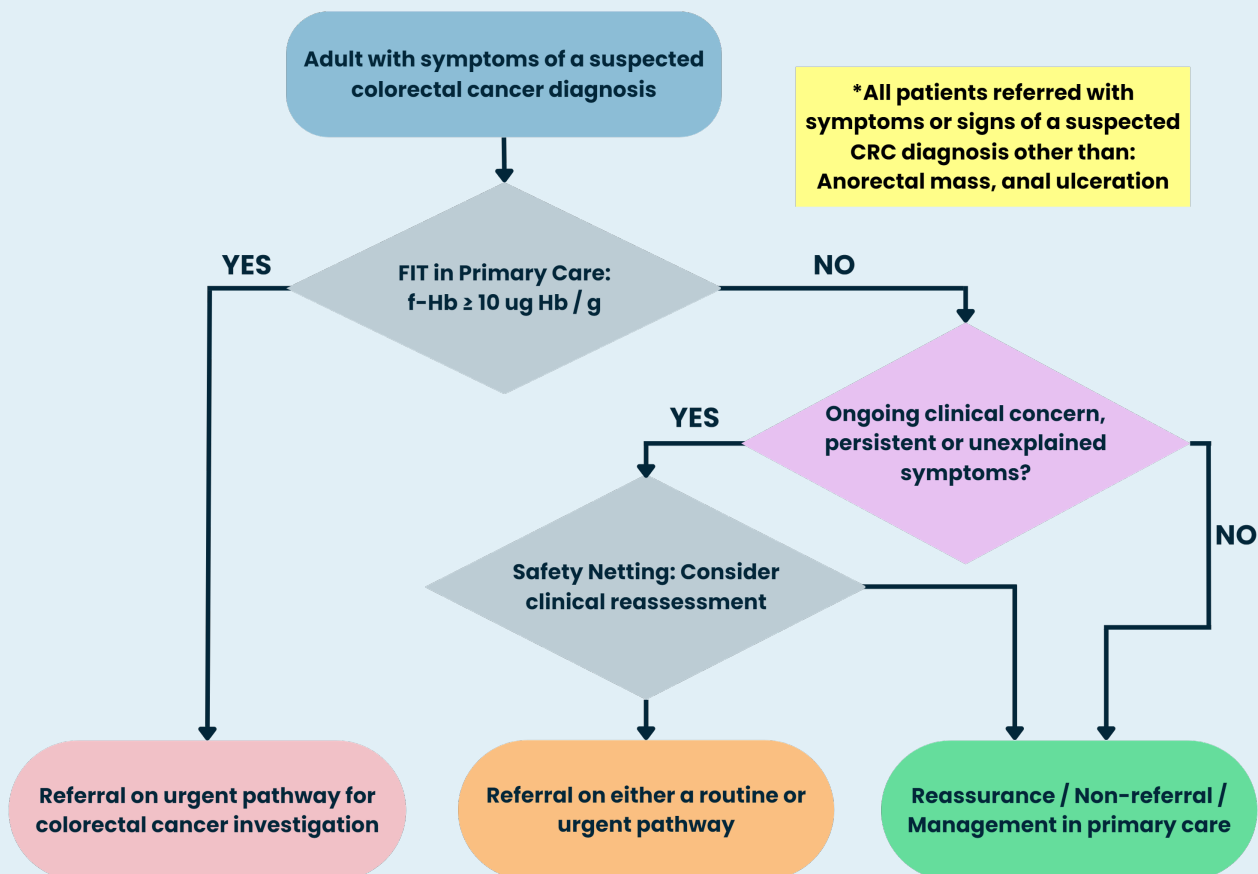
Based on [ACPGBI & BSG Guideline 2022](#); [BJGP2022;72:366](#); [NHSE 2022](#); [Scottish Guideline 2022](#)

■ **FIT testing is a quantitative test of mcg of Hb per g of faeces**

- Crucially, the threshold for a positive FIT test when used for diagnostic purposes in symptomatic patients is much lower (10mcg/g) compared to when it is used as a screening test in an asymptomatic patient (e.g. 120mcg/g in England)
- If a patient has symptoms that may be suggestive of colorectal cancer, and a recent negative screening FIT test then do not be reassured by this and repeat a symptomatic FIT test
- The evidence base is strong that FIT testing in primary care improves early diagnosis of colorectal cancer, and this technology should be 'adopted with confidence' in primary care

■ **2022/3 recommendations** from [ACPGBI & BSG 2022](#), [NHSE 2022](#) and [NHS Scotland 2022](#)

- **FIT testing should be used in primary care in all patients with possible clinical features of colorectal cancer to prioritise referrals to primary care**
 - see flowchart below; *patients with a positive test prioritised for urgent referral*
- **FIT testing can be used in patients with rectal bleeding**
 - if patients have a positive FIT, refer them urgently
 - if they have a negative FIT, the bleeding is likely to be benign and-rectal bleeding and these patients may be referred for flexible sigmoidoscopy
- **Robust safety netting is needed**
 - If patients have a negative FIT test, but persistent and unexplained symptoms and the GP has an ongoing clinical concern they should be referred to secondary care for evaluation
 - *Consider a second FIT test in these patients*
 - Patients need to be advised why we are doing the FIT test, and why it is important to return it
 - *Systems need to be in place so that patients who do not return a FIT test are followed up*



NB Learning Points: FIT testing

- The evidence base is strong that FIT testing in primary care improves early diagnosis of colorectal cancer, and we should 'adopt with confidence' this technology in primary care
- A positive FIT test has greater sensitivity for the detection of colorectal cancer than any single or combination of symptoms,
- Crucially the cut-offs are set higher in the screening programme (typically 120mcg/g) than the diagnostic test (typically 10 mcg/g) so do not be falsely reassured by a symptomatic patient who reports a recent 'normal' screening test.
- A significant change is that FIT testing can also be used in patients with rectal bleeding
- At the current threshold of 10 mcg/g for a symptomatic test, FIT has a sensitivity of 89% for colorectal cancer
 - This means that 11% of cancers may have a negative FIT test at this threshold and be missed, hence the need for safety netting and follow-up and consider a second repeat test if ongoing symptoms
- **Bottom lines:**
 - We need to embrace FIT testing in symptomatic patients including those with rectal bleeding, and refer those urgently who are positive
 - We need to make sure patients understand that this is a test for cancer and follow up those who do not return a test and have robust safety netting in place
 - We need to make sure that patients with a negative test but ongoing symptoms and clinical concern are not denied further investigation, and that a second repeat test may be useful in these patients

PANCREATIC CANCER

If there is one statistic that all GPs are only too painfully aware of, it is this: *pancreatic cancer has the worst survival rate of all cancers in the UK*. Recent data [Lancet 2020](#) show that late diagnosis is the norm, with 85% of cases having advanced and non-resectable disease at diagnosis. As a result of this, there is a dismal survival rate of 3% to 15% reported for 5 years, and 5% for 10 years. Despite significant recent advances in early diagnosis and treatment translating into improved outcomes for other gastrointestinal malignancies, there has been no improvement in the rates of early diagnosis or survival of pancreatic cancer. Furthermore, the incidence of pancreatic cancer has risen by 17% over the last 30 years. By 2030 it is predicted that pancreatic cancer will overtake breast cancer as the 4th most common cancer killer in the UK, and approximately a third of deaths are in adults aged under 65. We all currently have a 1.3% lifetime chance of developing the disease, significantly greater if a positive family history or familial syndrome such as Lynch syndrome.

As every GPs knows, early diagnosis of pancreatic cancer is incredibly difficult because of the vague and protean ways it presents compounded by the lack of an early test to help identify possible cases. Over half of cases are diagnosed at emergency hospital presentations, and research shows that patients have a median of 3 GP visits before diagnosis. We have known for a long time that a new diagnosis of type 2 diabetes is associated with pancreatic cancer and is a potential indicator, but T2DM is very common and pancreatic cancer is rare so we need additional symptoms or markers. In the absence of a screening test or programme, a lot of research is going on looking at numerous bio-markers that could assist earlier diagnosis (including the [GALLERI test](#) looking at abnormalities in DNA that could indicate the presence of multiple cancers, see our [NB blog](#) on this) but until these are available we need to rely on early clinical detection based on symptoms. The problem of course is that symptoms are so vague and non-specific.

To help us, this new research has been published - the largest study to date systematically examining the symptomatology of new diagnoses of pancreatic cancer in primary care. There has also been new research looking at the potential role of the tumour marker CA 19-9 as a screening test.

► *Identifying symptoms associated with pancreatic neoplasms* [BJGP2021;71:e836](#)

- The aim of this study was to investigate symptoms associated with a new diagnosis of pancreatic cancer. Patients diagnosed with pancreatic cancer were matched with 10 controls from the same general practice
- Alarm symptoms associated with pancreatic cancer were not surprisingly jaundice and weight loss
- Other symptoms which may be less commonly attributed to pancreatic cancer but were associated with it included *new gastrointestinal bleeding, abdominal pain, thirst, dark urine, appetite loss, dyspepsia, nausea and steatorrhoea within the last year*
- **Conclusion:** Pancreatic cancer presents in primary care with a broad range of clinical symptoms, most of which are non-specific and do not qualify for urgent referral. *In the absence of clear red flag symptoms such as jaundice and weight loss, GPs should be vigilant of patients presenting with several concurrent non-specific symptoms, especially in the presence of co-morbidities such as type 2 diabetes.* Asking patients about steatorrhoea, dark urine and thirst is important. These patients should be referred for an urgent CT scan, or if not available on a suspected cancer pathway.

► *Does testing for CA 19-9 have a role for diagnosis in primary care?*

CA 19-9 is a protein that is a well-established tumour marker for pancreatic cancer. It is used as part of the work up with new cases of pancreatic cancer, to monitor response to treatment and to check for recurrence. However, because it can be raised in healthy people and in other conditions, including other GI cancers and benign conditions of the liver and pancreas such as biliary disease or chronic pancreatitis (i.e. false positives due to low specificity) and because some people with pancreatic cancer have normal CA 19-9 (i.e. false negatives with low sensitivity) it has not been recommended as a test that we should do in primary care.

Some recent research challenged that assumption [Gastroenterology 2021;160;1373](#) suggesting that it may be useful as a screening test in higher risk patients. However, it was a screening study so is not applicable to a UK primary care population and the positive predictive value of a test was incredibly low (approx 2%) meaning that there would be large numbers of false positives and some false negatives.

So, the bottom line is that we should continue to NOT use CA 19-9 as an initial test in primary care for possible pancreatic cancer due to its very low positive predictive value. If we suspect clinically based on symptoms and other bloods that pancreatic cancer is possible, we should order an urgent CT scan or refer to a rapid diagnostic centre.

NB Learning Points & Quality Improvement Ideas

- Be vigilant to the risk of patients having pancreatic cancer when they present with several of these concurrent non-specific symptoms
- Consider the possibility of the diagnosis if patients present with abdominal pain, thirst, dark urine, appetite loss, dyspepsia, nausea and steatorrhoea within the last year
- Be aware of the association with recently diagnosed Type 2 Diabetes
- CA 19-9 is not an appropriate diagnostic test for pancreatic cancer in primary care due to high numbers of false positives and some false negatives; if you suspect it, order urgent CT scan or refer to rapid diagnostic centre
- In the near future, cell free DNA testing offers real hope of earlier clinical diagnosis

OVARIAN CANCER

Along with pancreatic cancer, ovarian cancer is another covert malignancy associated with late diagnosis and poor prognosis. It is the most deadly of all gynaecological cancers [Lancet 2021;397:2182](#), with most patients (58%) diagnosed at an advanced stage (stage 3 or 4) and with an overall 5-year survival of 43%. Survival is strongly linked to stage at diagnosis:

Ovarian cancer stage at diagnosis and prognosis [Lancet 2021;397:2182](#)

Stage	5 year survival
1	> 90%
3	27%
4	13%

Recent papers and an editorial in the BJGP have explored how we may improve earlier detection in primary care.

► ***Will screening improve early diagnosis of ovarian cancer?*** [Lancet 2021;397:2182](#)

Sadly, it appears not. The long-awaited **UK Collaborative Trial of Ovarian Cancer Screening** has recently been published in the [Lancet 2021;397:2182](#). Ovarian cancer would appear to fulfil most of the criteria for a screening programme, with earlier diagnosis linked to improved survival and CA-125 and transvaginal ultrasound (USS) available as easy-to-perform tests. So, the results of this large trial with long follow-up have been eagerly awaited.

- This was a large RCT of 200,000 postmenopausal women aged 50 to 74 in the UK
- They were randomised to annual transvaginal ultrasound (USS), annual multimodal screening (MMS) which consisted of CA-125 +/- USS based on the risk of ovarian cancer ROCA score and no screening
 - The [ROCA](#) score uses serial CA-125 measurements together with age, menopausal status and risk status which had been shown in earlier trials to be more accurate at detecting ovarian cancer at an earlier stage
- At a median follow-up of 16 years, 1% of women in all 3 groups were diagnosed with ovarian cancer
- In the MMS group compared with no screening, there was a 47% increase in stage 1 diagnoses and a 25% decrease in stage 4 diagnoses, which seems encouraging but...
- ***There was no difference in overall or cancer-related mortality (0.6%) between all 3 groups***
- ***Conclusion: A reduction in the incidence of stage 3 and 4 ovarian cancer was seen in the multimodal screening group, however, this was insufficient to translate into a reduction in ovarian cancer or overall mortality. Given that screening did not reduce mortality general population screening cannot be recommended.***

► ***So, how can we detect ovarian cancer earlier in primary care?*** [BJGP2022;72:312](#)

The editorial points out that with the disappointing results of this large screening trial, we shall have to continue to rely on earlier clinical detection. Promising new ovarian cancer-specific tests and multi-cancer panels are being developed (including DNA tests, see the early diagnosis of cancer chapter) but time will be needed for them to be rigorously evaluated.

For the next decade or so we shall have to continue to rely on earlier clinical detection in symptomatic patients. Ensuring primary care assessment, triage and referral is as accurate and timely as possible will be the key to reducing diagnostic delay and improving outcomes. However, the problem we face in primary care of course is the non-specific nature of symptoms in women with ovarian cancer. Delays in diagnosis are common, especially in women who have an initial CA-125 test < 35 (below the NICE threshold to prompt an urgent USS) who take twice as long to be diagnosed with ovarian cancer compared to women with a CA-125 > 35.

► *How then should we approach investigation in women with possible early symptoms?*

The current [NICE NG12](#) guidelines for early cancer detection recommend a CA-125 test for women aged over 50 with persistent abdominal distention or bloating, early satiety or loss of appetite, pelvic or abdominal pain, increased urinary urgency and/or frequency or new symptoms suggestive of IBS. They also recommend a CA-125 test if a woman reports unexplained weight loss, fatigue or changes in bowel habits. If the CA-125 is ≥ 35 , then to arrange an urgent USS. By contrast, the [Scottish guidelines](#) recommend a **parallel approach** of CA-125 AND a transvaginal USS as initial investigations.

The editorial [BJGP2022;72:312](#) references recent data which has elucidated the performance of CA-125 in primary care [PLOS 2020](#). This was a very useful cohort study:

- CA-125 tests were requested in primary care for women presenting with a broad range of symptoms including abdominal pain, bloating, change in bowel habit, fatigue, pelvic pain, weight loss, urinary symptoms and appetite loss
- **Overall, of women found to have CA-125 levels ≥ 35 , 10% were diagnosed with ovarian cancer and a further 12% with another form of cancer**
 - Of women aged over 50 with a CA-125 ≥ 35 , 15% were diagnosed with ovarian cancer
 - Of women aged under 50 with a CA-125 ≥ 35 , 3.4% were diagnosed with ovarian cancer
- In women aged over 50 with a raised CA-125 who did NOT have ovarian cancer, 20% were diagnosed with other non-ovarian cancers
 - The most common non-ovarian cancers were pancreas, lung, gastrointestinal and uterus
- Overall CA-125 with a cut-off value of 35, had a sensitivity of 77% for all ovarian cancers and 85% for the invasive subtypes responsible for most ovarian cancer mortality.
- **Conclusion:** *CA-125 is a useful test for ovarian cancer detection in primary care, particularly in women aged over 50. Clinicians should also consider non-ovarian cancers in women with a raised CA-125 if ovarian cancer is excluded*

So, while this data shows that CA-125 is a useful test in primary care for investigating possible cases of ovarian cancer, a sensitivity of 77% means that *a significant proportion of cases will be in the normal range and may be missed as under NICE guidelines they would not be referred for a scan*. The paper, therefore, argues that the NICE NG12 recommendations, which were developed a decade ago, are now out of date and in light of this new evidence a review of them should be prioritised. They suggest that a lower CA-125 cut-off and less stringent criteria for an urgent 2WW referral may be needed (subject to an updated health economic assessment). The paper states that performing CA-125 and USS in parallel rather than sequentially, as in Scotland, should improve the sensitivity of the diagnostic tests and may shorten primary care diagnostic intervals. Of course, the potential for earlier diagnosis must be balanced against the potential harms of over investigation and there are resource implications with scanning availability.

The authors also point out that the risk of ovarian cancer varies by age and also by CA-125 level, so using a single CA-125 cut-off across all ages is an over simplification. So, another potential strategy is to use an individual risk-based triage tool based on CA-125 and the patient's age based on this published research [PLOS 2020](#). The probability of ovarian cancer at a given CA125 level varied markedly age. For example, the CA-125 level to reach the 3% probability of ovarian cancer (which is the recommended risk threshold to trigger an urgent referral by NICE) fell from 100 in a woman aged 40 to 32 in a woman aged 70. The authors propose that an estimate of cancer risk based on age and CA-125 could be provided along with the blood test result to help inform referral decisions. As a significant proportion of women with high CA-125 levels have other cancers (20% of women aged over 50 with a CA-125 ≥ 35) referral to one of the new rapid diagnostic centres (where investigations are performed for multiple potential cancers simultaneously) would be appropriate.

NB Learning Points & Quality Improvement Ideas:

- The prognosis for ovarian cancer remains poor, and is strongly linked to early-stage at diagnosis
- Sadly a large, long-term UK screening trial using CA-125 and USS has been shown NOT to reduce ovarian cancer mortality, so we shall have to continue to rely on earlier clinical detection
- Early ovarian cancer is linked with a broad range of non-specific symptoms including abdominal pain, bloating, change in bowel habit, fatigue, pelvic pain, weight loss, urinary symptoms and appetite loss
- Delays in diagnosis are common, *especially in women who have a CA-125 level below the current NICE threshold of 35 to trigger an urgent referral*
- Recent primary care research has shown that CA-125 is a useful primary care test in patients with a broad range of non-specific pelvic, abdominal and general symptoms
 - Overall a CA-125 level > 35 has a sensitivity of 77% for all ovarian cancers, and 85% for invasive subtypes
 - This means that whilst CA-125 will detect most ovarian cancers in primary care, a significant proportion have a CA-125 < 35 and will be missed
 - A raised CA-125 is also linked with other non-ovarian cancers; 20% of women with a CA-125 aged over 50 with a CA-125 > 35 who have ovarian cancer excluded are subsequently diagnosed with other cancers; these are most commonly pancreatic, lung, gastrointestinal and uterus; referral of these women to a rapid diagnostic clinic for simultaneous testing for different cancers could facilitate earlier diagnosis
- The NICE guidelines for ovarian cancer need revision in light of this new evidence, and the BJGP paper suggests that in the meantime a parallel testing strategy (as used in Scotland) with CA-125 AND ultrasound should improve the sensitivity of primary care testing and shorten diagnostic intervals

MYELOMA

Myeloma has one of the longest diagnostic delay intervals of all cancers, and yet early diagnosis is crucial in limiting complications such as lytic bone disease, pathological fractures kidney injury and infection risk. Earlier diagnosis has further been hampered by the COVID-19 pandemic and confused further still by the increasing diagnosis of MGUS (monoclonal gammopathy of uncertain significance). A recent BJGP editorial entitled *Diagnosing myeloma in general practice: how might earlier diagnosis be achieved* has explored this further [BJGP 2022;72: 462](#), and there has been primary care research on appropriate blood tests to rule in and rule out myeloma [BJGP2018;e586](#).

Background points: Myeloma & MGUS

Based on [BJGP 2022;72:462](#), [NICE, NG35](#), [CKS 2022](#) & [BJGP2018;e586](#)

Myeloma

- Is caused by a clonal proliferation of monoclonal plasma cells in the bone marrow which secrete immunoglobulins (abnormal monoclonal immunoglobulins, known as M proteins or paraproteins).
- These paraproteins increase plasma viscosity and cause renal damage and the proliferation of plasma cells leads to bone marrow suppression and can cause hypercalcaemia.
- The median age of diagnosis is 70; only 15% of patients are aged under 60.
- A positive family history increases the likelihood of the diagnosis.
- 'Symptom awareness' for both patients and GPs is key to earlier diagnosis
- The typical features of myeloma give rise to the symptoms of:
 - Fatigue from anaemia and/or renal failure.
 - Bone pain from skeletal involvement, often lower back.
 - Headache and cognitive impairment from hyperviscosity.
 - Nausea, from hypercalcaemia or uraemia.
 - Confusion, constipation, thirst, and muscle weakness due to hypercalcaemia.
 - Dyspnoea from anaemia.
 - Infections due to immune suppression and leucopenia.
 - Weight loss due to advanced multifocal disease.
- Patients often present late with complications from hypercalcaemia, pathological fractures or renal failure; spinal cord compression is a late complication, beware with sensory loss, paresthesia, limb weakness, walking difficulty and sphincter disturbance.
- Prognosis is very variable, with survival times varying from a few weeks to 20 years; most respond to initial treatment and have a period of stability before relapse after 2 to 5 years; subsequent relapses become less responsive to treatment; younger fitter patients who have high dose therapy and stem cell transplantation can expect to survive for a median of 7 years from diagnosis.

MGUS = monoclonal gammopathy of uncertain significance

- MGUS is a non-malignant condition with an abnormal paraprotein or M-protein; it is common, with a prevalence of 3% to 5%
- It is an asymptomatic, usually chance finding on blood or urine electrophoresis
- Patients have a small **chance of progression to myeloma or lymphoma**, at a rate of ~1% per year (so younger people have a higher lifetime risk of progression), and the risk of progression to myeloma is also dependent on both the level and the type of M-protein (e.g. IgA and IgM MGUS more likely to progress than IgG MGUS).

► *Diagnosing myeloma in GP: how might earlier diagnosis be achieved* [BJGP 2022;72: 462](#)

The editorial discusses the challenges of earlier diagnosis in patients with non-specific symptoms such as back and bone pain, fatigue, and blood test abnormalities such as anaemia, renal impairment and raised inflammatory markers all of which are common in an ageing population. The COVID-19 pandemic resulted in a significant reduction in referrals for suspected myeloma which have not recovered, suggesting a pool of patients with undiagnosed disease yet to be discovered, and the subsequent changes in triage and consulting such as e-consults may not prioritise vague symptoms such as fatigue and bone pain.

Symptom awareness is key to earlier diagnosis, and a 'fundamental challenge' is for clinicians to consider the diagnosis and initiate appropriate investigations with serum protein electrophoresis, immunofixation and serum-free light chain assays are sensitive and specific initial tests for the majority of patients followed by referral for bone marrow biopsy.

► *Early detection of myeloma in primary care* [BJGP2018:e586](#)(11)

Because of the non-specific nature of the presenting symptoms of myeloma, we obviously have to rely on blood tests, especially inflammatory markers. This original research set out to answer the important questions: *what are the best tests for the initial investigation of possible myeloma, and when can we be confident that we have ruled it out?*

- This was a UK-based primary care case-control study; symptoms and blood tests were analysed in 3,000 cases of myeloma in the 5 years pre-diagnosis and matched with 12,000 controls
- **Results:**
 - **Symptoms:** non-specific symptoms such as fatigue, back pain, rib pain, chest infections and nose bleeds became more common in myeloma cases than controls up to 2 years before diagnosis
 - **Blood tests:** In the year before diagnosis 85% of patients had an abnormal ESR, compared to 46% of controls; 81% had an abnormal plasma viscosity (PV) compared to 41% of controls; 46% had an abnormal CRP compared to 37% of controls
 - No single test was able to rule out myeloma, but the best 'rule out' test combinations were when Hb, ESR and Ca are all normal
 - A negative calcium or creatinine result does NOT rule out myeloma as these rise late in the disease process, whereas Hb starts to drop earlier in the disease process
 - In the year before diagnosis, 95% of patients with myeloma have an abnormal FBC and/or ESR
- **Conclusion:** ESR and PV are better than CRP for both ruling in and ruling out the disease; a combination of a normal ESR or PV and a normal haemoglobin is a simple rule-out approach in primary care

What are the implications of this study for primary care?

The authors propose that the combined use of FBC, ESR or PV and calcium 'appears to be sufficient to select those who may have the disease'. Those with abnormalities on these tests can then proceed to more specific testing for protein electrophoresis, light chain assays and urine Bence Jones protein in primary care. Those who have normal initial tests can be reassured and another potential cause should be sought for their symptoms. The finding is important that ESR and PV in primary care have a better diagnostic performance for myeloma, and that CRP has very little value in myeloma diagnosis. Importantly a negative calcium or creatinine result does NOT rule out myeloma as these rise late in the disease. However, the finding of hypercalcaemia and raised creatinine is highly suggestive of late disease and should prompt protein electrophoresis and urine Bence Jones protein prior to urgent referral.

NB Learning points: Myeloma

- Myeloma has one of the longest diagnostic delays of all cancers and yet earlier diagnosis significantly improves outcomes
- Symptom awareness and to consider the diagnosis and trigger appropriate investigations is key
- A normal FBC, ESR and Calcium is sufficient to rule out myeloma in most patients presenting in primary care; CRP is NOT a useful rule-out test
- Abnormality in FBC, ESR or calcium or significant clinical suspicion should prompt testing for protein electrophoresis, serum-free light chain assays and urine Bence Jones protein; these are sensitive and specific initial tests to prompt urgent referral for bone marrow biopsy
- Calcium and creatinine rise late in the disease process, whereas haemoglobin falls at an earlier stage

KISS: Myeloma and MGUSBased on [BJGP 2022;72:462](#), [NICE, NG35](#), [CKS 2022](#) & [BJGP2018:e586](#)**Myeloma**■ **Diagnosis in primary care:**

- Consider possible myeloma in adults (especially those aged over 60) presenting with non-specific symptoms e.g. fatigue, bone pain, recurrent infection, headaches etc.
- Check FBC, ESR, renal function and calcium studies in all patients:
 - A normal FBC, ESR and Ca is sufficient to rule out the disease in most patients presenting in primary care with non-specific symptoms.
- If FBC, ESR, creatinine or calcium are abnormal or if index of suspicion is higher arrange serum electrophoresis, serum-free light chain assays and Bence-Jones protein urine tests:
 - Serum electrophoresis and Bence-Jones protein are negative in 2% of people who have a non-secretory form of the disease, so refer if significant clinical suspicion persists.
- Arrange XRays of symptomatic areas of bone pain to exclude pathological fracture.
- Arrange urgent admission if hypercalcaemia ($\geq$ corrected calcium 3.0 mmol/L) or acute kidney injury.
- If serum and/or urine protein electrophoresis suggest myeloma urgent 2WW haematology referral.

■ **Secondary care management:**

- Diagnostic confirmation via further blood tests, bone marrow aspirate and biopsy and skeletal MRI.
- Secondary care management depends on disease stage, prognosis and co-morbidities but may include:
 - Bisphosphonates; corticosteroids; chemotherapy; immune-modulating drugs; VTE prophylaxis.
 - High-dose chemotherapy and stem cell transplant may be offered.

■ **Primary care management:**

- On-going **support and holistic care**, including lifestyle advice e.g. maintaining good hydration status.
- Identification and prompt investigation of **possible new complications**, including pathological fractures, spinal cord compression, impaired immunity, anaemia, bleeding disorders, acute kidney injury, cognitive impairment and stroke due to hyperviscosity.
- **Prompt treatment of infections** with broad-spectrum antibiotics.
- **Vaccination**, seasonal influenza and also pneumococcal.
- **Pain control** with paracetamol +/- opiates eg codeine (avoid NSAIDs due to renal risk).
- Management of **depression and anxiety**.
- **Supportive and palliative care**.

MGUS

- If in doubt haematology advice should be sought but for **patients at low risk of progression** (no symptoms and either IgG M-protein <15g/L or IgA/IgM M-proteins <10g/L, and generally older patients) **primary care follow-up may be appropriate:**
 - Clinical review with FBC/U&E/calcium/serum electrophoresis every 3-4 months for 1st year then 6-12 monthly thereafter.
- **Refer back to haematology** if any of the following develop:
 - If M-proteins increase by >25% (min absolute increase 5g/L).
 - If symptoms compatible with myeloma or lymphoma develop.
 - If unexplained anaemia, cytopenia, abnormal renal function or hypercalcaemia develop.
- Excellent information on MGUS for patients (and clinicians!) from [Myeloma UK here](#)

PSA TESTING AND PROSTATE CANCER

Prostate cancer remains a true diagnostic quandary for us in primary care. We and our patients face a number of difficult dilemmas:

- Prostate cancer is a serious disease and a major cause of cancer death, indeed the second most common cause of cancer death in men accounting for 13% of all male cancer deaths, thus causing a lot of media attention and understandable patient anxiety; and yet it is a disease with very variable rates of clinical progression, with slow-growing and relatively indolent tumours being common and many men facing an unnecessary diagnosis of prostate cancer with associated anxiety, medicalisation and risk of iatrogenic harm from investigation and treatment.
- Prostate cancer mortality [Prostate cancer mortality](#) has dropped by >20% in the last 15-20 years yet [incidence](#) has gone up by close to 50% over a similar time frame; this means we are diagnosing a lot of men with prostate cancer without altering their outcome.
- Lower urinary tract symptoms are so common that they are almost universal in men over a certain age, and yet they correlate poorly with underlying disease (see information below on LUTS in men).
- The PSA test has both poor sensitivity and (especially) specificity.
- As a result of these problems, teasing out who has clinically significant prostate cancer without over-diagnosing the majority who don't, remains, as yet, an unsolved puzzle.

These dilemmas and diagnostic challenges for us are of course not new, but NICE have undertaken a review of the evidence on prostate cancer and recently published updated guidance to both the [prostate cancer specific guideline NG131](#) and to the [suspected cancer referral guideline NG12](#). The [Prostate Cancer Risk Management Programme \(PCRMP\)](#) guidance has also been updated in May 2022 with some changes to the wording and guidance:

KISS: Asymptomatic Men who ask about PSA testing

[Prostate Cancer Risk Management Programme 2022](#)

- **Who is this guidance for?**
 - This guidance is for men who do NOT have symptoms of prostate disease (although as we will discuss towards the end of the chapter this may be a spurious distinction) who are aged >50.
 - Use clinical judgement and consider for men aged <50 with additional risk factors (see below).
- **Test or not to test?**
 - There remains insufficient evidence to recommend population screening, and it is not recommended by the [UK National Screening Committee](#)
 - Men are at higher risk if they have a close relative with the disease or are of black ethnic origin (lifetime risk is 1 in 4, as opposed to 1 in 8 for white men).
 - There is an updated [PSA Patient Information Leaflet](#) to help men make an informed choice.
 - PSA tests do not distinguish between aggressive and more indolent cancers.
- **Before a PSA test**, to avoid falsely raised levels, men should not have:
 - Had UTI within the last 6 weeks.
 - Ejaculated within the previous 48 hours.
 - Exercised vigorously (esp cycling) in the previous 48 hours.
 - Had urological intervention within the last 6 weeks (e.g. prostate biopsy).
- **The results and what to do with them:**
 - Most men will have a PSA level < 3 ng/ml.
 - ~3/4 of men with a raised PSA level ≥ 3 ng/ml will NOT have prostate cancer (v poor specificity).
 - Previously the guidance was quite clear that if PSA level was ≥ 3 ng/ml we should refer.
 - However, **now the guidance simply advises that if the PSA level is raised** (with no clarification on what that level is, but presumably the implication is if ≥ 3) **we should 'use clinical judgement to decide whether to refer'**.
 - But it is possible that many referral pathways will now use the NICE age-adjusted PSA levels...



► **NICE updates December 2021** [Prostate cancer NG131](#) [Referral for suspected cancer NG12](#)

NICE have undertaken a review of the evidence on prostate cancer and very recently published updated guidance to both the [prostate cancer-specific guideline NG131](#) and to the [suspected cancer referral guideline NG12](#).

The prostate cancer-specific guidance has little relevance to primary care and deals more with risk assessing those with confirmed prostate cancer and treatment options, but there *has been some subtle but important changes in NG12 which will have direct relevance for us in primary care.*

Suspected Prostate Cancer, Referral Guidance

NICE, NG 12 Update December 2021

- Refer people on a suspected cancer pathway if their prostate feels malignant (hard, irregular or nodules) on digital rectal examination (DRE)
- **Consider** a PSA test and DRE to assess for prostate cancer in people with:
 - Any lower urinary tract symptoms (LUTs), or erectile dysfunction, or visible haematuria
- **Consider** referring people if their PSA levels are above the threshold for their age in the table below, taking the person's preferences and co-morbidities into account
- NB: *the wording of NICE recommendations is important. 'Offer' is used where there is clear and strong evidence of benefit, and 'consider' where the benefit is less certain*

Age	PSA threshold (micrograms/litre or ng/ml)
Below 40	Use clinical judgement
40 to 49	More than 2.5
50 to 59	More than 3.5
60 to 69	More than 4.5
70 to 79	More than 6.5
Above 79	Use clinical judgement

- The [Scottish Referral Guidelines for Suspected Cancer](#) have lower thresholds
 - Less than 60 years, PSA > 3
 - Aged 60-69, PSA > 4
 - Aged 70-79, PSA > 5
 - For older men, use clinical judgement and the principles of [Realistic Medicine](#) to decide on referral
 - Aged 80-85, PSA >10
 - Aged > 86, PSA > 20

Why have NICE made these changes?

Despite the fact that we have been dealing with PSA testing for years **NICE conclude that the diagnostic accuracy of PSA thresholds remains uncertain** because most studies are based on populations that had already been referred to secondary care. They also noted a number of the issues with PSA testing e.g. that some cancers are slow-growing and may not impact a person's life expectancy and given that, some people might choose not to be referred for invasive investigations and treatment. In the previous version of the guideline, NICE recommended referral if PSA levels were above the age-specific reference range, however, **they now recommend that referral should be *CONSIDERED* based on PSA thresholds**. They did not make a stronger recommendation because of the uncertainty in the evidence and the likely low positive predictive value of the PSA test.

This subtle change in wording to CONSIDER referral (rather than just 'refer') based on age-adjusted thresholds reflects a change of direction from NICE. *The implication for us is to take a more patient-centred approach to referral, and that we should take the person's preferences and any comorbidities into account before considering referral.*

NICE also highlight the important risk factors of ethnicity and family history and thus we should take those factors into account when assessing someone's risk. This is **particularly important for men of black ethnicity.**

NICE also looked specifically at the evidence for referrals based on fixed PSA thresholds vs age-adjusted thresholds. There was no strong evidence to differentiate between the two, including no cost-effectiveness evidence. *But the conclusion was that fixed PSA thresholds would likely pick up more prostate cancers but would also lead to significant over-diagnosis and over-treatment, and would put substantial pressures on NHS resources.*

This gives us a dilemma as the PCRMP guidance mentioned above is to refer if PSA is ≥ 3 ng/ml. Technically the PCRMP guidance refers to asymptomatic men and the NICE guidance refers to symptomatic men, but given the uncertainty as to whether there is a genuine association between LUTS and clinically significant prostate cancer, it is clinically difficult to differentiate between the two. *However, it would seem sensible to refer based on the more recent NICE guidance obviously taking into account other risk factors, examination findings and patient preference.*

► ***Is knowing a baseline PSA in middle-aged men useful to guide future PSA testing?***

Yes, according to this new research [JAMA 2020](#)

- There is a growing evidence base that PSA levels in mid-life are associated with an increased risk of developing clinically significant prostate cancer in the future, and *using this information could help us to offer a more targeted future testing strategy for patients with higher baseline levels (but below referral threshold), whilst reassuring patients with low baseline levels that future testing may not be needed*
- This was a secondary analysis of data from the large PLCO (prostate, lung, colorectal & ovarian cancer) screening trial
- They examined data from 11,000 men aged 55 to 60, looked at their baseline PSA levels and followed them up over the next 13 years to see what the incidence of clinically significant prostate cancer was
 - Clinically significant was defined as metastatic disease, disease with metastatic potential or Stage 2 or Gleason score > 7
- **Results:**

PSA Level aged 55 to 60	13 year incidence of clinically significant prostate cancer
< 1	1.5%
1-1.9	5.4%
2 - 2.9	10.6%
3-3.9	15%
≥ 4	29.5%

- **Conclusions:**
 - Higher baseline levels of PSA in men aged 55 to 60 are associated with an increased risk of clinically significant prostate cancer over the next 13 years.
 - These findings can be used to implement a more targeted testing strategy e.g. discontinued in men with a baseline < 1 , less frequent future testing with baseline of 1-2, and more regular testing in men with a baseline level > 2 .
 - Using a more targeted testing strategy tailored to individual's baseline needs is likely to reduce overdiagnosis of indolent and harmless tumours and improve detection of clinically significant disease.

► *LUTS in men: should I have a PSA test doctor?*

[BMJ 2018;361:k1202](#) [Br J Gen Pract 2018; 68 \(676\)](#)

Lower urinary tract symptoms (LUTS) are commonly thought to be associated with prostate cancer by patients. But this is generally NOT the case, as highlighted by the excellent information by [Cancer Research UK](#). So should we '*consider a PSA test in men with any lower urinary tract symptoms*' as recommended by NICE NG12? This important clinical question was reviewed in both the uncertainties section in the BMJ and the BJGP in 2018. The key points from these reviews are as follows:

- LUTS in men are very common and estimated to be present in ~80% of men >60.
- Indolent prostate cancer (one that will not cause morbidity or mortality) is also common, autopsy studies estimate the rate of indolent prostate cancer to be ~60% in men >79 years.
- Given the high prevalence of LUTS and indolent prostate cancer in older men, a link between the two is unavoidable leading to a high risk of overdiagnosis if LUTS are investigated with PSA.
- Most international guidelines (including NICE suspected cancer guidelines) recommend doctors consider doing a PSA in men with LUTS for prostate cancer detection, but it is uncertain if men with LUTS have an increased risk of clinically significant prostate cancer or not.
- The current evidence is mainly limited to observational studies of PSA-screened populations, and although some evidence shows a positive association with LUTS and prostate cancer, this is mostly due to the detection of localised prostate cancers many of which will turn out to be indolent.
- ***Currently, there is no significant evidence to suggest that men with self-reported LUTS are at an increased risk of advanced or potentially fatal prostate cancer when compared to men without LUTS although the data is heterogeneous and incomplete.***

So, what should we do in light of this uncertainty with a lack of evidence to guide us? The authors of these papers offer the following conclusions and strategies:

- Early prostate cancer is unlikely to have any symptoms and in most cases, LUTS will be due to causes other than prostate cancer.
- PSA testing to investigate LUTS may lead to the detection of prostate cancer that would not have gone on to cause harm; this may lead to overtreatment, side effects, and psychological distress.
- ***Do not routinely offer PSA testing in men with LUTS and no risk factors for prostate cancer.*** Explore why the patient is concerned about prostate cancer and discuss with them the lack of evidence that LUTS in isolation suggests prostate cancer, explain limitations of the PSA test including risks of overdiagnosis and related treatment as part of shared decision making.
- Assess risk factors, and take these into account: age, family history of prostate cancer, ethnicity, BMI; offer digital rectal examination and do a urine dipstick test.

NB Learning Points: Prostate Cancer

- We have newly updated guidance on PSA testing from the [Prostate Cancer Risk Management Programme \(PCRMP\)](#) in May 2022 - in theory **this guidance is for men with no symptoms** of possible prostate disease but given the ubiquitous nature of LUTS as men get older it is uncertain whether this is a spurious distinction; clinical judgement will be needed as to when to apply the PCRMP guidance or the updated NICE NG12 'symptomatic' guidance.
- The PCRMP guidance on when to refer has changed quite significantly - previously the guidance was clear that we should refer if PSA was ≥ 3 ng/ml where as now it states that if the PSA is raised (not specified but the implication is that ≥ 3 ng/ml is raised) **we should 'use clinical judgement to decide whether to refer'**.
- GPs should not be proactively recommending PSA testing for men but can offer the test in men >50 who request it, and in younger men with risk factors (family history, black ethnic origin) after a discussion about the pros/cons based on an informed choice. There is an updated [PSA Patient Information Leaflet](#) to help men make the choice.
- Be aware many men want the PSA test because it is seen as responsible health behaviour to prevent prostate cancer and some men may accept PSA testing without clearly understanding the potential harms.
- NICE have updated their suspected prostate cancer referral guidelines in December 2021. They advise to **consider referral based on age-related thresholds**, taking clinical judgement and patient preference into account.
- There is evidence that a baseline PSA level in men aged 55 to 60 helps to predict future risk of developing clinically significant disease;
 - These findings can be used to implement a more targeted testing strategy e.g. discontinued in men with a baseline < 1 , less frequent future testing with baseline of 1-2, and more regular testing in men with a baseline level > 2
 - Using a more targeted testing strategy tailored to individual's baseline needs is likely to reduce overdiagnosis of indolent and harmless tumours and improve detection of clinically significant disease

NB Quality Improvement Ideas: PSA Testing

- If asymptomatic men request prostate cancer screening/PSA testing discuss the pros and cons of PSA testing - make them aware that PSA testing may reduce prostate cancer-specific mortality but does not reduce overall mortality and leads to increased risk of sepsis, urinary incontinence and erectile dysfunction.
- Use the [patient information leaflet](#) from the PCRMP to help guide patients requesting PSA.
- Be aware of the increased risk with family history and black ethnicity.
- Further patient guidance can be found here - [Prostate Cancer Risk Calculator](#) (developed in conjunction with the ERSPC study); this calculator helps men assess their background risk (both with and without a PSA result) and whether to undergo PSA testing

EARLY DIAGNOSIS OF CANCER

Since 2020 early diagnosis of cancer in the UK has become even more of a Hot Topic, especially in the light of the COVID-19 pandemic. Improving early diagnosis of cancer was made a key part of the 2019 [NHS Longterm Plan](#). Then coronavirus arrived and had a massive impact on cancer referrals and diagnosis. As we exit the pandemic, early diagnosis of cancer has now become even MORE of a priority than it was before. We also discuss the latest data around GP referral rates and outcomes, and how the 2-week wait system fits into that, as well as the emergence of rapid diagnostic clinics. In this chapter, we also briefly review the Galleri test, which has caused considerable excitement both in the medical press as well as the [lay press](#).

► *FIT tests for LGI 2-week waits Network Contract DES* [Impact and Investment Fund DES 2022/23](#)

There have been some particularly bad cancer 'losers' regarding whose investigation pathways have been disproportionately affected by COVID, **one of which is lower GI cancers**. COVID put a complete halt to all endoscopy services for a considerable period of time and the waits for colonoscopy for suspected LGI cancers have never recovered. In England, there has been an eightfold increase in the colonoscopy backlog for patients waiting over six weeks for a colonoscopy during the pandemic. LGI patients now account for more than a quarter of all patients who do not meet the 62-day wait standard, with almost all breaching because of colonoscopy waits. The idea of using FIT tests to accompany LGI 2-week waits to help risk stratify those in most urgent need of colonoscopies has been floated, and indeed that some people referred under the 2-week wait may be triaged to not have a colonoscopy at all if they have low FIT levels.

This strategy is being reviewed urgently by NICE, based on data from a large study (see chapter on FIT tests for more detail), but as part of the Impact and Investment Fund GPs for 2022/23 GPs in PCNs are being incentivised to do a FIT test either before, or after having done, a LGI 2 week wait - [see P50-53 of the IIF document](#) for detail.

► *The Galleri test for early cancer diagnosis* [SYMPLIFY study](#) & [NHS Galleri trial](#)

Although we have a number of cancer tests for both symptomatic and asymptomatic (screening) people, the role of blood tests in this mix has been broadly speaking disappointing. The role of PSA testing even in 'symptomatic' men is highly uncertain (see the prostate cancer chapter in the urology section for more on this), Ca125 for ovarian cancer has been shown to be ineffective for screening in the [UKCTCOS trial](#) (although there is evidence for it's use in symptomatic women and remains part of our diagnostic workup as per the NICE and SIGN cancer guidelines) and the other range of tumour markers have been shown to be so poorly sensitive and specific for primary care populations and are not recommended primary care tests.

However, there is potential excitement in the world of cancer blood tests with **two trials that are underway to evaluate the diagnostic accuracy of a new blood test - the Galleri test**, which has caused [considerable publicity](#). The Galleri test is a type of Multi Cancer Early Detection (MCED) test, designed to detect over 50 types of cancer. Using innovative sequencing technology, Galleri has the potential to complement existing screening programmes and current tests to enhance early-stage diagnosis, when cancers can be treated more successfully. The test identifies cell-free DNA (cfDNA) which is specific to different cancer types. This type of test has the potential to accelerate cancer diagnoses, reduce investigations and improve cancer care through increasing the rate of curative treatment and survival. Such tests may have a role to play in patients with vague, non site-specific symptoms that may be a result of cancer, or even as a screening test. It has potential significant advantages over our current plethora of tests in that **early data suggests it can pick up cancers at an earlier stage, but also that it can specify the likely cancer type**, allowing those with non-specific symptoms to be more accurately and specifically investigated further in the hope of earlier diagnosis and treatment. Currently, it is claimed to have a low false-positive rate of <1%. If these claims are borne out in the current trials underway it really would be a substantial step forward for early cancer diagnosis.

The aim of the [Oxford-based SYMPLIFY](#) study is to demonstrate how the test could be used to increase cancer detection rates and simplify diagnostic pathways. Around 6000 patients have been recruited. Recruitment criteria was that patients needed to be over 18y, symptomatic of possible gynaecological, lung, upper or lower GI cancer and referred from primary care via the 2WW pathway. In addition to usual investigations, each participant will have blood taken for the Galleri test. Patients will be followed up for 12m, with their Galleri test result and any subsequent diagnosis recorded. The [NHS-Galleri trial](#) is considering how the test performs when combined with standard cancer testing in asymptomatic people (i.e. screening). It aims to recruit 140,000 people aged 50-77y and results may influence screening programmes in the future. Cancer Research UK have further information [here](#).

► ***GP referral rates and cancer outcomes*** [BJGP 2020;70:e389](#) [BJGP 2022; 72 \(714\)](#)

This important new study from 2020 has just been published looking at the association between GP referral rates and cancer outcomes. Traditionally, a mark of good practice for GP referrals was to have low referral rates but a 'high conversion rate' i.e. a high proportion of patients referred with a possible diagnosis of X, as subsequently being diagnosed with X (i.e. the GP was 'right' to refer), and a low conversion rate was associated with unnecessary over referral and considered a marker of less good practice. Traditionally a low referral but high conversion rate has been seen as a surrogate marker of good practice. Avoiding over-referral is increasingly also seen as a marker of good practice, to protect patients from 'overdiagnosis' i.e. unnecessary investigation and potential iatrogenic harm. *New research, however, challenges this with an inconvenient truth: a high referral and low conversion rate is associated with better patient outcomes.* Previous research [BMJ2015;351:h5102](#) had suggested that patients registered with practices that have low suspected cancer referral rates have greater excess cancer-related mortality compared to practices with higher referral rates. This ***newly published research*** [BJGP 2020;70:e389](#) explored this further:

- This was a large, national cohort study of 1.4 million patients diagnosed with cancer.
- There is significant variation between individual practices in referral rates and the study shows a significant association between higher use of urgent cancer referral pathways and lower mortality for all cancers combined and the four most common cancers (CRC, lung, breast, prostate).
- Reductions in late-stage cancers at diagnosis were found from practices with higher referral rates.
- **Conclusion:** *higher practice use of referrals for suspected cancer is associated with fewer late-stage diagnoses and lower mortality, and this evidence supports a policy of lowering referral thresholds from primary care and increased use of subsequent cancer referral pathways.*

But referring more people via the 2-week wait (2WW) is all well and good in principle, but how does that impact service provision more widely, are there any differences between cancer type pathways, and are there specific cancer types that do better than others via the 2WW system? Some of these questions were addressed in new data and analysis in [BJGP 2022; 72 \(714\)](#) which reviewed 2WW data from the decade 2009/10 to 2019/20 (i.e. right up to the start of the pandemic) also discusses another important metric - detection rates.

The aims of the 2WW system when introduced were relatively simple - have defined pathways for suspected cancers, aim to get as many cancers as possible through the 2WW system in a rapid fashion and as few as possible through emergency routes, as combining these two factors leads to better outcomes. Changes to the 2WW pathways over the years and **particularly the change in referral thresholds in 2015** (when the 'bar' for referral was lowered to a standard 3% PPV across all adult cancers) has led (as was predicted) to a massive increase in referrals via the 2WW. **The total number of referrals for all cancers in England via 2WW has increased from 902,943 in 2009/10 to 2,374,718 in 2019/20 - more than doubling in 10 years.**

Has this helped? Well, the data from the above study suggests it has, in that higher 2WW referral rates improves outcomes. The data analysed in this recent BJGP supports the increased referral rates too, albeit in an indirect fashion. The paper discusses **the importance of detection rates** - this is the number of new cancers that *need treatment* (i.e. it does not include incidental or 'low grade' cancers that don't need active treatment, that could be thought of as potential 'over diagnoses') that have come through the 2WW system as a percentage of the total number of cancers needing treatment (that may have come through other referral rates). E.g. if you have 100 new cancers that need treating and 40 came via the 2WW system then the detection rate is 40% - **the higher the detection rate the better as it means more clinically significant cancers have been diagnosed and treated through the 2WW system** (therefore more promptly and with resultant better outcomes in general). Detection rates overall improved over this decade 2009/10 to 2019/20 from 42.3% to 53.5%. However, this **still means that despite this improvement about 50% of all clinically relevant cancers are still picked up through routes other than the 2WW system** - the worry is this means an inequality has developed where those that go through the 2WW system get prompt treatment, and those that don't may have even longer waits through routine pathways. This has been one of the drivers for the RDCs as we know many cancers do not present with typical 'red flag' or '2WW' symptoms. (see below).

The other important point to come through from this latest 2WW data is the cancer-specific data. Whereas detection rates have improved overall in some cancers they have not. **The notable exception is lung cancer. The detection rate for lung cancer is lower than for many cancers and has actually decreased over the period studied.** The result is many lung cancers are still picked up through emergency routes at later stages, with poorer outcomes.

The paper highlights a few points for consideration:

- Overall referrals via the 2WW system have increased dramatically (more than doubling over 10 years) - this has led to **improvements in one regard i.e. better detection rates, but at a cost of large pressures on secondary care and investigation services**, and may widen the inequality between cancers diagnosed via 2WW systems and those picked up through other pathways.
- However, **substantial variations are still apparent between different cancer types** - the paper calls for cost-effectiveness evaluation and assessment of capacity implications for each individual cancer.
- Finally, **the paper calls for better access to investigations for GPs** to help ease pressures on secondary care - this may also improve detection rates in some cancers e.g. would use of CT chest for ?lung cancer in primary care give better detection rates given we know CXR has relatively poor sensitivity?

► ***The 2019 NHS Long-term plan and early cancer diagnosis*** [NHS Longterm Plan](#)

Improving cancer diagnosis and outcomes are key parts of the [NHS Long-term Plan](#) (LTP) published in 2019, which plans over the next decade to save thousands of lives each year through earlier diagnosis and treatment. A target has been set that the percentage of all cancers diagnosed at an early stage 1 or 2, will increase from the current 50% figure to 75% by 2028. The NHS LTP estimates that if achieved this will translate into 55,000 more people in England each year surviving their cancer for more than 5 years.

To achieve this it is proposed that:

- There will be changes to **screening** e.g. lowering the age for bowel screening, introducing 'lung cancer health checks' in high-risk groups (see lung cancer screening chapter for more on this) and improved cervical screening with primary HPV testing.
- There will be new [Rapid Diagnostic Centres](#) across the country so that patients with possible cancer can be assessed and diagnosed in a single 'one-stop-shop' visit; these will initially be rolled out to refer patients who have non-specific symptoms that are concerning but do not 'fit' with a specific diagnosis, but the longer-term plan is for ALL referrals with suspected cancer to go through this single point of access to prevent patients falling through the gaps between different pathways (see below for recent research on the outcomes from an RDC).
- Patients with cancer to receive personalised care.
- Broaden the range of diagnostic tests available.

Responsibilities for GPs will include:

- Analysis of practices and networks' own data related to screening, referral and diagnosis.
- Improving screening uptake.
- Active case finding for identifying people with suspected cancer.
- Improve referral processes across GP practices for early and timely referral for people with suspected cancer including more standardised systems and processes for referral management and safety netting.

► *Diagnosing cancer in primary care: results from the National Cancer Diagnosis Audit*

The [National Cancer Diagnosis Audit](#) (NCDA) was started in England a decade ago back in 2009-2010 and is now collecting data across the UK from primary care. The NCDA is a collaborative project between the NHS, the RCGP and Cancer Research UK and Macmillan. It looks at primary and secondary care data relating to patients diagnosed with cancer to help understand referral pathways and delays and to inform improvements in care. Participating practices are encouraged to use their data to highlight and share good practice and identify areas for improvement. The aim of the NCDA is to generate a more detailed understanding of the diagnostic processes, and delays, in cancer diagnosis in primary care. It is, therefore, a Quality Improvement tool specific for primary care, which enables practices to receive personalised feedback on their data to support reflection, learning and quality improvement within their organisation. For GPs in England, participation in the NCDA will therefore 'tick a lot of boxes' for the QI initiatives in the GP Contract 2020/21 (see separate section). The results from the NCDA looking at 2014 data have been recently published [BJGP2018;68:e63](#). Data were collected on 17,000 patients with a new diagnosis of cancer during 2014 from 439 practices.

Key findings:

- Most patients (72%) diagnosed with cancer first present to their GP, and only 7% via A&E
- The median diagnostic interval for all patients was 40 days, with a median primary care interval of 5 days from first presentation to referral
- 74% of patients were referred to a specialist after only one or two consultations
- 52% were referred via the urgent two-week wait system
- 45% of patients had investigations, such as blood tests or scans before referral
- Safety netting was recorded for 44% of patients
- **Avoidable diagnostic delays were found in 22% of patients** and these delays were due to a mix of patient (26%), clinician (28%) or system (34%) factors
- Multimorbidity was common; 76% had at least one co-morbid condition, 21% had three or more

Conclusion: the study reveals that most patients who are diagnosed with cancer in the UK present first to their GPs, most patients are referred promptly from their GP but avoidable delays were found for 22% of patients due to a mix of patient, clinician or system factors indicating areas for quality improvement.

For Scotland, specific data see [National Cancer Audit Scotland](#)

► *Early cancer diagnosis, and the GPs' 'gut feel'* [BJGP2020;70:e612](#)

Over the last decade, the direction of travel with an early cancer diagnosis has been 'tick box led', either by looking for specific red flags that trigger concern and a need for referral (as typified by [NICE NG12](#) suspected cancer referral guidelines) or by calculating cancer risk using decision support tools such as [QCancer](#) or the [Macmillan Cancer Decision Support \(CDS\) Tool](#). However we all have experience of patients that we are worried about, our 'gut feel' tells us that this patient may have an undiagnosed malignancy, and yet they do not fulfil the criteria for an urgent referral.

This study [BJGP2020;70:e612](#) was a *systematic review of the evidence exploring GP's gut feelings in the diagnosis of cancer*. Not surprisingly, there is a lack of evidence but from the evidence, we do have they conclude that when gut feelings are recorded the pooled odds ratio of a cancer diagnosis were four times higher. GP's gut feelings became more predictive of cancer as clinical experience and familiarity with the patient increased. However, GPs were often reluctant to record gut feelings as some specialists questioned their diagnostic value. *In conclusion, GP a cancer diagnosis is more likely in patients for whom the GP had a gut feeling* compared to patients for whom the GP did not have a gut feeling. The predictive value of gut feelings increases with clinical experience and knowledge of the patient. Urgent referral based on GP gut feelings is now a referral criteria to the new and emerging NHS Rapid Diagnostic Centres...

► *Rapid cancer diagnosis centres*

The [ACE programme](#) was introduced in 2014 as part of a 3-year initiative to drive the development of new diagnostic pathways to support earlier and faster diagnosis of cancer in the UK. The key objectives of the ACE programme were three-fold: Improve the early diagnosis (stage 1 & 2) of cancer, reduce the number of cancers presenting as an emergency and improve the patient experience overall.

The ACE programme aimed to provide evidence to support system-wide changes in referral processes. It recommended and supported the development of **Multidisciplinary Diagnostic Centres (MDCs)** across England. As discussed above development of MDCs (or **Rapid Diagnostic Centres, RDCs**) are part of the NHS Long Term Plan. RDCs are designed to investigate patients with non-specific but concerning symptoms that could be indicative of several cancers. Referrals are made from GPs and a multidisciplinary team organises the necessary investigations at the same time and in one place. But do they work and what is the evidence? Important data has been published in [Brit J of Cancer 2020;123:722–729](#), reviewing the evidence from December 2016 to July 2018 from 5 RDCs in the UK.

Key points:

- This cohort study covered 2961 patients with 241 confirmed diagnoses of cancer.
- 27% of those referred had moderate/severe levels of comorbidity.
- 28% of those referred had had ≥3 previous GP consultations.
- **8% of those referred were diagnosed with cancer** (rates comparable to current traditional organ-specific urgent cancer referral pathways), and *'a substantial proportion of early-stage lung, upper GI tract and haematological cancers were diagnosed'*.
- The most **common cancers** picked up were: UGI cancers (53 in total, including 25 pancreatic cancers), lung cancer (52), haematological, lower GI and urological (all ~30 each); encouragingly, several of the diagnoses were rare and less common cancers.
- 7% of those diagnosed with cancer had recurrent disease and **all of these patients presented without any recognised 'traditional' alarm symptoms**.
- The **commonest reasons for referral** were: Weight loss (66% of referrals), 'clinical suspicion' (36% of referrals), pain (32% of referrals); 61% of referrals had ≥1 symptom.
- Patient age was found to be the most important factor associated with a cancer diagnosis ($p < 0.001$), and the **only other significant factor associated with a diagnosis of cancer was GP 'clinical suspicion'** ($p = 0.006$). This highlights the importance of GP 'gut feel' in diagnosis of cancer (see below).
- **Conclusion:** The RDC referral pathway for patients with non-specific but concerning symptoms could be a valuable addition to the referral options for suspected cancer, across a broad range of cancers.

It is important to note that all patients had to have presenting symptoms *'not sufficiently clear to indicate an appropriate tumour-specific urgent referral pathway'*. This suggests that the **vast majority of cancers diagnosed in the RDCs would be missed or delayed under current referral pathways**. It is also interesting to note that in this study there was a *reduced* association between a cancer diagnosis and patients with more than three GP consultations prior to referral, leading to the authors suggesting that *'Within the context of this study, it may be that where they have a strong suspicion of cancer, GPs are more likely to pursue available options such as the MDC to achieve earlier referral for these patients'*. As always there are caveats and it is important to be aware that this is cohort data only, with no comparator data set, meaning that we **cannot assess the over-diagnosis rate** (the authors rightly pointing out that *'It is possible that a number of cancers, including those at an early stage, were diagnosed 'incidentally', in that the symptom or symptoms leading to referral were not caused by the cancer'*) or indeed **potential harms**.

A separate study looked at 'less common cancer' diagnoses in 5 MDC pilot projects, considering whether the MDC approach improves the diagnosis of these cancers [BJGP 2021; 71\(712\);e846-e853](#). Results confirmed that a non-specific symptomatic referral route diagnoses a broad range of less common cancers and can support primary care management of patients with possible cancer that do not fit the criteria for a site-specific urgent referral. A more **extensive rollout of these RDCs is now a key part of the NHS Plan**, and the longer-term plan is for ALL referrals with suspected cancer to go through this single point of access rather than down individual pathways where patients frequently fall through the gaps. Cancer Alliances are working hard to deliver the RDC service model for the management of all cancer pathways by 2023/24. As discussed in a recent editorial [BJGP 2021 71\(712\); 487-88](#), there is massive potential for RDCs to improve earlier diagnosis of cancers (especially those that present with non-specific symptoms), through clinical activity and facilitating new research. Input from primary care will be vital in determining the effectiveness of the model.

► ***Safety netting for primary care, evidence from a literature review*** [BJGP2019;69:e70](#)

Ever since Roger Neighbour first used the term in his consultation model back in 1987 (!), safety netting has been an integral part of GP consulting to help ensure patient safety. However, although safety netting is widely taught and recommended in guidelines, there has been no clear consensus on when it should be used and what advice or actions it should contain. For practices in England, more formalised safety netting is a key part of the QI initiatives in the GP Contract 2020/21 (see separate section). This research paper [BJGP2019;69:e70](#) was a useful review of the evidence from 47 studies, from which they were able to conclude that the definitions of what safety netting is and its key components are fairly universal. From this, they were able to produce a useful evidence-based definition of what safety netting is and to conceptualise what its key components are. **Key points:**

■ ***Definition:***

- Safety netting can be defined as an essential process to help manage uncertainty in the diagnosis and management of patients, by providing appropriate information and organising follow up
- Safety netting aims to empower patients and protect health professionals

■ ***Key components of the safety netting process can be identified:***

- Communication of the uncertainty with patients, to reduce the risk of false reassurance
- Specific advice on worrying or 'red flag' symptoms that patients should look out for
- Advice on the likely time course of the illness, and therefore when to re-present
- How and where to seek future medical care, including arranged planned follow up
- The management of investigations and communication of results

■ ***Investigations***

- Safety netting around investigations is a crucial part of cancer safety netting
- Patients may be falsely reassured by normal blood tests, or assume 'no news is good news'
- Defining responsibility for the checking of results and how the patient is informed is key

■ ***Organisational components***

- These are also key to safety netting, for example ensuring there is an administrative system in place to ensure abnormal results are dealt with and also to document safety netting advice in the notes
- Management of hospital letters and follow-up of action points from them is also key

Although this evidence review was able to produce useful consensus on what safety netting is, the authors highlight that *there is a lack of empirical evidence on the efficacy of safety netting as a whole and its component parts*. The findings are largely based on logic and expert opinion. Further research is needed to evaluate the effectiveness of safety netting and its component parts. **Conclusion: *safety netting is more than the communication of uncertainty, and requires specific advice and plans for follow up including administrative action such as management of test results and hospital letters***

Further recent research from an observational study [BJGP2019;69:e878](#) of 23 GPs looking at 318 recorded GP consultations, found that GPs do commonly give safety netting advice (65% of all consultations). However, there was wide variation and when safety netting advice was given it was only recorded in the notes 45% of the time, and the contingencies or actions required on the patient's part were often not specific. The authors conclude that *'if GPs are going to be held to clinical standards that they should give safety netting advice, further guidance is required on exactly what type, when and how'*. Furthermore, GPs may be putting themselves at medico-legal risk by not documenting safety netting advice. See KISS below on Safety Netting. CRUK have produced useful guidance for [cancer safety netting during the COVID-19 pandemic](#).

NB Learning Points & Quality Improvement Ideas: Early Diagnosis of Cancer

- The COVID-19 pandemic is leading to significant diagnostic delays for cancer, which will have a major adverse effect on cancer outcomes; this has been particularly felt for LGI referrals due to a large backlog of waits for colonoscopy - New recommendations in the [2022/23 Impact and Investment Fund](#) is incentivising GPs in England (via PCNs) to do FIT tests to accompany LGI 2WW referrals to help risk stratify/triage use of colonoscopy.
- Improving cancer diagnosis and outcomes are key parts of the [NHS Long-term Plan](#) (LTP) published in 2019, which plans to save thousands of lives through earlier diagnosis and treatment.
- Practices can demonstrate participation and engagement in QI activities by signing up for the [National Cancer Diagnosis Audit](#) (NCDA) which is taking place in England, Scotland and Wales, which will provide them with personalised data and feedback on which to implement change.
- There will be more GP **direct access referral for tests**; research shows this performs as well, and on some measures better than, consultant triaged testing.
- New multidisciplinary **Rapid Diagnostic Centres** are now being rolled out across the UK, and evidence on their use is very positive - the data suggests that these RDCs will pick up many patients with cancer who present with non-specific symptoms, and that many of these people would not have 'qualified' under the 2WW system; a large number of 'hard to pick up' cancers are diagnosed more rapidly through RDCs, including UGI cancers (inc pancreatic cancers), lung and haematological cancers.
- We have **increasing evidence on the importance of 'GP Gut Feel'** when it comes to diagnosis of cancer and in the RDC data this was particularly apparent - apart from age of patient, the only other significant associated factor in cancer diagnosis was 'GP clinical suspicion'.
- New research reveals that practices with higher **referral rates via the 2WW system** have improved cancer outcomes and mortality.
- **Detection rates** are the number of cancers treated via a 2WW referral as a % of the total number of treated cancers - the higher the number the better as these people will have prompt investigation and treatment via the 2WW system and are known to have better outcomes; this is another useful metric to benchmark your practice.
- A new **multi-cancer detection blood test (the Galleri test)** is being investigated through two UK studies; Using cell-free DNA analysis, it is hoped this test may be able to identify specific cancers through a simple blood test for both asymptomatic people (for screening) and for those with symptoms.
- **Quality Improvement Ideas:**
 - Sign up for the [National Cancer Diagnosis Audit](#)
 - Benchmark your practice's referral rates and detection rates using the [NHS Fingertips data](#)
 - Introduce a more formalised process of safety netting (see KISS below)

KISS: Safety Netting for Cancer

[CRUK Safety Netting Guidance 2019 BJGP2019;69:e70](#)

[CRUK guidance: safety netting patients during the COVID-19 pandemic](#)

[CRUK cancer safety netting during the COVID-19 pandemic](#)

- Safety netting can be defined as an essential process to help manage uncertainty in the diagnosis and management of patients, by providing appropriate information and organising follow up
- CRUK *'Safety netting is a diagnostic monitoring strategy that ensures that patients are monitored throughout the diagnostic process until their signs or symptoms are explained'*
- Safety netting is more than solely the communication of uncertainty but should include specific plans for follow-up and administrative areas such as communication of test results and management of hospital letters
- For GPs in England, more formalised safety netting processes and documentation (a new SNOMED code is being developed) is a key part of the new cancer-related contractual requirements for the QOF and PCN DES modules (see separate KISS)
- **Key components of the safety netting process can be identified:**
 - Communication of the uncertainty with patients, to reduce the risk of false reassurance
 - Specific advice on worrying or 'red flag' symptoms that patients should look out for
 - Advice on the likely time course of the illness, and therefore when to re-present
 - How and where to seek future medical care
 - Arranged planned follow up
 - The management of investigations and communication of results
- **Investigations**
 - Safety netting around investigations is a crucial part of cancer safety netting
 - Patients may be falsely reassured by normal blood tests, or assume 'no news is good news'
 - Defining responsibility for the checking of results and how the patient is informed is key
- **Organisational components**
 - These are also key to safety netting, for example ensuring there is an administrative system in place to ensure abnormal results are dealt with and also to document safety netting advice in the notes
 - Management of hospital letters and follow-up of action points from them is also key
- **Be specific.** Inform patients of:
 - Anticipated time frame of symptom resolution
 - Offer patients with symptoms a review in an agreed time frame, even if initial tests are negative
 - The process for being informed of test results
 - Potential alarm symptoms which should trigger re-consultation
 - How, when and where to re-consult
 - The presence of diagnostic uncertainty
 - Give written instructions and document in the notes
- Code symptoms and examination findings (NB risk assessment tools are dependent on coding)
- Check patient understanding of advice and time frames
- If symptoms do not resolve or persist intermittently, conduct further investigations even if previous tests were negative and consider referral (e.g. 'three strikes and you are in' approach) e.g. to one of the new **Rapid Diagnostic Centres** being set up for patients with worrying but uncertain symptoms being set up by the [ACE \(Accelerate, Coordinate, Evaluate\) Programme](#)
- Document safety netting in the notes

LUNG CANCER

Improving earlier diagnosis in lung cancer is a major Hot Topic. It was the subject of a BJGP editorial in December 2020 [BJGP2020;70:572](#). As they point out, the '*current approach to the diagnosis of lung cancer is ineffective*'. By the time typical symptoms become apparent, the disease has usually spread beyond the primary site. It is a cancer that is disproportionately associated with areas of socio-economic deprivation, and late diagnosis is often another example of the inverse care law.

As further recent papers illustrate [Thorax 2019](#), BMJ review [BMJ2019;365:1725](#) and [BJGP2020;70:e193](#) & [BJGP 2019;69:90](#) points out, lung cancer outcomes have improved only marginally over the last 40 years. They remain 'dismally poor' compared to outcomes for other cancers, such as colon or breast, which have significantly improved. *Only 18% of women and 13% of men in the UK diagnosed with lung cancer survive more than 5 years* and lung cancer is now the most common cause of cancer mortality in the UK. Our survival rates for lung cancer in the UK, lag significantly behind other countries with advanced economies [Lancet 2018](#).

There are many reasons for this, but the poor survival is largely attributable to low rates of treatable disease at diagnosis with 75% of lung cancers are diagnosed at a late stage and early diagnosis being strongly linked with improved survival. The prognosis for lung cancer is strongly linked to the stage at diagnosis and is better for patients diagnosed in primary care than via emergency routes. So, improving early diagnosis of lung cancer in primary care will be a major Hot Topic over the next few years. Recent research has looked at how presenting symptoms of lung cancer are changing, relatively *poor sensitivity of CXRs* and *targeted screening with low radiation dose CT scanning*. [Lung health checks](#) with CT scans are now being introduced into the NHS.

► *The changing presenting symptoms of lung cancer* [BJGP 2020;70:e193](#)

Most of us were taught that haemoptysis was the cardinal symptom of lung cancer. Most patients diagnosed with lung cancer present with symptoms, but this new original research, an observational study from primary care, found that the pattern of presenting symptoms of lung cancer has changed over the last 20 years. They found that:

- Since 2010, the proportion of patients diagnosed with lung cancer presenting with index symptoms of cough or breathlessness increased whilst the proportion of those with an index symptom of haemoptysis decreased
- Cough and dyspnoea are now the most common presenting symptoms of lung cancer
- Haemoptysis is now a very rare presenting symptom of lung cancer
- ***These findings have implications for patient and clinician awareness.***
 - Clinicians need to be aware that although haemoptysis remains an important presenting symptom of lung cancer, its presentation is increasingly rare
 - Clinicians need to place as much emphasis on the more common symptoms of cough and dyspnoea as they do on haemoptysis
 - For patients, symptom awareness campaigns should target cough and dyspnoea

Conclusion: *Cough and dyspnoea are the most common presenting symptoms of lung cancer in UK primary care, and haemoptysis and appetite loss are rare presenting symptoms.*

► *Sensitivity of CXR for detecting lung cancer in symptomatic patients* [BJGP2019;69:e827](#)

CXR remains the first-line investigation for lung cancer in primary care and the most common radiological route to diagnosis. We have been aware for a long time of the potential for false-negative results with CXR for early lung cancer, but this is the first systematic review to examine the evidence for the sensitivity for CXR in patients presenting to their GP with symptoms. Results:

- A total of 21 studies were examined, almost all of which were 'poor quality'
- Looking at the highest-quality studies suggest that the sensitivity of CXR for symptomatic lung cancer is only 77% to 80%

- **Conclusion:** *CXRs will miss lung cancer in at least 20% of patients; in symptomatic high-risk patients, GPs should consider further investigation even if the CXR is reported as negative*

The author's highlight that compared to comparable countries that have much better lung cancer outcomes, GPs in the UK have relatively poor direct access to CT scanning. As earlier diagnosis is strongly linked with improved survival it is possible that our reliance on CXR in practice may delay diagnosis in some patients. If CXR were a novel technology 'it is debatable whether the evidence would support its implementation as a test for lung cancer'. These results, therefore, support calls to increase open access to CT scans for GPs.

However, *a more recent study on estimating lung cancer risk from CXR and symptoms is more reassuring* [BJGP 2021;71:e280](#). This cohort study agrees that CXR has limited sensitivity at 75%, however, in low prevalence settings the negative predictive value of a negative CXR is still very high at 99%. The authors argue that for most patients the high negative predictive value means that lung cancer is very unlikely after a negative result, however careful safety netting and surveillance is required as a diagnosis of lung cancer is still possible. For patients with haemoptysis, however, the data do support immediate referral for scanning as the PPV of lung cancer with a normal CXR is significant at 3%.

► **Early diagnosis of lung cancer: is rapid access CT scanning the answer?** [BJGP 2019;69:90](#)

Given the poor sensitivity of CXR and the need for early diagnosis, this is an important question to answer. The relatively poor early diagnosis and survival rates in the UK compared to other comparable countries has been partly attributed to less direct access to CT scans. This discussion and analysis paper reviewed the evidence and concludes that *direct access CT scanning for patients who have had a normal CXR but where the GP is concerned about the possibility of lung cancer can potentially expedite the diagnosis*.

► **Lung cancer screening with CT scanning**

Currently, Canada, the US and a number of European countries have implemented screening programmes for lung cancer with low-dose CT scanning. A recent EU position statement has recommended that screening for lung cancer with CT should be implemented across Europe.

In 2020 new evidence for screening has published with the **NELSON Trial** [NEJM 2020;382:503](#)

- This high-quality RCT was from the Netherlands and Belgium, and was independently funded by the Netherlands Organisation of Health Research
- 13,000 men and 2,600 high-risk women aged 50 to 74 were randomly assigned to low-dose CT screening at baseline, year 1, year 3, year 5.5 or no screening
- A minimum follow-up of 10 years was completed for all participants and there was 90% adherence
- **Results**
 - The overall referral rate for evaluation of suspicious nodules was 2.1%
 - At 10 years of follow-up, the incidence of lung cancer was 5.6 cases per 1,000 person-years in the screened group and 4.9 cases per 1,000 person-years in the control group
 - *Lung cancer mortality was reduced by 24% in men and 33% in women*
 - In absolute terms, lung cancer mortality was reduced from 3.3 deaths per 1,000 patient-years in the control group to 2.5 deaths per 1,000 patient-years in the screened group
 - The authors estimate that 60 deaths were prevented as a result of 4 rounds of screening in 7,900 patients
- **Conclusion:** in this trial of high-risk people, lung cancer screening significantly reduced lung cancer mortality

► **What about improving diagnosis using a panel of biomarkers? The EarlyCDT-Lung test**

Another approach to earlier diagnosis of lung cancer may be looking at biomarkers, such as an autoantibody biomarker that could predict increased lung cancer risk.

This biomarker called the **EarlyCDT-Lung test** has been tested in an evaluation randomised controlled trial in [Scotland Early Diagnosis of Lung Cancer Scotland Study 2020](#). Patients with a positive test were offered low-dose CT scanning at 6 monthly intervals for up to 2 years. There was no significant difference at 2 years in lung cancer incidence and mortality, but in the intervention arm, there were fewer late-stage diagnoses (60%) compared to the control arm (73%). An interesting finding, but more research is needed to see if it gives any advantage over standard CT screening.

► **The UK approach: NHS Lung Health Checks**

In the UK, there is no system-wide lung cancer screening but *targeted community-based screening* has been introduced with the NHS Lung Health Check programme with mobile CT scanners.

Although lung cancer screening has been shown to be effective, smoking and lung cancer are strongly linked with socio-economic deprivation and reduced uptake of screening. To overcome these barriers in the UK, pilot projects in the UK have looked at community-based lung cancer screening using mobile scanners in supermarket car parks in areas of socio-economic deprivation to minimise barriers to attendance and maximise uptake. This also offers the opportunity of health promotion, and smoking cessation support (there is evidence [BMJ 2017;359:j5742](#) that lung cancer screening is a good opportunity to improve smoking cessation). There is evidence that smoking cessation rates in screened groups are at least double that of the general population, and importantly that screening does not give 'false reassurance' to those with negative screening

So, what is the NHS Lung Health Check Programme?

- This programme is being rolled out across England from Autumn 2019 and aims to be a 'holistic lung health check' for people at risk
- The programme is not currently available in other countries in the UK
- The protocol is based on a number of pilots, including the Manchester Lung Health Pilot (results below)
- Patients are being invited to take part in certain areas if
 - They are aged 55 to 75
 - Are registered with a GP
 - Are recorded as ever having smoked
- The check involves spirometry and a discussion based on the smoking history of their lung cancer risk; if they are considered lower risk they receive advice but do not have a CT scan if at higher risk they are offered a low radiation dose CT scan which is usually performed in a mobile truck
- Patients will also receive stop smoking advice

On what evidence is the NHS Lung Health Check based? [Manchester Pilot, Thorax 2019](#)

Since 2017 a number of pilot projects have taken part in the UK, looking at a community-based targeted screening in areas with a high incidence of lung cancer using mobile units. *The results from this important 'lung health pilot' in Manchester have just published:*

- Mobile CT scanners were set up in car parks in deprived areas of North Manchester, which has the highest premature mortality rates from lung cancer in the UK
- Patients aged 55-74 who were either current or ex-smokers and registered with a GP were invited to attend a lung health check
 - Demand was higher than anticipated, as 30% of all those invited booked appointments and the DNA rate was only 8%
 - 75% of attendees were in the lowest socio-economic deprivation quartile
- Patients had an individual lung cancer risk assessment in the pop-up units and those deemed high risk (56%) underwent immediate CT scanning
- *3% of patients screened were diagnosed with lung cancer*
- Of those diagnosed with lung cancer, 80% were at stage 1 or 2 and only 10% at stage 4
 - 65% of patients had a surgical resection, four times higher than the national average
 - With current normal practice and pathways, only 20% of lung cancers are diagnosed at stage 1 or 2 with 50% at stage 4
- 13% of those screened had an initial indeterminate scan which was repeated 3 months later, of these 12% were then discharged and 1% referred to the cancer clinic
- **Conclusion:** *taking lung cancer screening into communities with the lung health check approach is effective and engages populations in deprived areas*

NB Learning Points & Quality Improvement Ideas

- Early diagnosis of lung cancer is a major Hot Topic. For more see our [NB Blog Lung Cancer Diagnosis 2021](#) by our cancer lead Kate Digby, and we have a FREE on-demand [NB webinar on Lung Cancer Diagnosis 2021](#) supported by Cancer Research UK
- Lung cancer outcomes have improved only marginally over the last 40 years and remain 'dismally poor' compared to outcomes for other cancers, such as colon or breast
- Only 18% of women and 13% of men in the UK diagnosed with lung cancer survive more than 5 years and lung cancer is now the most common cause of cancer mortality in the UK.
- There is a major push to improve early diagnosis rates for lung cancers
- Recent research shows that in 20% of patients who present to their GP with symptoms may have a false negative CXR; these results, therefore, support calls to increase open access to CT scans for GPs
- In symptomatic at-risk patients, GPs should consider referral or direct access CT scan if available even if the CXR is reported as negative
- There is new evidence from the European NELSON trial that screening in high-risk patients with low radiation dose CT scans improves mortality
- In the UK targeted community-based screening is being introduced, as part of a holistic NHS 'Lung Health Check' for ever smokers aged 55 to 74
- Results from a Lung Health check pilot in a deprived area of Manchester showed a high take-up rate and a 3% prevalence of lung cancer detected; of these lung cancers 80% were at an early stage

BLADDER CANCER

Bladder cancer rarely seems to feature in the GP literature, so an interesting new primary care paper [BJGP2020;70:e629](#) is welcome and has some interesting, clinically important learning points. First, a reminder on the recommended referral criteria for suspected bladder cancer...

Suspected Bladder Cancer Referral Criteria

[NICE NG 12](#) and [Scottish Referral Guidelines for Suspected Cancer 2019](#)

- Refer urgently using an urgent 2-week wait pathway for bladder cancer if
 - Patients aged ≥ 45 and have either
 - unexplained visible haematuria without urinary tract infection
 - visible haematuria that persists or recurs after successful treatment of UTI
 - Patients aged ≥ 60 and have unexplained non-visible haematuria and either dysuria or a raised white cell count on a blood test
 - (NB comment: *unexplained persistent leukocytosis can be a paraneoplastic phenomenon thought to be due to production of granulocyte colony-stimulating factor, and is associated with a number of cancers including bladder and is linked to a poorer prognosis* [J Clinical oncology](#))
- Consider non-urgent referral for bladder cancer in people aged over 60 with recurrent or persistent unexplained UTI

► *Effect of pre-existing conditions on bladder cancer at diagnosis* [BJGP2020;70:e629](#)

Bladder cancer is the 10th most common cancer in the UK and the 9th most common cause of cancer death. Like most cancers, early diagnosis is strongly linked with survival but in the UK *25% of all bladder cancers are diagnosed at a late stage*. Previous research has found significant gender disparities in bladder cancer diagnosis and outcomes.

- Although women have a lower incidence of bladder cancer than men, they are more likely to be diagnosed with a late disease and have a worse survival rate
- Of all the cancer types, bladder cancer has the largest sex difference in one-year survival rates
- Compared with men, if women present with haematuria they are 25% more likely to experience a delay of >9 months in bladder cancer diagnosis
- Women are more likely than men to have ≥ 3 pre-referral consultations for bladder cancer

This study was a prospective cohort study using electronic primary care records in the UK. They examined 1500 patients (75% men) aged over 40 with an incident diagnosis of bladder cancer, they looked at the stage of diagnosis and looked at presenting symptoms and pre-existing conditions that may have been a barrier to early-stage diagnosis and referral. **Results:**

- Women were more likely than men to be diagnosed with advanced-stage bladder cancer
- 36% of men and 63% of women diagnosed with bladder cancer had alternative explanations for their signs and symptoms, most notably urinary tract infection (UTI)
- Alternative explanations for signs and symptoms, such as UTI, were strongly associated with advanced late-stage diagnosis for both men and women
- The most common alternative explanation was UTI with haematuria
- **Conclusions:**
 - *Alternative explanations, such as UTI, for the presenting symptoms of bladder cancer, are associated with later-stage diagnosis in men and women*
 - *Women are more likely than men to be diagnosed with late-stage disease, suggesting that in women we are less likely to investigate and refer at an earlier stage*

This is the first research study to show that pre-existing conditions, such as UTI, providing plausible alternative diagnoses to bladder cancer are associated with advanced-stage at diagnosis of bladder cancer. The effect was the same in men and women and did not account for the observation that women are more likely to be diagnosed with advanced-stage bladder cancer.

A clear clinical implication for this research is that we are missing an earlier diagnosis of bladder cancer by attributing presenting symptoms of haematuria and dysuria as being due to UTI. **The 'alternative explanation' hypothesis** is a diagnostic cognitive bias that we all recognise. Existing conditions, such as UTI, that provide a plausible diagnostic alternative for the presenting cancer symptom may delay investigation and cancer diagnosis. Some of these UTIs may be a misdiagnosis, and some may be true UTI occurring as a complication of bladder cancer.

This paper is a clear reminder to us, in line with the suspected cancer referral guidance above, to consider undiagnosed bladder cancer in patients aged over 40 with unexplained visible haematuria, visible haematuria that persists or recurs after successful treatment of UTI and in patients aged over 60 with unexplained non-visible haematuria and either dysuria or a raised WCC. We should also 'consider' non-urgent referral for people aged over 60 with recurrent or persistent unexplained UTI.

Further studies are needed for the optimal period of 'watchful waiting' for patients with haematuria and a UTI that minimises the risk of late-stage diagnosis.

NB Learning Points & Quality Improvement Ideas

- Bladder cancer survival is strongly linked to stage at diagnosis, and 25% of all bladder cancers are diagnosed at late-stage
- Women are more likely than men to be diagnosed late and consequently have a worse prognosis
- *This new research shows that attributing symptoms to an alternative diagnosis, such as UTI, delays referral and investigation and contributes to late diagnosis of UTI*
- We need to remember to recall patients aged over 45 with haematuria and/or dysuria after they have been diagnosed and successfully treated for a UTI and to re-check their urine
- Consider non-urgent referral for people aged over 60 with recurrent or persistent unexplained UTI
- In patients aged over 60, a persistent unexplained leukocytosis can be a sign of bladder cancer and is associated with a worse prognosis

PLATELETS AND CANCER

The [NICE NG12 suspected cancer guidelines](#) introduced significant changes in practice, one of which was the recommendation to consider a raised platelet count as a potential trigger for referral when seen in certain clinical situations. A seminal paper back in 2017 [BJGP2017;67:e405](#) (see below for discussion) showed that a finding of unexplained thrombocytosis in primary care was a risk marker for cancer and should prompt further investigation. *The Hot Topic in 2021 is that the same group have now published new research suggesting that a 'high normal' platelet count in older men may also be indicative of underlying cancer.*

► **Cancer incidence following a high-normal platelet count** [BJGP2020;70:e622](#)

We know from the 2017 study below, that a raised platelet count of >400 is associated with an increased incidence of cancer. *This new study looked at outcomes in patients who were found in primary care to have a high normal platelet count in the range of 325 to 400.*

- This was a primary care prospective cohort study of 300,000 patients aged over 40 with no prior cancer diagnosis who had an FBC performed for any reason
- The subsequent cancer incidence was compared in those patients who had a high normal count (325 to 400) compared to those with counts <325 and examined to see if it met the UK's agreed 3% risk threshold for urgent investigation
- **Results**
 - One year cancer incidence increased with age, male sex and also higher platelet count
 - Men aged over 60 with a high normal count had a one-year cancer incidence of 4.2%
 - Men aged over 80 with a high normal count had a cancer incidence of 6.7%, compared to 3.6% in those aged over 80 with a platelet count of < 325
 - In women with high normal platelet counts, the cancer incidence at one year was <3% in all age groups
 - The high normal platelet counts in men were most predictive of lung and colorectal cancers
 - **Conclusion:** *platelet counts at the high normal range in men aged over 60 may be indicative of underlying cancer and referral for further investigation should be considered*

In summary then, in men aged over 60, the finding of a platelet count in the high normal range of 325 to 400 is associated with a one-year incidence of cancer above the UK's agreed urgent referral threshold for 3%. In women, this effect is not seen with high normal platelet counts. Women naturally have higher platelet counts than men, primarily due to oestrogenic effects on platelet-forming processes in the bone marrow.

So, what are the clinical implications of this finding? Should we refer all men with a count of > 325?

As the authors point out in the discussion, it is probably an observed change in the platelet count that is a more clinically important measure and this should be a focus of future research. In the meantime, the authors suggest:

- A platelet count of > 325 in men, and an unexplained increase in platelets to > 400 in women, should prompt clinicians to consider possible cancer
- We should consider why the FBC was done, and investigate or refer based on any symptoms or signs that may point towards a specific cancer
- In the absence of specific symptoms, the higher risk ratios associated with lung and colorectal cancer mean that FIT testing and CXR would be appropriate first investigations
- This clinical finding should NOT be interpreted as evidence that a low normal platelet count will exclude cancer

► *The clinical relevance of thrombocytosis in primary care* [BJGP2017;67:e405](#)

This was a primary care-based, prospective cohort study based on the UK Clinical Practice Research Datalink analysed by researchers at the University of Exeter. Thrombocytosis is known to be present in 2% of the consulting population in primary care aged over 40.

- The cohort studied was a random cohort of 50,000 people aged over 40 who had an FBC performed in primary care. They looked prospectively at the one-year incidence of cancer in 40,000 identified with a platelet count of >400 (thrombocytosis) and 10,000 age-matched controls with a normal platelet count
- **12% of men and 6% of women with raised platelets were diagnosed with cancer in the next year, compared to 4% of men and 2% of women in the control group**
 - The risk was increased significantly if a second raised platelet count was recorded within 6 months, to 18% for men and 10% for women
 - The incidence of cancer rose both with age and platelet count
- Lung and colorectal cancer were the most commonly diagnosed cancers. One-third of these had no other symptoms indicative of malignancy. However, a wide range of other cancers were diagnosed e.g. breast, ovary, prostate, oesophagus, stomach etc.
- **Conclusion:** thrombocytosis in primary care is a risk marker of cancer in adults and requires further investigation. The positive predictive value is significant: 12% for men and 6% for women, rising to 18% and 10% if it is found on a second FBC within 6 months.

So, what should we do with an unexplained raised platelet count? [BJGP2017;67:e405](#)

- These positive predictive values are well above the 3% risk level suggested by NICE for urgent investigation of possible cancer and should prompt further investigation and/or referral.
- The PPV of thrombocytosis for cancer in middle age is greater than for a woman presenting with a new breast lump: a woman in her 50s presenting with a new breast lump in primary care has a PPV of 8.5% of being diagnosed with cancer, and 10% if she is found to have two raised platelet counts.
- The range of possible cancers is wide, **so GPs will have to seek additional features from the history and examination to select the most appropriate referral route.** The patients included would have had these blood tests done for a clinical reason and this should help inform further investigation.
- Whilst the range of possible cancers are wide, lung and colorectal are most likely so a CXR and faecal blood (FIT) test are obvious starting points for further investigation.

► *Investigating thrombocytosis* [BMJ2019;366:l4183](#)

This recent paper gave useful guidance for how we should investigate a raised platelet count. Key points:

- Thrombocytosis is usually reactive (e.g. to acute inflammation, blood loss, anaemia or tissue damage) or caused by clonal myeloproliferative disorders
- An initial assessment should include repeat history and examination, a blood smear exam and determining iron status and acute phase reactants
- If no cause of inflammation is found, then consider investigation for an occult malignancy; based on clinical assessment consider a CXR in adults over 40, a faecal FIT test for colorectal cancer and a pelvic USS for women over 55 with unexplained discharge or haematuria
- Refer to haematology if clonal cause suspected (and consider testing for the JAK-2 mutation)

NB Learning Points & Quality Improvement Ideas

- This primary care research confirms a raised platelet count > 400 as an important new indicator marker of possible cancer.
- That patients identified in primary care with a raised platelet count of >400 have a high one-year positive predictive value of developing cancer (12% for men, 6% for women rising to 18% and 10% respectively if repeated within 6 months)
- New research shows that in men aged over 60, a high normal platelet count (325-400) is also associated with an increased cancer risk > 3% of the suggested UK threshold for urgent investigation
- Colorectal and lung are the most commonly diagnosed cancers, but the range of possible cancers is wide
- GPs should use additional features from the history and examination to guide further referral and investigation
- **Practice audit:** identify patients aged over 40 with a raised platelet count: have they been investigated?

MICROCYTOSIS & CANCER RISK

Do you ever spend much time looking at an MCV on an FBC result? If the MCV is low but the haemoglobin is normal, do you ignore it or investigate it further?

In July 2020 the BJGP published an interesting new paper on the finding of microcytosis (a low mean cell volume MCV <85) in primary care and the subsequent risk of developing cancer [BJGP 2020;70;332](#). The research was done by the team at Exeter led by Prof Willie Hamilton who have produced a lot of cancer-related primary care research looking at early diagnosis, including the practice-changing paper on thrombocytosis and cancer [BJGP2017;67:e405](#) (see platelets and cancer chapter).

As this paper points out, although cancer survival rates are improving the UK still lags behind comparable countries in cancer survival in part due to delays in diagnosis. Previous research has taught us that finding hypercalcaemia and thrombocytosis may indicate underlying cancer, and some studies have shown an association of a low MCV with cancer independent of anaemia. This large study based on primary care records sought to examine this association further.

► *Microcytosis as a risk marker for cancer in primary care* [BJGP 2020;70;332](#)

Very similar in design to the seminal thrombocytosis and cancer paper, this was a primary care-based cohort study looking at the association in primary care records of finding microcytosis on a full blood count, with and without anaemia, and then the subsequent incidence of cancer compared to matched controls.

- They examined a large cohort of 13,000 patients aged over 40 with the finding of microcytosis and then compared outcomes over the next year, compared to 73,000 controls with no diagnosis of cancer
 - Microcytosis was defined as MCV < 85 fL, with 85 to 101 being normal
- Each case of cancer found in the microcytosis group was matched to five controls with no record of cancer; controls had had an FBC performed in primary care for any other reason, and therefore are likely to have had symptoms and *could* have had had cancer or any other pathology that was not known or recorded
- **Results**
 - 4% of patients in the microcytosis group had a diagnosis of cancer over the next year, compared to 2% of controls
 - In men with low MCV, the association was stronger with 6.2% developing cancer compared to 2.7% of controls
 - In people aged over 70 with a low MCV, 5% developed cancer over the next year compared to 2.5% in controls
 - In women with low MCV, 2.7% developed cancer vs. 1.4% of controls
 - Menstruation and menopausal status was not recorded, but 2.9 % of women with microcytosis developed cancer aged over 70 (significant difference from controls) compared to 1.4% of women aged under 70 (no significant difference to controls)
 - In the microcytosis group, 45% of men and 33% of women also had anaemia at the index date of the study; not surprisingly, these patients were more likely to have cancer with 10% of men and 4% of women diagnosed with cancer
 - Looking at the people with a low MCV but normal haemoglobin, 3.3% of men and 2% of women were diagnosed with cancer
 - Cancer sites were most commonly colorectal (24%), lung (12%), lymphoma (5%) and stomach (3%)
- **Conclusion:** *The finding of a low MCV is a predictor of underlying cancer, even if haemoglobin is normal*

What are the implications for practice?

The paper concludes that although a benign cause of microcytosis is much more likely, clinicians in primary care should consider further simple, primary care-based testing for possible cancer if they find an unexplained low MCV (particularly in men). A low MCV has of course long been recognised as being comorbid with iron deficiency (and haemoglobinopathies), and iron deficiency is a well-known feature of some cancers, especially gastrointestinal. ***MCV is always accompanied by a haemoglobin value and if anaemia is found this should trigger further clinical enquiry and investigation.***

So, the main implication here is for patients who have a low MCV but normal haemoglobin. 3.3% of these men were diagnosed with cancer, which is over the 3% suggested NICE referral limit. The vast majority, of course, will have a benign cause and we have to find the right balance between over and under investigation.

So, what should we do if we find a low MCV?

- The paper suggests we should ask about possible symptoms and risks, especially for colorectal and lung cancer. Why did we do the test in the first place?
- If iron deficiency is found, then further GI evaluation is warranted e.g. by testing for gastrointestinal blood loss with FIT (faecal immunochemical test) testing and considering referral dependent on other symptoms and risk factors
- The paper suggests offering patients a CXR if symptoms suggest possible lung cancer (NB but note separate recent research pointing out the low sensitivity of CXR for early lung cancer, see lung cancer section).

The paper concludes that using this approach the small number of patients whose low MCV is due to cancer could receive a quicker diagnosis using simple primary care investigations, whilst protecting the majority from over-referral and unnecessary investigation.

NB Learning Points & Quality Improvement Ideas

- This new research shows an association between the finding of a low MCV (<85) in primary care and the subsequent development of cancer even if the haemoglobin is normal
- This association is stronger in men, especially aged over 70
- The association was with a broad range of cancers, but especially colorectal and lung
- If a low MCV is accompanied with anaemia, this should trigger an appropriate investigation for the anaemia
- If the low MCV is accompanied with a normal haemoglobin, consider clinical enquiry and evaluation for possible cancer (especially colorectal and lung) with some simple primary care tests such as for iron deficiency, faecal blood loss using faecal immunochemical tests and CXR if respiratory symptoms
- Advice for GPs from Prof Willie Hamilton (personal communication): 'Ask yourself 'Why did I do this test?' and then act on the Why'.

LYNCH SYNDROME

Growing awareness of Lynch Syndrome, a drive for earlier diagnosis, new original research showing the benefit of aspirin for cancer prevention and new guidance from NICE in 2020 very much make Lynch Syndrome a Hot Topic!

As new guidance from [NICE DG 42, October 2020](#) tells us:

- Lynch syndrome is an inherited clinical condition with an increased risk of several cancers, particularly endometrial and colorectal cancer
- It is caused by mutations in the DNA sequence of mismatch repair genes that repair damaged DNA
- If a person has Lynch syndrome, these mutations are present in every cell of their body
- *It is now recommended by NICE that all new cases of endometrial and colorectal cancer are tissue tested for Lynch syndrome* leading to a rapid increase in patients who are identified
- Identifying Lynch syndrome earlier is important to
 - Help prevent that person from developing other cancers through increased surveillance and strategies to reduce risk
 - Help to identify relatives (it is autosomal dominant) to reduce their cancer risk
 - Help relatives diagnosed at an early age to consider family planning and if they wish to have risk-reducing procedures, such as a hysterectomy

A recent Lancet paper [Lancet 2020;395:1817](#) tells us that Lynch syndrome is among the most common form of inherited cancer susceptibility and predisposes to high risk of colorectal cancer, endometrial and ovarian cancer and other malignancies. *It used to be thought of as being a rare condition, but a recent review of the evidence UpToDate2022 quotes an estimate that 1 in 279 people carry Lynch mutations, and that 8% of bowel cancers under the age of 50 are due to Lynch syndrome.* The vast majority of people with Lynch are unaware that they have it, but with increasing testing in hospital incidence and awareness of the syndrome is increasing so we GPs are increasingly being asked about it. Although it carries a high risk of cancer, diagnosis of Lynch syndrome and subsequent regular screening and surveillance for colorectal and gynaecological cancers greatly improves outcomes so there is a drive for earlier diagnosis. See the KISS summary below.

► Cancer prevention with aspirin in hereditary colorectal cancer (Lynch syndrome)

In 2020 NICE produced new guidance [NICE 2020, NG151](#) that people with Lynch syndrome should consider taking aspirin for more than two years to prevent colorectal cancer. The optimal dose is not clear but NICE state 'commonly used doses in current practice are 150mg or 300mg', however, since then newly published research from the CAPP2 study suggests 600mg for at least two years may be optimal.

The CAPP2 Study [Lancet 2020;395:1855](#) looked at cancer prevention with aspirin in hereditary colorectal cancer (Lynch syndrome) with an associated editorial [Lancet 2020;395:1817](#). This new research was a long-term follow-up of the CAPP2 study, comparing 600mg of aspirin vs placebo in 937 patients with Lynch syndrome (mean age 45) taken for two years with long-term follow-up. **Results:**

- 600mg aspirin daily for two years results in a significant reduction in colorectal cancer incidence in Lynch syndrome (HR 0.56, 95% CI 0.3-0.9)
- However, the therapeutic effect was delayed and did not become apparent until 5 years
- In absolute terms, 24 people who are Lynch syndrome carriers need to be treated with aspirin for two years to prevent one colorectal cancer
- A significant 'legacy effect' was observed, in that patients taking aspirin for 25 months had a sustained reduction in colorectal cancer incidence lasting up to 20 years
- Fewer endometrial cancers were reported among female Lynch carriers (7 vs 17 cases in the placebo group) but the numbers were too small to reach statistical significance
- Major bleeding and significant adverse events did not differ between placebo and aspirin groups, however, the patients were young (mean age 45) and the numbers were relatively small to show significant adverse effects

- **Conclusion:** *The CAPP2 study data support recommendations that adult carriers of Lynch syndrome should be advised to take 600mg of aspirin for at least two years to reduce the risk for cancer*

► What about cancer prevention with aspirin generally? [Lancet 2020;395:1855](#)

- Over 100 observational studies have shown an association between regular aspirin use and reduced incidence of cancer; studies looking at data from large-scale cardiovascular trials show a significant reduction in cancer incidence and mortality among those randomly assigned aspirin following a delay of 3 to 5 years
- Just two randomised controlled trials have looked at cancer outcomes as a primary endpoint with aspirin
 - the Women's Health Study showed no effect of low dose (81mg aspirin on alternate days) at 10 years, but a reduction in colorectal cancer incidence in the next decade
 - The CAPP2 study, see above
- An updated meta-analysis and systematic review of observational studies in 2018 [PLoS One 2018](#) provide limited evidence for a role of aspirin in cancer treatment. However, the authors are rightly cautious in their conclusion stating the 'evidence of benefit from aspirin is limited'. There was considerable heterogeneity in studies, risk of bias and little evidence on the risks of aspirin.

So, we have a wealth of observational evidence but everyone agrees that to definitively answer the question on the efficacy of aspirin for cancer prevention and treatment, good quality prospective randomised trials are now needed. In the UK The [Add-Aspirin](#) clinical trial started in 2015. This is part NHS funded and will be the world's largest-ever clinical trial to answer definitively whether daily aspirin prevents cancer recurrence.

NB Learning Points & Quality improvement Ideas: Lynch Syndrome

- There is a growing awareness of Lynch syndrome as it is now recommended that all new cases of colorectal and endometrial cancer are tested for it
- It is thought to be greatly underdiagnosed, with only 5% of people who carry the genes being aware of it
- Lynch syndrome can also cause non-colorectal cancers, especially endometrial and ovarian
- It can be easily missed as a diagnosis when there different cancers across generations in the same family
- **Quality Improvement Idea:**
 - Search for cases of LS in the practice, are they having appropriate screening and surveillance and have their relatives been offered testing?
 - Consider LS as a possible diagnosis and referral to clinical genetics when there are ≥ 3 cancers in the family from the typical Lynch phenotype, especially if aged under 50
 - New research from the CAPP2 study confirms the efficacy of aspirin 600mg daily for ≥ 2 years, but the benefits do not become apparent for at least 4 years

KISS: Lynch Syndrome

[Lancet 2020;395:1817](#) & [NICE 2020 NG151](#) & [NICE DG 42, Oct 2020](#) & [NEJM2018;379:764](#)

- The Lynch syndrome is also known as hereditary non-polyposis colorectal cancer (HNPCC) and is the most common inherited syndrome associated with colorectal cancers
- It is greatly under-recognised and estimated that only 5% of cases are diagnosed
- It is autosomal dominant: people identified with the genetic mutation have a 50% chance of passing the mutation onto their offspring regardless of whether cancer has developed in them or not
- Lynch syndrome accounts for 3% to 5% of all bowel cancers, and NICE recommend *testing for Lynch syndrome should be offered to all people with colorectal and endometrial cancer* when they are first diagnosed leading to an increase in diagnosis rates and awareness
- Lynch syndrome is associated with extra-colonic cancers, especially endometrial and ovarian cancer in women but also an increased risk of renal tract, hepato-biliary and other GI cancers thus heightened clinical awareness of the risks in people with Lynch syndrome is key
- The diagnosis is often missed, especially when non-colorectal Lynch-related cancers appear in different family members across generations
- Diagnostic 'Amsterdam' criteria are ≥ 3 relatives with histologically verified Lynch syndrome-associated cancer, cancer involving ≥ 2 generations and ≥ 1 case diagnosed before the age of 50
- Lynch syndrome is caused by a mutation in a group of genes called mismatch repair (MMR) genes which usually repair DNA damage in cells, leading to an increased risk of malignant change; there are four main Lynch-associated genes (MLH1, MSH2, MSH6, PMS2) which have different penetration & expression
- The lifetime risk of bowel cancer for people with Lynch syndrome is 50-80%, which is greatly reduced by colonoscopies surveillance
- Women have a 30% to 60% risk of uterine cancer and 10% risk of ovarian cancer
- **Screening** for confirmed carriers of the mutation
 - Colonoscopic surveillance is recommended from age 20 to 25 every 2 years
 - Frequent colonoscopies significantly improve mortality but some Lynch syndrome-associated colorectal cancers bypass a precursor adenoma phase making tumours impossible to prevent via colonoscopies polypectomy thus clinical awareness is crucial even in screened patients
 - For women regular screening with ultrasound, hysteroscopy and blood CA-125 tests are recommended generally from age 35, and hysterectomy and oophorectomy are surgical options women may choose once families have been completed
- **Chemoprevention with aspirin**
 - NICE advise to 'consider' taking aspirin for chemoprevention
 - The optimal dose and duration is uncertain, but NICE advise at least two years and long-term treatment will be appropriate in most but not all cases (e.g. people with a history of peptic ulcers)
 - The recent [CAPP2 trial](#) shows that 600mg aspirin daily taken for two years significantly reduces the risk of colorectal cancer, but the benefit does not become apparent for 4 years
- **Further resources**
 - [Macmillan Lynch Syndrome](#)
 - [Lynch Syndrome UK](#)



LONG-TERM CANCER CARE

There has been a lot of emphasis in recent years on increasing early diagnosis of cancer in primary care, and rightly so. The good news is that more people are surviving cancer than ever before, and with improved treatments for many patients even with advanced stages of disease cancer has become more of a long-term condition. This means that more patients living with and beyond cancer need support, help and guidance from their primary care teams. *So, what help do they need and how should this be delivered?* A recent paper has explored this further.

► *Delivering long-term cancer care in primary care* [BJGP 2020;70:226](#)

- Ten-year survival for all cancers has more than doubled in the UK from 24% to 50% over the last 4 decades due to a combination of earlier diagnosis, better diagnostic technology, screening programmes and better treatments
- It is estimated that the number of people living with cancer in the UK will rise from 2.5 million to 4 million over the next decade with half of all people diagnosed with cancer today are predicted to survive for 10 years or more
- Recovery after cancer and its treatment presents new challenges for patients in all biological, psychological and social domains (see box below)
- There is no clear consensus where the responsibilities for care for cancer survivors lie between primary and secondary care
- Patients want their GPs to be better engaged with cancer care
 - *'Anecdotally, the offer of a proactive appointment and ongoing support can legitimise patient concerns and initiate the start of or continuation of an ongoing therapeutic relationship with the GP long after completion of treatment'*
- In the UK the [Holistic Needs Assessment](#) (HNA) has been developed as an individualised tool to provide tailored health interventions and personalised care planning. Typically initiated at the end of treatment it includes
 - physical concerns e.g. fatigue, pain
 - practical concerns e.g. work, finances, travel, education etc
 - family and relationship concerns
 - emotional concerns
 - lifestyle or information needs, including support for smoking cessation, activity etc
- The most prevalent concerns raised by patients in the HNA include psychological problems, concerns around the diagnosis, sleep problems, fatigue, pain and weight changes
- Despite this, currently, only 1 in 4 people in the UK living with and beyond cancer have a HNA in the UK
- The 'Cancer Care Review' (see below) was introduced into primary care in the UK (and linked to financial payments via the QOF Quality and Outcomes Framework) to provide a proactive model of care, but *'there is a danger of it being a tick box exercise without acknowledgement of the personal impact of the cancer diagnosis and current treatment'*
- In the UK guidelines exist for GPs caring for those living with and beyond cancer [RCGP Consequences of Cancer and Treatment](#) but awareness and implementation of this guidance is unknown

Conclusion: Many more patients are surviving cancer than ever before, but the consequences of cancer and its treatment are significant and ongoing. It is unclear how much support and help patients receive. There is a need to deliver better, more effective long-term cancer care in primary care and this will need to be effectively resourced

► *Cancer Care Review (CCR)* [Macmillan Cancer Care Review](#)

Cancer Care Reviews are perhaps the part of the [Personalised Care and Support Plan](#) (PCSP) that we are most familiar with in Primary Care, not least because they form part of the QOF cancer requirements. The changes made to QOF recognise key times of need, when patients may feel more vulnerable and benefit from more support.

KISS: Cancer Care ReviewMacmillan Cancer Care Review

- A Cancer Care Review (CCR) is an opportunity for patients to talk about their cancer experiences, understand what support is available in the community and receive the information they need to self-manage their condition.
- CCRs can be carried out by any member of the primary care team and is an opportunity to involve members such as practice nurses, social prescribers or care coordinators in delivering cancer care to patients.
- Asking the patient to complete an electronic HNA ([eHNA](#)) beforehand can be a helpful way in recording the information needed.
- **Macmillan highlights the following areas that should be discussed:**
 - The diagnosis.
 - Cancer treatment and the possible consequences.
 - Prescription entitlements (plus a medication review).
 - Patient and carers' information needs.
 - Physical activity advice and signposting to local support services.
 - Signposting to Macmillan and other organisations.
- **Top Tips for carrying out a CCR:**
 - Face-to-face or virtual? Be guided by the patient, what is their preference.
 - Consider using non-GP members of the team to conduct the CCR.
 - Use an extended appointment slot if possible.
 - Encourage a friend or relative to attend too, to gain insight into life at home/impact on others and increase retention of information. Also makes patients feel more supported.
 - Use [eHNA](#) to increase the chance of a well-structured review. Also acts as a template to follow during the CCR.
 - Send patients information in advance, so they know what to expect / what topics are appropriate to cover. Ask them to complete a HNA beforehand.
 - Check their understanding of their treatment and possible late consequences of this. Find out what treatment they are having/have had, including site of radiotherapy. This helps predict future possible consequences.
 - Discuss diagnosis, treatment, consequences. Review medication. Establish what support network they have, ask if they are a carer. Allow them time to ask any questions they may have.
 - Be aware of local services/support who may be well-placed to help with specific requests. Signpost to Macmillan website / Support Line Services on 0808 808 00 00 to speak to a professional.
 - Use the opportunity for health promotion advice. Encourage lifestyle changes & physical activity, for the positive impact on physical and psychological health and to reduce the risk of recurrence/treatment consequences.
 - Consider sharing the CCR record with the secondary care team to improve communication, to link in with any HNA that has already been completed. This will improve the personalisation of each patient's care.

QoF 2022/23

- CAN004: % patients with cancer, diagnosed within the preceding 24m, who have a CCR using a structured template within 12m of diagnosis (6 pts)
- CAN005: % patients with cancer, diagnosed within the preceding 12m, who have had the opportunity for a discussion and been informed of the support available from primary care, within 3m of diagnosis (2 pts)
- Consider using the Macmillan CCR template [here](#)



KISS: Consequences of Cancer

[BJGP 2020;70;226](#) & [BJGP2016;66;348](#)

- *Survivors of cancer and its treatment report poorer health and well-being than the general population, have a higher prevalence of common medical problems and long-term conditions and are at increased risk of a second primary cancer*
- **Physical problems**
 - Overwhelming fatigue
 - Weight changes
 - Chronic pain
 - Urinary and bladder dysfunction
 - Including incontinence, urgency and irritable bladder symptoms
 - Gastrointestinal dysfunction
 - Upper, including nausea, swallowing, weight loss etc.
 - Lower, including diarrhoea, incontinence, urgency, flatus etc.
 - Menopausal and hormonal symptoms
 - Early menopause in women
 - Similar symptoms in men on hormonal treatment
 - Endocrine problems include thyroid disease, androgen deficiency, metabolic syndrome and ovarian failure
 - Increased incidence of other chronic long-term conditions e.g. osteoporosis, heart disease and endocrine disorders
 - Cancer recurrence
 - And increased risk of a second primary cancer, especially after chemotherapy and radiotherapy
 - **Psychological problems and mental health**
 - Anxiety, depression and PTSD
 - Poor sleep
 - Secondary health anxiety
 - Fear of recurrence
- **Social problems**
 - Relationship challenges and family problems
 - Loss of employment and income
 - Sexual difficulties & dysfunction

NB Learning Points & Quality Improvement Ideas

- More patients are living with and beyond cancer than ever before, and it is estimated that over half of all people diagnosed with cancer now will live beyond 10 years and over the next decade the number of people in the UK living with cancer will increase from 2.5 million to 4 million
- Whilst this is good news, survivors of cancer and its treatment report poorer health and well-being than the general population, have a higher prevalence of common medical problems and long-term conditions and are at increased risk of a second primary cancer
- The consequences of cancer and its treatment affect all biological, psychological and social domains
- There is a system in the UK of a primary care cancer care review, but *'there is a danger of it being a tick box exercise without acknowledgement of the personal impact of the cancer diagnosis and current treatment'*
- *A personal offer of an appointment, a review and ongoing support and care is hugely supportive for patients and can help to legitimise and ease their concerns*

Quality Improvement Idea

- Take a more proactive approach to ongoing cancer care with an offer of an appointment to discuss patients' needs across the biological, psychological and social domains with appropriate support and signposting
- Are you carrying out an effective cancer care review? Audit current reviews, introduce change and re-audit
- Utilise these resources
- How to [Carry out an effective cancer review in primary care](#)
- [MacMillan Top Tips](#) for how to manage patients in primary care after a cancer diagnosis
- [RCGP Consequences of Cancer Guidelines and Toolkit](#)
- [RCGP Cancer Recovery Package](#)
- The [Holistic Needs Assessment](#) for patients

NICE SUSPECTED CANCER GUIDELINES

The NICE guidance [NICE NG12](#) from 2015 had a major impact on primary care, with urgent referral pathways now radically changed. *The guideline was updated in 2021* with clarification on the role of FIT testing for occult blood in patients with suspected colorectal cancer and also for suspected prostate cancer. See separate chapters on FIT tests and prostate cancer for more on those topics.

In Scotland, the [Scottish referral guidelines for suspected cancer](#) have been updated in 2019, and we cover these separately.

► NICE Suspected Cancer: recognition and referral [NICE NG12, 2021](#)

When considering the guideline, a crucial point is that *all the included evidence came from primary care studies*. Patient selection, clinical responsibility and setting should all have been conducted in primary care to be eligible for inclusion, which makes it (the authors believe) the most evidence-based primary care applicable guideline. **Key points** from the guideline:

- **Recommendations** are organised both by cancer site (as is traditional) but also, for the first time, by presenting symptoms.
- **The guideline explicitly encourages GPs to use their clinical judgement.** They state: 'these recommendations are recommendations, not requirements, and they are not intended to override clinical judgement'. The guideline asks clinicians to continue to trust their clinical experience and it assumes a full history, examination and primary care-appropriate initial blood tests have been done. Recommendations are for symptoms or signs, which are 'persistent and unexplained' i.e., you have been unable to make a diagnosis after your initial primary care assessment.
- The positive predictive values (PPV) referral threshold for cancer, has been lowered from 5% to 3% (less for children). The authors trawled through the primary care literature to find *which clinical features are associated with a >3% positive predictive value (PPV) of cancer* and recommends that these should be referred urgently. A 3% threshold will obviously lead to many more referrals.
- There will be **more direct access** to investigations for GPs, such as scanning and endoscopy (including brain MRI and abdominal CT).
- We are encouraged to consider using a wider range of **diagnostic tests** in certain clinical situations (e.g. faecal occult blood testing as a diagnostic test, PSA tests with LUTS) and to re-consider how we interpret some tests (e.g. platelet counts)
- **Safety netting and referral.** The guideline makes these points, which have important implications:
 - The healthcare professional who orders the investigation takes responsibility for reviewing and acting on the results appropriately, or of passing on responsibility for this
 - Be aware of the possibility of false-negative results (e.g. for CXRs and faecal occult blood tests) and refer if symptoms continue
- Discussion with specialists is recommended when there is uncertainty about symptoms, signs, results or urgency of referral
- Once a decision to refer has been made, make sure that the referral is made within one working day

So, what are the recommendations?

There are recommendations for 37 different cancer types and well over 100 different symptoms over 378 pages! The quick reference PDF is nearly 100 pages long. (NB a criticism of the guideline is, therefore, its complexity, compared to a single electronic tool such as [QCancer](#)). The recommendations can be viewed and used online on the pathways pages:

[By Symptoms: NICE pathways](#)

[By Cancer Site or Type: NICE pathways](#)

NICE & Cancer Research UK have also produced an [Interactive Desk Easel](#) you can download to help navigate the guidance

To make it easier for you, we have synthesised the site and symptoms pathways into a single KISS guide to the most common presentations and cancer sites and **highlighted some of the clinical implications**.

The words used in the guideline have specific meanings:

- **Unexplained** means that a diagnosis has not been made in primary care after appropriate history, examination and primary care investigation.
- **Offer** denotes a stronger level of recommendation than **consider**, based on stronger evidence.
- Patient **values and preferences** should, of course, be taken into consideration when discussing referral options
- **Urgent** means within 2 weeks (i.e. consistent with 2-week wait, 2WW), **very urgent** within 48 hours and **immediate** is what it says on the tin!
- **Children and young people** means birth to 24

KISS: Upper GI cancers

Oesophagus and stomach NB Implication: when checking FBC, we need to take into account not just Hb levels but also platelet counts The change in recommendations to 'consider non-urgent endoscopy' in these situations should result in fewer urgent 2WW referrals. Dysphagia, upper abdominal mass or weight loss + reflux/dyspepsia aged over 55 are now the only indications for urgent referral. However, clearly, the clinical judgement here is key and many of these patient scenarios here we will still want to urgently refer.	■ Offer urgent (2-week wait, 2WW) direct access upper GI endoscopy if • Dysphagia, or • Age ≥ 55 with weight loss and any of upper abdominal pain, reflux or dyspepsia ■ Consider urgent referral if upper abdominal mass consistent with stomach cancer ■ Consider non-urgent endoscopy in people aged ≥ 55 with : • Treatment resistant dyspepsia • Upper abdo pain with low Hb • Raised platelets with any of nausea, vomiting, weight loss, reflux, dyspepsia or upper abdo pain • Haematemesis • Nausea or vomiting with any of weight loss, reflux, dyspepsia, upper abdo pain or thrombocytosis
Pancreatic, Liver and Gall bladder NB Implication: New-onset diabetes with weight loss over 60? Urgent CT scan.	■ Refer urgent 2WW if aged 40 or over and jaundice ■ Consider urgent direct access CT scan (or USS if not available) in aged 60 or over with weight loss and any GI symptoms (including pain, nausea, unexplained diarrhoea or constipation), or back pain or new-onset diabetes ■ Consider urgent direct access USS if an upper abdominal mass consistent with liver or GB

KISS: Lower GI Cancers

Lower GI

A significant change in practice is that NICE now advise that the FIT test should be used in primary care to guide referral for suspected colorectal cancer in patients with symptoms without overt bleeding but who do not meet the urgent cancer referral criteria. (DG30)

NG12 was updated in 2021 and now recommends FIT test for adults *without* rectal bleeding who:

- Are aged over 50 with unexplained abdominal pain, *or* weight loss *or*
- Are aged under 60 with changes in their bowel habit *or* iron deficiency anaemia, *or*
- Aged over 60 with anaemia, even in the absence of iron deficiency.

▪ Refer urgent 2WW if

- ≥ 40 with unexplained weight loss and abdominal pain, *or*
- ≥ 50 with unexplained rectal bleeding, *or*
- ≥ 60 with
 - Iron deficiency anaemia *or*
 - Change in bowel habit, *or*
- Tests show occult blood in their faeces

▪ Consider urgent referral if:

- Abdominal, rectal or anal mass or unexplained anal ulceration
- Adults aged under 50 with rectal bleeding and any of the following unexplained symptoms or findings
 - Abdominal pain
 - Change in bowel habit
 - Weight loss
 - Iron deficiency anaemia

KISS Lung and Pleural Cancers

Lung and mesothelioma

NB Implications:

We need to remember to routinely ask about asbestos exposure as the incidence of mesothelioma is rising rapidly

We need to consider unexplained fatigue as a potential presenting symptom of lung cancer, and so a CXR should be considered in patients over 40 who are 'Tired all the Time' if they have any of these additional features. We should again also look at platelet counts as thrombocytosis increases the risk of lung cancer.

NICE acknowledge that *false-negative CXRs* may occur. Consider discussing with radiology for a CT if high risk and continuing symptoms. [Scottish Guidelines](#) suggest urgent chest clinic referral if a CXR is normal but symptoms persist for more than 6 weeks.

Haemoptysis has a high PPV (2.4% to 17% increasing with age), thus the recommendation to refer urgently irrespective of CXR findings.

▪ Offer an urgent CXR if aged ≥ 40 and 2 or more of: (symptoms unexplained)

- Ever smoked or exposed to asbestos
- Cough
- Fatigue
- Breathlessness
- Chest pain
- Weight or appetite loss

▪ Consider an urgent CXR if ≥ 40 and any of:

- Persistent or recurrent chest infection
- Finger clubbing
- Supraclavicular or persistent cervical lymphadenopathy
- Chest signs consistent with lung cancer or pleural disease
- Thrombocytosis

▪ Refer urgent 2WW if

- CXR suggests lung cancer
- ≥ 40 with unexplained haemoptysis



KISS Breast and Gynaecological Cancers

Breast	▪ Urgent 2 WWW referral if • ≥ 30 with unexplained breast lump • ≥ 50 with unilateral nipple discharge, retraction or other changes of concern ▪ Consider urgent referral if • Suggestive skin changes • ≥ 30 with unexplained axillary lump ▪ Consider non-urgent referral ≤ 30 with unexplained breast lump
Ovarian NB Implication: the ovarian cancer guidelines recommending much wider use of CA-125 in primary care are unchanged from 2011. Of note, is the recommendation to check CA-125 in new onset of over-active bladder symptoms in women aged over 50.	▪ Urgent 2WW if examination finds ascites or a pelvic mass which is not obviously uterine fibroids ▪ Measure CA-125 in primary care in women (especially if ≥ 50) if they have persistent or frequent (particularly >12 times per month) • Persistent distension/bloating • Early satiety • Loss of appetite • Or abdominal pain • Increased urinary urgency and frequency • New symptoms suggestive of irritable bowel syndrome aged over 50 ▪ If CA-125 is ≥ 35 IU/ml arrange pelvic USS • Urgent 2WW referral if USS suggests ovarian cancer • If USS normal re-assess and investigate if appropriate
Endometrial and cervical NB Implication: note that there are no specific recommendations made for <i>inter-menstrual bleeding</i> , this is because the vast majority (95%) of cases of endometrial cancer occur in post menopausal women. However, it can occur pre-menopausally and BMJ2011;343:d3954 suggests referral for USS and endometrial biopsy if aged over 40 with a recent onset of an intermenstrual bleeding pattern. Scottish Guidelines suggest referral if persistent IMB despite normal pelvic exam, especially with other risk factors and for any woman on tamoxifen who develops unscheduled bleeding.	▪ Urgent 2WW if • Post-menopausal bleeding (unexplained vaginal bleeding >12 months after last menstruation) aged ≥ 55 ◦ Consider in women aged under 55 with PMB • Appearance of cervix is consistent with cervical cancer ▪ Consider a pelvic USS in women aged over 55 with • New unexplained vaginal discharge and have thrombocytosis or haematuria • Visible haematuria (see also urology section) and low Hb, raised platelets or high blood glucose levels
Vulval & vaginal	▪ Consider urgent 2WW if unexplained vulval lump, ulceration or bleeding or unexplained palpable mass in or at entrance to vagina

KISS Urological Cancers

Prostate cancer Age related PSA thresholds to 'consider' referral (2021 Update) ■ Under 40 years, clinical judgement ■ 40 - 49 years > 2.5 ■ 50 - 59 years > 3.5 ■ 60 - 69 years > 4.5 ■ 70 - 79 years > 6.5 ■ Over 79, use clinical judgement	■ Consider a PSA and DRE to assess for prostate cancer in men with • Any lower urinary tract symptoms (LUTs), or • Erectile dysfunction, or • Visible haematuria ■ Refer urgent 2WW if: • Prostate feels malignant on DRE (hard, irregular or nodules) ■ Consider refer urgent 2WW if: • PSA levels are above the age-specific reference range
Bladder and renal cancer NB Implication: the recommendation for those over 60 with non-visible haematuria seems odd! This is because the presence of either dysuria or a raised WCC raised the PPV of cancer above the 3% threshold. So, in a patient over 60 with non-visible haematuria ask about dysuria and check a FBC! The Scottish guidelines are simpler: urgent referral for painless visible haematuria and non-urgent if asymptomatic, unexplained persistent non-visible haematuria Scottish Guidelines	■ Refer urgent 2WW if • Aged 45 and over with unexplained visible haematuria without UTI • Visible haematuria that persists or recurs after successful treatment OF UTI • ≥ 60 with unexplained non-visible haematuria and either dysuria or a raised WCC on blood test ■ Consider non-urgent referral with recurrent or persistent unexplained UTI
Testicular and penile	■ Consider a direct access USS if unexplained or persistent testicular symptoms ■ Consider urgent 2WW appointment if • Non-painful enlargement or change in shape or texture of the testis • Penile mass or ulcerated lesion and an STI has been excluded • Unexplained or persistent symptoms affecting the foreskin or the glans

KISS: Skin Cancers

Malignant melanoma Weighted 7-point checklist Major features (2 points each) ▪ Change in size ▪ Irregular shape ▪ Irregular colour Minor features (1 point each) ▪ Largest diameter ≥ 7mm ▪ Inflammation ▪ Oozing ▪ Change in sensation NB implication: nodular melanoma, when there is vertical growth downwards through the skin, melanoma is a difficult diagnosis as they do not fulfil classic criteria and furthermore, 30% of them are not pigmented. Nodular melanomas account for 15% of cases melanoma. Consider with a rapidly growing nodule over weeks or months, often dome-shaped, firm and symmetrical. They are most often black, red or skin coloured. For more information and images see nodular melanoma	▪ Urgent 2WW referral if a suspicious pigmented lesion and • Weighted 7-point checklist score of ≥ 3, OR • Dermoscopy suggests melanoma ▪ Consider urgent referral if a pigmented or non-pigmented skin lesion suggests nodular melanoma
Squamous cell carcinoma Typical features: enlarging scaly or crusted lumps, grow over weeks and months, may ulcerate; often tender or painful and located on sun-exposed sites	▪ Consider urgent 2WW if lesion raises suspicion of SCC
Basal cell carcinoma Typical features: a 'pearly' or 'waxy' skin nodule, ulcer with raised rolled edge, prominent fine blood vessels around a lesion	▪ Routine referral if skin lesion raises the suspicion of basal cell carcinoma ▪ Consider urgent if particular concern about lesion site or size

KISS: Head, Neck & Brain cancers

Laryngeal NB Implication: how long is persistent? NICE don't specify. The Scottish Guidelines suggest if hoarseness or an unexplained neck lump persists for more than 3 weeks. They also recommend referral for persistent pain in the throat lasting >3 weeks, which the NICE guidelines do not.	■ Consider urgent 2WW if • Hoarseness, persistent and unexplained ≥ 45 • Unexplained lump in neck ≥ 45
Oral NB implication: Oral cancer rates are rising, largely thought to be due to greater prevalence of HPV. Urgent GP referral to a dentist is an interesting idea, which reflects the 'different level of expertise' between GPs and dentists. (This should be an NHS 'no cost' referral, which is easier said than done! We recommend that if this is likely to create a barrier or significant delay, to refer straight to oral surgery) Erythroplakia is an unexplained patch of erythema on a mucous membrane and erythroleukoplakia with red and white patches. This may represent carcinoma in situ and may transform into invasive SCC. Erythroplakia is much higher risk than leukoplakia. Scottish Guidelines suggest urgent referral if unexplained ulceration or red or red+ white patches persist for >3 weeks.	■ Consider urgent 2WW if • Unexplained oral ulceration lasting > 3 weeks • Persistent and unexplained lump in the neck ■ Consider an urgent referral (within 2 weeks) for assessment by a dentist in people who have either • A lump on the lip or oral cavity or • A red or red & white patch in the oral cavity consistent with erythroplakia or erythroleukoplakia • Dentist or GP will then arrange an urgent suspected cancer referral if dentist thinks signs are consistent with possible oral cancer
Thyroid	■ Consider urgent 2WW referral in people with unexplained thyroid lump
Neck lumps NB implication: we commonly see neck lumps, most often harmless reactive nodes. There is no guidance as to how 'persistent' is defined, but Scottish Guidelines suggest if > 3weeks.	■ Consider urgent 2WW if • Unexplained neck lump ≥ 45 • Neck lump persistent and unexplained
Brain NB implication. Interesting that <i>headache</i> does not feature. In primary care the risk of a brain tumour with a headache presentation is 0.09% BJGP2008;58:880 . Scottish Guidelines suggest urgent referral if recent-onset non-migrainous headache with features of raised intracranial pressure e.g. papilloedema. A recent paper BJGP 2017;67:278 has explored how difficult it is for GPs to diagnose brain tumours, as the positive predictive value of individual symptoms is so low.	■ Consider urgent direct access brain MRI in adults with progressive, sub-acute loss of central neurological function ■ Very urgent (<48 hours) in children and young people with newly abnormal cerebellar or other CNS function

KISS: Haematological Cancers & Sarcomas

Leukaemia	Arrange very urgent FBC if any of: ■ Pallor, persistent fatigue, generalised lymphadenopathy or hepatosplenomegaly ■ Unexplained fever, persistent or recurrent infection, bruising, bleeding or petechiae and persistent or unexplained bone pain in children and young people ■ Refer children & young people for immediate assessment if unexplained petechiae or hepatosplenomegaly
Myeloma NB Implications: Myeloma is 'easily missed' BMJ2012;344:d7953 in primary care due to its protean clinical characteristics. Consider with <i>bone pain, fatigue and weight loss</i> especially if also anaemia, renal impairment, hypercalcaemia and raised ESR. The presence of a monoclonal paraproteinaemia in urine or blood is a key feature. Early diagnosis helps prevent debilitating complications such as pathological fractures and spinal cord compression and improves survival.	■ Offer a FBC, Calcium blood tests & ESR for people over 60 with persistent bone pain, particularly back pain, or unexplained fracture ■ Offer very urgent protein electrophoresis and urine Bence Jones protein test in patients if the ESR and presentation are consistent with possible myeloma (including hypercalcaemia or leukopenia in patients over 60) ■ Refer urgently via 2WW if the results suggest myeloma
Non-Hodgkins & Hodgkins Lymphoma	■ Consider urgent 2WW referral in adults (very urgent < 48 hours in children & young people) with unexplained splenomegaly or lymphadenopathy. Take into account any associated symptoms e.g. fever, night sweats, dyspnoea, pruritus, weight loss and alcohol-induced lymph node pain
Bone Sarcoma NB Implication: for most GPs, this will be a 'once a career' diagnosis. Pain and loss of function are thought to be the main presenting features. Note unexplained pain and swelling means Osgood-Schlatter disease has been excluded, see section on Osgood-Schlatter CKS 2015	■ Consider urgent X-Ray with unexplained bone swelling or pain (very urgent <48 /24 in children and young people) ■ If X-Ray suggests bone sarcoma urgent 2WW, very urgent in children & young people
Soft Tissue Sarcoma NB Implication: we commonly see soft tissue lumps, we rarely refer them for urgent USS so this is a significant change in practice and one radiology departments may struggle to deliver. Scottish Guidelines also suggest urgent referral if a lump is increasing in size, or is >5cm or fixed deep to fascia or there is regional LN enlargement.	■ Consider urgent direct access USS to assess for soft tissue sarcoma with an unexplained lump that is <i>increasing in size</i> (very urgent <48 /24 in children & young people) ■ If the USS findings are suggestive of sarcoma or uncertain & clinical concern persists, arrange urgent 2WW referral

KISS: Musculoskeletal Symptoms

Persistent back pain, or bone pain or unexplained fracture aged ≥ 60 NB Implication: surprisingly, the guideline does not suggest an urgent MRI for new onset of persistent back pain in older adults but clinical experience would suggest a low threshold for this if the pain remains unexplained after these initial blood tests, as well as assessment of prostate and breast. See below for MSCC.	- Check FBC, calcium studies and ESR for possible myeloma - Offer very urgent protein electrophoresis and urine Bence Jones protein test if the ESR and presentation are consistent with possible myeloma - Refer urgently via 2WW if the results suggest myeloma
Back pain with weight loss aged ≥ 60	Urgent access abdominal CT scan for possible pancreatic cancer
Possible Metastatic Spinal Cord Compression This is not addressed in this NICE guideline, but was last year in NICE QS56. It was covered in BJGP 2014;64:e596 and is in the Scottish Guidelines. Whilst most patients diagnosed with MSCC have an established diagnosis of cancer, in 10% to 20% of cases MSCC is the first presentation of malignancy. Note that a normal neurological examination does NOT preclude epidural disease or evolving MSCC. The definitive test is MRI of the whole spine. <i>All cancer centres should have an MSCC co-ordinator</i> to urgently discuss possible case with, and patients with cancers that spread to bone should receive an MSCC symptoms card warning them about early symptoms to look out for.	Scottish Guidelines - Urgent referral for patients with known cancer (particularly prostate, breast, lung or multiple myeloma) with a history of cancer and any of the following symptoms - Significant localised back pain, especially thoracic - Severe, progressive pain - Spinal pain aggravated by straining - Nocturnal pain, especially if preventing sleep - Radicular pain e.g. round chest, down front or back of thighs - Limb weakness or difficulty walking - Sensory loss, including perineal anaesthesia, bladder or bowel dysfunction

KISS: Non-site-specific Symptoms

Unexplained weight loss	Carry out an assessment for additional signs, symptoms or findings that may clarify which cancer is most likely and offer urgent 2WW investigation
Unexplained appetite loss	Carry out an assessment for additional signs, symptoms or findings that may clarify which cancer is most likely and offer urgent 2WW investigation
Deep Vein Thrombosis	Carry out an assessment for additional signs, symptoms or findings that may clarify which cancer is most likely and consider urgent investigation or urgent 2WW investigation

KISS: Cancers in Children and Young People

The BMJ summary [BMJ2015;350:h3036](https://www.bmj.com/lookup/doi/10.1136/bmj.2015.350.h3036) of this part of the guideline has a colourful 'infographic' you may find useful. Children are up until aged 16, and young adults to 24. This part of the guideline uses a lower threshold (<3% cancer risk) for further investigation.

Abdominal Features	Possible Cancer	Investigation & Referral
▪ Splenomegaly unexplained +/- symptoms	▪ Lymphoma	▪ Very urgent <48 /24 referral
▪ Hepatosplenomegaly unexplained	▪ Leukaemia	▪ Immediate referral
▪ Abdominal mass	▪ Wilm's tumour or neuroblastoma	▪ Very urgent <48 /24 referral
Lumps		
▪ Lymphadenopathy + symptoms of fever, sweats, dyspnoea, pruritus, weight loss	▪ Lymphoma	▪ Very urgent <48 /24 referral
▪ Lymphadenopathy generalised	▪ Leukaemia	▪ Urgent FBC ▪ Immediate referral if suggests leukaemia
▪ Soft tissue lump, unexplained and growing	▪ Soft tissue sarcoma	▪ USS within 48 hours
Bleeding		
▪ Haematuria visible & unexplained	▪ Wilm's tumour	▪ Very urgent <48 /24 referral
▪ Petechiae unexplained	▪ Leukaemia	▪ Immediate referral if suggests leukaemia
▪ Bleeding, pallor, bruising unexplained	▪ Leukaemia	▪ Urgent FBC ▪ Immediate referral if suggests leukaemia

KISS: Cancers in Children and Young People (continued)

Abdominal Features	Possible Cancer	Investigation & Referral
Neurological		
▪ Abnormal CNS function	▪ Brain	▪ Very urgent <48 /24 referral
▪ Absent red reflex	▪ Retinoblastoma	▪ Immediate
Skeletal		
▪ Unexplained bone swelling or pain	▪ Bone Sarcoma or leukaemia	▪ Urgent X-ray within 48 hours & then very urgent referral if X-ray is suggestive ▪ Urgent FBC if persistent or unexplained bone pain
Non-specific ▪ Persistent fatigue, unexplained fever, unexplained or persistent infection		▪ Urgent FBC

NB Quality Improvement Ideas: Early diagnosis of cancer

- Include some recent cancer diagnoses as significant events. Could any have been diagnosed at an earlier stage?
- Organise some in-house training based around **these new guidelines**
- Are FIT (faecal immunochemical tests) available to you? Do you have direct access to abdominal CT or brain MRI? Liaise with your local providers and commissioning group to make these investigations available.
- Do you need to revise local referral pathways?

Learning Points: Early diagnosis of cancer

- We really do have a problem with late diagnosis of cancer...it is estimated implementation of these guidelines will save 5,000 lives per year
- Improving early diagnosis via earlier and more streamlined referral is a key part of [Achieving World Class Cancer Outcomes 2016 Report](#) in England
- New guidelines from NICE and in Scotland encourage earlier referral across a broad range of different cancers and symptoms. These introduce some key changes to practice and contain multiple learning points.
- Testing for faecal occult blood with FIT test as part of the clinical assessment of patients presenting with abdominal symptoms but without overt bleeding should now be routine.
- NICE now advise using the newer **FIT test to help aid colorectal cancer referral decisions in patients presenting with abdominal symptoms but without overt rectal bleeding, if they are:**
 - Are aged over 50 with unexplained abdominal pain, or weight loss or
 - Are aged under 60 with changes in their bowel habit or iron deficiency anaemia, or
 - Aged over 60 with anaemia, even in the absence of iron deficiency.
- Consider a DRE and PSA for all men presenting in primary care with LUTS or erectile dysfunction, with appropriate patient counselling. NICE have updated and clarified thresholds to **consider referral under the 2WW if PSA above the following age-adjusted thresholds:**
 - Under 40 years, clinical judgement
 - 40 - 49 years > 2.5
 - 50 - 59 years > 3.5
 - 60 - 69 years > 4.5
 - 70 - 79 years > 6.5
 - Over 79, use clinical judgement
- NICE criteria:
 - [By Symptoms: NICE pathways](#)
 - [By Cancer Site or Type: NICE pathways](#)
 - [Interactive Desk Easel](#)
- [Scottish Guidelines](#)

SCOTTISH CANCER REFERRAL GUIDELINES 2019

In 2019 the [NHS Scotland Referral Guidelines for Suspected Cancer 2019](#) were published.

These represent a significant update from the previous guidelines in 2014, with some important implications for primary care. The guidance is best viewed [online](#) and has an easy-to-navigate quick reference guide (and an associated app). **Significant changes** from the previous guideline include:

■ Thrombocytosis

- In line with NICE guidance and also recent research discussed elsewhere on this course (see Platelets & Cancer chapter) thrombocytosis is listed as a strong risk factor for cancer, with a higher PPV than a breast lump or haemoptysis
- For cancers, most associated with raised platelets think **LEGO-C**
 - Lung
 - Endometrial
 - Gastric
 - Oesophageal
 - Colorectal

■ Lung cancer

- Thrombocytosis as a risk marker
- Age range for urgent referral for persistent haemoptysis in smokers/ex-smokers reduced to age ≥ 40
- Check CXR for fatigue in smokers aged ≥ 40 , appetite loss and persistent or recurrent chest infections

■ Breast Cancer

- Age range for referral for new discrete lumps is reduced to 30 from 35
- Refer new asymmetrical nodularity in patients aged over 35 that persists after 2-3 weeks
- Refer new isolated axillary lymph node in women persisting after 2-3 weeks

■ Colorectal

- Thrombocytosis as a risk marker
- Faecal immunochemical test (FIT) for occult blood in symptomatic patients is being rolled out (currently in pilot projects across different health Boards), again in line with NICE guidance (see chapter on FIT tests for colorectal cancer) - with a new indication for FIT test if abdominal pain and weight loss.

■ Upper GI cancer

- Urgent endoscopy for unexplained weight loss with other alarm symptoms aged ≥ 55
 - Dyspepsia with no red flags at any age should be managed in primary care
- As with NICE new-onset diabetes and weight loss, esp in people >55 is now an urgent referral.

■ Prostate cancer

- A rough guide to PSA levels
 - Less than 60 years: PSA < 3
 - Aged 60 to 69y: PSA < 4
 - Aged 70 to 79y: PSA < 5
- Note PSA test may be raised within 3 days of ejaculation, 6 weeks of a UTI, catheterisation or prostate biopsy but digital rectal examination effect is not significant
- Pragmatic, individualised referral decisions based on 'realistic medicine' should be applied for patients aged over 80 and in older men, routine or no referral may be appropriate; however, referral may be considered
 - Aged 80 to 85 if PSA > 10
 - Aged ≥ 86 , if PSA > 20

■ Urological

- Age range now ≥ 45 for unexplained visible haematuria
- Note raised white cell count on FBC associated with bladder cancer (in line with NICE guidance)

- **Children, teenagers and young adults**

- Unexplained petechiae or purpura need an emergency referral (was urgent)
- Unexplained haematuria (new)
- Failure to thrive, persistent pallor and weight loss or clinical concern then discuss with the paediatrician
- Brain tumours, new red flags include
 - Loss of balance, increasing head circumference, failure of closure of fontanelles, new neurological signs especially if associated with behaviour change

Realistic Medicine

Importantly this new guideline also emphasises the importance of considering the principles of [Realistic Medicine](#) when deciding on referral decisions. Whichever health system we work in it is likely to be resource-limited and we have to tread the difficult path between not missing likely cases of cancer and also not over-referring. Avoiding over-referral is also important to protect our patients from the danger of 'overdiagnosis', including all the associated anxiety that a cancer referral brings.

Good patient communication, shared decision making and managing risk are the key to this. They propose we and our patients ask:

- Is this test, treatment or procedure really needed?
- What are the possible benefits and risks?
- What are the possible side effects?
- Are there simpler, safer alternatives?
- What would happen if I did nothing?

NB Learning Points

- The new [NHS Scotland Referral Guidelines for Suspected Cancer 2019](#) have introduced some significant changes, particularly looking at platelets as possible cancer markers for LEGO-C (lung, endometrial, gastric, oesophageal and colorectal) cancers
- FIT testing is being rolled out across different Scottish Health Boards
- Consider the principles of [Realistic Medicine](#) when making referral decisions

COLORECTAL CANCER IN YOUNGER ADULTS

You see a young adult with some 'blood on the paper'; do you always do a rectal examination? Should you do an FBC? What about a new case of probable irritable bowel syndrome, should you check their faeces for calprotectin or occult blood?

A recent BMJ editorial has highlighted the *rising rates of colorectal cancer in younger adults*, aged under 50 [BMJ2019; 365:l4280](#). Recent research [BMJ2019;365:l2249](#) has stressed that whilst the absolute number remains low, there has been a significant increase in incidence, especially between the ages of 20 to 39 leading to calls to lower the age of screening programmes. In response to this increase in incidence, in the US the screening programme age has been lowered from 50 to 45.

The effect is seen across high-income industrialised countries, and although it has been observed for a number of years it is only just starting to receive attention. It is thought the main driver are lifestyle factors with worsening diet, obesity and declining exercise however other theories have been postulated from the effect of antibiotics on gut flora to plastics.

Another recent study [BJGP2017](#) highlighted *the increasing incidence of colorectal cancer and inflammatory bowel disease in younger adults* and has given us a [new risk tool](#) to help us to assess patients and make appropriate referral decisions. There is also an international campaign to raise awareness called [Never Too Young](#).

This is a hugely important issue for us given that it is estimated that one in twelve of our consultations is for gastrointestinal symptoms. There has also been a significant and highly relevant update to the [NICE NG12 Suspected Cancer guidance in 2017](#) which recommends wider use of occult blood testing as a diagnostic aid using the FIT test (faecal immunochemical test, see separate section on this).

Research [BJGP2017](#) indicates that younger adults are more likely to have multiple GP presentations before diagnosis, are more likely to present as an emergency with advanced disease and have a worse 5-year survival. Delayed diagnosis is common for younger patients with 20% of new cases having presented to primary care 5 or more times before referral.

So, how should we manage young adults presenting with GI symptoms?

This new research [BJGP2017](#) was a large case-control study. The authors looked at new cases of colorectal cancer (CRC) and inflammatory bowel disease (IBD, which is also increasing in incidence) in adults aged under 50 and then looked at their presenting symptoms compared to matched controls. They found that *rectal bleeding and change in bowel habit were strongly predictive of CRC and IBD when combined with abnormal haematology*, including low haemoglobin, raised platelets or white cells, low MCV and raised inflammatory markers. For individual symptoms, the positive predictive values (PPV) are low, but in combination, the PPVs of significant pathology rise significantly. The researchers produced a useful table in the paper with the PPVs for different symptoms e.g. rectal bleeding alone the PPV is just 1.2% for cancer or IBD, but with rectal bleeding plus low MCV it rises to 3.2% (NB Macmillan are developing an online version of this risk assessment tool, in the meantime it is freely available in the paper).

On the basis of these findings, we should manage patients according to a new proposed [risk assessment tool](#) based on this research. Patients at a higher level of risk (>3%) should be referred for urgent colonoscopy or specialist assessment. Patients with an indeterminate level of risk (1% to 3%) should have their faecal calprotectin measured, as IBD is the more likely of the two diagnoses and can be excluded by a normal calprotectin level. Those with a lower level of risk (<1%) can be managed expectantly and appropriately safety-netted.

The paper concludes that the results support a diagnostic strategy outlined in the [NICE IBS Guidance](#), which has been updated in 2017. *Patients presenting with possible IBS symptoms* should be assessed and clinically examined for suspected cancer 'red flag' indicators, have an inflammatory marker for IBD (i.e. faecal calprotectin) and bloods (FBC, ESR, CRP and coeliac antibodies).

Does that mean we should always do an FBC when assessing patients with 'piles'?

Clinical judgement here is key. It certainly means we should always examine younger adults with rectal bleeding. Then it very much depends on clinical judgement, but if the bleeding is unexplained (i.e. not obviously due to haemorrhoids, fissure or local cause) or associated with other symptoms then this research does support doing an FBC because an abnormal result greatly increases the likelihood of significant pathology.

What does the updated NICE suspected cancer guidance say about FIT tests?

This is very relevant to this issue of assessing bowel symptoms in adults aged under 50. Initially, a new NICE diagnostic guidance came out in 2017 ([DG30](#)) recommending:

- Consider tests for faecal occult blood using the FIT test for people without rectal bleeding but with unexplained symptoms that do not meet the criteria for an urgent suspected cancer pathway, and refer if positive.

Then [NG12](#) (suspected cancer guideline) was updated in 2021 and now recommends FIT test for adults without rectal bleeding who:

- Are aged over 50 with unexplained abdominal pain, or weight loss or
- Are aged under 60 with changes in their bowel habit or iron deficiency anaemia, or
- Aged over 60 with anaemia, even in the absence of iron deficiency.

So now *we should consider using the test, based on our clinical judgement, in any patient of any age presenting with lower-risk symptoms not warranting an urgent (2WW) referral.*

NB Learning Points:

- The incidence of colorectal cancer in adults aged under 50 is increasing
- There is an international Never Too Young campaign to raise awareness
- New research shows that rectal bleeding and altered bowel habit are strongly predictive of CRC or IBD when combined with abnormal haematology
- Consider using this new risk assessment tool to aid referral decisions. The latest guidance suggests lowering the threshold for checking an FBC and also a FIT test in patients presenting with lower-risk GI symptoms which do not warrant an urgent referral
- Research suggests abnormal haematology and clinical symptoms significantly increase the PPV of significant pathology
- NICE now say a FIT test can be done on any patient at any age presenting with lower-risk symptoms which do not warrant an urgent referral
- NICE also advise that patients with symptoms likely to be IBS should be clinically assessed for suspected cancer 'red flags', have a faecal calprotectin and also bloods (FBC, inflammatory markers and coeliac antibodies)

FAMILIAL BREAST CANCER

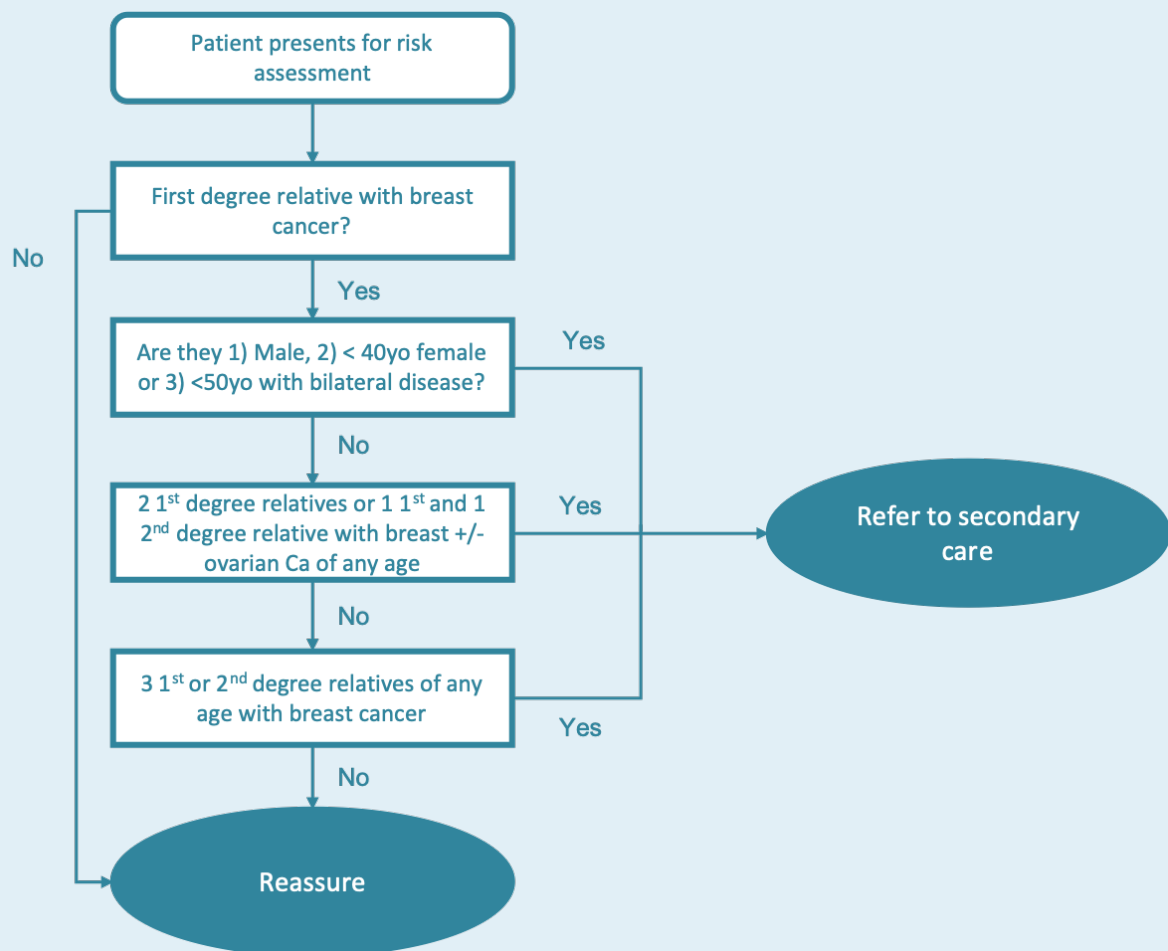
KISS: Assessment of Breast Cancer Risk in Primary Care

[NICE 2013 CG 164](#), [BMJ 2013;346:f3829](#)

Tamoxifen and related medications can reduce the risk of developing breast cancer and women at higher risk may be offered chemoprophylaxis or increased surveillance by secondary care. Our role is to identify those at higher risk when they present with concerns.

If a woman with no personal history of breast cancer presents with concerns:

- Take a first and second degree family history (including paternal side)
 - First degree = parents/sibling/child
 - Second degree = grandparent/aunts & uncles/nephews & nieces/half-siblings
 - Ask about age, type of cancers, multiple cancers (e.g. bilateral), Jewish ancestry
- Follow the algorithm below to assess when a referral is recommended:



- People without any of the above can be managed in primary care – but seek advice if any of the following are present in a woman with an FHx of breast cancer not meeting the above criteria:
 - History of glioma or childhood adrenal cortical carcinoma, sarcoma in a relative <45yo or Jewish origin in the presence of 1 1st or 2nd-degree relative with breast cancer
- Patient information can be found on lots of sites: [Cancer Research UK](#), [MacMillan](#) and a good one specifically on hereditary breast cancer on [patient.co.uk](#)

BREAST CANCER SURVIVORSHIP

There are currently over 500,000 breast cancer survivors in the UK. As this recent article in the BJGP ([2017;67:140](#)) notes it is estimated there will be 2 million survivors by 2040. 80% of women with breast cancer now survive >10 years.

Most of these women will not require extensive secondary care follow-up - indeed it seems unlikely this would be even possible - and there is an ongoing conceptual shift in the long-term management of cancer patients acknowledging the importance of both self-management and primary care input.

Cancer centres are now giving patients a “recovery package” which consists of:

- **Cancer care review:** by GP within 6 months, helps share knowledge on local services
- **End of treatment summary:** explains ongoing surveillance, red flags to watch for
- **Health needs assessment:** a comprehensive questionnaire to identify ongoing concerns
- **Health and wellbeing events:** group activities promoting health education, e.g. maintaining a healthy lifestyle, managing long-term consequences of treatment

► Medical Issues in Survivors

Clinics are becoming more geared towards patient self-referral, but there are many common problems the paper highlights which may be managed in primary care. Equally, there are a number of longer-term implications from the disease +/- its treatments of which we should be aware.

▪ Menopausal symptoms

- Commonly caused by tamoxifen and aromatase inhibitors (the latter is particularly associated with urogenital symptoms such as atrophic vaginitis), but also may result from various chemotherapies and surgery involving the removal/destruction of the ovaries
- May be more severe than physiological menopause
- HRT should be avoided
 - NICE ([CG80](#)) recommends HRT should not be given except in exceptional circumstances where women have severe symptoms after a discussion about the potential risks
 - The paper comments that topical oestrogen preparations “are best avoided where possible” due to the potential for systemic absorption, although data is lacking on its safety or otherwise and NB notes that many oncologists use this treatment
- Various alternative approaches may be helpful:
 - Antidepressants, gabapentin, acupuncture, psychological therapies
 - SSRIs can be effective, but due to their inhibition of CYP2D6 (required to convert tamoxifen to its active metabolite) fluoxetine, duloxetine, bupropion and paroxetine should be avoided in women on tamoxifen.,
 - Sertraline, citalopram, escitalopram and venlafaxine exert less inhibition and are considered reasonable choices ([BMJ 2017;354:i5309](#))
- For vaginal symptoms, non-hormonal topical treatments are recommended 1st line
- Symptoms may be so troubling that patients wish to cease hormonal treatment - they should be referred back to their oncologist for discussion on risk/benefit/alternatives (e.g. tamoxifen is usually less symptomatic than aromatase inhibitors)

▪ Bone health

- All women with early breast cancer are at increased risk in later life, but especially those on aromatase inhibitors



- The BJGP paper recommendations have arguably been superseded by international guidelines on the management of aromatase inhibitor-related bone loss ([J Bone Oncology June 2017](#)):
- ***Aromatase inhibitors significantly increase the risk of osteoporosis and fracture***
 - All patients should to perform weight-bearing exercises and receive calcium/ vitamin D supplementation (BJGP recommends a calcium-rich diet and vit D supps at 1-2000u daily)
 - **Patients need a baseline DEXA scan. Bone-sparing agents are advised if:**
 - T score <2, or
 - T score <1.5 and 1 additional risk factor, or
 - 2 risk factors
 - Risk factors include: age >65, smoker, BMI <20, FHx hip fracture, personal fragility fracture history>50yo, 6/12+ oral steroid use
 - Data shows FRAX will under-estimate risk in women using aromatase inhibitors and fracture risk may be significantly increased in this group if risk factors are present even if a normal bone mineral density
 - Denosumab or bisphosphonates (oral or IV) are recommended options
 - Women should have BMD checked at 1-2years whether on treatment or not
 - The paper also discusses the potential cancer benefits of bisphosphonates - there a survival benefit is seen in post-menopausal women taking AIs with a reduction in the risk of bone mets (Lancet [2015;386:1353](#) and [2015;386:13319](#)) although whether this is seen with oral as well as IV treatment is somewhat uncertain
- **Tamoxifen in premenopausal women is less problematic** - bone loss is only accelerated by 1-2% per annum - and is not routinely an indication for bone protection or surveillance
- **Tamoxifen has little effect on post-menopausal bone density or fracture risk**
- ***Fatigue***
 - Very common - affects 30-50% within 5 years of treatment - poorly understood, mechanism is unknown
 - Risk factors include chemotherapy, depression, anxiety, catastrophising personality
 - Encouraging physical activity, acupuncture, CBT, yoga, massage may help
 - Herbal treatments such as guarana and ginseng are reported as helpful but lack RCTs
- ***Mental health***
 - Good communication is essential to identify psychological problems, with self-management helpful or talking therapies for more troubling symptoms
 - Chemotherapy can lead to cognitive changes and may persist - data shows a range of non-pharmacological therapies (e.g. exercise, yoga, mindfulness) can be helpful
- ***Breast changes***
 - Physical changes to a woman's breast(s) are common with surgery and radiotherapy
 - Early counselling is vital, but if there is any doubt regarding a change review in breast clinic is appropriate
 - Lymphoedema affects 15-20% women after surgery - but is 3x more likely after axillary node dissection than with sentinel node biopsy. Radiotherapy further increases risks.
 - Patients are given advice on self-management and prevention and treatment (e.g. manual drainage, compression therapy and good skincare).
 - Venepuncture should be avoided on the affected side
- ***Long-term complications from breast cancer treatment***
 - Infertility may result from chemotherapy;
 - Cardiotoxicity is seen with some chemos, with possible long-term increased risk such as LVF. More common in those with pre-existing CVD risk.
 - Patients and doctors need to remain vigilant for recurrence (local or metastatic) and other primary malignancies, which are more common
 - e.g. endometrial cancer > tamoxifen, lung cancer > chest wall radiotherapy, blood cancers > chemotherapy exposure

NB Learning Points: Breast Cancer Survivorship

- 80% of women diagnosed with breast cancer will live for >10 years
- Self-management is key in long-term care, with easy access to clinics for patients
- Menopausal symptoms are very common with hormonal treatment - a range of interventions may help ease troubling symptoms
- Bone health should be monitored regularly in those on aromatase inhibitors and bisphosphonates or denosumab started in those with significant bone density loss
- Women are at increased risk of recurrence and new primary cancers
- MacMillan provides some excellent resources for healthcare professionals and patients:
 - The Recovery Package - includes how to sign up for an electronic version of a Holistic Health Needs Assessment
 - Also a useful Top 10 Tips for a cancer care review

NB Quality Improvement Ideas: Breast Cancer Survivorship

- Audit breast cancer survivors on tamoxifen for antidepressant use
 - Are any receiving fluoxetine, paroxetine, duloxetine or bupropion?
 - If so, invite them in to discuss alternative treatments
- Audit breast cancer survivors receiving tamoxifen or aromatase inhibitors
 - Are they receiving appropriate bone health follow up, e.g. annual DEXAs if on AI, and not annual DEXAs if they are on tamoxifen?

CHRONIC KIDNEY DISEASE

There have been significant changes in CKD. NICE published an important update of their CKD Guideline [NICE NG203, 2021](#) in November 2021. This new guideline has important implications for us in primary care, especially in terms of assessment of risk, referral guidance and the fact that ethnicity has been taken out of the eGFR equation. The guideline was summarised in the [BMJ 2021;374:n1992](#) and discussed in an editorial [BMJ 2021;374:n2159](#).

The other Hot Topic with CKD is the emerging role of the seemingly ubiquitous SGLT2 inhibitors. With CKD we know that improved blood pressure control, diabetic control and also ACEi or ARBs in patients with CKD and microalbuminuria improve outcomes. There is now ***emerging evidence that SGLT2 inhibitors also improve outcomes in patients with proteinuric and non-proteinuric CKD and in patients both with and without diabetes***, based on the [DAPA-CKD](#) and [EMPA-KIDNEY](#) trials and a subsequent [meta-analysis](#) - all discussed below. **The SGLT2i dapagliflozin is now recommended as a treatment option for CKD by NICE in certain situations, even in patients without diabetes** [NICE TA775, 2022](#).

► ***Dapagliflozin in patients with CKD: The DAPA-CKD Trial*** [NEJM 2020;383:1436](#)

Recent studies looking at SGLT2 inhibitors have shown improvements in cardiovascular and renal outcomes in large trials in patients with type 2 diabetes (e.g. CREDENCE Trial, see diabetes section), and more recently in patients with chronic heart failure both with and without diabetes (DAPA-HF and EMPEROR-Reduced Trials, see Chronic Heart Failure chapter). **The DAPA-CKD Trial was designed to test the hypothesis that SGLT2i may improve clinical outcomes in patients with CKD.**

- This was a prospective double-blind RCT in 4,300 patients with CKD *and* proteinuria
- Patients had an eGFR between 25 and 75 ml/min and a raised albumin creatine ratio (ACR 22.6 mg/mmol to 565 mg/mmol; NB the ACR units in the original publication are mg/g but we have converted them to mg/mmol which are the standard UK units and reflect the subsequent NICE guidance below).
- 70% of patients had type 2 diabetes, 30% did not
- They were randomised to dapagliflozin 10mg or placebo, added to standard background therapy including ACEi or ARBs. The primary outcome was a composite of sustained decline in eGFR of at least 50%, end-stage renal disease or death from renal or cardiovascular causes
- **Results:**
- The trial was stopped after a median of 2.4 years, with the primary outcome occurring in 14.5% of the placebo group vs. 9.2% of the dapagliflozin group, *NNT to prevent one primary outcome was 19*
- Overall mortality was reduced from 6.8% to 4.7%
- The effects of dapagliflozin were similar in patients with and without Type 2 Diabetes
- The incidence of adverse events and serious adverse events were similar in both groups
- **Conclusion:** Dapagliflozin improves outcomes in patients with CKD and proteinuria, including mortality, whether patients have diabetes or not. The authors state that their trial '*confirms that the kidney protective effects of SGLT2 inhibitors extend to a broader range of patients with CKD without type 2 diabetes*'.

► ***Empagliflozin for CKD: EMPA-KIDNEY*** [N Engl J Med 2023; 388:117-127](#)

Further evidence for the renal benefits of SGLT2i for people with and without diabetes has recently come in the form of the **EMPA-KIDNEY trial in January 2023**. This further suggests a class effect for renal protection in CKD even for people without diabetes, this time with empagliflozin:

- RCT of 6609 patients with a broad range of CKD randomised to empagliflozin 10mg daily or placebo
 - Patients had either eGFR 20-45 ml/min OR eGFR >45 with raised ACR > 22.6 mg/mmol
- Primary outcome was a composite of CKD progression (end-stage kidney disease, sustained decrease in eGFR to <10 ml/min, sustained decrease in eGFR of ≥40 ml/min, or death from renal causes) or death from CVD

- The trial was stopped early after a median of 2 years as significant efficacy was established
- There was a **significant reduction in the primary endpoint in the empagliflozin group** (HR 0.72, 95% CI = 0.64-0.82) and like the DAPA-CKD trial the **results were consistent irrespective of diabetes status**
 - Relative risk reduction 28%, absolute risk reduction over 2 years from 16.9% placebo, to 13.1% empagliflozin over 2 years = NNT of 25
- Alongside this there was a **significant reduction in all-cause hospitalisation in the empagliflozin group** (HR 0.86, 95% CI = 0.78-0.95)
- Risk factors (blood pressure, body weight and urine ACR) were lower in the empagliflozin group.
- The rates of serious adverse events were similar
- **Conclusion:** Empagliflozin improves clinically meaningful CKD endpoints including cardiovascular mortality in a wide range of CKD patients (including those *without* raised ACR and eGFR 20-45) irrespective of diabetes status and with no increase in serious adverse events

An associated editorial [NEJM 2023;388;179](#) concludes that EMPA KIDNEY is another step forward in the management of CKD, and that the results support prescribing SGLT2i to prevent CKD progression in a broader range of patients than previously studied. It is important to note though that most of the patients were receiving renin angiotensin system (RAS) inhibitors (i.e. ACEi or ARBs), and in a sub-group who were not the beneficial effect of empagliflozin was not as evident. They conclude that the trial adds to the evidence that SGLT2i improve outcomes when they are added to ACEi or ARB.

► ***Meta-analysis of renal outcomes with SGLT2 inhibitors*** [Lancet 2022; 400: 1788–801](#)

On the back of the EMPA-KIDNEY study an updated meta-analysis was conducted to incorporate new data on the renal benefits of SGLT2i. Whilst the benefits in those with type 2 diabetes is now well established, the **real added value of this study was the new data on CKD outcomes for those *without* type 2 diabetes**, based mainly on the EMPA-KIDNEY study and DELIVER study (see heart failure section). Using a standardised definition of kidney disease progression they were able to **confirm a class effect for the renal benefits of SGLT2i** as well as giving reassuring data on a variety of potentially serious side effects:

- The study identified 13 trials involving 90 413 participants of whom 17.3% did not have diabetes.
- Compared with placebo, **SGLT2 inhibitors reduced the risk of kidney disease progression by 37%** (RR 0.63, 95% CI = 0.58-0.69) with **similar RRs in patients with and without diabetes**.
- **SGLT2 inhibitors reduced the risk of acute kidney injury by 23%** (RR 0.77, 95% CI = 0.70-0.84) and the risk of cardiovascular death or hospitalisation for heart failure by 23% (RR 0.77, 95% CI = 0.74-0.81), again with similar effects in those with and without diabetes.
- For all outcomes, **results were broadly similar irrespective of trial mean baseline eGFR**.
- In those people without diabetes there was no increased risk of either DKA or lower limb amputation.
- **Conclusion:** This evidence supports the use of SGLT2i for modifying risk of kidney disease progression and acute kidney injury, not only in patients with type 2 diabetes at high cardiovascular risk, but also in patients with chronic kidney disease or heart failure irrespective of diabetes status, primary kidney disease, or kidney function.

The [linked editorial](#) summarises where we are now with SGLT2i and CKD and suggests that '*SGLT2 inhibition should be a foundational therapy for chronic kidney disease, including in patients without diabetes*'. The editorial highlights the **important findings that SGLT2i reduce the risk of AKI and that in the non-diabetic group there was no increase in DKA or lower limb amputations**. Given the reassuring safety profile and ease of administration of SGLT2i, with no dose titration or risk of hyperkalaemia it is not a large step to the authors opinion that for CKD '*earlier prescription of SGLT2 inhibition should be considered, either before or in parallel with RAS inhibitor initiation and in preference to prolonged dose titration of RAS inhibitors*.'

► **NICE 2021 Guideline on CKD** [NICE NG203, 2021](#)

This guideline is an important update, with significant clinical implications for us and our patients. The guideline was summarised in the [BMJ 2021;374:n1992](#) and discussed in an editorial [BMJ 2021;374:n2159](#). We have summarised it in the KISS below, but the most important clinical implications are:

- ***Ethnicity has been removed from the equation for measuring eGFR***
 - Previously eGFR calculations have had an adjustment made for people from black ethnic groups. This was based on previous research on African-American patients which showed that they had a higher serum creatinine, thought to be due to greater muscle mass, than white patients with the same renal function. However the validity of this has long been disputed, and as the BMJ editorial [BMJ 2021;374:n2159](#) points out ethnicity and race are social constructs, and with such heterogeneity, in all 'ethnic' groups it is not valid to take ethnicity into account.
 - *So for all patients now eGFR will be calculated on the same criteria of creatinine, age and sex*
- ***Broadening the use of regular testing for CKD in people in at-risk groups, using eGFR and urine albumin creatinine ratio (ACR) in patients with risk factors***
 - Diabetes
 - Hypertension
 - Acute kidney injury
 - Any cardiovascular disease
 - Any structural renal tract disease (including benign prostatic hypertrophy)
 - Any multi-system disease with potential kidney involvement (e.g. SLE)
 - A family history of CKD or incidental finding of haematuria or proteinuria
- ***Assessment of prognosis and stage requires eGFR and urinary ACR, and can be calculated by the Kidney Failure Risk Equation***
 - The classification of CKD requires the GFR stage **and** the degree of proteinuria as measured by the urinary ACR. Urinary ACR is a crucial part of any clinical assessment for CKD.
 - They have a synergistic effect on prognosis and outcome - as seen on a visual [heat map](#) of prognosis based on the two categories.
 - A new tool has been developed called the ***Kidney Failure Risk Equation***. This was developed in Canada but has been validated in a UK population. This is a risk score based on age, eGFR and ACR which calculates a person's 5-year risk of developing kidney failure. The test is freely available, quick and easy to use
 - [Kidney Failure Risk Equation](#). It is not yet integrated into UK GP computer systems but hopefully, this will follow.
- ***When should we refer patients?***
 - It has always been difficult to know at what point should we refer patients to renal medicine
 - The [Kidney Failure Risk Equation](#) now makes this decision easy, as NICE recommend referral *if the score gives a 5-year risk of needing renal replacement therapy of > 5%*
 - For other referral criteria see the KISS summary below
- ***Management***
 - For frequency of monitoring, BP targets and treatments please KISS box below
 - Bear in mind that the CVD risk is nearly always greater than the end-stage renal disease risk and that reducing BP to targets and statins are important interventions to improve outcomes
 - ACEi and ARBs should NOT be used together; when starting either expect a fall in eGFR and also a rise in K. A reduction in eGFR of up to 25% is acceptable.
 - The **SGLTi dapagliflozin** is recommended as an option [NICE TA775, 2022](#) for CKD in adults if
 - It is an add-on to optimised standard care including the highest tolerated dose of ACEi or arb unless these are contraindicated, and
 - People have an eGFR of 25 to 75 at the start of treatment, and have
 - Type 2 diabetes, or
 - A urine albumin-creatinine ratio of > 22.6 mg/mmol

■ Anaemia and CKD

- A common challenge for us in primary care is finding that our patient with CKD has anaemia, and then uncertainty as to whether or not the anaemia can be attributed to the CKD or should be investigated further. NICE undertook a new evidence review of the eGFR and haemoglobin thresholds that should trigger an investigation. This is an area where individualised clinical decisions need to be made (e.g. rate of change, any change in symptoms etc) but in general, they conclude:
 - ***Consider investigating and managing anaemia in adults with CKD if*** their Hb falls to ≤ 110 g/L
 - If eGFR is > 60 ml/min investigate other causes of anaemia as it is unlikely to be due to CKD
 - If eGFR is $30 - 60$ ml/min investigate other causes of anaemia, but use clinical judgement to decide how extensive those investigations should be because the anaemia may be caused by CKD
 - If eGFR is < 30 ml/min think about other causes of anaemia, but note that anaemia is often caused by CKD

KISS: Chronic Kidney Disease DiagnosisBased on [NICE NG203, 2021](#) and [NICE TA775, 2022](#)

CKD is classified according to estimated GFR and proteinuria using albumin-creatinine ratio (ACR)

G is used to denote the GFR category, G1-G5 and A for the ACR category, A1-A3

Increased ACR and decreased GFR are associated with adverse outcomes

Increased ACR *and* decreased GFR multiply these risks e.g. G2A1 = low risk of adverse outcomes, but G3bA2 = high risk and G3bA3 = very high risk (see [heat map](#) for risk according to GFR and A category)

GFR Category	GFR	Terms
G1	>90	Normal or high
G2	60-89	Mildly decreased relative to young adult level (NB: there is NO CKD for G1&2 in the absence of markers of kidney damage e.g. proteinuria)
G3a	45-59	Mild to moderate decrease
G3b	30-44	Moderate to severe decrease
G4	15-29	Severely decrease
G5	<15	Kidney Failure
CKD: urine Albumin Creatinine Ratio (ACR) categories		
ACR category	ACR (mg/mmol)	Terms
A1	<3	Normal
A2	3-30	Moderate increase; clinically important with increased risk of adverse outcomes
A3	>30	Severely increased

- **eGFR testing**

- If first discovery of low eGFR/raised creatinine, clinically assess carefully and if well repeat the test within 14 days to exclude acute kidney injury. If eGFR stable, repeat again at 3 months to see if sustained. If low eGFR sustained >90 days, CKD is diagnosed.
- Interpret eGFR with caution in those with extremes of muscle mass e.g. reduced muscle mass will lead to over-estimation of eGFR this has implications when interpreting levels in frailty

- **Proteinuria Testing**

- Do not use reagent strips to identify proteinuria, test for CKD with eGFR and ACR
- If the initial ACR is between 3 and 70, this should be confirmed with an early morning sample
 - A confirmed ACR > 3 is considered clinically important
- If the initial level is over 70, a repeat is not needed

- **Indications for renal ultrasound**

- Accelerated progression of CKD
- Persistent invisible haematuria
- Symptoms of urinary tract obstruction
- Family history of polycystic kidney disease
- GFR category G4 or G5

KISS: Chronic Kidney Disease Management Based on [NICE NG203, 2021](#) & [BJGP2018;68:356](#)

1 Inform

- Optimise risk factors e.g. lifestyle, hypertension, diabetes etc & direct patients to [Information Sources](#)

2 Monitor for progression

- Regular monitoring of eGFR
 - G1-3a at least annually, G3b-G4 at least 6 monthly, G5 3 monthly
- Check for accelerated progression = a change in G category & a sustained decrease in eGFR of >25%, or a decrease in eGFR of 15 per year
- Assess the 5-year risk of needing renal replacement [Kidney Failure Risk Equation](#)

3 Reducing cardiovascular risk, anaemia and bones

- Lifestyle advice**
- Lower BP: Target BP < 140/90; lower target <130/80 if ACR > 70**
 - Follow usual hypertension guidelines if ACR < 30 (i.e. groups A1 and A2)
 - Offer ACEi or ARB if ACR > 30 (A3)
 - Be aware that ACEi and ARBs may be associated with a temporary fall in eGFR after starting ACEi or ARBs of up to 25%, repeat after 1-2 weeks & seek specialist advice if sustained and significant change. Do not start if serum K >5, and stop if serum K >6
- Proteinuria**
 - If CKD and diabetes ([NG28 Nov 2021](#)) offer ACEi or ARB if ACR > 3,
 - If CKD without diabetes, refer for nephrology assessment if ACR > 70
- SGLTi dapagliflozin** is recommended as an option [NICE TA775, 2022](#) for CKD in adults if
 - It is an add-on to optimised standard care including highest tolerated dose of ACEi or ARB
 - People have an eGFR of 25 to 75 at the start of treatment, and have
 - Type 2 diabetes, or
 - A urine albumin-creatinine ratio of > 22.6 mg/mmol
- Statins:** offer atorvastatin 20mg to all patients with CKD G3&G4 ([NICE 2014](#))
- Antiplatelets,** offer only for secondary prevention of CVD
- Anaemia: Consider investigating and managing anaemia in adults with CKD if Hb ≤ 110 g/L**
 - If eGFR is > 60 ml/min investigate other causes of anaemia as it is unlikely to be due to CKD
 - If eGFR is 30 - 60 ml/min investigate other causes of anaemia, but use clinical judgement to decide how extensive those investigations should be as the anaemia may be caused by CKD
 - If eGFR is < 30 ml/min think about other causes, but note anaemia is often caused by CKD
- Bones**
 - Serum calcium, phosphate and PTH levels need to be measured in patients with category G4 or G5 (i.e. eGFR<30) and obtain specialist advice. Offer cholecalciferol to treat Vit D deficiency if found

4 Referral criteria

- 5-year risk of needing renal replacement therapy > 5% using the [Kidney Failure Risk Equation](#)
- Stage A2 (ACR>30) plus haematuria OR ACR >70, unless secondary to diabetes and on treatment
- Accelerated progression
 - Sustained decrease in GFR of 25% or more and a change in category, OR
 - Sustained decrease in absolute value of eGFR of 15 ml/min/1.73m² or more within 12 months
- Hypertension that poorly controlled despite at least 4 drugs at therapeutic doses
- Suspected renal artery stenosis, or suspected rare or genetic cause

ACUTE KIDNEY INJURY

NICE updated their AKI guideline in December 2019 [NICE NG148](#). Acute kidney injury (AKI) remains a very Hot Topic for primary care and has been the subject of a recent editorial *Acute kidney injury in primary care* [BJGP2017;67;394](#).

► *What do we mean by acute kidney injury?* [NICE 2019, NG148](#) [BJGP2017;67;394](#)

The NICE guideline from 2013 was updated in December 2019 [NICE 2019, NG148](#). Acute kidney injury (AKI) was previously known as acute renal failure, but the definition has changed to reflect a wider spectrum of injury to the kidneys. Even small reversible deteriorations in kidney function, which may not previously have been considered as acute kidney failure, are associated with significantly worse health outcomes including hospitalisation. It is more common in older patients, especially with co-morbidities particularly chronic kidney disease and AKI often starts in the community when a vulnerable patient develops an acute intercurrent illness such as diarrhoea, vomiting or infection.

The RCGP strongly endorsed the guideline as *GPs will play a key role in preventing, identifying and treating AKI with our generalist skills looking after an ageing and multimorbid population*.

AKI: Background Points [NICE NG148](#) and [BJGP2017;67;394](#)

- Acute kidney injury (AKI, previously known as acute renal failure) costs the NHS up to £620M a year, more than is spent on breast, lung and skin cancer combined)
- Up to 20% of hospital admissions have AKI, with a 30% mortality rate
- 65% of cases of AKI start in the community
- Incidence is increasing with an ageing population, frailty and multimorbidity
- Early detection is key, and 30% of cases are preventable
- Key risk is the presence of risk factors plus an acute dehydrating illness (due to fever, fluid loss or reduced intake)
- Key risk factors are CKD stage 4 and 5, age, frailty, multimorbidity, past history of AKI, known renal disease and suspected outflow obstruction

► *How should GPs react to e-alerts warning of possible AKI?* [BJGP2017;67;394](#)

Useful guidance has been produced for primary care from [Think Kidneys NHS Guidance 2016](#), which we have summarised in a KISS below. The e-alerts are automatically generated in response to an increase in creatinine levels from baseline. If the last measured creatinine is more than 7 days ago, the algorithm uses the median value from the last year as the baseline reference.

The e-alert is NOT a diagnostic label, but a prompt to a clinical review and appropriate response. *The key thing is to interpret these e-alerts in the clinical context*. For example, if it is a routine blood for chronic disease monitoring or a routine check-up, then it may well represent a deterioration in otherwise stable CKD. If it is in the context of an acute illness, however, the patient is at risk of AKI and needs further evaluation. Clinical judgement here is key.

KISS: Responding to e-alerts of possible Acute Kidney Injury

[Think Kidneys Guidance 2016](#) and [BJGP2017;67;394](#)

- **Always consider results in clinical context and clinical judgement always prevails**
 - **LOW RISK of AKI** = stable clinical context (creatinine rise in a stable clinical context may reflect worsening CKD and not AKI)
 - **HIGH RISK of AKI** = context of acute illness and risk factors e.g. frailty, multimorbidity, CKD stage 4 or 5, past history of AKI, poor oral intake, suspected urinary obstruction
- **Recommended primary care response times**
 - **AKI Warning Stage 1** = creatinine > 1.5 x baseline, or rise >26 in last 48 hours
 - LOW-RISK context: clinical review within 72 hours
 - HIGH-RISK context: clinical review within 24 hours, likely Stage 1 AKI
 - **AKI Warning Stage 2** = current creatinine > 2 x baseline
 - LOW-RISK context: clinical review within 24 hours
 - HIGH-RISK context: clinical review within 6 hours, likely Stage 2 AKI
 - **AKI Warning Stage 3** = current creatinine > 3 x baseline, or >1.5 x baseline and >354
 - LOW-RISK context: clinical review within 6 hours consider admission
 - HIGH-RISK context: consider immediate admission, likely Stage 3 AKI
- **Recognising and responding to AKI for adults in primary care**
 - **Think CAUSE**
 - Hypotension? Sepsis? Urinary tract obstruction?
 - Urinalysis? intrinsic kidney disease e.g. nephritis
 - **Think MEDICATION** [Medicines Optimisation Guidance in AKI](#)
 - Consider withholding NSAIDs, diuretics, antihypertensives
 - Any new medication that may cause AKI
 - Any current medication which may accumulate and cause harm in AKI
 - **Think FLUIDS**
 - Assess patient's volaemic status
 - When did they last pass urine? What is fluid intake like? Do they need admission for i.v. fluids?
 - **Think REVIEW**
 - Does the patient need acute admission? If not, arrange early review
 - Manage as per KISS: AKI below...

KISS: Acute Kidney InjuryBased on [NICE NG148 Dec 2019](#) & [Think Kidneys 2016](#) & [BJGP2017;67;394](#)**1 Avoiding acute kidney injury**

- Pay close assessment to **hydration status and BP** in at-risk patients (e.g. with CKD, elderly, frail, multimorbidity & polypharmacy) when they develop a dehydrating illness (e.g. fever, diarrhoea, vomiting or reduced oral intake)
- **Sick day guidance.** Consider suspending medication which may precipitate AKI when at-risk patients have an acute illness e.g. the DAMN drugs (Diuretics, Angiotensin drugs, Metformin, NSAIDs), other BP-lowering drugs, SGLT2 inhibitors and sulfonylureas. Avoid nitrofurantoin & trimethoprim in the case of infection. **Resume medication when well**, 24 to 48 hours after return of normal fluid intake.
- **Check and monitor serum creatinine** in at-risk patients & compare to baseline
- Avoid NSAIDs wherever possible in elderly and at-risk patients.
- Do not prescribe nitrofurantoin for UTI in elderly people unless you know the eGFR is >45, and avoid for upper UTI or UTI with suspected sepsis

2 Recognition and diagnosis

- Consider in at-risk patients with an acute illness when a rapid decline in functional status
- Based on acute rise in creatinine or fall in urine output
- Check hydration status and serum creatinine in at-risk patients (e.g. with CKD, elderly, frail, multimorbidity & polypharmacy)

AKI STAGE	Serum Creatinine	Urine Output
1	Increase of ≥ 26 mmol/l within 48 hours, OR a 1.5 to 2-fold increase from baseline, OR a rise by >50% in 7 days	< 0.5ml/kg/hour for > 6 hours
2	Increase in serum creatinine to > 2 or 3 fold from baseline	< 0.5 ml/kg/hr for >12 hours
3	Increase to > 3-fold from baseline OR serum creatinine >354 with an acute	< 0.3 ml/kg/hr for 24/24 or anuria for >12 hours

3 Management

- Dipstick urine (looking for blood, protein which may indicate nephritis)
- Optimise fluid status: does the patient need admission for i.v. fluids?
- Obtain specialist advice. Patients with AKI 2 & 3 will likely need admission.
- AKI 1 may be possible to manage in the community with stopping at-risk drugs and improving hydration status with careful monitoring
 - Suspend drugs which may worsen AKI or which may accumulate and be dangerous in AKI (e.g. NOACs) until well. For detail see [Medicines Optimisation Guidance in AKI](#)
 - Resume reno-protective drugs e.g. ACEi/ARB when safe to do so

AKI: Learning points and Quality Improvement Ideas

- AKI is preventable with early detection and active management in primary care
 - AKI causes 100,000 deaths per year and is responsible for 20% of hospital admissions
 - 65% of cases arise in the community, and 30% are potentially preventable through early detection and active management
 - Most cases of AKI arise in the community in the context of at-risk patients developing an acute intercurrent illness
- From April 2016, GPs and primary care team will receive e-alerts from labs highlighting possible AKI based on rises in serum creatinine
- We have new guidance as to how to respond to these alerts [Think Kidneys NHS Best Practice Guidance](#), summarised in our KISS
- Clinical judgement is vital, and these alerts should be considered in the context of whether or not this is low or high risk for AKI
- For patients in whom it is safe for them to stay at home, suspending exacerbating drugs and optimising fluid intake with careful follow-up is key

Quality improvement ideas:

- Raise AKI awareness within your practice, including the importance of responding to the e-alerts
- Take a pro-active approach to detecting possible AKI early in at-risk patients and delivering simple interventions
- Undertake some recent case reviews of acute kidney injury or unplanned admissions, was AKI implicated?
- Audit of recent e-alerts, have they been acted on?
- Search on patients in high-risk groups (e.g. multimorbidity, CKD stage 4 or 5) and undertake medication reviews to minimise risk and provide patients with information (below) on how to protect their kidneys

Further resources:

- [Sick day guidance](#) on medication
- [Medicines Optimisation Guidance in AKI](#)
- [Patient information leaflet](#)
- 10-minute video [Acute Kidney Injury in Primary Care](#)

GYNAECOMASTIA

Whilst gynaecomastia is commonly idiopathic or physiological, it is commonly a side effect of prescribed medications and it may also be the first presentation of serious pathology including breast and testicular cancer and endocrine disorders including hypogonadism. It is common, affecting up to a third of men in their lifetime, so a recent BMJ review paper highlighting when we should be concerned [BMJ 2022;379:e069771](https://doi.org/10.1136/bmj.2022.379.e069771) is welcome.

KISS: Gynaecomastia

Based on [BMJ 2022;379:e069771](https://doi.org/10.1136/bmj.2022.379.e069771)

- ***Gynaecomastia or is it just fat i.e. pseudogynaecomastia?***
 - Gynaecomastia has a firm rubbery disc of sub-areolar tissue; pseudo feels like subcutaneous fat
 - Pain or tenderness can occur but is not universal with gynaecomastia; pseudo is always painless
 - Gynaecomastia is usually bilateral but may be unilateral; pseudo is always bilateral
- ***Breast cancer*** is important to exclude, most common in men aged over 50: it is unilateral, with hard fixed irregular mass and may have skin changes, be away from the nipple/areola and have local LN involvement; check for risk factors, such as family history or Klinefelter's syndrome
- **Common causes of gynaecomastia**
 - **Physiological 25%**
 - Common in newborns, adolescents and older people
 - in adolescents, resolves spontaneously in 90% of cases
 - Middle-aged and older men: greater adiposity increases peripheral aromatase activity, converting testosterone to oestrogen which is associated with age-related declining free testosterone is a common cause of physiological gynaecomastia, often exacerbated by alcohol
 - **Idiopathic 25%**
 - **Medication and drug use 20-25%**
 - Multiple potential culprits including PPI, antihistamines, anti-psychotics, ACEi, CCB, TCA antidepressants, spironolactone, anti-retrovirals, opioids, amiodarone, digoxin, metronidazole, finasteride and ketoconazole
 - Non-prescription drugs include anabolic steroids, cannabis, amphetamines, heroin
 - when these are stopped or switched, expect improvement over weeks and months
 - **Hypogonadism 10%**
 - **Liver cirrhosis and renal failure 8%**
 - **Endocrine disorders** including primary and secondary hypogonadism, hyperthyroidism, hyperprolactinaemia and cushings
 - **Testicular cancer** gynaecomastia is a presenting symptom in approx 10% of cases
 - *Ask about a history of lump, pain or swelling and if so perform a testicular examination*
- **Further testing or referral**
 - ***If testicular or breast examination are suggestive of cancer, urgent suspected cancer referral***
 - **Investigate further if the following**
 - Not physiological or pseudo gynaecomastia
 - Not due to medication or drugs
 - Rapid enlargement, recent onset in men aged over 20, persistent and painful
 - Persistent or massive gynaecomastia in adolescents
 - First line **blood tests** include TSH, LFT, U&E, 9 am testosterone, estradiol, alpha-fetoprotein (AFP) and beta HCG. Consider in addition prolactin, LH/FSH, SHBG
 - ***Urgent referral to urology and testicular USS if b-HCG or AFP deranged***
 - Refer to endocrinology if abnormal hormone profile
 - Routine breast USS not indicated
 - Referral is indicated to breast clinic if painful gynaecomastia for > 6 months with normal bloods, or unilateral breast lumps with no clear cause or genetic risk factors e.g. Klinefelter's

LOWER URINARY TRACT SYMPTOMS

Such a common problem...and sometimes so hard to help!

Way back in 2010 NICE published their [LUTs Guideline CG97](#). At the time it was quite radical for cementing a symptom-based approach to the syndrome (i.e. calling it LUTs rather than a presumed pathology such as BPH) and for clinically differentiating between voiding and storage symptoms.

But, that was over a decade ago and the BJGP have recently published an excellent paper on the management of male LUTs in primary care [BJGP2021;71;388](#). We have had evidence reviews on the primary care management of overactive bladder (OAB) in men and women ([BJGP2017;67;187](#)) and specifically in men ([Br J Gen Pract 2018; 68 \(671\)](#)). The role of desmopressin in nocturnal polyuria has also been reviewed [DTB2017;55;30](#).

► **Primary Care Management of OAB** [Br J Gen Pract 2018; 68 \(671\)](#) [BJGP2017;67;187](#)

OAB is defined as: urinary urgency, usually associated with frequency and nocturia, with or without incontinence in the absence of urinary infection or other pathology.

OAB is highly prevalent, with symptoms of overactive bladder (OAB) affecting 16% of all adults over 40, rising further sharply with age. It is a typical primary care syndrome in that it is common and generally benign, but it can have a significant negative impact on the quality of life and has been associated in older people with an increased risk of falls, fractures and skin breakdown. And whilst it is usually harmless it may be the first presentation of a serious underlying pathology such as cancer so excluding red flags is an important part of the assessment. In men symptoms of OAB are often attributed to bladder outflow obstruction (BOO) which can result in misdiagnosis and delays in treatment, although in practice many men may have overlap in their symptoms. These papers give us a simple guideline to management (**see KISS below**), as well as a very useful **screening tool for OAB** (the **OAB-V3 score**, where a score of ≥ 4 is positive):

How bothered have you been by...	Not at all	A little bit	Somewhat	Quite a bit	A great deal	A very great deal
Frequent urination in daytime hours	0	1	2	3	4	5
A sudden urge to urinate with little or no warning	0	1	2	3	4	5
Urine loss is associated with a strong desire to urinate	0	1	2	3	4	5

In terms of medication, the beta-3 agonist **mirabegron** has been around for a while now as an alternative to antimuscarinics in those in whom they are contraindicated, not tolerated or ineffective. The new relevant research is a new study exploring **combination therapy with mirabegron and antimuscarinics** [Eur Urology 2016;70;146](#)

- RCT of 2,000 patients who had OAB with incontinence, who remained incontinent after 4 weeks of solifenacin 5mg
- They were randomised to a combination of solifenacin 5mg + mirabegron 50mg, versus solifenacin alone in a 5mg or 10mg dose
- After 12 weeks the combination was superior to solifenacin 5mg or 10mg alone and it was well tolerated
- **Conclusion:** In patients with OAB who remained incontinent despite initial solifenacin 5mg, combining with mirabegron 50mg was superior in treating symptoms of incontinence and urinary frequency and was well tolerated.

KISS: Overactive Bladder in Men & Women

[BJGP2017;67;187](#) & [NICE Suspected cancer, 2015](#) & [Br J Gen Pract 2018; 68 \(671\)](#)

Overactive bladder is defined as urinary urgency, usually associated with frequency and nocturia, with or without incontinence in the absence of urinary infection or other pathology (in contrast to **bladder outflow obstruction** which generally presents with weak/intermittent stream, hesitation, straining or incomplete emptying)

- **Exclude 'red flag' features, and refer immediately if present**
 - Haematuria
 - Previous surgery or radiotherapy
 - Neurological disease
 - Recurrent UTI
 - Difficulty with bladder emptying
 - Constant leak suspicious of fistula
- **Examination** should include the pelvis in women and a digital rectal exam in men
- **Initial investigations**
 - Urinalysis
 - **Women:** Consider CA-125 if persistent and new onset of OAB, especially if age >50
 - **Men:** Consider a PSA in men based on risk factors, clinical judgement & pt choice
 - Keep a bladder diary - [click here](#)
- **Management**
 - **Conservative first-line and should be the cornerstone of treatment:**
 - Altering fluid intake, avoid caffeine, alcohol and irritants ([click here](#) for PIL on bladder irritants to avoid)
 - Weight loss
 - Bladder re-training - [click here](#) for an excellent leaflet
 - Start pelvic floor exercises, especially if associated urinary incontinence
 - See here for excellent general [Patient Information](#) Leaflet
 - Review & alter potential **exacerbating medications** e.g. diuretics
 - **Medication**, if the above interventions are ineffective & the impact is significant
 - **Intravaginal oestrogens** can be beneficial in post-menopausal women
 - **Antimuscarinics** (e.g. oxybutynin, tolterodine, solifenacin, darifenacin, trospium etc.) first line, but caution in elderly especially if cognitive impairment or falls; may take 4 weeks to get a clinical response; start with the lowest dose; try a second one if first ineffective and titrate up to maximum dose; risk of acute retention low
 - **Mirabegron** (selective beta 3 agonist) is a non-anticholinergic alternative if antimuscarinics are contraindicated, not tolerated or at least two are ineffective. Check blood pressure as may cause hypertension (contraindicated if severe) [MHRA Guidance](#)
 - **Combination therapy:** there is some evidence to support solifenacin 5mg daily + mirabegron 50mg daily if monotherapy has failed
 - **Refer** if symptoms persist



► **LUTS in men: *should I have a PSA test doctor?***

[BMJ 2018;361:k1202](#) [Br J Gen Pract 2018; 68 \(676\)](#)

One scenario we see a lot in General Practice is the man who presents with LUTS who is worried he may have prostate cancer. ***Should we offer a PSA?*** This important clinical scenario was reviewed recently in both the [uncertainties section in the BMJ in May 2018](#) and the [BJGP in November 2018](#).

The key points from these reviews are as follows:

- LUTS in men are very common and estimated to be present in ~80% of men >60
 - Indolent prostate cancer (one that will not cause morbidity or mortality) is also common, autopsy studies estimate the rate of indolent prostate cancer to be ~60% in men >79 years
- Given the high prevalence of LUTS and indolent prostate cancer in older men, a link between the two is unavoidable leading to a high risk of overdiagnosis if LUTS are investigated with PSA
- As we know so well from practice, studies confirm that men with LUTS are often worried about prostate cancer and consult a doctor to seek reassurance
- Most international guidelines (including NICE suspected cancer guidelines) recommend doctors consider doing a PSA in men with LUTS for prostate cancer detection, but it is uncertain if men with LUTS have an increased risk of clinically significant prostate cancer or not
- The current evidence is mainly limited to observational studies of PSA-screened populations, and although some evidence shows a positive association with LUTS and prostate cancer, this is mostly due to the detection of localised prostate cancers many of which will turn out to be indolent
 - Currently, there is ***no significant evidence to suggest that men with self-reported LUTS are at an increased risk of advanced or potentially fatal prostate cancer when compared to men without LUTS*** although the data is heterogeneous and incomplete
- The BMJ authors searched for any ongoing relevant studies examining whether LUTS are predictive of advanced prostate cancer, or studies offering PSA testing in men with LUTS, but they identified none - a real pity as this is such an important clinical question

So, what should we do in light of this uncertainty with a lack of evidence to guide us? The authors of these papers offer the following conclusions and strategies:

- Early prostate cancer is unlikely to have any symptoms and in most cases, LUTS will be due to causes other than prostate cancer
- PSA testing to investigate LUTS may lead to the detection of prostate cancer that would not have gone on to cause harm; this may lead to overtreatment, side effects, and psychological distress
- ***Do not routinely offer PSA testing in men with LUTs and no risk factors for prostate cancer***
 - Explore why the patient is concerned about prostate cancer and discuss with them the lack of evidence that LUTS in isolation suggest prostate cancer, explain limitations of the PSA test including risks of overdiagnosis and related treatment as part of shared decision making
- Assess risk factors, and take these into account: age, family history of prostate cancer, ethnicity, BMI; offer digital rectal examination and do a urine dipstick test

KISS: LUTS in Men

[BJGP2021;71;388](#) & [BMJ2017;357;J1493](#) & [Br J Gen Pract 2018; 68](#)

1 Initial Assessment

- Are the symptoms predominantly of
 - **Storage** (suggests overactive bladder): frequency, urgency, nocturia, urge incontinence
 - **Voiding** (suggests obstructed outflow, usually BPH): hesitancy, poor stream, straining
 - **Post-micturition** (suggest chronic retention): incomplete emptying, terminal dribble
 - **Mixed pattern is common with benign prostatic enlargement**
- **Examination:** including the abdomen and digital rectal exam (DRE)
- **Tests**
 - Dipstick urine and send an MSU
 - Renal function if suspect significant obstruction or renal impairment
 - Ultrasound if suspect obstruction and/or chronic retention
 - **PSA test?** Cancer referral guidelines advise to *consider* a PSA test, based on clinical judgement and pt preference [NICE NG12, 2021 Update](#)
 - Information for men on PSA testing [PCRMP 2022, PIL](#), however, note this information is technically for men *without* LUT symptoms, but as discussed above the distinction between men with and without symptoms with respect to prostate cancer is probably somewhat spurious
- Give the patient a symptom diary and score to complete for the follow-up appointment, to help differentiate the symptoms and assess severity: [Bladder Diary](#)

2 Management

- **Assess** impact with the International Prostate Symptom Score (IPPS) - [pdf](#) or [online](#)
- **Initial management of uncomplicated PUTs should be conservative, based on information, reassurance and lifestyle advice**
 - **Lifestyle** advice in all (see also OAB KISS above if OAB present) - alter volume and timing of fluids (e.g. reduce evening fluids if nocturia); reduce caffeine and alcohol
 - Treat **constipation** if present
 - Review & alter potential **exacerbating medications** e.g. diuretics
- **BPH: Base medication decisions on prostate size & symptom impact.** Only continue medications if there is a significant improvement in symptoms:
 - **Moderate to severe LUTs (IPPS ≥ 7) & normal-sized prostate:** alpha-blocker (e.g. tamsulosin, doxazosin etc)
 - **Mild LUTs (IPPS < 8) & enlarged prostate:** 5 alpha-reductase inhibitor e.g. finasteride
 - Advise patient need 6 months to assess response. Expect a 25% reduction in prostate size and a 50% fall in PSA levels
 - **Moderate to severe LUTs (IPPS ≥ 7) & enlarged prostate:** consider a combination of alpha-blocker & 5 alpha-reductase inhibitor
- **OAB:** see OAB KISS above for medication options
- **When should we refer?**
 - DRE or PSA are suggestive of prostate cancer
 - Failure of medical treatment or very severe symptoms
 - Urinary retention
 - Haematuria, any visible or persistent invisible haematuria

Learning Points & Quality Improvement Ideas: Overactive Bladder & LUTs

- Overactive bladder syndrome generally only ever needs appropriate lifestyle advice. However, if medication is needed we have new guidance on the role of antimuscarinics and also mirabegron. In resistant cases, there is some emerging evidence for combination treatment
- Don't forget the role of intravaginal oestrogens in post-menopausal women
- All men should have a DRE to help exclude cancer, but also to assess prostate size which will help guide treatment decisions
- Cancer referral guidelines from NICE & Scotland recommend a PSA test in selected men with LUTs, based on clinical judgement and patient preference, although the evidence for this is weak; assessing risk factors for prostate cancer may be a better predictor of developing clinically significant prostate cancer than LUTs
- Take a symptom-based approach to management: are symptoms predominantly storage, voiding or mixed? The OAB-V3 is a useful and quick screening questionnaire for OAB
- Commonly used drugs often have marginal benefit, do not continue without clear benefit
- Use the IPPS score to assess severity and response to drug treatment in BPH - [click here](#)
- New, less invasive procedures, are now available for BPH instead of surgical TURP, including day case 'laser therapy' and prostate artery embolisation - consider the lower threshold for a referral if patients are very symptomatic
- **Audit** of recent cases, have they been appropriately managed as per the KISS e.g. have they all been appropriately examined and had a urinalysis performed?

RENAL COLIC

NICE produced a guideline on renal and ureteric stones in 2019 ([NG118](#)), the European Association of Urology (EAU) updated their [guideline on urolithiasis in 2019](#) also, and we had a useful review of GP management of kidney stones in the [BJGP recently BJGP 2020; 70 \(693\)](#).

Renal Colic: Background Points

Based on & [NICE 2019, NG118](#) & [EAU guideline 2019](#)

- The incidence (and therefore intervention) of kidney stones is increasing and nephrolithiasis is now recognised as a chronic and systemic condition; reducing recurrence is a management priority.
- 5% to 10% of the population get renal colic, often young working-age adults, and it is strongly linked with dehydration and a fluid intake of <1L per day (peak incidence in the summer).
- Peak incidence 20 to 50, male:female 3:1, 20% have a positive FH.
- Medical costs have increased dramatically with the widespread use of lithotripsy, ureteroscopy and invasive techniques (hence NICE push to reduce recurrence with better lifestyle advice etc).
- Small ureteral stones (<10mm) often pass spontaneously, but this is highly variable between individuals; EAU suggest to consider conservative management if ureteral stones <6mm with no underlying renal disease/signs of obstruction and pain adequately controlled with analgesia.
- 50% of stones have just one-lifetime recurrence, highly recurrent disease in 10%.
- Diseases associated with stone formation include:
 - Hypercalcaemic disorders e.g. hyperparathyroidism
 - GI disease (e.g. Crohn's) & malabsorptive conditions inc. bariatric and bypass/resection surgery
 - Recurrent UTI
 - Gout
 - Anatomical abnormalities e.g. polycystic kidneys, horseshoe kidney, vesico-ureteric reflux

► Medical Expulsive Therapy using Alpha-blockers

There is increasing evidence for the use of alpha-blockers as a non-surgical option for ureteral stones in some circumstances, and is endorsed as an option by both NICE and EAU:

- A systematic review of 55 RCTs [BMJ2016;355:i6112 \(1a\)](#) found moderate-quality evidence that alpha-blockers facilitate stone passage compared to controls (in absolute terms, 76% stone passage with alpha-blocker and 48% with controls NNT= 4) and significantly reduce time to stone passage (mean 4 days), pain and need for surgical intervention (64%RRR).
- All alpha-blockers e.g. tamsulosin, doxazosin seem equally effective.
- NICE recommend a-blockers for distal stones <10mm; EAU recommend them for distal stones>5mm.

► What about asymptomatic stones? [BJGP 2020](#)

Not an uncommon scenario for use in general practice, most likely an incidental finding on USS or CT, and the BJGP article gives useful advice. **Refer if any of the following present: Stones >5mm in size, Single kidney, Chronic obstruction, Recurrent UTIs.** Asymptomatic stones not requiring referral can be actively monitored with annual follow-up with imaging using the modality that the stone was initially picked up on (being mindful of radiation exposure if using CT). If stones grow >5mm or new complications develop (pain, infection, obstruction) then refer.

► Trouble-shooting in GP after intervention for stones [BJGP 2020](#)

Again the BJGP gives some useful advice for the common scenario of patients presenting to primary care after stent insertion with haematuria, urgency, frequency, dysuria and flank pain. These are common symptoms and we can reassure the patient and watch/wait unless symptoms are persistent +/- severe or associated with systemic symptoms, urine retention or +ve urine culture, in which case stent migration needs to be considered. **Alpha-blockers +/- anticholinergics can be tried for stent colic symptoms.**

KISS: Management of Renal Colic
[NICE NG118 2019](#) & [EAU guideline 2019](#) & [BJGP 2020](#)

- **Typical symptoms** are loin or flank pain radiating to groin or genitalia, vomiting and sometimes fever but stones may be asymptomatic.
 - Pain usually severe, predominantly loin, groin or both; typically restless with the pain.
 - Urinary symptoms: distal stones often cause frequency, dysuria and urgency.
- **Exclude important differential diagnoses** e.g. Ectopic, torsion, appendicitis, cholecystitis, pancreatitis, peritonitis ruptured AAA (in people >50) and **Assess for sepsis**.
- **Dipstick urine to look for evidence of haematuria or UTI:**
 - Invisible haematuria seen in 80% to 85% of renal stones, so absence does not rule it out.
 - If signs UTI, send urine culture and start empiric antibiotics.
- **Bloods** - check renal function urgently if not being admitted.
- **Pain management:**
 - NSAIDS first-line - route will depend on pain levels and if nausea/vomiting use PR route.
 - Second-line (or NSAIDS contraindicated) use opioids e.g. tramadol, but they are less effective.
 - Do NOT use antispasmodics - [Cochrane 2015](#) concluded best evidence for NSAIDS and addition of antispasmodics did not improve pain levels.
- **Anti-emetics if vomiting.**
- **Refer immediately to hospital if:**
 - Systematically unwell or febrile, or a history of fever and rigors.
 - Pain that does not settle within an hour of analgesia.
 - Persistent nausea and vomiting, especially if dehydrated.
 - If they have a solitary functioning kidney, have had a kidney transplant, or are pregnant.
- **Imaging:**
 - If immediate referral not necessary, refer for urgent CT-KUB ideally within 24 hrs.
 - Refer for urgent ultrasound instead of CT in pregnancy, children and young people.
- **Stone expulsive therapy:**
 - Consider alpha-blockers e.g. tamsulosin 400mcg daily.
 - NICE recommend for distal stones <10mm, EAU recommend for distal stones >5mm; in practice, we will not know size/position of stone but given low-risk profile of alpha-blockers treatment in primary care alpha-blockers should be considered unless contraindicated.
- **Preventing stone recurrence** ([click here](#) for patient information from BAUS):
 - Measure serum calcium.
 - Good fluid intake 2.5-3 L/day; consider adding lemon juice to drinking water; avoid fizzy drinks.
 - Restrict salt intake to <6g /day, maintain normal calcium intake, avoid excess dietary protein.
 - Calcium oxalate stones: Potassium citrate +/- thiazides may be considered.
 - Stone analysis if possible; if stones are predominantly calcium oxalate, consider potassium citrate
 - If recurrent stones are predominantly calcium oxalate and hypercalciuria, consider thiazides.

Learning Points: Renal Colic

- **Always dipstick urine** but if negative for blood this does NOT rule out renal colic.
- **CT-KUB is the imaging of choice** now for renal colic, ideally within 24 hours.
- NSAIDS are first-line analgesia; efficacy of opiates not as good; antispasmodics NOT recommended.
- We have **good evidence to support the use of alpha-blockers** for patients appropriate for conservative management with distal stones.



LOW TESTOSTERONE IN MEN (HYPOGONADISM)

For what should seemingly be quite a straightforward topic, I think many of us in general practice find investigating and managing low testosterone in men and knowing what our role is, difficult. And it appears there is a good reason for this, as highlighted in an excellent BJGP 'guide for primary care' on the subject [BJGP 2020; 70 \(696\)](#) - there is no single UK consensus on how to manage male hypogonadism. We have the [British Society for Endocrinology consensus statement 2018](#), the British Society for Sexual Medicine [BSSM guideline 2017](#) and the [European Association of Urology guidelines](#) for starters. Although broadly similar there are differences so this brief guide in the BJGP was welcome to put in context what our role is, as GPs. We have a full KISS below summarising management but first some key points to be aware of:

► *How does testosterone deficiency (TD) present in men?*

- **Psychological:**
 - Mood changes e.g. irritability, anger, depression.
 - Reduced cognitive function e.g. impaired concentration.
- **Cardiometabolic:**
 - Increased BMI with visceral obesity and metabolic syndrome.
 - Insulin resistance and Type 2 diabetes.
- **Physical:**
 - Decreased body hair.
 - Gynaecomastia.
 - Decreased muscle mass and strength.
 - Hot flushes, sweats, sleep disturbance and fatigue.
 - Osteoporosis/fractures.
- **Sexual:**
 - Infertility.
 - Low sexual desire and reduced sexual thoughts.
 - Erectile dysfunction and delayed ejaculation.
 - Decreased morning or nighttime erections.

Although it is clear from the list above the presentations can be vague and varied the **three commonest symptoms of TD are erectile dysfunction, loss of early morning erections and low sexual desire.**

► *Pitfalls in diagnosis*

Whilst it may be stating the obvious, making a correct diagnosis of TD is absolutely crucial, as subjecting men who don't have true TD to potentially long-term treatment may not help any of their symptoms, and puts them at risk of harm. As the paper points out the growing demands and expectations of men and their health and well-being **presents a risk of overdiagnosis especially if testing is not optimal.** On the flip side, untreated TD can lead to long-term health problems including osteoporosis, depression and sexual dysfunction. So a few things we should bear in mind:

- **Do not screen asymptomatic men** apart from in a few very specific circumstances (e.g. low impact fractures/osteoporosis, infertility, HIV-associated weight loss).
- **Diagnosis of TD** requires presence of clinic features of TD with unequivocally low testosterone on 2 separate tests, at least 4 weeks apart.
- Testosterone has a diurnal variation and is suppressed post-prandially so **blood tests should be done between 7-11am after an overnight fast.**
- **Drugs can affect/cause TD** - these include alcohol, opioids, and exogenous steroids (both glucocorticoids for medical treatment but also the increasingly used anabolic steroids).
- Whilst the diagnosis of primary hypogonadism should (in theory) be clear from blood parameters, distinguishing between true secondary hypogonadism and non-gonadal illness (NGI) triggering TD (e.g. obesity, T2DM) can be very difficult - although not spelt out in black and white the implication is clear that **we should always refer to a specialist before considering testosterone treatment**, as those with reversible/functional causes of their TD are unlikely to benefit from testosterone replacement.



► ***What tests should we do and how do we interpret them?***

- Initially, **early morning total testosterone (TT) between 7-11am after overnight fast, along with sex hormone-binding globulin (SHBG)** - most testosterone is protein bound so unbound free testosterone (FT) only accounts for 2-4% of the total; however, if SHBG is low or high it may give a misleading TT level so in these circumstances, FT is a better measure, otherwise if SHBG normal TT can be used.
- If testosterone is low or low/normal in the presence of other characteristic features (e.g. osteoporosis) **repeat the fasting testosterone levels 4 weeks apart together with LH/FSH, prolactin, thyroid function, cortisol and iron stores (inc ferritin)**; we should also at this stage do a **baseline PSA**, as well as **screening for CVD risk factors** if not already done - lipids, Hba1c, BP etc.
- **Primary hypogonadism (PH)** is diagnosed by low testosterone and raised LH/FSH. For most people with PH testosterone therapy is indicated.
- **Secondary hypogonadism (SH)** is characterised by low testosterone and low/normal LH/FSH. However, the biochemical profile of SH can be indistinguishable from non-gonadal illness/NGI (where there is a physiological suppression of LH/FSH e.g. obesity, T2DM), so **referral is essential for clarification of diagnosis** and consideration of pituitary MRI, especially if prolactin raised. If NGI is thought to be the cause optimisation of medical comorbidities rather than testosterone therapy should be the focus (inc PDE inhibitors for ED).
- **Blood levels** are not discussed in the BJGP review but the BSSM guide gives useful guidance:
 - Normal testosterone = TT >12 nmol/L or FT >0.225 mol/L.
 - Borderline/low testosterone = TT 8-12 mol/L or FT 0.180-0.225 mol/L.
 - Low testosterone = TT <8 nmol/L or FT <0.180 mol/L.

► ***Monitoring once on therapy***

This is possibly our most important role as once diagnosis is confirmed and treatment initiated most people will be discharged to primary care for ongoing treatment and monitoring:

- **Testosterone levels yearly aiming for the reference range.** (and ideally at 3 and 6 months after initiation)
 - if on injectables do 'trough' level just before an injection; if on gels do levels at least 4 hours after application.
- **FBC annually checking haematocrit** - if ≥54% stop testosterone.
- **PSA annually in men > 40 years old** (as well as baseline and at 3-6 months after initiating) - although the general consensus is that testosterone does not cause de novo prostate cancer, it may unmask occult prostate cancers, so rising PSA levels warrant urology referral; the BSSM guideline recommends that an increase of >1.4ng/L over 1 year or a velocity of >0.4 ng/L/year over 2 years should prompt referral.
- As part of annual check, we should also **monitor other CVD risk factors** - BP, weight, lipids, Hba1c etc.
- **Contraindications to testosterone therapy/stop therapy** - Erythrocytosis (Haematocrit >54%); desire to father children; active prostate or breast cancer; decompensated cardiac or liver disease.

NB Learning Points & Quality Improvement Ideas: Low Testosterone in Men

- **Diagnosis of testosterone deficiency (TD)** requires the presence of clinical features of TD with unequivocally low testosterone on 2 separate tests, at least 4 weeks apart.
- Testosterone has a diurnal variation and is suppressed post-prandially so **blood tests should be done between 7-11am after an overnight fast**.
- Differentiating between secondary hypogonadism and non-gonadal illness can be difficult based on blood levels, and those with non-gonadal illness causing the low testosterone may not benefit from testosterone treatment - **refer all patients with suspected testosterone deficiency for specialist assessment before starting testosterone**.
- **QI Idea:**
 - Audit all men on testosterone therapy. Are they having annual blood tests?
 - As a minimum they should be having **FBC (check haematocrit and if >54% stop treatment)** and if **aged >40 a PSA**; consider also testosterone levels and review CVD risk factors.



KISS: Low Testosterone in Men (Hypogonadism)

[BJGP 2020; 70 \(696\)](#) [BSSM 2017](#) [EAU guideline](#) [Summary of BJGP recommendations](#)

Assessment:

- Untreated testosterone deficiency (TD) can lead to long-term health problems including osteoporosis, depression and sexual dysfunction, however, there is a risk of over (and under) diagnosis if testing is not optimal (diurnal variation of testosterone and it is suppressed post-prandially).
- **Diagnosis of TD** requires presence of clinical features of TD with unequivocally low testosterone on 2 separate tests, at least 4 weeks apart.
- Typical clinical features include:
 - **Psychological:** Mood changes; reduced cognitive function.
 - **Cardiometabolic:** ↑ BMI, visceral obesity, metabolic syndrome, Type 2 diabetes.
 - **Physical:** Decreased body hair; gynaecomastia; decreased muscle mass and strength; hot flushes, sweats; sleep disturbance and fatigue; osteoporosis/fractures.
 - **Sexual:** Infertility; decreased libido; erectile dysfunction; decreased morning or night time erections.
- The **3 commonest symptoms of TD** are erectile dysfunction, loss of early morning erections and low sexual desire.
- **Do not screen asymptomatic men** apart from in a few very specific circumstances (e.g. low impact fractures/osteoporosis, infertility, HIV-associated weight loss).
- **Review medications** that can cause TD - opiates, dopamine antagonists, alcohol, anabolic steroids, 'vitamin' or 'protein' supplements may contain undeclared androgens.

Diagnosis:

- **Testosterone blood tests should be done between 7-11am after overnight fast.**
- Initially - total testosterone (TT) and SHBG; if SHBG low or high then use free testosterone (FT), if SHBG normal TT can be used.
- If TT (or FT) low or low/normal in the presence of other characteristic features (e.g. osteoporosis) repeat blood levels 4 weeks apart together with LH/FSH, prolactin, thyroid function, cortisol and iron stores (inc ferritin); also do baseline PSA (if aged >40) and do CVD screen.
- Typical blood levels (but check your local lab):
 - Normal testosterone = TT >12 nmol/L or FT >0.225 mol/L.
 - Borderline/low testosterone = TT 8-12 mol/L or FT 0.180-0.225 mol/L.
 - Low testosterone = TT <8 nmol/L or FT <0.180 mol/L.
- If TD suspected based on clinical features/blood levels refer for specialist assessment before starting testosterone therapy; initiation of testosterone therapy usually specialist-led/initiated.
- Primary hypogonadism (PH) = low TT/FT with high LH/FSH; testosterone therapy usually indicated.
- Secondary hypogonadism (SH) = low TT/FT with low or inappropriately 'normal' LH/FSH - needs consideration of MRI; testosterone therapy usually indicated.
- Non-gonadal illness (NGI) e.g. obesity, T2DM - can have very similar profile to SH but management may not be with testosterone with more focus on underlying factors/causes.

Ongoing management/monitoring in primary care:

- **Contraindications to testosterone therapy** - Erythrocytosis (Haematocrit >54%); desire to father children; active prostate or breast cancer; decompensated cardiac or liver disease.
- Testosterone levels & PSA (in men >40) should be checked 3-6 months after initiation of testosterone.
- Then **yearly check - testosterone levels, PSA (in men >40), FBC/Hct, CVD risk factors.**
- If on injectables do 'trough' level just before an injection (aim for low end of normal); if on gels do levels at least 4 hours after application (aim for mid-range).
- Good patient information from the Sexual Advice Association - [click here](#)



PROSTATITIS

Acute bacterial prostatitis is a diagnosis that is easy to miss, and can lead to the potentially serious consequences of urosepsis and chronic prostatitis. Chronic non-bacterial prostatitis, increasingly known as **Chronic Pelvic Pain Syndrome**, is easy to diagnose clinically but can be very difficult to manage. Fortunately, we have new evidence-based guidelines to help with both.

1 Acute Prostatitis

In January 2019 we had a newly updated guideline on acute bacterial prostatitis from [NICE-CKS 2019](#). In late 2018, NICE also produced joint guidelines with Public Health England [Managing Common Infections NICE/PHE 2018](#) which is incredibly useful and includes guidance on a broad range of infections discussed elsewhere in this book including bacterial prostatitis.

Key points from the guidelines:

- Acute bacterial prostatitis is accompanied by an infection of the urinary tract, occasionally there may be associated with epididymitis or urethritis
- It is usually caused by typical urinary pathogens, but it may be secondary to sexually transmitted infections such as chlamydia or gonorrhoea
- It can occur secondary to instrumentation and procedures (e.g. catheter) or bladder outflow obstruction
- **Complications** [NEJM2016;374:562](#)
 - 25% of men are estimated to have bacteraemia and 10% can develop a localised prostatic abscess
 - 10% of men will develop chronic prostatitis
 - The risk of urosepsis is unknown but increases with age and co-morbidities
- **Diagnosis:**
 - Suspect acute bacterial prostatitis in a man who presents with
 - Sudden onset of fever, may be accompanied by rigors, arthralgia or myalgia
 - Irritative voiding symptoms (dysuria, frequency, urgency)
 - Difficulty passing urine (retention may occur)
 - Perineal or suprapubic pain
 - Tender prostate on digital rectal examination
 - Confirm diagnosis with a urine culture
 - There is no need to test prostatic secretions
- **Management**
 - Admit urgently if severely ill, suspected sepsis, in acute urinary retention (do not attempt inserting a urethral catheter as this may cause an abscess or trigger sepsis, refer for supra-pubic) or the possibility of an abscess (fluctuant swelling on DRE, especially if recent biopsy, catheter or procedure)
 - Consider urgent referral if immune-compromised or diabetes
 - **Primary care**
 - Start antibiotics immediately pending urine culture result
 - A quinolone (ciprofloxacin 500mg bd or ofloxacin 200mg bd) for 14 days; review antibiotic choice when culture results are available and change according to susceptibility results if the bacteria is resistant using a narrow spectrum wherever possible
 - If quinolone is contraindicated then trimethoprim 200mg bd for 14 days pending MSU results
 - Treat the pain with paracetamol and/or ibuprofen +/- codeine
 - Arrange a review after 24 to 48 hours and advise him to seek urgent medical advice beforehand if he deteriorates
 - Review with MSU result and ensure on the correct antibiotic
 - If clinical deterioration, admit
 - If fails to respond to antibiotics, refer to urology to exclude abscess
 - If a sexually transmitted infection is suspected, refer to GUM

- Review at 14 days and either stop or continue for another 14 days based on clinical assessment
- Following recovery refer for an ultrasound to exclude a structural abnormality of the urinary tract
- **Conclusion:** suspect acute bacterial prostatitis and manage pro-actively any man presenting with acute onset of fever and symptoms of a lower UTI, especially with pain and irritative LUTs

What about recent quinolone concerns?

We have been discouraged from prescribing quinolones for years because of all the risks associated with broad-spectrum antibiotics. In November 2018 there was a [European Medicines Agency review](#) which has recommended restricting their use (see Prescribing section for more on this) following reports of long-lasting and disabling side effects affecting muscles, tendons or bone. So, the recommendations to use ciprofloxacin for acute prostatitis may seem surprising. NICE are aware of this, but they and the [European Medicines Agency](#) state that quinolones can still be used to treat severe infections such as acute bacterial prostatitis or pyelonephritis.

GISS Acute Bacterial Prostatitis

Based on [NICE-CKS 2019](#) & [NICE/PHE 2018](#)

- **Acute bacterial prostatitis is a serious infection, with a significant risk of sepsis and abscess formation**
 - UTI + pain + fever: think acute bacterial prostatitis and not 'bad UTI'
- **Diagnosis:**
 - Suspect acute bacterial prostatitis in a man who presents with
 - Sudden onset of fever, may be accompanied by rigors, arthralgia or myalgia
 - Irritative voiding symptoms (dysuria, frequency, urgency)
 - Difficulty passing urine (retention may occur)
 - Perineal or suprapubic pain
 - Tender prostate on digital rectal examination
 - Confirm diagnosis with an MSU urine culture
- **Management:**
 - Admit urgently if severely ill, suspected sepsis, in acute urinary retention or suspected abscess (fluctuant swelling on DRE, especially if recent biopsy, catheter or procedure)
 - Consider urgent referral if immune-compromised or diabetes
 - Start antibiotics immediately pending urine culture result
 - A quinolone (ciprofloxacin 500mg bd or ofloxacin 200mg bd) for 14 days; review antibiotic choice when culture results are available and change according to susceptibility results if the bacteria is resistant using a narrow spectrum wherever possible
 - If quinolone is contraindicated then trimethoprim 200mg bd for 14 days pending MSU results
 - Treat the pain with paracetamol and/or ibuprofen +/- codeine
 - Arrange a review after 24 to 48 hours and advise him to seek urgent medical advice beforehand if he deteriorates
 - Review with MSU result and ensure on the correct antibiotic
 - If clinical deterioration, admit
 - If fails to respond to antibiotics, refer to urology to exclude abscess
 - If a sexually transmitted infection is suspected, refer to GUM
 - Review at 14 days and either stop or continue for another 14 days based on clinical assessment
 - Following recovery refer for an ultrasound to exclude a structural abnormality of the urinary tract
- **Prognosis:** 10% of men will go on after acute bacterial prostatitis to develop chronic bacterial prostatitis and a further 10% will develop chronic non-bacterial prostatitis/chronic pelvic pain syndrome

NB Learning Points

- Acute bacterial prostatitis is a serious infection, with a significant risk of sepsis and abscess formation
- UTI + pain + fever: think acute bacterial prostatitis and not 'bad UTI'
- Start immediate antibiotics (quinolone first choice, but be aware of recent concerns and restrictions on their use) and review after 48 hours with MSU results
- Refer urgently if patients have signs of sepsis, abscess, do not respond to oral antibiotics or clinically deteriorate

2 Chronic Prostatitis

This is a very common and difficult-to-manage problem in primary care, so an excellent new guideline in 2019 is very welcome [NICE-CKS Chronic Prostatitis 2019](#).

Patients with chronic non-bacterial prostatitis (which is increasingly called chronic pelvic pain syndrome) often end up on long, harmful and useless courses of antibiotics and more effective treatments such as alpha-blockers are often not offered. Research shows that it has a significant negative effect on the quality of life, and can often lead to secondary depression and anxiety. Chronic prostatitis may follow acute prostatitis.

How do we define chronic prostatitis?

Chronic prostatitis is defined by:

- > 3/12 of pain in the perineum or pelvic floor associated with lower urinary tract symptoms (LUTs) and sexual dysfunction (erectile dysfunction, painful ejaculation or post-ejaculatory discomfort)
- There are two types of chronic prostatitis
 - **Chronic bacterial prostatitis (CBP)**
 - 10% of men with chronic prostatitis, characterized by recurrent or relapsing symptoms with lower UTI symptoms and a positive MSU culture result
 - This is the most common cause of recurrent bacterial UTI in men
 - **Chronic non-bacterial prostatitis (CP/CPPS)**
 - 90% of cases, there is no proven bacterial infection
 - Also known as chronic prostatitis/chronic pelvic pain syndrome CP/CPPS (internationally is also sometimes called Prostate Pain Syndrome)
- The good news is, observational research shows that symptoms do improve over time; in most men symptoms have improved after 6 months

How does chronic prostatitis present?

With pain and symptoms affecting urinary and sexual function, in men who are NOT systemically unwell, unlike acute prostatitis.

- **Pain:**
 - Pain or discomfort in multiple urogenital regions including perineum, testicles, penis (especially penile tip pain), suprapubic region
 - Also lower back, abdomen, groin, rectum
- **Lower urinary tract symptoms:**
 - Voiding symptoms with a weak stream, hesitancy
 - Storage symptoms with frequency, urgency +/- incontinence, nocturia and dysuria

■ Sexual dysfunction:

- Erectile dysfunction, ejaculatory dysfunction (premature, delayed or painful), erectile, decreased libido
- Haematospermia

How should we assess patients?

As well as our usual thorough history and examination to exclude other causes of the symptoms (including possible sexually transmitted infections) or significant pathology, we should perform a digital rectal examination. The prostate may be enlarged, tender or normal. The good news is that we are advised 'do not perform a prostatic massage to obtain prostatic secretions to test for infection'. Good news for all concerned...In terms of investigations, we should:

- Dipstick the urine and collect an MSU for culture
 - A positive urine culture indicates CBP
- Consider if sexually transmitted infection screen is needed in sexually active men
- Consider a PSA test only after discussing the pros and cons of testing in the usual way
- Assess renal function with a blood test if clinically indicated e.g. recurrent infections, previous stones or known structural abnormality

Management

- Explanation and reassurance
- **Chronic prostatitis/chronic pelvic pain syndrome CP/CPPS**
 - **Pain relief:** paracetamol +/-NSAIDs, do not prescribe opioids
 - Consider a stool softener e.g. docusate or lactulose if defecation is painful
 - **LUTs:** offer a 6-week trial of an alpha-blocker
 - **Antibiotics**
 - There is a lack of evidence that antibiotics are effective in CP/CPPS (except in one study), but expert opinion and international guidelines support a single course of antibiotics in antibiotic-naive patients who are in the early stages of the condition
 - Offer a 4 to 6-week trial of antibiotics if symptoms have been present for < 6 months
 - Do NOT offer antibiotics and alpha-blocker at the same time
 - Antibiotic options are
 - Quinolone e.g. ciprofloxacin 500mg bd or ofloxacin 200mg bd
 - Trimethoprim 200mg bd if quinolones are not tolerated or contraindicated
 - NB this guidance is in conflict with the recent [European Medicines Agency Quinolone review in 2018](#), which advises NOT to use quinolones in non-bacterial chronic prostatitis
 - **Refer to urology if symptoms persist**
- **Chronic bacterial prostatitis**
 - Refer to urology for specialist assessment
 - While awaiting referral, prescribe a single course of antibiotics for 4 to 6 weeks
 - **Antibiotic options are**
 - Quinolone e.g. ciprofloxacin 500mg bd or ofloxacin 200mg bd
 - Trimethoprim 200mg bd if quinolones are not tolerated or contraindicated

KISS: Chronic Prostatitis

Based on [NICE-CKS Chronic Prostatitis 2019](#) & [BJU Consensus Guideline 2015](#)

Definition: > 3/12 of pain in the perineum or pelvic floor associated with lower urinary tract symptoms (LUTs) and sexual dysfunction (erectile dysfunction, painful ejaculation or post-ejaculatory discomfort). There are two types of chronic prostatitis

- **Chronic bacterial prostatitis (CBP)**

- 10% of men with chronic prostatitis, characterized by recurrent or relapsing symptoms with lower UTI symptoms and a positive MSU culture result

- **Chronic non-bacterial prostatitis (CP/CPPS)**

- 90% of cases, there is no proven bacterial infection; also known as chronic prostatitis/chronic pelvic pain syndrome CP/CPPS (internationally is also sometimes called Prostate Pain Syndrome)

Assessment: After full history and examination, check

- Digital rectal examination: prostate may be enlarged, tender or normal
- Urine dipstick and culture
- Consider based on clinical judgement and patient choice the need for STI testing, PSA and/or renal function

Management:

- **Reassurance & explanation.** The good news is, observational research shows that symptoms do improve over time in most men. [Patient information](#)
- **Chronic prostatitis/chronic pelvic pain syndrome CP/CPPS**
 - **Pain relief:** paracetamol +/-NSAIDs, do not prescribe opioids
 - Consider a stool softener e.g. docusate or lactulose if defecation is painful
 - **LUTs:** offer a 6-week trial of an alpha-blocker
 - **Antibiotics**
 - Offer a 4 to 6-week trial of antibiotics if symptoms have been present for < 6 months; antibiotic options are
 - Quinolone e.g. ciprofloxacin 500mg bd or ofloxacin 200mg bd
 - (NB in conflict with this guideline recommendation, the [Euro Meds Agency 2018](#) advise to avoid quinolones in CP/CPPS)
 - Trimethoprim 200mg bd if quinolones not tolerated or contraindicated
 - **Refer to urology if symptoms persist**
- **Chronic bacterial prostatitis**
 - Refer to urology for specialist assessment
 - While awaiting referral, prescribe a single course of antibiotics for 4 to 6 weeks
 - Quinolone e.g. ciprofloxacin 500mg bd or ofloxacin 200mg bd
 - Trimethoprim 200mg bd if quinolones are not tolerated or contraindicated

HAEMATOSPERMIA

Doctor, there is blood in my semen...should I worry?

...is a question that we have never been sure how to answer! Despite the fact that haematospermia presents not infrequently, it never seems to appear in guidelines (NB including the NICE suspected cancer guidelines). So, it was very useful to see it recently reviewed in the BMJ [BMJ2016;353:i5124](#) and also in [CKS Haematospermia 2022](#)

What causes haematospermia?

It is usually benign, most commonly due to infection or inflammation of the prostate, urethra or seminal tract and it normally resolves spontaneously. Interestingly, prolonged abstinence from ejaculation is often cited as a cause although the evidence base for this is weak.

Whilst infection, particularly prostatitis is the most common cause ***it can be a presenting cause of urological cancer***. We do not know how often haematospermia is due to a malignant cause but case-control studies suggest that prostate cancer presents with haematospermia between 4% and 14% of the time. However, an isolated episode in men under 40 is highly unlikely to have a malignant cause and much more likely to be due to infection. Often in younger men it is idiopathic, benign and resolves spontaneously and once an infection has been excluded these men can be reassured.

Associated urinary symptoms suggest infection and perineal discomfort and ejaculatory pain point to prostatitis. A sexual history is important as it can be the presenting feature of a new sexually transmitted infection, including chlamydia.

So, how should we manage the problem?

The paper and CKS gives us clear guidance on assessment and referral, which we have summarised in a KISS below. The headlines however are:

- Low-grade infection and inflammation is the most common cause and it usually resolves spontaneously
- In men aged under 40, it is usually benign and self-limiting
- It is important however to take a full sexual history, examine, exclude infection and also to check simple bloods
- In men aged over 40 in whom it is unexplained, a urological referral is recommended
- Men aged over 40, it is important to exclude cancer and to refer urgently if cancer is suspected e.g. because of a raised PSA or DRE

KESS: Haematospermia**CKS Haematospermia 2022****Causes:**

- Infection, most common cause especially in men aged under 40:
 - Sexually transmitted infection e.g. HSV, chlamydia, gonorrhoea.
 - UTI.
 - Prostatitis, acute or chronic.
 - Orchitis or epididymitis.
- Trauma:
 - Including testicular or perineal trauma during sex.
 - Urological intervention e.g. biopsies (may take 4 weeks to resolve).
- Benign prostatic hyperplasia.
- Calculi or cysts in ejaculatory ducts or prostate.
- Systemic disease e.g. uncontrolled hypertension, bleeding disorder etc.
- Cancer (more common in men >40):
 - Prostate, bladder, urethral, testicular or seminal vesicle cancer.

Assessment:

- Careful history (including sexual history/symptoms of STI, LUTS, and ?any haematuria) and examination, and consider digital rectal exam.
- Check urinalysis, and MSU in all.
- Do PSA in all men >40 or those with family history of prostate cancer.
- Other investigations as guided by clinical findings:
 - Consider semen analysis (microscopy & culture) and bloods - FBC & clotting, U&E, LFT.

Management:

- **Single episode in a man aged under 40 with no underlying cause found:**
 - Reassure that a single episode is likely to be benign and self-limiting, advise to return if recurs.
- Reassure men who have had **recent prostate procedure** that haematospermia should settle in 3-4 weeks.
- If STI is suspected refer to the GUM clinic for assessment, treatment and contact tracing.
- If UTI is suspected treat as per local/national guidelines based on MCS.
 - For acute prostatitis prescribe ciprofloxacin 500mg BD or ofloxacin 200mg BD, initially 14 days.
- **Refer:**
 - **Urgent referral to urology via 2WW** (see [NICE NG12](#) for details) if any other associated signs and symptoms suggestive of cancer e.g. suspicious DRE/raised PSA ?prostate cancer, testicular lumps/abnormalities ?testicular cancer, unexplained haematuria ?bladder/renal cancer.
 - **All men aged >40 with no identifiable cause found** to urology.
 - Recurrent/persistent haematospermia with no cause, or that recurs after treatment of identified cause, irrespective of age.

Patient information: [What should I do if I have blood in my semen?](#)

MIGRAINE

Migraine is a common and distressing condition affecting 15% of the population (up to 25% of women) and is estimated to cause 100,000 people to miss school or work every day. The Global Burden of Disease study ranks migraine as the **third most common cause of disability** in people under 50. Given the impact of this common condition new treatments gather a lot of interest. [SIGN 155](#) has been recently updated (Sept 2022) with evidence on botulinum toxin A, acupuncture and a new class of drugs **CGRP monoclonal antibodies**. NICE has recently released a series of technology appraisal guidance on the use of CGRP monoclonal antibodies in migraine ([TA659](#), [TA682](#), [TA764](#)), whilst the clinical guideline on headache ([CG150](#)) was last updated with limited change in December 2021. [BASH guidelines published in 2019](#) give an excellent, simple, evidence-based overview of headache management. In addition, we have some interesting primary research suggesting high omega-3 diets may reduce migraine burden [BMJ 2021;374:n1448](#) and an RCT providing an evidence base for use of acupuncture [BMJ 2020;368:m697](#).

KISS: Migraine - Clinical Features and Diagnosis

[BASH 2019](#), [NICE 2021](#), [CG150](#)

- Recurrent headache associated with sensitivity to light, nausea or a reduced ability to function is most likely to be migraine regardless of other characteristics (NB severe disabling 'tension-type' headache is much more likely to be migraine).
- Migraine typically has the **following features**:
 - Unilateral or bilateral pain, typically lasting 4 to 72 hours.
 - Pulsating, throbbing or banging pain of moderate or severe intensity.
 - Associated with **other symptoms** e.g. sensitivity to light or sound or motion, nausea/vomiting.
- Typical **premonitory symptoms** include fatigue, difficulty concentrating, stiff neck. After the headache has ended most experience similar postdrome symptoms.
- The **3 most sensitive and specific features** of migraine are:
 - Nausea; Disability/functional impairment; Photophobia.
 - In a primary care setting, 2/3 of these features gives a PPV for migraine of 93%.
- Be aware a number of features can be **present in both migraine and cluster headache**: nausea/vomiting, photophobia, aura and autonomic symptoms (see Headache chapter for more info)
- Typical **aura** (affects ~1/3) symptoms (can occur with or without headache):
 - Visual (flashing lights, zig-zags, spots, lines or partial loss of vision).
 - Sensory (pins and needles, numbness).
 - Speech disturbance.
 - Aura symptoms are fully reversible, develop over 5 minutes and last 5-60 mins.
- Episodic migraine is defined as headache < 15 days per month, chronic migraine ≥15 days per month for 3 months
- **Atypical features requiring further investigation** include:
 - Motor symptoms, double vision, visual symptoms affecting only one eye, poor balance or reduced consciousness
 - Migraine lasting > 72 hours.
 - Aura symptoms which last more than an hour.

► *What are the most common triggers for migraine?* [BASH 2019](#)

Skipped meals, irregular caffeine intake, irregular sleep and stress are the most commonly patient-identified triggers. However, whilst a search for triggers is often recommended, a recent review [N Engl J Med 2020; 383:1866-1876](#) reported that when patients were exposed to self-identified triggers only 3/27 had an attack. They concluded that 'triggers' are often misattributed to migraine episodes.

Medication overuse headaches are common. BASH state that **2/3 of chronic migraine sufferers have associated medication overuse**. Acute medications should be limited to a maximum of 2 days/week (particularly triptans which can cause medication overuse headaches most rapidly). See separate Headache chapter for more on medication overuse headaches. Other medications linked to migraine include oral contraceptives, HRT, nasal decongestants, SSRIs and PPIs. In some patients, the frequency and severity of attacks can be reduced by adjusting or discontinuing these medications. Note [FSRH](#) guidance on the safety of COCP in sufferers of migraine: UKMEC 4 for migraine with aura due to increased CVA risk. Migraine without aura is UKMEC 2 for initiation and UKMEC 3 for continuation of COCP.

Migraines are more common before and during periods. **Menstrual migraines** occur between days -2 and +3 of the first day of menstruation in $\geq 2/3$ cycles. Of note, most women with menstrual migraines also have migraines at other times of the cycle.

► *Dietary omega-3 fatty acids and migraine* [BMJ 2021;374:n1448](#)

An interesting RCT may give some hope to migraine sufferers wanting to avoid medication. This **study focused on two dietary fatty acids, omega-3 and omega-6**. Both are precursors to oxylipins which are involved in the regulation of pain and inflammation. The oxylipins derived from omega-3 are associated with *anti-nociceptive* and *anti-inflammatory* effects whereas omega-6 derivatives have been shown to *worsen* pain and provoke migraines in experimental models. This RCT was a 3 arm study looking at a control group (normal levels of omega-3/omega-6) and two intervention groups - high omega-3 (H3) or high omega-3 *and* low omega-6 (H3-L6).

- 182 people were randomised to one of the 3 groups and reviewed over 16 weeks.
- The observed improvements in the **main primary endpoint** of a change in the **headache impact test (HIT-6)**, a 6-item questionnaire assessing headache impact and quality of life, were **not statistically significant**.
- However **secondary endpoints were statistically significant**:
 - Both H3 and H3-L6 groups had **lower total headache hours/day** (-1.3 and -1.7 respectively) and decreased moderate-severe headache hours/day (-0.7 and -0.8 respectively).
 - Both H3 and H3-L6 groups had **reduced numbers of headache days per month** (-2.0 and -4.0 respectively).
- **Conclusion:** Both H3 and H3-L6 diets reduce frequency and severity of headache but do not significantly improve quality of life.

So is this clinically meaningful? The associated editorial ([BMJ 2021;374:n1535](#)) makes the **case that these are significant results**. First, the trial population were a 'difficult to treat' group, $\sim 2/3$ met the criteria for chronic migraine (>15 headache days/month) and almost half met the criteria for medication overuse headache. The evidence suggests that a group such as this is likely to be resistant to many of the standard treatments. Second, the **reduction of headache days/month of 2-4 compares very favourably with the CGRP monoclonal antibody treatments** (discussed below) which give a reduction in headache days/month of $\sim 2-2.5$. Are these diets realistic in daily life? The intervention groups had to eat oily fish pretty much every day of the week (salmon, trout, tuna and sardines mostly) - a weekly menu that may be neither affordable nor palatable for many. However, for those patients really struggling with migraine and wanting to investigate non-drug options, a switch to a high omega-3 diet may be worth considering.

► *Acupuncture for episodic migraine without aura* [BMJ 2020;368:m697](#)

This blinded RCT gives some high-quality evidence that **acupuncture is effective in migraine without aura**. Robust controls included a sham acupuncture arm using a non-penetrating needle (important as penetrating sham acupuncture is not inert) and a 'usual care' arm (important as the therapeutic effects of clinician contact and placebo response is typically very high in migraine studies). The effects of acupuncture were modest at **2.1 fewer migraine days/month**. Follow-up was limited to 20 weeks, so whether the effects are maintained is unknown. The participants had 20 sessions of acupuncture (or sham), which is quite substantial and equated to 10 hours of acupuncture in total, so whether this is cost-effective is questionable. Nevertheless, the associated editorial ([BMJ 2020;368:m1096](#)) concludes this data *moves 'acupuncture from having an unproven status in complementary medicine to an acceptable evidence-based treatment'*. The latest NICE guidelines ([CG150](#)) suggest considering a course of up to 10 sessions of acupuncture if prophylaxis with topiramate or propranolol is unsuitable or ineffective.

► *CGRP monoclonal antibodies* [NEJM 2020; 383:1866-1876](#) [BJGP 2020; 70 \(697\)](#) [SIGN 155](#)

CGRP monoclonal antibodies (MABs) are a new class of drugs showing promise for migraine prophylaxis, as discussed in a recently updated [SIGN guidance](#), an [NEJM](#) review, as well as a [BJGP editorial](#). Calcitonin gene-related peptide (CGRP) is widely distributed neurotransmitter involved in nociception and vasodilation. Whilst its involvement in migraine pathophysiology is incompletely understood, levels are raised during a migraine attack and exposure to CGRP molecules triggers migraine in sufferers.

Erenumab targets the CGRP receptor and Fremanezumab and Galcanezumab target the CGRP ligand. They are given as injections, usually monthly. RCT evidence has been published showing Erenumab, Fremanezumab and Galcanezumab **reduce migraine frequency compared to placebo**. They seem generally well tolerated, work quicker than standard treatments and may be more effective in those with concurrent medication overuse headache. Expert opinion suggests ~1/3 will have significant improvement, ~1/3 will have modest improvement and ~1/3 no improvement. There is emerging data showing benefits are sustained over longer time frames (5 yr data for erenumab). However, there are butts (aren't there always..) some of which are big!

1. The overall effectiveness is not likely to be as high as initial hype has suggested with the BJGP review highlighting outcomes similar to current best practice.
2. There are currently few head to head trials with current first-line treatments. However, the available data is promising. When compared to topiramate in an RCT, erenumab was more effective in reducing monthly migraine days (-5.86 vs 4.02) and better tolerated (SIGN 155)
3. These drugs work in part by causing vasoconstriction so there are **theoretical concerns about long-term cardiovascular safety**, and indeed most of the studies excluded people with established vascular disease.
4. These drugs are **eye-wateringly expensive at >£5000/year**, meaning use will be restricted

Despite these caveats both NICE ([TA659](#), Nov 2020) ([TA682](#), March 2021) ([TA764](#), Feb 2022) and [SIGN SIGN 155](#) have recently approved use in selected cases, with similar criteria:

- Recently updated **SIGN guidance Sept 2022**:
 - Fremanezumab and galcanezumab are options for episodic or chronic migraine, and erenumab as an option for chronic migraine where medication overuse has been excluded AND at least 3 oral preventative treatments have failed.
- **NICE criteria**:
 - Fremanezumab and galcanezumab and erenumab are options for people with ≥ 4 migraine days/month AND At least 3 preventive drug treatments have failed AND It should be stopped if the migraine frequency does not reduce by at least 30% (for chronic migraine) and at least 50% (for episodic migraine) after 12 weeks of treatment.

KISS: Migraine in Pregnancy

[BMJ 2018;360:k80](#) [BASH 2019 Drug and Therapeutics Bulletin 2021;59:152-156](#) [SIGN 155](#)

Background:

- **Reassure women migraine often improves in pregnancy** - 50-75% will have improvement mainly in the second/third trimester, although those with aura have a more variable course.
- There is an increased risk of secondary headaches in pregnancy e.g. cerebral venous thrombosis and pre-eclampsia, particularly after the first trimester.

Assessment: Rule out red flags/secondary causes (see Headache chapter also):

- In women with a history of migraine check if the current headache is similar to previous ones or are there any differences, particularly relating to aura.
- Do neuro exam, specifically assessing for neck stiffness; test eye movements, visual fields, and pupillary responses; perform fundoscopy to rule out papilloedema.
- Check blood pressure and dip urine.

Management: ([click here for good info from UK Teratology Information Service](#))

- **Non-pharmacological** - keep well hydrated, don't skip meals; reduce caffeine but avoid abrupt withdrawal; good sleep hygiene; regular exercise; consider acupuncture.
- **Acute pharmacological treatments:**
 - Paracetamol first line; ibuprofen can be considered in 1st/2nd trimester (avoid in 3rd trimester when risks of premature ductus closure and oligohydramnios increase).
 - Sumatriptan can be considered in all trimesters in refractory cases - reasonable safety evidence for sumatriptan. Meta-analysis showed not associated with major congenital malformation or prematurity. Conflicting data on spontaneous abortion rates- some small studies showed increase, recent large cohort study (432 patients) showed no increase. Avoid in those with hypertension/ pre-eclampsia
 - Use **anti-emetic together with analgesia** (even in absence of nausea/vomiting - can improve gastric motility/drug absorption) - cyclizine or prochlorperazine 1st line; ondansetron, domperidone metoclopramide may be considered 2nd line.
- **Prophylaxis (should not be initiated in primary care - seek specialist guidance):**
 - Consider stopping prophylactic treatments before pregnancy
 - If required aspirin 75mg OD up to 36/40, B-blockers (e.g. propranolol 10-40mg TDS) or amitriptyline 10-25mg nocte may be considered.
 - Do NOT use valproate or topiramate (high teratogenicity risk), or ARBs (high risk in 2nd/3rd trimester).

What about breastfeeding?:

An important question given that it is relatively common for women to get migraine flares after delivery (due to abrupt fall of oestrogen). Paracetamol, NSAIDs, sumatriptan, propranolol and amitriptyline (low dose) are all thought to be safe with breastmilk concentrations not thought to be high enough to affect babies. The [breastfeeding network](#) has a handy summary.